DEPARTMENT
OF
ELECTRONICS & COMMUNICATION ENGINEERING

Semester-V LAB MANUAL

ELECTROMAGNTICS & RADIATING SYSTEMS
(22EC52)- *An Integrated Course*

Prepared by:

Dr. Sudha K L., Dr. Kumar P. and Prof. Ravikumar S.

Lab In-charge: Dr. Kumar P and Prof. Ravikumar S

Name of the Student	:	
Semester /Section	:	
USN	:	
Batch	:	

Dayananda Sagar College of Engineering

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(An Autonomous Institute Affiliated to VTU, Approved by AICTE & ISO 9001:2008 Certified)
(Accredited by NBA, National Assessment & Accreditation Council (NAAC) with 'A' grade)

Dayananda Sagar College of Engineering

Department of Electronics & Communication Engineering

Name of the Laboratory : Electromagnetics & Radiating Systems.

Semester/Year : V / 3rd Year

No. of Students/Batch : 20 to 24

	LAB - 1	**LAB - 2**
Room No.	**Room No.17216**	**Room No. 17219**
Major Equipment available	**PCs - 26**	**PCs - 25**
Operating System & Applications	**Windows 10, MATLAB 2024a, Ansys Electronics Desktop Student 2024**	**Windows 10, MATLAB 2024a, Ansys Electronics Desktop Student 2024**
Area in square meters	**66.93 sq. mtr.**	**66.93 sq. mtr.**
Total Cost	**₹ 12,62,000/-**	**₹ 12,32,000/-**

Lab In charges : Dr. Kumar P and Prof. Ravikumar S.

Instructors : Mrs. Gayatri H. & Mrs. Veena H S.

HOD : Dr. Shobha K R.

DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institution affiliated to Visvesvaraya Technological University, Belagavi)

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING,

BENGALURU-560111.

Vision of the Institute

To impart quality technical education with a focus on Research and Innovation emphasizing on Development of Sustainable and Inclusive Technology for the benefit of society.

Mission of the Institute

- To provide an environment that enhances creativity and Innovation in pursuit of Excellence.
- To nurture team work in order to transform individuals as responsible leaders and entrepreneurs.
- To train the students to the changing technical scenario and make them to understand the importance of Sustainable and Inclusive technologies.

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DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING,

BENGALURU-560111.

Vision of the Department

To achieve continuous improvement in quality technical education for global competence with focus on industry, societal needs, research and professional success.

Mission of the Department

- Offering quality education in Electronics and Communication Engineering with effective teaching learning process in multidisciplinary environment.
- Training the students to take up the projects in emerging technologies and work with team spirit.
- To imbibe professional ethics, development of skills and research culture for better placement opportunities.

Program Specific Outcomes (PSOs)

Students will be able to:

PSO1: Design, develop and integrate electronic circuits and systems using current practices and standards.

PSO2: Apply knowledge of hardware and software in designing Embedded and Communication systems.

DO's

- All the students should come to LAB on time with proper dress code and identity cards
- Keep your belongings in the corner of laboratory.
- Students have to enter their name, USN, time-in/out and signature in the log register maintained in the laboratory.
- All the students should submit their records before the commencement of Laboratory experiments.
- Students should come to the lab well prepared for the experiments which are to be performed in that particular session.
- Students are asked to do the experiments on their own and should not waste their precious time by talking, roaming and sitting idle in the labs.
- Observation book and record book should be complete in all respects and it should be corrected by the staff member.
- Before leaving the laboratory, students should arrange their chairs and leave in orderly manner after completion of their scheduled time.
- Prior permission to be taken, if for some reasons, they cannot attend lab.
- Immediately report any sparks/ accidents/ injuries/ any other untoward incident to the faculty /instructor.
- In case of an emergency or accident, follow the safety procedure.
- Switch OFF the power supply after completion of experiment.

DON'Ts

- The use of mobile/ any other personal electronic gadgets is prohibited in the laboratory.
- Do not make noise in the Laboratory & do not sit on experiment table.
- Do not make loose connections and avoid overlapping of wires.
- Don't switch on power supply without prior permission from the concerned staff.
- Never point/touch the CRO/Monitor screen with the tip of the open pen/pencil/any other sharp object.
- Never leave the experiments while in progress.
- Do not insert/use pen drive/any other storage devices into the CPU.
- Do not leave the Laboratory without the signature of the concerned staff in observation book

List of experiments

Electromagnetics & Radiating Systems

(22EC52) - An Integrated Course

Expt. No.	Module No.	List of Experiments	No. of Sessions	CO Level
Electromagnetics Experiments Using MATLAB				
1.	**1**	Area Integral and Volume Integral of a Solid.	01	CO4
2.	**1**	Computation of electric field intensity at an observation point due to a point and a line charge.	01	CO4
3.	**1**	Flux passing through a given area from a point charge.	01	CO4
4.	**2**	Verification of Ampere's law/Curl in differential form.	01	CO4
Antenna Experiments using Ansys HFSS				
5.	**3**	Design and analysis of a Dipole Antenna.	01	CO4
6.	**3**	Design and analysis of a Monopole antenna.	01	CO4
7.	**3, 5**	Design of coaxial probe fed rectangular microstrip patch antenna and study of its parameters.	01	CO4
8.	**3, 5**	Design of inset fed rectangular microstrip patch antenna and study of its parameters.	01	CO4
9.	**4**	Design and analysis of Microstrip patch antenna array.	01	CO4
10.	**3, 5**	Design and analysis of Yagi-Uda antenna.	01	CO4
11.	**3, 5**	Design and analysis of Helical antenna.	01	CO4
12.	**3,4,5**	Open-Ended Experiment on Antennas.	01	CO4

Index

Sl. No	Experiments	Page No.
1.	Area Integral and Volume Integral of a Solid.	1
2.	Computation of electric field intensity.	10
3.	Flux passing through a given area from a point charge.	15
4.	Verification of Ampere's law/Curl in differential form.	20
5.	Design and analysis of a Dipole Antenna.	29
6.	Design and analysis of a Monopole antenna.	45
7.	Design of coaxial probe fed rectangular microstrip patch antenna and study of its parameters.	60
8.	Design of inset fed rectangular microstrip patch antenna and study of its parameters.	80
9.	Design and analysis of Microstrip patch antenna array.	100
10.	Design and analysis of Yagi-Uda antenna.	120
11.	Design and analysis of Helical antenna.	136
12.	Open-Ended Experiment on Antennas.	--

Continuous Internal Assessment

Sl. No	Experiments	Date	Record (5M)	Condu. (10M)	Sign (Condu.)	Assig. (10M)	Sign (Assig.)
1.	Area Integral and Volume Integral of a Solid.						
2.	Computation of electric field intensity.						
3.	Flux passing through a given area from a point charge.						
4.	Verification of Ampere's law/Curl in differential form.						
5.	Design and analysis of a Dipole Antenna.						
6.	Design and analysis of a Monopole antenna.						
7.	Design of coaxial probe fed rectangular microstrip patch antenna and study of its parameters.						
8.	Design of inset fed rectangular microstrip patch antenna and study of its parameters.						
9.	Design and analysis of Microstrip patch antenna array.						
10.	Design and analysis of Yagi-Uda antenna.						
11.	Design and analysis of Helical antenna.						
12.	Open-Ended Experiment on Antennas.						

Experiment 1

Area Integral and Volume Integral of a Solid

Example: The open surfaces $\rho = 2.0$ m and $\rho = 4.0$ m, $z = 3.0$ m and $z = 5.0$ m, and $\phi = 20^\circ$ and $\phi = 60^\circ$ identify a closed surface. Find a) the enclosed volume, b) the total area of the enclosed surface. Write a MATLAB program to verify your answers.

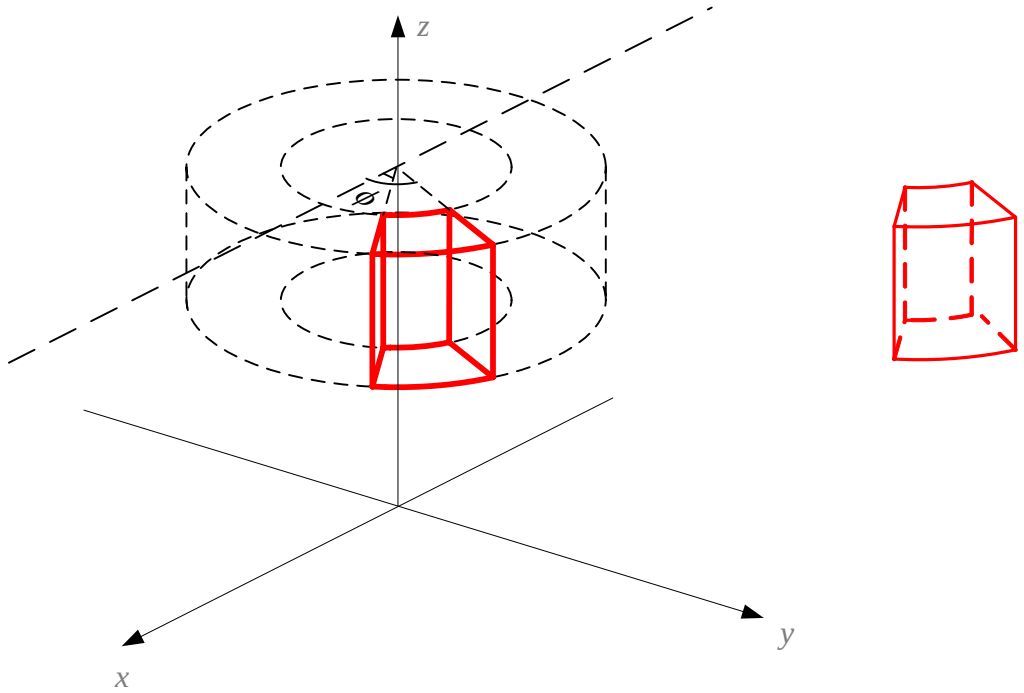


Figure 1: The enclosed volume for the example

Analytical Solution:

The closed surface in this problem is shown in Figure 2.1 and Figure 2.2. To find the volume v of a closed surface we first find out dv, the volume element. In cylindrical coordinates, dv is given by $dv = \rho d\phi d\rho dz$ as shown in Figure 2.2**.** Once we get the expression of dv, we integrate dv over the entire volume.

$$dv = \rho d\phi d\rho dz$$

$$v = \iiint_v dv$$

$$= \iiint_v \rho d\phi d\rho dz$$

$$= \int_{\rho=2}^{\rho=4} \int_{\phi=20^\circ}^{\phi=60^\circ} \int_{z=3}^{z=5} \rho d\phi d\rho dz$$

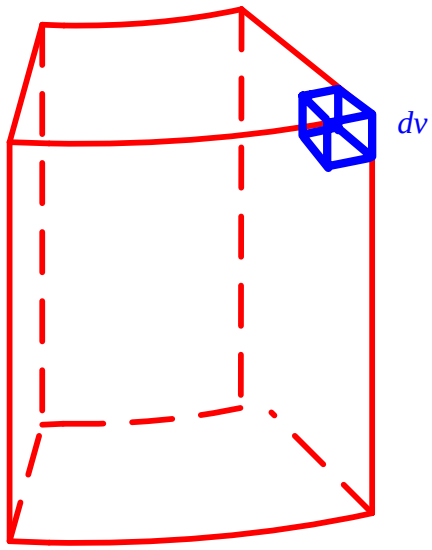
$$= \int_{\rho=2}^{\rho=4} \rho d\rho \int_{\phi=\frac{20}{180}\pi}^{\phi=\frac{60}{180}\pi} d\phi \int_{z=3}^{z=5} dz$$

$$= \frac{1}{2}\rho^2\Big|_{\rho=2}^{\rho=4} \times \phi\Big|_{\phi=\frac{2}{18}\pi}^{\phi=\frac{6}{18}\pi} \times z\Big|_{z=3}^{z=5}$$

$$= \frac{1}{2}\times(4^2-2^2)\times(\frac{6}{18}\pi-\frac{2}{18}\pi)\times(5-3)$$

$$= \frac{8}{3}\pi = 8.378$$

When evaluating an integral, we have to convert degree to radian for all angles, otherwise this will result in a wrong value for the integral.



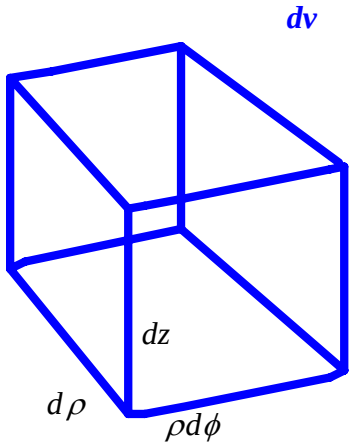


Figure The unit volume.

The area of the closed surface is given by

$S_{enclosed} = S_1 + S_2 + S_3 + S_4 + S_5 + S_6$

We need to find dS_1 , dS_2 , …, and dS_6 and to integrate them over their boundary. It is obvious that $S_3 = S_4$ and $S_5 = S_6$. The surfaces S_1 and S_2 have similar shapes as we can see from the following expressions,

$$dS_1 = \rho d\phi dz\Big|_{\rho=2} = 2\, d\phi dz \quad \text{and}$$

$$dS_2 = \rho d\phi dz\Big|_{\rho=4} = 4\, d\phi dz$$

The steps to evaluate the area of each surface are executed as follows:

$$dS_1 = \rho d\phi dz$$

$$S_1 = \iint_s dS_1$$

$$= \iint_s \rho d\phi dz$$

$$= 2\int_{\phi=\frac{20}{180}\pi}^{\phi=\frac{60}{180}\pi} d\phi \int_{z=3}^{z=5} dz$$

$$= 2\phi\Big|_{\phi=\frac{2}{18}\pi}^{\phi=\frac{6}{18}\pi} \times z\Big|_{z=3}^{z=5}$$

$$= 2\times(\frac{6}{18}\pi - \frac{2}{18}\pi)\times(5-3)$$

$$= \frac{8}{9}\pi \text{ m}^2$$

$$S_2 = \iint_S dS_2$$

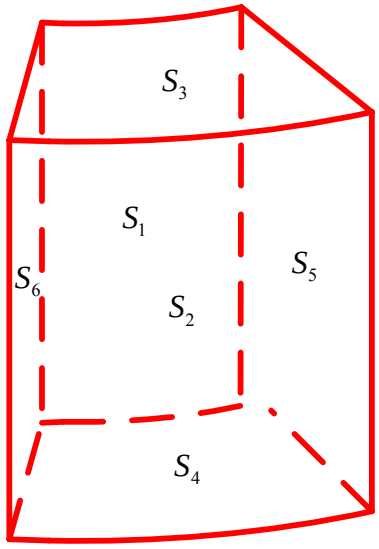
$$= \iint_S \rho d\phi dz$$

$$= 4\int_{\phi=\frac{20}{180}\pi}^{\phi=\frac{60}{180}\pi} d\phi \int_{z=3}^{z=5} dz$$

$$= 4\phi\Big|_{\phi=\frac{2}{18}\pi}^{\phi=\frac{6}{18}\pi} \times z\Big|_{z=3}^{z=5}$$

$$= 4\times(\frac{6}{18}\pi - \frac{2}{18}\pi)\times(5-3)$$

$$= \frac{16}{9}\pi \text{ m}^2$$



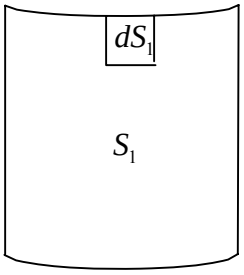


Figure The closed surface.

$$dS_3 = \rho\, d\phi\, d\rho$$

$$S_3 = \iint_S dS_3$$

$$= \iint_S \rho\, d\phi\, d\rho$$

$$= \int_{\rho=2}^{\rho=4} \rho\, d\rho \int_{\phi=\frac{2}{18}\pi}^{\phi=\frac{6}{18}\pi} d\phi$$

$$= \frac{1}{2}\rho^2\Big|_{\rho=2}^{\rho=4} \times \phi\Big|_{\phi=\frac{2}{18}\pi}^{\phi=\frac{6}{18}\pi}$$

$$= \frac{1}{2} \times (4^2 - 2^2) \times (\frac{6}{18}\pi - \frac{2}{18}\pi)$$

$$= \frac{4}{3}\pi \ \mathrm{m}^2$$

$$dS_5 = d\rho\, dz$$

$$S_5 = \iint_S dS_5$$

$$= \iint_S d\rho\, dz$$

$$= \int_{\rho=2}^{\rho=4} d\rho \int_{z=3}^{z=5} dz$$

$$= \rho\Big|_{\rho=2}^{\rho=4} \times z\Big|_{z=3}^{z=5}$$

$$= (4-2) \times (5-3)$$

$$= 4 \ \mathrm{m}^2$$

$$S_{closed} = S_1 + S_2 + 2S_3 + 2S_5$$

$$= \frac{8}{9}\pi + \frac{16}{9}\pi + 2 \times \frac{4}{3}\pi + 2 \times$$

$$= 24.755 \ \mathrm{m}^2$$

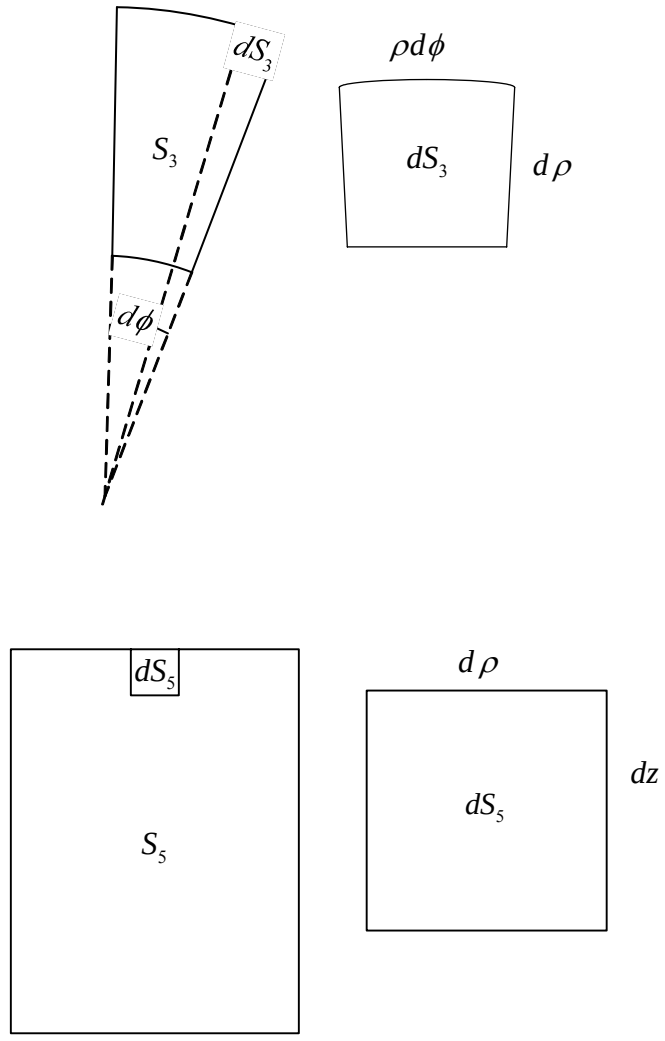


Figure The surfaces S_3 and S_5 and their incremental elements.

MATLAB SOLUTION:

As shown in the figure to the right, the approximate value of the enclosed volume is

$$v \doteq \sum_{k=1}^{p}\sum_{j=1}^{m}\sum_{i=1}^{n}\Delta v_{i,j,k} = \sum_{k=1}^{p}\sum_{j=1}^{m}\sum_{i=1}^{n}(\rho_{i,j,k}\Delta\phi)\times(\Delta\rho)\times(\Delta z)$$

We write a MATLAB program to evaluate this expression. To do this, our program evaluates all element volumes $\Delta v_{i,j,k}$, and increase the total volume by $\Delta v_{i,j,k}$ each time. We cover all elements $\Delta v_{i,j,k}$ through 3 loops with counters i in the inner loop, j in the middle loop and k in the outer loop. The approach used to evaluate the surfaces is similar to that of the volume. The MATLAB code is shown in the next page.

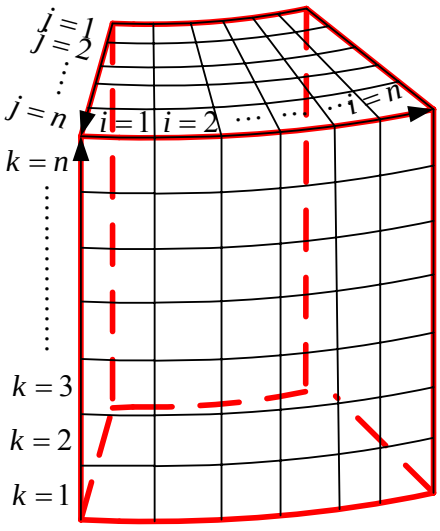


Figure The discretized volume.

MATLAB code:

```
clc; %clear the command line
clear; %remove all previous variables

V=0;%initialize volume of the closed surface to 0
S1=0;%initialize the area of S1 to 0
S2=0;%initialize the area of S1 to 0
S3=0;%initialize the area of S1 to 0
S4=0;%initialize the area of S1 to 0
S5=0;%initialize the area of S1 to 0
S6=0;%initialize the area of S1 to 0
rho=2;%initialize rho to the its lower boundary
z=3;%initialize z to the its lower boundary
phi=pi/9;%initialize phi to the its lower boundary

Number_of_rho_Steps=100; %initialize the rho discretization
Number_of_phi_Steps=100;%initialize the phi discretization
Number_of_z_Steps=100;%initialize the z discretization

drho=(4-2)/Number_of_rho_Steps;%The rho increment
dphi=(pi/3-pi/9)/Number_of_phi_Steps;%The phi increment
dz=(5-3)/Number_of_z_Steps;%The z increment

%%the following routine calculates the volume of the enclosed surface
for k=1:Number_of_z_Steps
   for j=1:Number_of_rho_Steps
      for i=1:Number_of_phi_Steps
         V=V+rho*dphi*drho*dz;%add contribution to the volume
      end
      rho=rho+drho;%p increases each time when z has been traveled from its lower boundary to its upper
boundary
   end
   rho=2;%reset rho to its lower boundary
end
```

```
%%the following routine calculates the area of S1 and S2
rho1=2;%radius of S1
rho2=4;%radius of s2
for k=1:Number_of_z_Steps
   for i=1:Number_of_phi_Steps
      S1=S1+rho1*dphi*dz;%get contribution to the the area of S1
      S2=S2+rho2*dphi*dz;%get contribution to the the area of S2
   end
end
%%the following routing calculate the area of S3 and S4
rho=2;%reset rho to it's lower boundaty
for j=1:Number_of_rho_Steps
   for i=1:Number_of_phi_Steps
      S3=S3+rho*dphi*drho;%get contribution to the the area of S3
   end
   rho=rho+drho;%p increases each time when phi has been traveled from it's lower boundary to it's upper
boundary
end
S4=S3;%the area of S4 is equal to the area of S3

%%the following routing calculate the area of S5 and S6
for k=1:Number_of_z_Steps
   for j=1:Number_of_rho_Steps
      S5=S5+dz*drho;%get contribution to the the area of S3
   end
end
S6=S5;%the area of S6 is equal to the area of S6
S=S1+S2+S3+S4+S5+S6;%the area of the enclosed surface
```

Running Result



By comparing, we see that the result of our analytical solution is close to the result of our MATLAB solution.

Exercise: The surfaces $r = 0$ and $r = 2$, $\phi = 45^\circ$, $\phi = 90^\circ$, $\theta = 45^\circ$ and $\theta = 90^\circ$ define a closed surface. Find the enclosed volume and the area of the closed surface S. Write a MATLAB program to verify your answer.

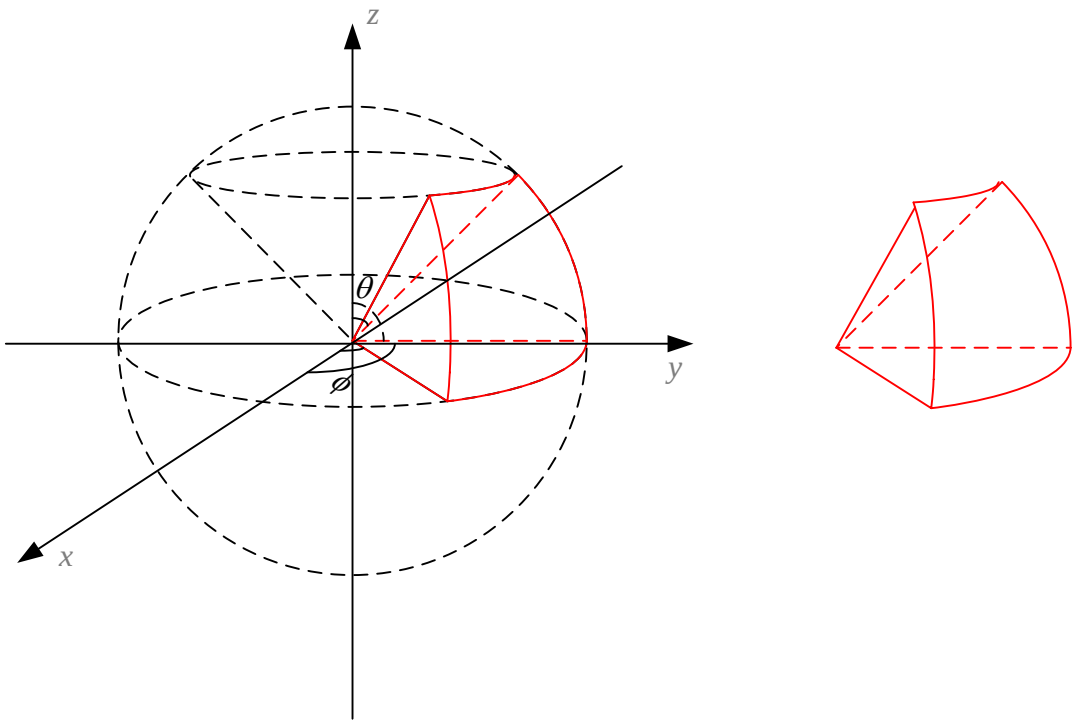


Figure The surface of the exercise

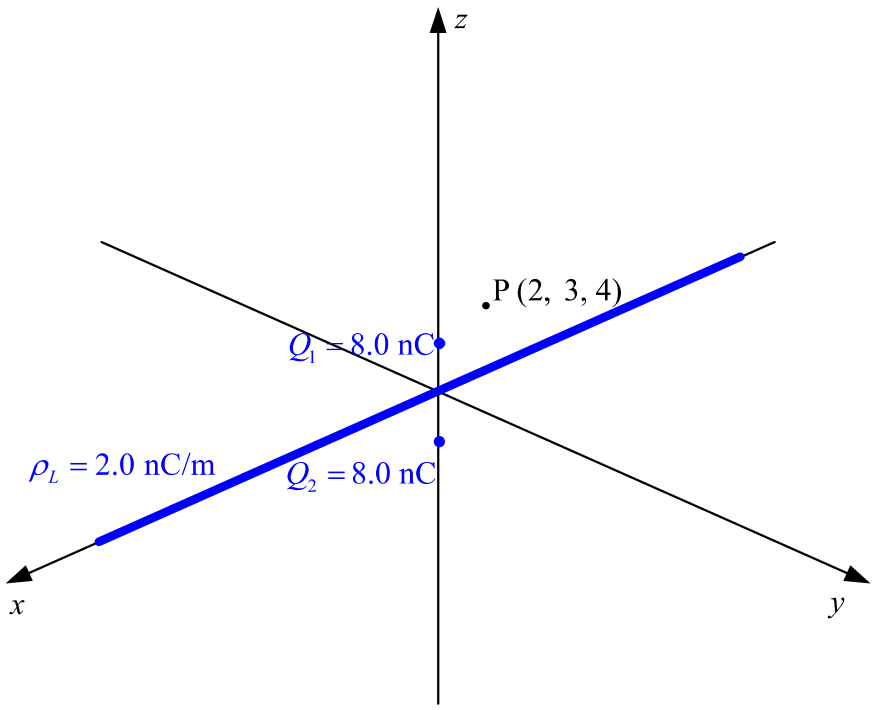
Assignment

Assignment

Experiment 2

Electric Field Intensity due to Point charge and Line charge

An infinite uniform linear charge $\rho_L = 2.0$ nC/m lies along the x axis in free space, while point charges of 8.0 nC each are located at (0, 0, 1) and (0, 0, -1). Find **E** at (2, 3, 4). Write a MATLAB program to verify your answer.



The charges of the example of Set 3.

Analytical Solution:

Based on the principle of superposition, the electric field at P (2, 3, 4) is $\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 + \mathbf{E}_L$ where $\mathbf{E}_1$ and $\mathbf{E}_2$ are the electric fields generated by the point charges 1 and 2, respectively, and $\mathbf{E}_L$ is the electric field generated by the line charge. The electrical field generated by a point charge is given by

$$\mathbf{E}_{\text{point}} = \frac{Q}{4\pi\varepsilon_0 |\mathbf{R}|^3}\mathbf{R}$$

where $\mathbf{R}$ is the vector pointing from the point charge to the observation point as shown in Figure 3.2 (a).

For the point charge Q_1:

$$\mathbf{R}_1 = (2\mathbf{a}_x + 3\mathbf{a}_y + 4\mathbf{a}_z) - (\mathbf{a}_z) = 2\mathbf{a}_x + 3\mathbf{a}_y + 3\mathbf{a}_z$$

$$\mathbf{E}_1 = \frac{Q_1}{4\pi\varepsilon_0 |\mathbf{R}_1|^3}\mathbf{R}_1$$

$$= \frac{8\times10^{-9}}{4\pi\times\frac{1}{36\pi}\times10^{-9}\times\left(\sqrt{2^2+3^2+3^2}\right)^3}(2\mathbf{a}_x + 3\mathbf{a}_y + 3\mathbf{a}_z)$$

$$= 1.395\mathbf{a}_x + 2.093\mathbf{a}_y + 2.093\mathbf{a}_z$$

For the point charge Q_2:

$$\mathbf{R}_2 = (2\mathbf{a}_x + 3\mathbf{a}_y + 4\mathbf{a}_z) - (-\mathbf{a}_z) = 2\mathbf{a}_x + 3\mathbf{a}_y + 5\mathbf{a}_z$$

$$\mathbf{E}_2 = \frac{Q_2}{4\pi\varepsilon_0 |\mathbf{R}_2|^3}\mathbf{R}_2$$

$$= \frac{8\times10^{-9}}{4\pi\times\frac{1}{36\pi}\times10^{-9}\times\left(\sqrt{2^2+3^2+5^2}\right)^3}(2\mathbf{a}_x + 3\mathbf{a}_y + 5\mathbf{a}_z)$$

$$= 0.615\mathbf{a}_x + 0.922\mathbf{a}_y + 1.537\mathbf{a}_z$$

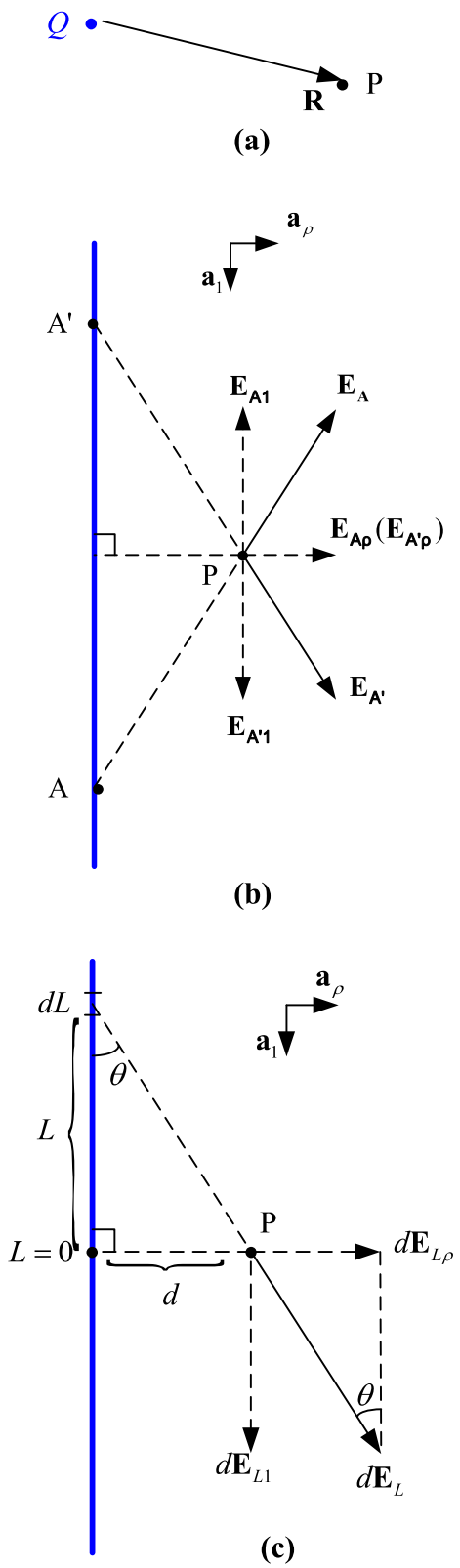


Figure 3.2. The different field components

As we can see from Figure 3.2(b), for any point A on the line charge we can always find one and only one point A' whose electric field at P has the same magnitude but the opposite sign of that of A in the direction which is parallel to the line charge. This is because the linear charge is infinitely long. Therefore we only need to find the electric field in the direction perpendicular to the line charge. As shown in Figure 3.2(c) each incremental length of line charge dL acts as a point charge and produces an incremental contribution to the total electric field intensity. The magnitude of $d\mathbf{E}_L$ is thus:

$$dE_L = \frac{dQ}{4\pi\varepsilon_0 |\mathbf{R}|^2} = \frac{\rho_L dL}{4\pi\varepsilon_0 (L^2 + d^2)}$$

therefore the magnitude of $d\mathbf{E}_\rho$ is

$$dE_{L\rho} = dE_L \sin\theta = \frac{\rho_L dL}{4\pi\varepsilon_0 (L^2 + d^2)} \frac{d}{\sqrt{(L^2 + d^2)}} = \frac{\rho_L d}{4\pi\varepsilon_0} \frac{dL}{(L^2 + d^2)^{3/2}}$$

and $E_{L\rho} = \int_{L=-\infty}^{L=\infty} dE_\rho = \frac{\rho_L d}{4\pi\varepsilon_0} \int_{L=-\infty}^{L=\infty} \frac{dL}{(L^2 + d^2)^{3/2}}$ (this integral is given in the formula sheet)

$$E_{L\rho} = \frac{\rho_L d}{4\pi\varepsilon_0} \times \frac{L}{d^2\sqrt{L^2 + d^2}} \Bigg|_{L=-\infty}^{L=\infty}$$

$$= \frac{\rho_L d}{4\pi\varepsilon_0} \times \left(\frac{1}{d^2\sqrt{1 + \frac{d^2}{L^2}}} \Bigg|_{L=\infty} - \frac{-1}{d^2\sqrt{1 + \frac{d^2}{L^2}}} \Bigg|_{L=-\infty} \right) = \frac{\rho_L d}{4\pi\varepsilon_0} \times \left(\frac{1}{d^2\sqrt{1+0}} - \frac{-1}{d^2\sqrt{1+0}} \right) = \frac{\rho_L}{2\pi\varepsilon_0 d}.$$

Then we have to find $\mathbf{a}_\rho$

$$\because \mathbf{R}_\rho = (2\mathbf{a}_x + 3\mathbf{a}_y + 4\mathbf{a}_z) - (2\mathbf{a}_x) = 3\mathbf{a}_y + 4\mathbf{a}_z$$

$$\therefore \mathbf{a}_\rho = \frac{\mathbf{R}_\rho}{|\mathbf{R}_\rho|} = \frac{3\mathbf{a}_y + 4\mathbf{a}_z}{\sqrt{3^2 + 4^2}} = \frac{3}{5}\mathbf{a}_y + \frac{4}{5}\mathbf{a}_z$$

therefore

$$\mathbf{E_L} = \mathbf{E}_{L\rho} = E_{L\rho}\mathbf{a}_\rho$$

$$= \frac{2\times10^{-9}}{2\pi \times \frac{1}{36\pi} \times 10^{-9} \times 5} \times \left(\frac{3}{5}\mathbf{a}_y + \frac{4}{5}\mathbf{a}_z \right)$$

$$= 4.32\mathbf{a}_y + 5.76\mathbf{a}_z$$

and

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 + \mathbf{E_L} = 2.01\mathbf{a}_x + 7.34\mathbf{a}_y + 9.39\mathbf{a}_z \text{ V/m}$$

MATLAB code :

```
clc; %clear the command line
clear; %remove all previous variables
Q1=8e-9;%charges on Q1
Q2=8e-9;%charges on Q2
pL=2e-9;%charge density of the line
Epsilono=8.8419e-12;%Permitivity of free space

P=[2 3 4];%coordinates of observation point
A=[0 0 1];%coordinates of Q1
B=[0 0 -1];%coordinates of Q2
C=[2 0 0];%coordinates of the center of the line charge
Number_of_L_Steps=100000;%the steps of L

%%the following routine calculates the electric fields at the
%%observation point generated by the point charges
R1=P-A; %the vector pointing from Q1 to the observation point
R2=P-B; %the vector pointing from Q2 to the observation point
R1Mag=norm(R1);%the magnitude of R1
R2Mag=norm(R2);%the magnitude of R1
E1=Q1/(4*pi*Epsilono*R1Mag^3)*R1;%the electric field generated by Q1
E2=Q2/(4*pi*Epsilono*R2Mag^3)*R2;%the electric field generated by Q2

%%the following routine calculates the electric field at the
%%observation point generated by the line charge
d=norm(P-C);%the distance from the observation point to the center of the line
length=100*d;%the length of the line
dL_V=length/Number_of_L_Steps*[1 0 0];%vector of a segment
dL=norm(dL_V);%length of a segment
EL=[0 0 0];%initialize the electric field generated by EL
C_segment=C-( Number_of_L_Steps/2*dL_V-dL_V/2);%the center of the first segment
for i=1: Number_of_L_Steps
    R=P-C_segment;%the vector seen from the center of the first segment to the
observation point
    RMag=norm(R);%the magnitude of the vector R
    EL=EL+dL*pL/(4*pi*Epsilono*RMag^3)*R;%get contibution from each segment
    C_segment=C_segment+dL_V;%the center of the i-th segment
end

E=E1+E2+EL;% the electric field at P
```

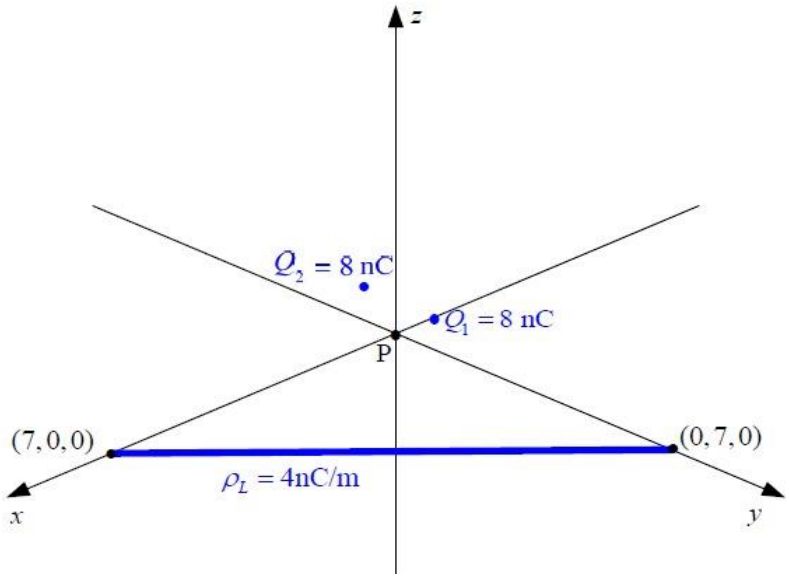
Running result



Comparing the MATLAB answer with the analytical answer we see that there is very little difference between them. This little difference is caused by the finite length of the steps of L and the finite line charge that we used to replace the infinite one.

Assignment

Exercise: A finite uniform linear charge $\rho_L = 4$ nC/m lies on the xy plane as shown in Figure 3.4, while point charges of 8 nC each are located at (0, 1, 1) and (0, -1, 1). Find **E** at (0, 0 ,0). Write a MATLAB program to verify your answer.



Assignment

Experiment 3

Electric Flux through a Plane

A point charge of 1.0 C is located at (0, 0, 1). Find analytically the total electric flux going through the infinite xy plane as shown in Figure 3.1. Verify your answer using a MATLAB program.

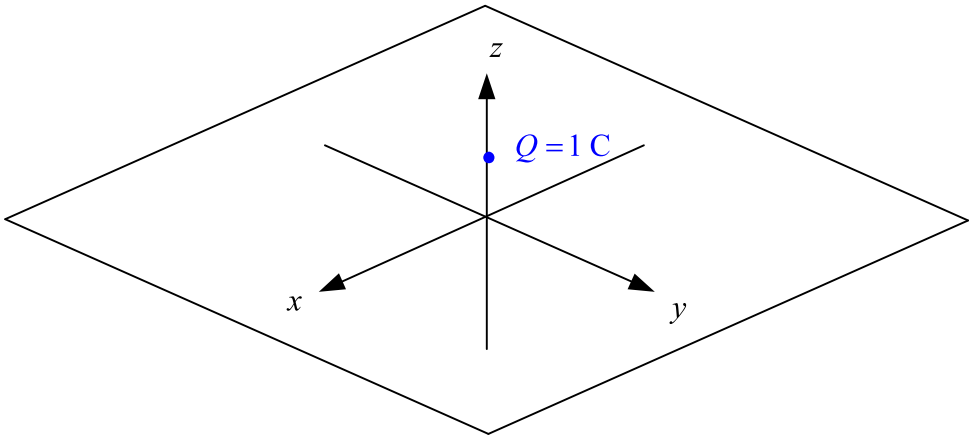


Figure 3.1 The point charge of Q = 1.0 C and the infinite x-y plane.

Analytical solution:

The total flux going through a surface is given by

$$\psi = \iint_S \mathbf{D}_S \bullet d\mathbf{s}$$

where $d\mathbf{s}$ is a vector whose direction is normal to the surface element ds and has a magnitude of ds, and $\mathbf{D}_S$ is the electric flux density passing through *ds*. In this problem (See Figure 2)

$$\mathbf{D}_S = \frac{Q}{4\pi R^2}\mathbf{a}_R$$

$$d\mathbf{s} = ds(-\mathbf{a}_z) = -dxdy\mathbf{a}_z$$

therefore the flux is given by

$$\psi = \iint_S \mathbf{D}_S \bullet d\mathbf{s} = \iint_S \left(\frac{Q}{4\pi R^2}\mathbf{a}_R\right) \bullet \left(-dxdy\mathbf{a}_z\right).$$

This is the general method to evaluate the electric flux passing through a surface. However, for certain problems we can create a Gaussian surface to find out the flux passing through the surface and avoid evaluating any integral. The Gaussian surface we created for this problem is shown in Figure 3, where S_{top} and S_{bottom} are two parallel planes symmetric relative to the point charge *Q*. Based on Gauss's law, the total flux passing through the enclosed surface is

$$\psi_{\text{total}} = \psi_{\text{top}} + \psi_{\text{bottom}} + \psi_{\text{side1}} + \psi_{\text{side2}} + \psi_{\text{side3}} + \psi_{\text{side4}} = \text{charge enclosed} = Q$$

since S_{top} and S_{bottom} are symmetric relative to the point charge Q,

$$\psi_{\text{top}} = \psi_{\text{bottom}}.$$

Using the same reason

$$\psi_{\text{side1}} = \psi_{\text{side2}} = \psi_{\text{side3}} = \psi_{\text{side4}}$$

and since

$$\psi_{\text{side1}} < \frac{Q}{4\pi L^2} \times (2Ld) = \frac{Qd}{2\pi L}$$

$$\frac{Qd}{2\pi L} \to 0 \text{ as } L \to \infty$$

we have

$$\psi_{\text{side1}} \to 0 \text{ as } L \to \infty$$

Hence as $L \to \infty$

$$\psi_{\text{total}} = \psi_{\text{top}} + \psi_{\text{bottom}} = 2\,\psi_{\text{bottom}} = Q$$

$$\psi_{\text{bottom}} = \frac{Q}{2} = 0.5 \text{ C}$$

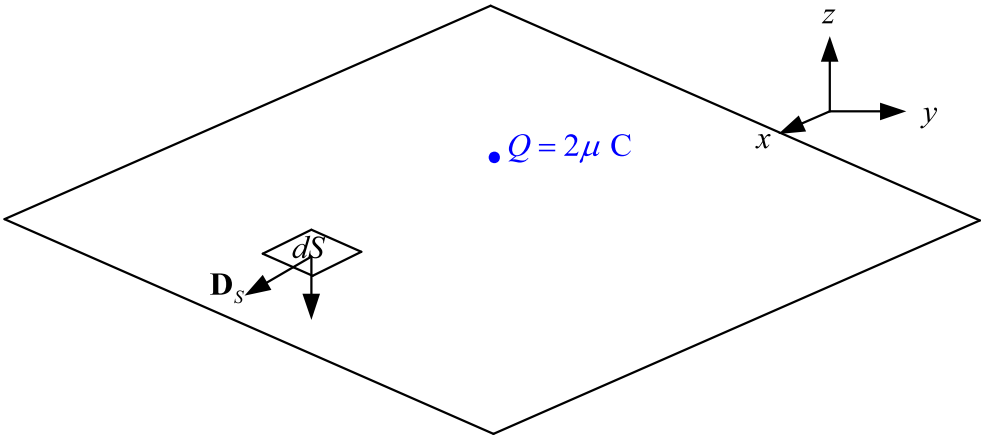


Figure 2 The electric flux through a surface element resulting from a point charge.

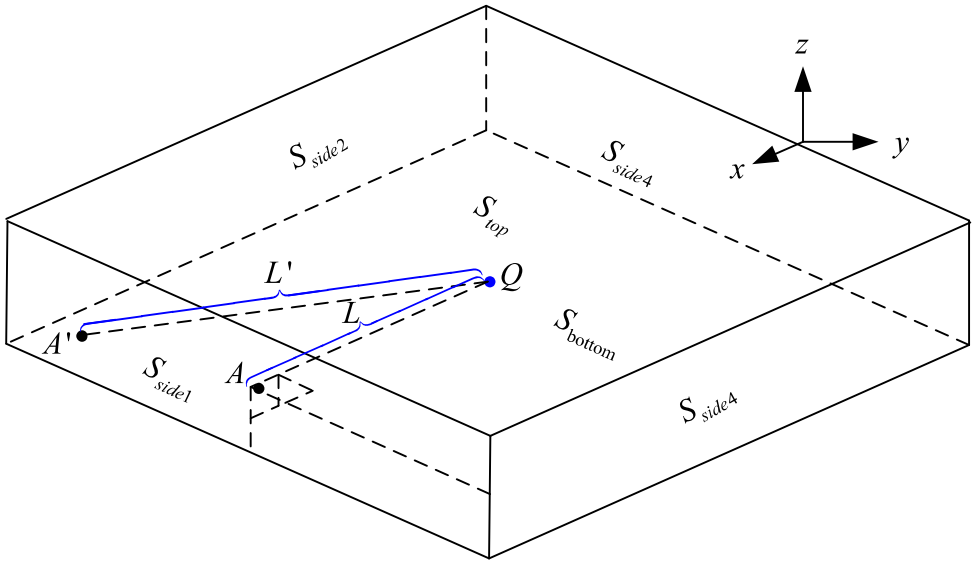


Figure 3 The electric field intensity at a point A is greater than any other point A' on the plane S_{side1} since $L<L'$. It follows that the flux through S_{side1}, ψ_{side1}, is smaller than (Area of S_{side1})× (|electric field density at point A|).

MATLAB Solution:

To write a MATLAB program, we replace the infinite plane with a finite one with a very large area. We equally divide this plane into a number of surface elements each has an area of ΔS. We then evaluate the flux $\Delta\psi$ passing through each cell and add all the $\Delta\psi$ together. This approach can be summarized by:

$$\Psi \doteq \sum_{j=1}^{m}\sum_{i=1}^{n}\Delta\Psi_{i,j} = \sum_{j=1}^{m}\sum_{i=1}^{n}\mathbf{D}_{i,j}\bullet\Delta\mathbf{S}_{i,j} = \sum_{j=1}^{m}\sum_{i=1}^{n}\left(\frac{Q}{4\pi R_{i,j}{}^{2}}\mathbf{a}_{Ri,j}\right)\bullet\left(-\Delta S_{i,j}\mathbf{a}_{z}\right)$$

where $\mathbf{R}_{i,j}$ is the vector pointing from the point charge to the center of the cell with indices *i* and *j*. Note that all $\Delta S_{i,j}$ have the same direction $-\mathbf{a}_z$.

MATLAB code

```
clc; %clear the command line
clear; %remove all previous variables

Q=1;%the point charge
C=[0 0 1];%location of the point charge;
az=[0 0 1];% unit vector in the z direction

x_lower=-100;%the lower boundary of x of the plane
x_upper=100;% the upper boundary of x of the plane
y_lower=-100;%the lower boundary of y of the plane
y_upper=100;%the upper boundary of y of the plane
Number_of_x_Steps=400;%step in the x direction
Number_of_y_Steps=400;%step in the y direction
dx=(x_upper-x_lower)/Number_of_x_Steps;%the x increment
dy=(y_upper-y_lower)/Number_of_y_Steps;%the y increment

flux=0;%initialize the flux to 0

for j=1:Number_of_y_Steps
    for i=1:Number_of_x_Steps
        ds=dx*dy;%the area of current element
        x=x_lower+0.5*dx+(i-1)*dx;%x component of the center of a grid
        y=y_lower+0.5*dy+(j-1)*dy;%y component of the center of a grid
        P=[x y 0];%the center of a grid
        R=P-C;%vector R is the vector pointing from the point charge to the center of a
grid
        RMag=norm(R);%magnitude of R
        R_Hat=R/RMag;%unit vector in the direction of R
        R_surface=-az;%unit vector of direction of the surface element
        flux=flux+Q*ds*dot(R_surface,R_Hat)/(4*pi*RMag^2);%get contribution to the flux
    end
end
```

Running result:

```
>> flux

flux =

    0.4955

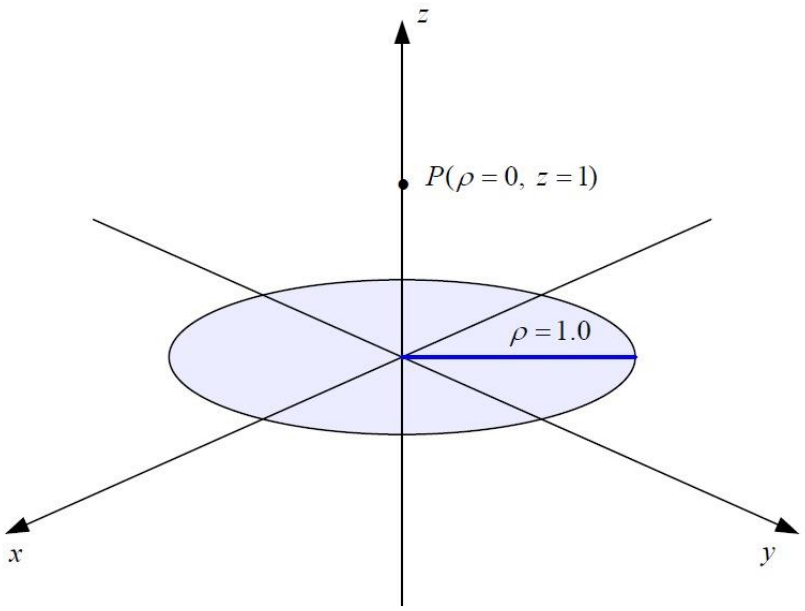
>>
```

Comparing the MATLAB answer and the analytical answer we see that there is a good agreement between them. The small difference between the two answers is attributed to the finite discretizations of the surface *S*, and to utilizing a finite plane instead of the actual infinite plane.

Assignment

Exercise: Given the surface charge density, ρ_s=2.0 μC/m^2, existing in the region ρ<1.0 m, z = 0, and zero elsewhere, find **E** at P (ρ = 0, z = 1.0) and write a MATLAB program to verify your answer.

.



Assignment

Experiment 4

CURL

Aim:-

To Verify Curl in differential form.

Theory:-

Curl measures the extent of rotation of the field about a point. Suppose that $\vec{F}$represents the velocity field of a fluid. Then, the curl of $\vec{F}$ at point P is a vector that measures the tendency of particles near P to rotate about the axis that points in the direction of this vector. The magnitude of the curl vector at P measures how quickly the particles rotate around this axis. In other words, the curl at a point is a measure of the vector field's "spin" at that point.

MATLAB code:-

```
%function curl()
syms x y z

% clear all

close all
clc

% request user function

prompt = 'Enter a i component of vector field: ' ;
x_in=input(prompt);
prompt='Enter a j component of vector field: ';
y_in=input( prompt);
prompt='Enter a k component of vector field: ';
z_in=input( prompt);
V= [x_in, y_in,z_in];

% plot figure 1 vector field
figure(1)
[X,Y] = meshgrid(-5:.5:5,-5:.5:5);

F1= subs(x_in, [x y] , {X,Y});
F2= subs(y_in, [x y] , {X,Y});

set(gcf,'units','normalized','position',[0.6 0.2 0.35 0.4]);
quiver(X,Y,F1,F2,2,'color',[0 0 1],'linewidth',2)
shading flat
hold on
xlabel('ix')
ylabel('jy')
title(sprintf('Vector Field = i[%s] + j[%s] + k[%s] \n' , V(1),V(2),V(3)))
hold off
```

```
%curl

%V= [x_in, y_in,z_in];
X= [ x y z];
cur= curl(V,X)

% Figure 2 plot of curl of field
figure(2)
fsurf(cur)
xlabel('ix')
ylabel('iy')
zlabel('kz')
```

Equation to find CURL of a vector

$$\nabla \times \underline{A} = \begin{vmatrix} \underline{a}_x & \underline{a}_y & \underline{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

$$= \underline{a}_x \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \underline{a}_y \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \underline{a}_z \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

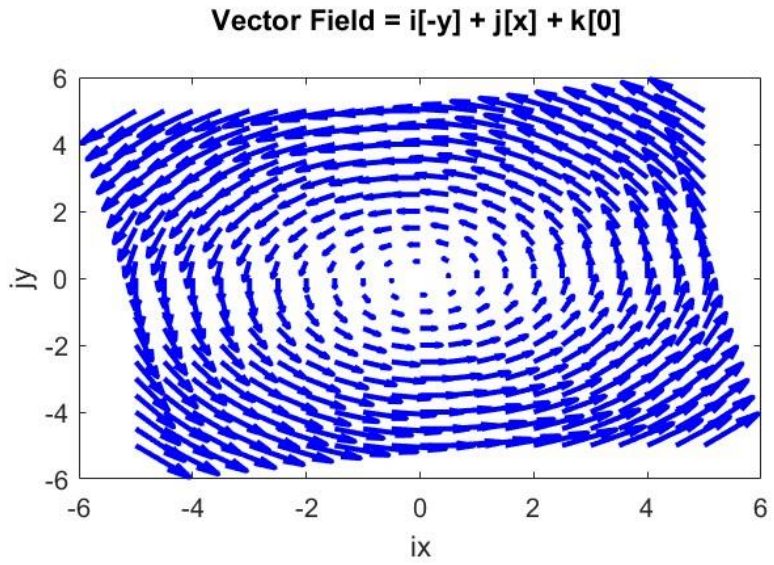
Input and output

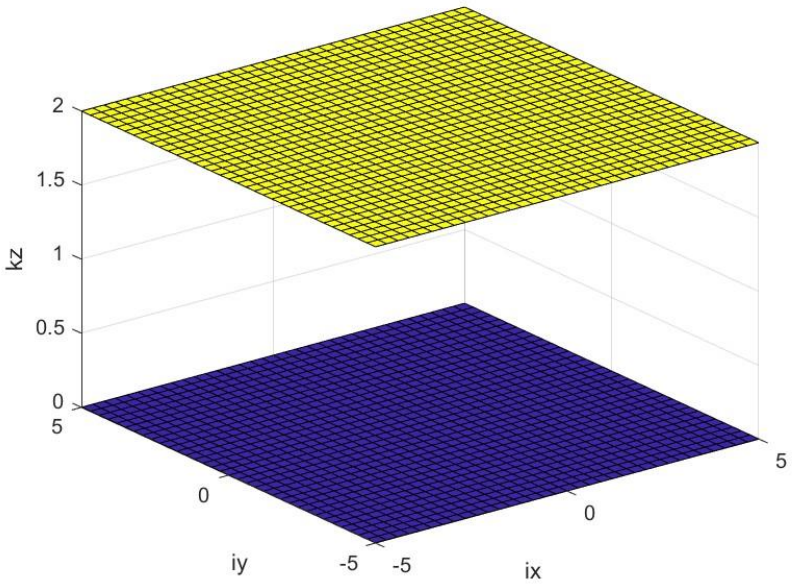
Example 1:

```
Enter a i component of vector field: -y
Enter a j component of vector field: x
Enter a k component of vector field: 0

cur =

0
0
2
```



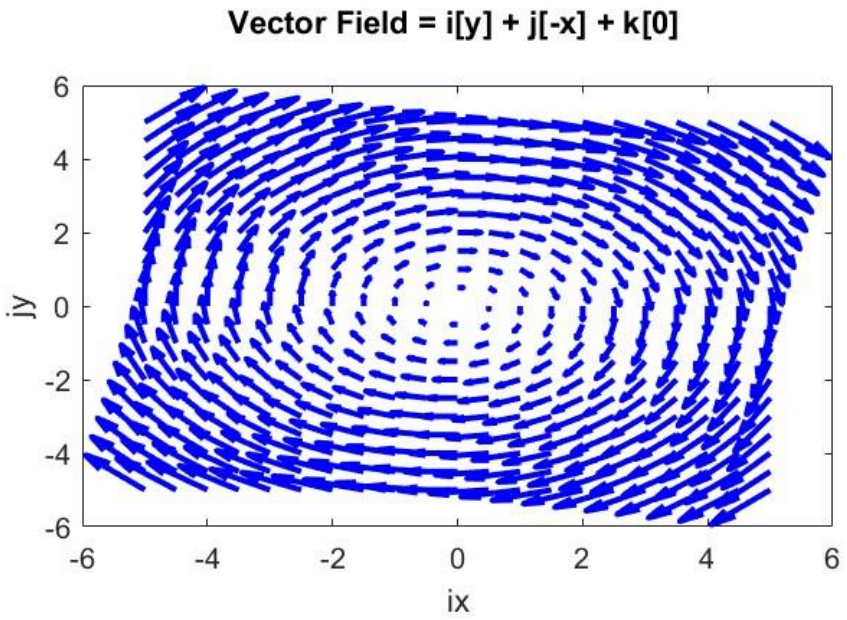


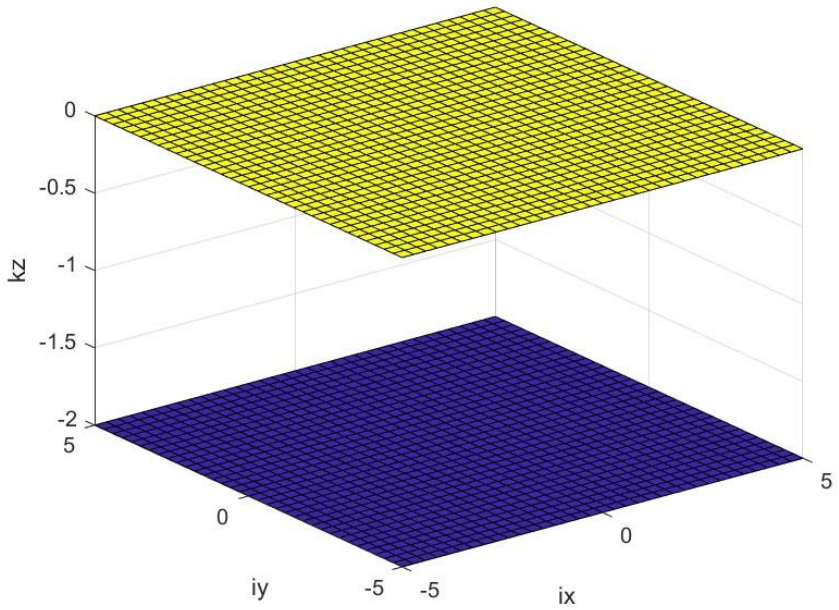
Example 2:

```
Enter a i component of vector field: y
Enter a j component of vector field: -x
Enter a k component of vector field: 0

cur =

 0
 0
-2
```





Assignment

Example 3:

```
Enter a i component of vector field: 0
Enter a j component of vector field: exp(-y^2)
Enter a k component of vector field: 0
```

Example 4:

```
Enter a i component of vector field: 0
Enter a j component of vector field: exp(-x^2)
Enter a k component of vector field: 0
```

Antenna Design and Simulation using HFSS

Antennas are an important element of the communication system. The communication system has two main components: transmitter and receiver. An antenna is a device used to transform current and voltage into electromagnetic waves and vice versa. In other words, the antenna is a traditional structure between free space and a guiding device. The Maxwell's equations characterize the transmitting and receiving functionalities of the antennas.

Around 1830 Faraday conducted experiments that showed a relationship between electricity and magnetism. Interestingly, the concept of electromagnetic waves had not even been thought up at this point. In 1887 the first experimental demonstration of wireless communication was carried out by Heinrich Hertz. He designed a transmitter dipole antenna and a receiver loop antenna. In 1901, the first experiment for wireless communication was carried out over a long distance across the Atlantic by Guglielmo Marconi.

Antennas were designed on practical approaches only until the late 1970s. Simple antenna geometries like horns, Yagi-Uda, dipoles, reflectors, etc. were popular. The antenna designers could vary parameters based on their requirements. Antenna development required extensive testing and experimentation and was, therefore, funded primarily by governments. However, there has been a drastic growth in computing speed and computational techniques in the recent years, making antenna designing a low-cost and efficient process. The growth of wireless communications and use of modern computational techniques have been the catalysts for the introduction of new low-cost antenna designs.

Types of Antennas

Let us look at a brief introduction to various types of antennas before we get into the details of design procedures.

Wire antennas

Wire antennas are the simplest and most basic types of antennas. They are widely used due to their simplicity of design. Wire antennas can be visually seen everywhere around us like on buildings or automobiles, aircrafts, ships, etc. They exist in various shapes such as straight wire antennas like dipoles and monopoles, loops of various shapes like rectangular, circular, etc., helix antennas, and so on.

Aperture Antennas

With the increasing demand in the field of antenna engineering, more sophisticated antenna designs have come into existence. Aperture antennas are used for high frequency applications.

There are a lot of geometrical configurations of aperture antennas. They can be waveguides or horns whose apertures may be rectangular, circular or elliptical in shape.

Microstrip Antennas

Microstrip antenna became famous in the 1970s due to their wide range of uses in spaceborne applications. These antennas consist of a substrate with a metallic patch. The metallic patch could be of various shapes. It's easy to fabricate microstrip patch antennas using the modern PCB technology. Installation of these antennas on aircrafts, satellites, cars is also convenient. Due to their advantages, microstrip antennas are one of the most widely used antennas.

Reflector Antennas

A very common type of reflector antenna is the parabolic dish antenna. These antennas can be easily spotted on top of buildings with large diameters. Reflector antennas have high gain properties which helps the signal to transmit over large distances.

Array Antennas

Certain radiation characteristics are hard to achieve using single element antennas. In such cases an aggregate of radiating elements called an array is made use of to get the desired results. The desired radiation pattern can be achieved my varying the arrangement of the elements.

Note: -

Heinrich Hertz, a German physicist, was the first one to demonstrate the existence of radio waves using a "dipole antenna".

High Frequency Structure Simulator (HFSS)

HFSS is a high-performance full wave electromagnetic (EM) field simulator for arbitrary 3D volumetric passive device modelling that takes advantage of the familiar Microsoft Windows graphical user interface. It integrates simulation, visualization, solid modelling, and automation in an easy to learn environment where solutions to your 3D EM problems are quickly and accurate obtained. Ansoft HFSS employs the Finite Element Method (FEM), adaptive meshing, and brilliant graphics to give you unparalleled performance and insight to all of your 3D EM problems. Ansoft HFSS can be used to calculate parameters such as S-Parameters, Resonant Frequency, and Fields.

Typical uses include:

- Package Modelling – BGA, QFP, Flip-Chip
- PCB Board Modelling – Power/ Ground planes, Mesh Grid Grounds, Backplanes
- Silicon/GaAs-Spiral Inductors, Transformers
- EMC/EMI – Mobile Communications – Patches, Dipoles, Horns, Conformal Cell Phone Antennas, Quadrafilar Helix, Specific Absorption Rate (SAR), Infinite Arrays, Radar Section (RCS), Frequency Selective Surface (FSS)
- Connectors – Coax, SFP/XFP, Backplane, Transitions
- Waveguide – Filters, Resonators, Transitions, Couplers
- Filters – Cavity Filters, Microstrip, Dielectric

HFSS is an interactive simulation system whose basic mesh element is a tetrahedron. This allows you to solve any arbitrary 3D geometry, especially those with complex curves and shapes, in a fraction of the time it would take using other techniques.

The name HFSS stands for High Frequency Structure Simulator. Ansoft pioneered the use of the Finite Element Method (FEM) for EM simulation by developing / implementing technologies such as tangential vector finite elements, adaptive meshing, and Adaptive Lancozos - pade Sweep (ALPS). Today, HFSS continues to lead the industry with innovations such as Modes to Nodes and Full wave Spice.

Experiment 5

Dipole antenna (λ/2 dipole)

Aim:-

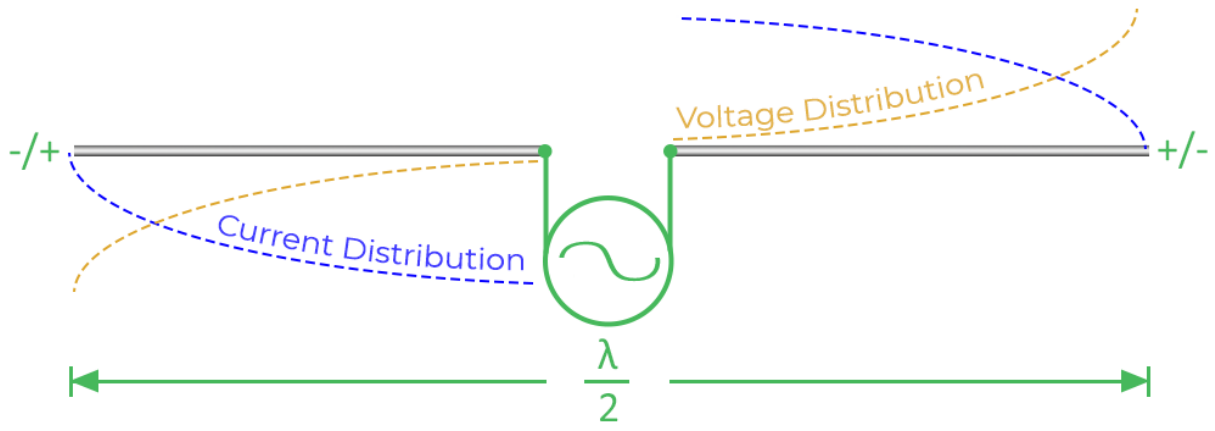
To design and analyze the performance of a Dipole antenna.

Theory:-

The word Dipole indicates “two poles”. It is also referred to as “doublet”. Dipole antenna has two conducting poles and is a balanced antenna because it is bilaterally symmetrical. Electromagnetic waves are generated due to flow of current in the poles.

The length of the total wire, which is being used as a dipole, equals half of the wavelength (i.e., l = λ/2). Such an antenna is called as half-wave dipole antenna. This is the most widely used antenna because of its advantages. Some of the advantages include its ease of design and analysis and cost-effectiveness. It is also known as Hertz antenna.

For a dipole antenna of length L oriented along the z-axis and centered at z=0 (assuming that the wire is a perfect electric conductor), the current flows in the z-direction.



Design Steps :-

Given

Operating Frequency – 1GHz

Necessary design parameters

Length of the dipole - λ/2

If frequency is 1GHz, then λ = c/f = 300 mm

Then L = 150 mm

So, L/2 = 75 mm

Consider radius of the dipole = 1mm

Consider feed gap = 1mm

Note: The radius, feed gap and length can be changed slightly to obtain the desired performance

Steps for simulation

Step 1: Starting HFSS

Step 2: Creating the antenna model

Step 3: Configurations (Settings)

Step 4: Validation and analysis

Step 5: Results

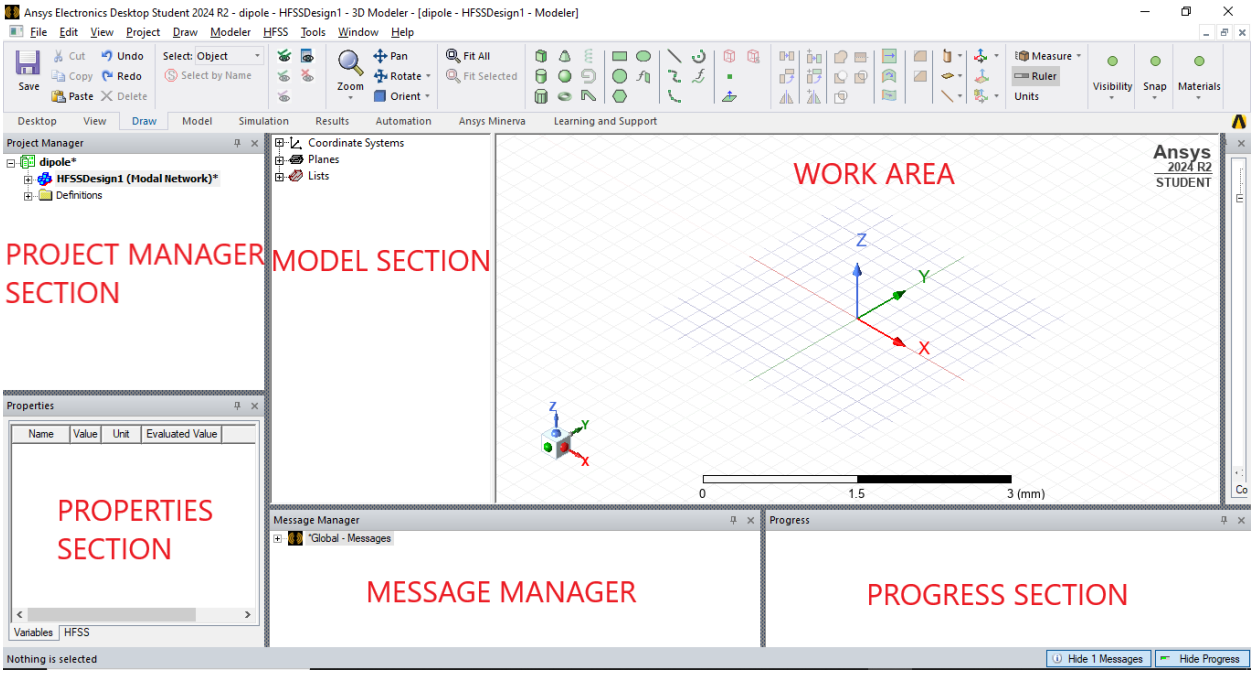
Step 1

Starting HFSS

- Double Click on Ansys Electronics Desktop Student icon



- A blank project with a blank HFSS design will be created as given below.



You can see the following sections on the screen

- Project manager section
- Model section
- Work area (Antenna construction)
- Message manager section
- Progress section
- Properties

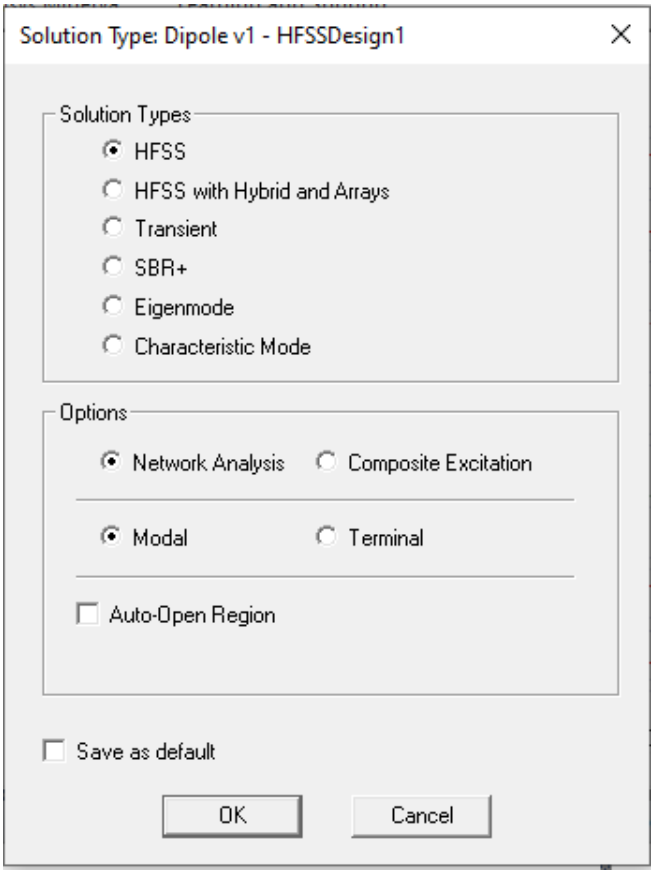
Creating the project

- Goto File, Save As
- Type name as dipole
- Click on Save

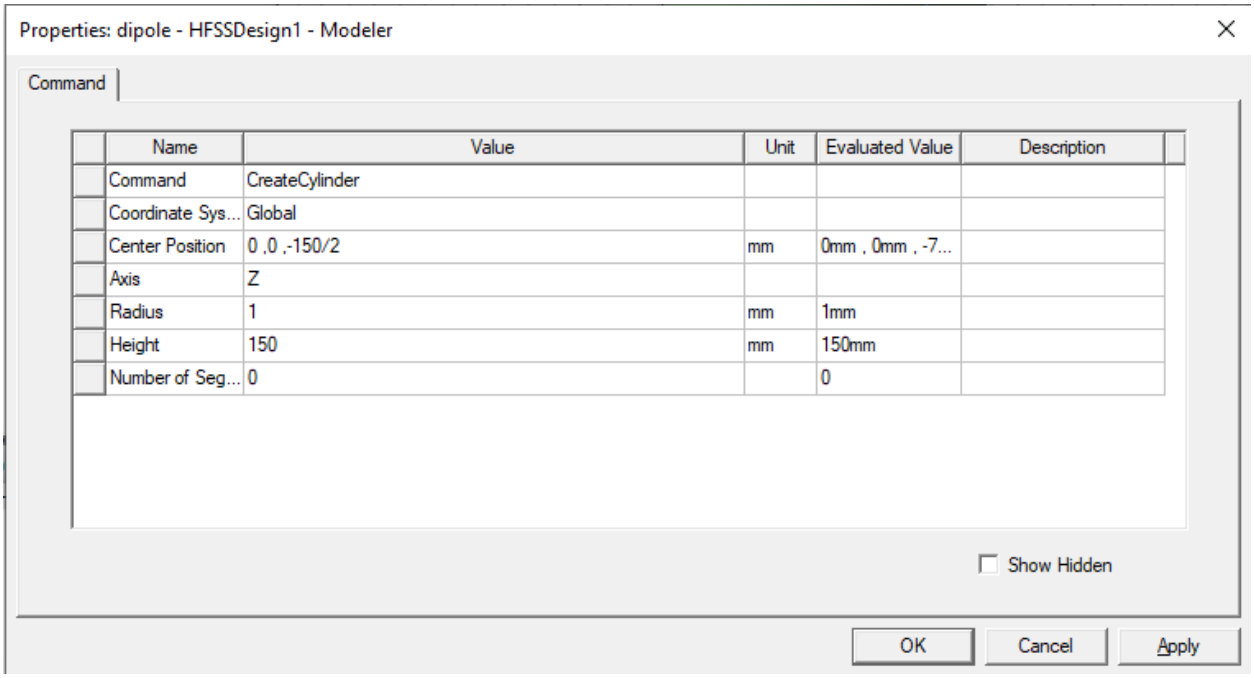
Step 2

Creating the antenna model

- From the HFSS menu select **Solution Type** as given below.

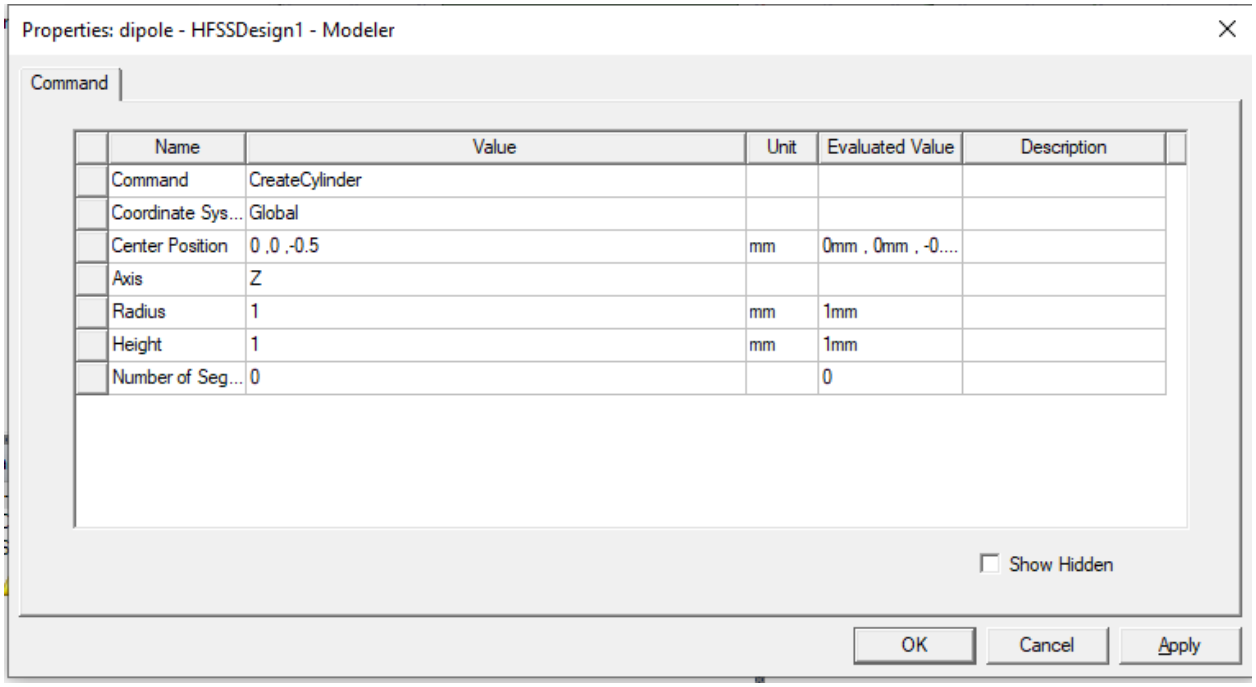


- Draw a cylinder(dipole) with following coordinates(Units - mm),

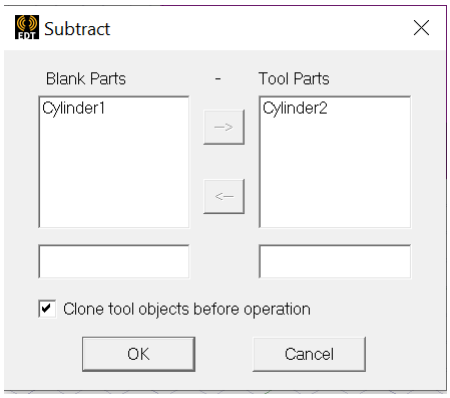


- Right click on cylinder1 in model section
- Select Assign material
- Type pec in the search by name space
- Select pec
- Click ok

- Draw another cylinder for creating a gap in the middle of the dipole with dimensions as given below,



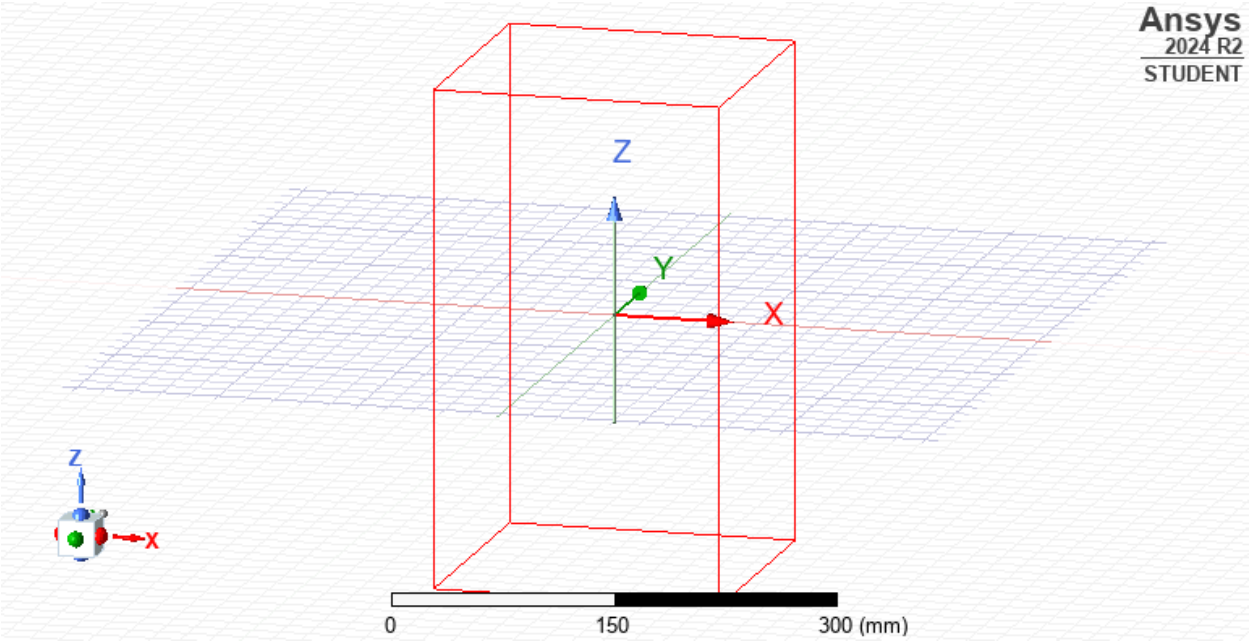
- To create space for giving excitation, select the two cylinders i.e. cylinder1(dipole) and cylinder2(feed)
- Go to Modeler, Boolean, Subtract
- The following window will appear,



- Check the box on clone tool objects before operation
- Click ok

- Select cylinder2
- Goto Modeler, Surface, Section, select YZ plane and press ok.
- A rectangle is now created in YZ plane within cylinder2(i.e. feed)

- Right click on cylinder2_section1 in model section, assign excitation, port, lumped port, resistance 50 Ohms, next, click on dropdown under integration line and select choose new line
- Draw a line from bottom center to the top center of the rectangle, next, finish
- Right click in the work area, create open region, set operating frequency as 1GHz, select radiation boundary, ok
- Now your model will look like the figure below with dipole antenna in the middle and radiation box enclosing the antenna.

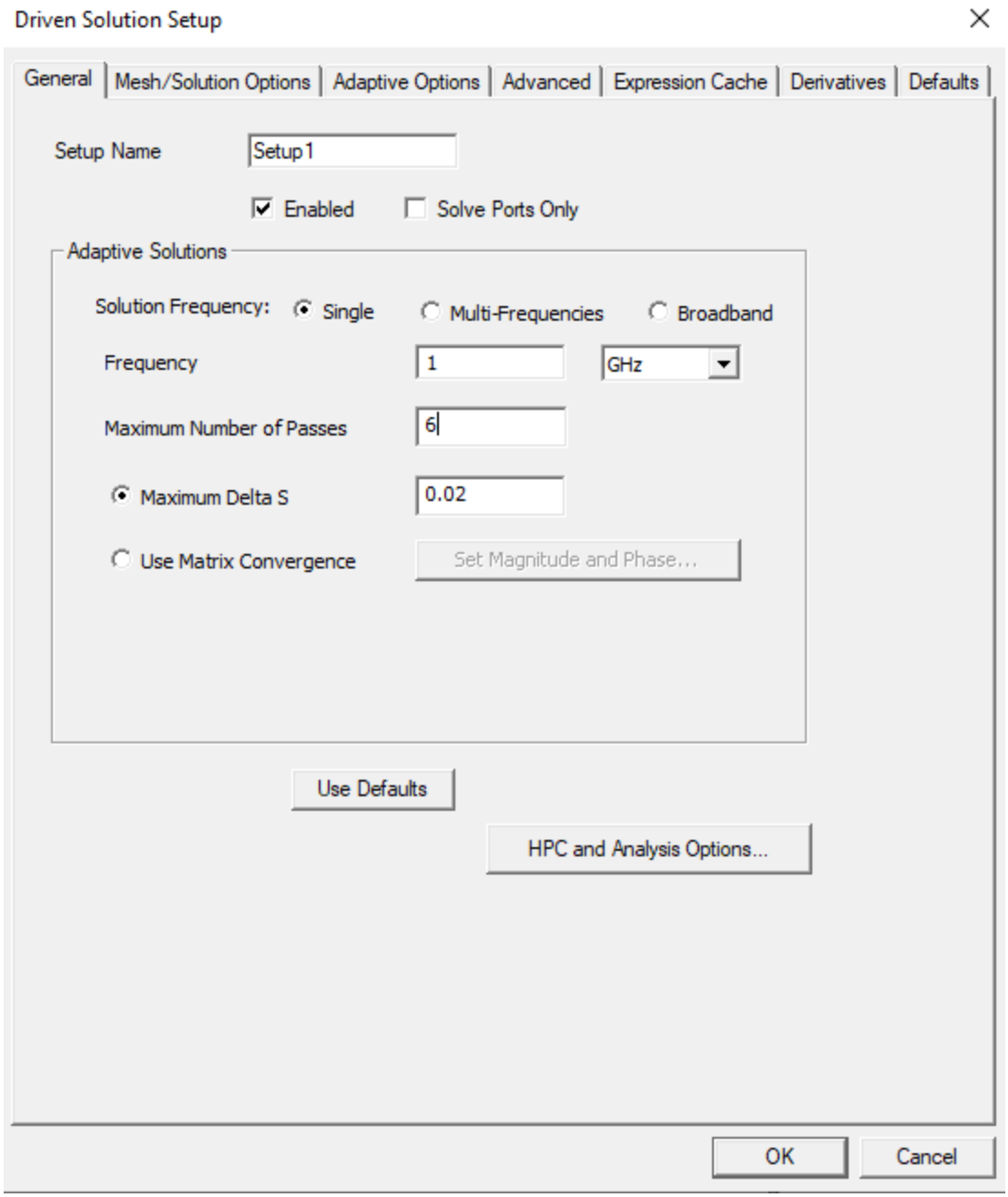


- Now your model is constructed along with radiation boundary.

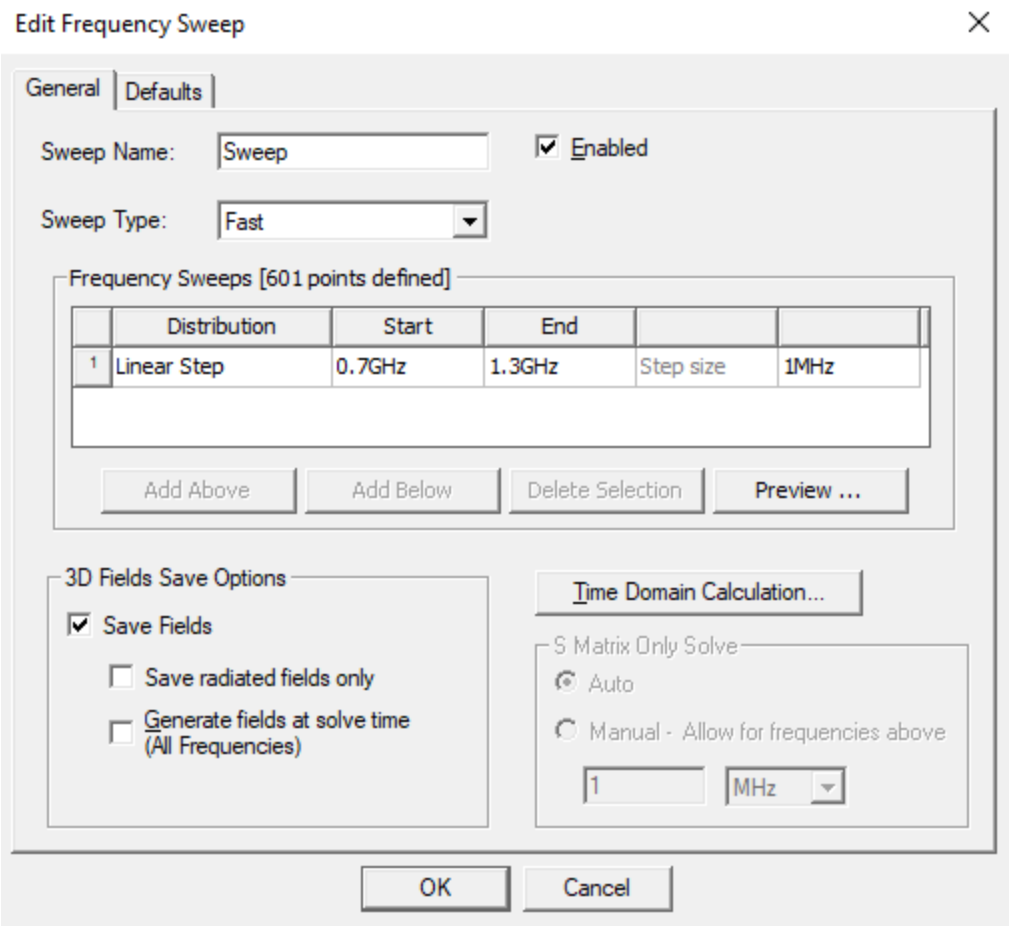
Step 3

Configurations (Settings)

- In the project manager section, Right click on analysis, add solution setup, advanced
- Enter the values as given below



- Click on ok
- In the next window that appears after clicking ok enter the values as given below

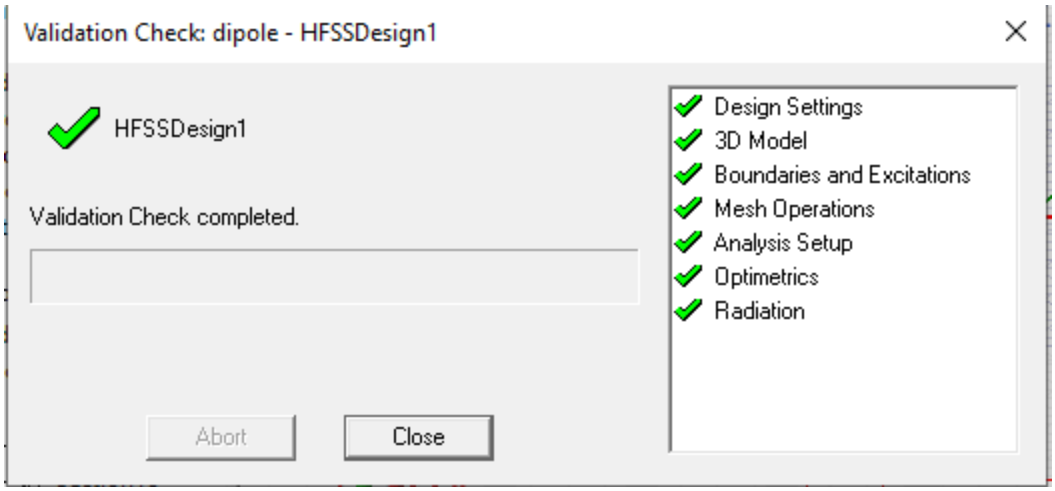


- Click ok

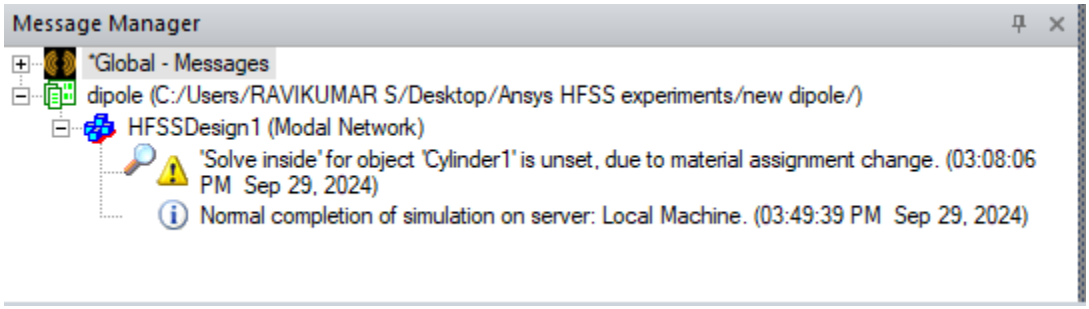
Step 4

Validation and Analysis

- Go to HFSS, Validation check
- The following window will appear if model created is correct



- Close the validation window
- Go to HFSS, Analyze all
- It will take some time to calculate results based on antenna complexity and computer configuration which is running the software
- If simulation goes well then you will receive a message in the message manager section as Normal completion of simulation on server:Local machine as given in figure below.

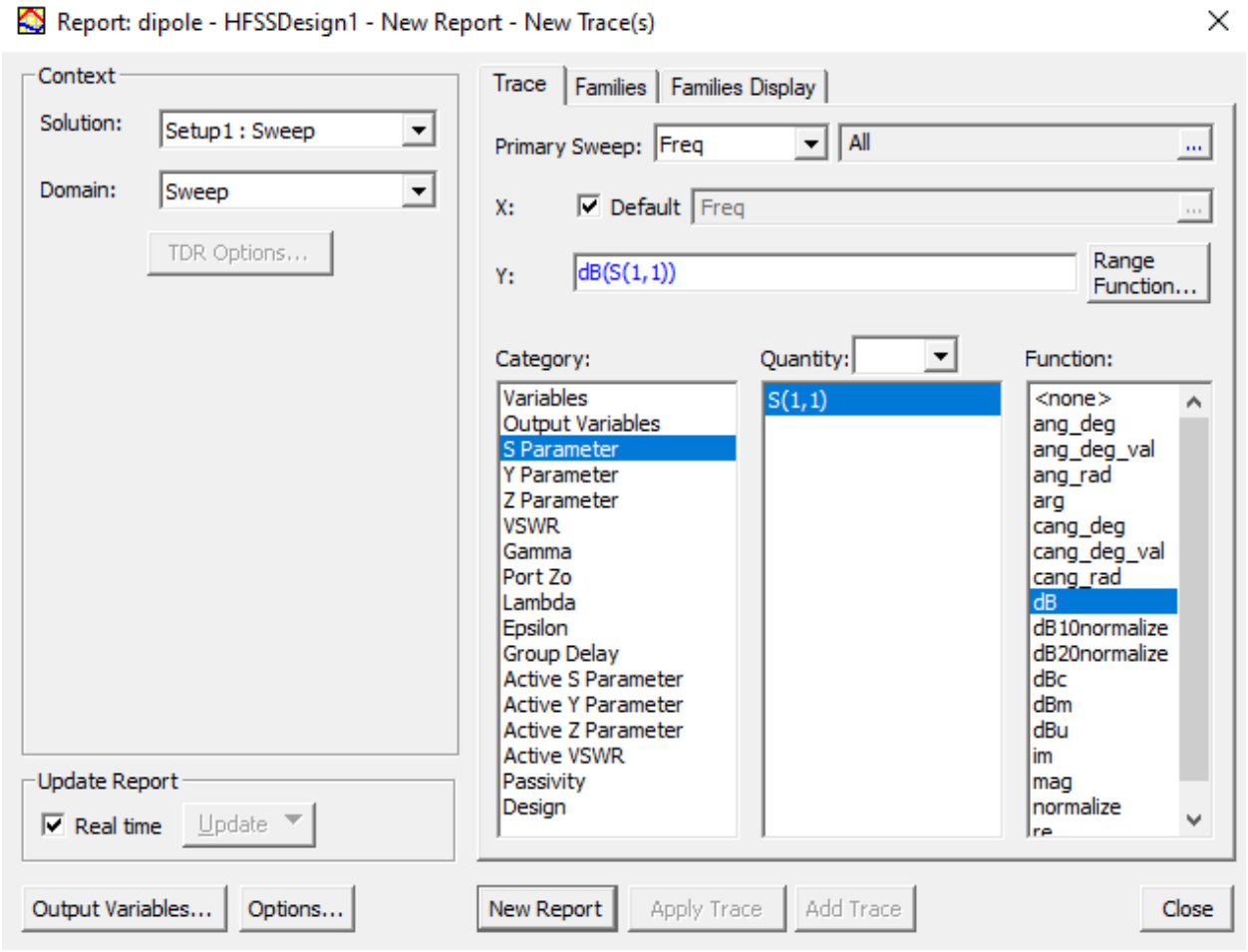


- If any errors are there then such error will be shown in message manager section. They have to be resolved.

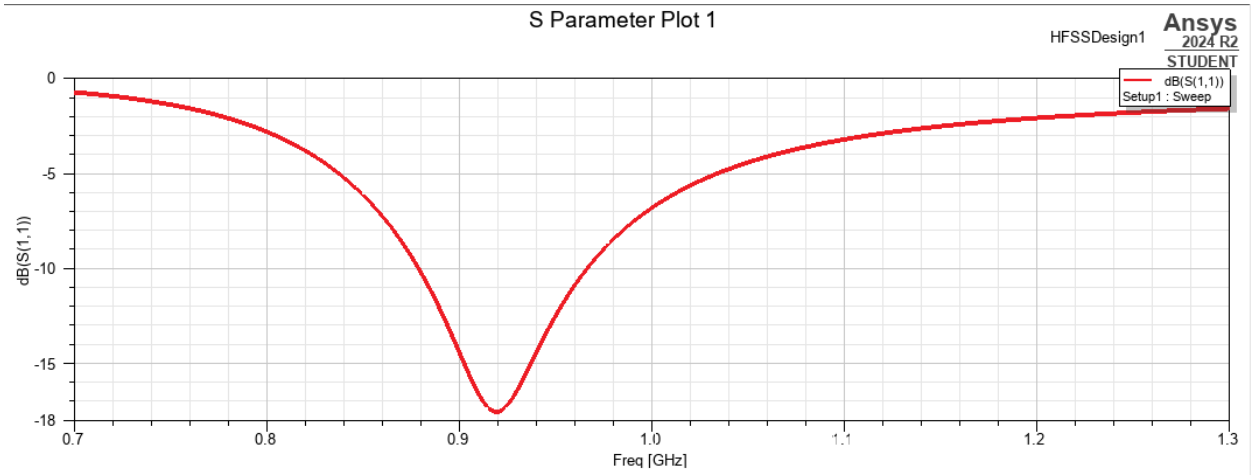
Step 5

Results

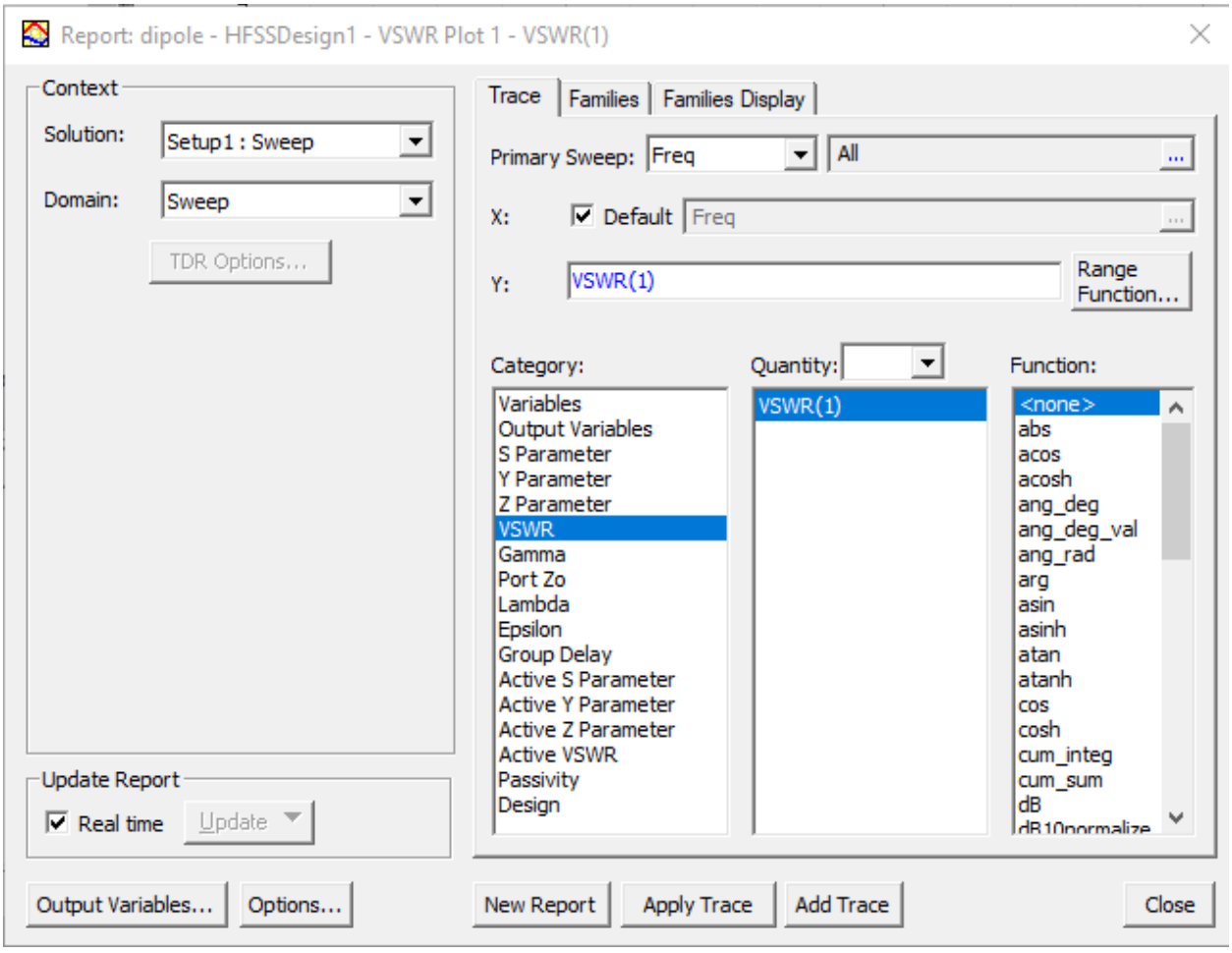
- Right click on results in project manager section, create modal solution data report, rectangular plot,
- The following window should appear



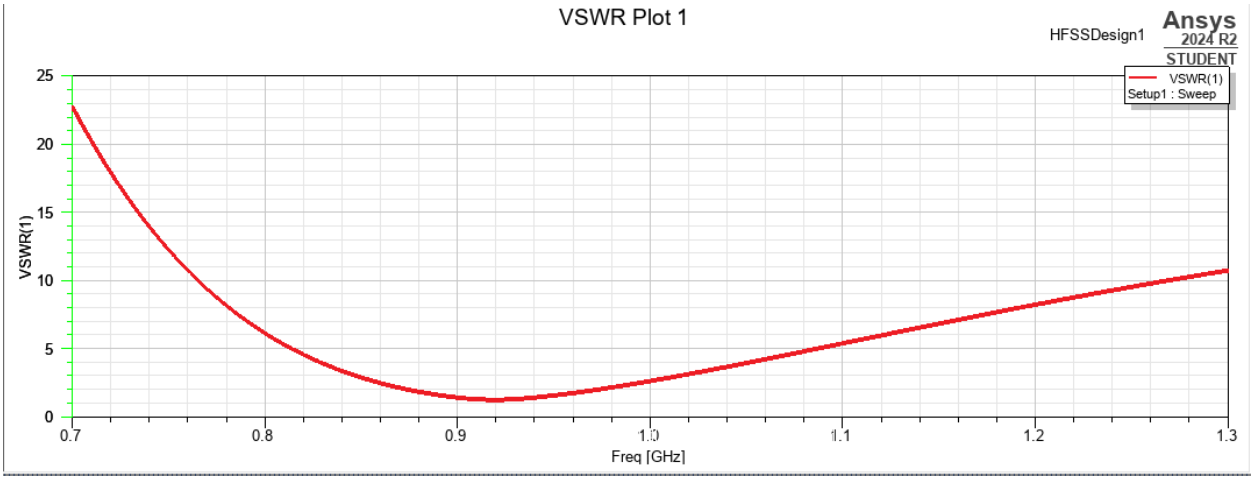
- Click on new report
- The following window will appear showing S11 parameter (Input reflection coefficient)



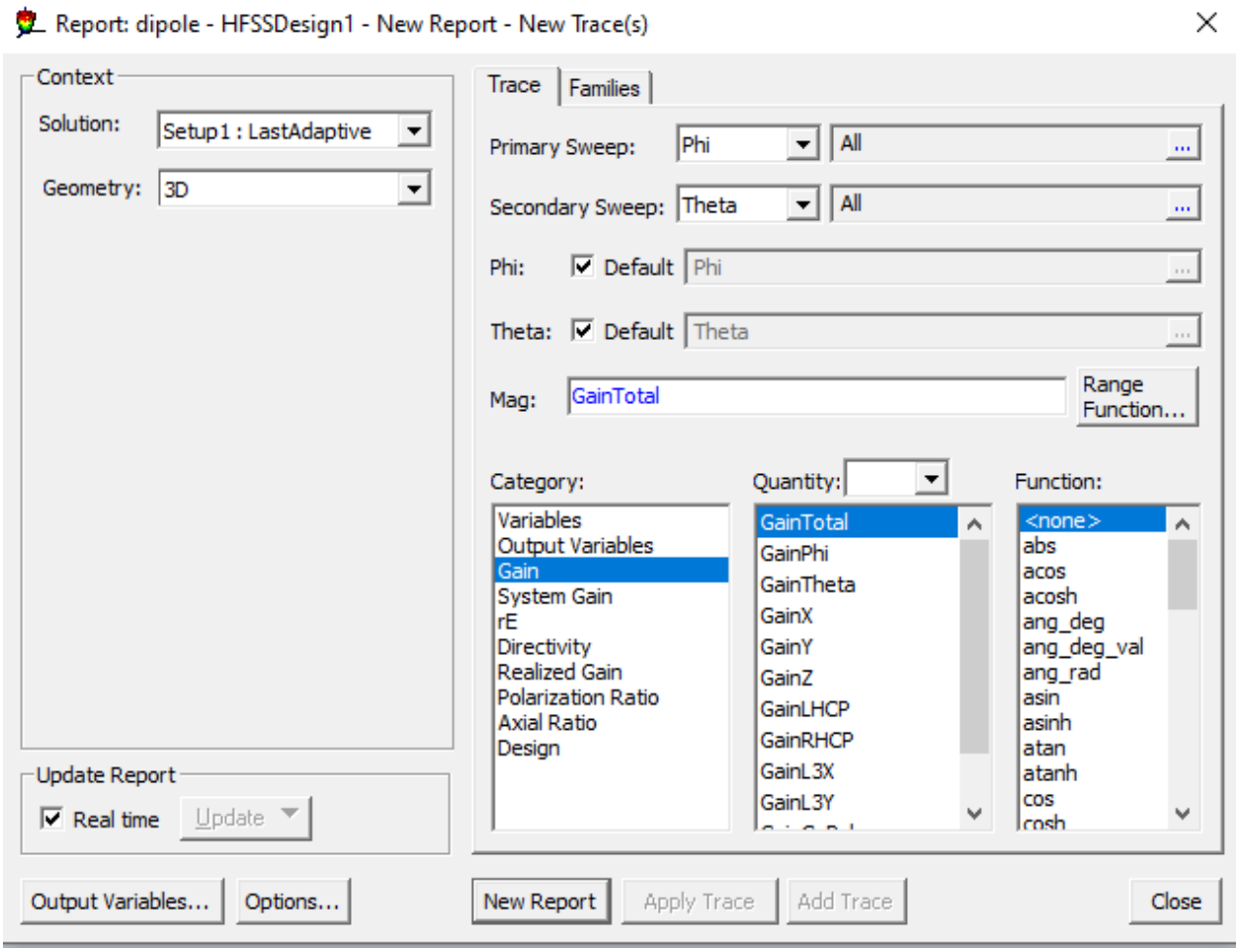
- In the same way Right click on results in project manager section, create modal solution data report, rectangular plot, select VSWR in category section, click on new report as given below



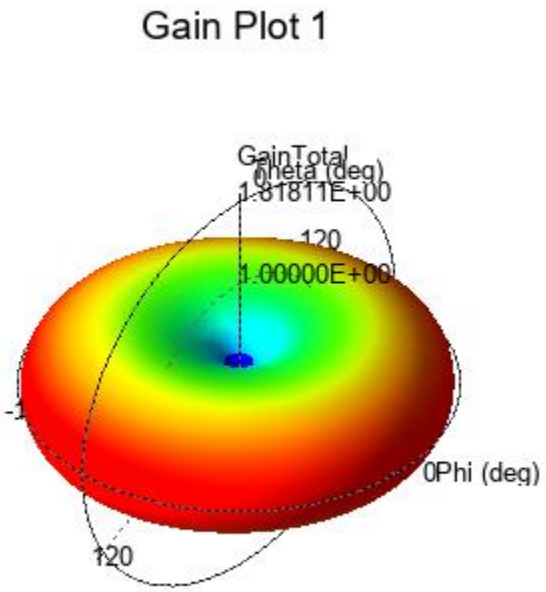
- The VSWR result will appear as given below,



- Right click on results, create far fields report, 3D polar plot
- The following window will open



- Click on new report
- The following 3D radiation pattern will appear



As seen in the S parameter graph there is a slight deviation in the operating frequency of the antenna from desired. The desired operating frequency is 1GHz and the current operating frequency as indicated by the S parameter graph is 0.92GHz. So, by slightly adjusting design parameters such as length, radius of wire or length of feed the operating frequency can be bought closer to the desired frequency.

Tabular Column

Sl No	Sl No	Before Tuning	After Tuning
1	Length of dipole	150mm	
2	S11(in dB) at 1GHz		
3	VSWR at 1GHz		
4	Bandwidth in Hz (at -10dB points of S11)		
5	Gain		

Note:

Length of the dipole can be reduced to get lowest S parameter value at 1GHz thereby making antenna operate at 1GHz.

Assignment

Assignment

Experiment 6

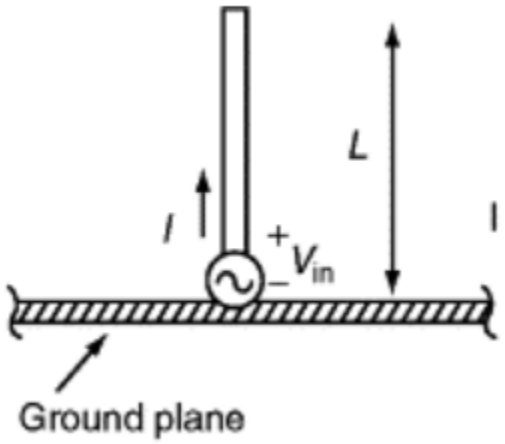
Monopole antennas (λ/4 monopole)

Aim:-

To design and analyze the performance of a Monopole antenna.

Theory:-

Monopole antennas come under the class of radio antennas that are rod-shaped straight conducting structures. They are mostly mounted over a conductive surface known as the ground plane. During his first experiments in the year 1895, Guglielmo Marconi invented the monopole antenna. After using the dipole antenna invented by Heinrich Hertz, he found out by experiments that if one of the two poles were connected to the ground, the antenna could transmit for longer distances. Hence monopole antenna is also often known as **Marconi antenna**. A very common example of monopole antennas are the ones mounted on the older cellphones. Figure shows a monopole antenna of length L mounted on top of an infinite conducting ground.



Hence, a monopole antenna is half the length of a dipole and is mounted on top of a ground plane. It is known from antenna theory that the radiation pattern of a monopole antenna above an infinite perfect electric conducting ground in the upper hemisphere is similar to that of a dipole antenna in free space. The presence of the ground plane allows the monopole antenna to operate as electrically equivalent to a dipole antenna. The main difference between the two antennas is the impedance values. The impedance of a monopole antenna is ½ times that of a full dipole. This is because the monopole antenna requires only half the voltage as compared to dipole antenna.

The radiation pattern of a monopole antenna is omnidirectional. In common practice monopole antennas are used on finite ground planes. The impedance of monopole antenna mounted on an infinite ground plane and on a finite ground plane is almost similar, but there is a vast difference in the radiation pattern of both the antennas.

Design Steps :-

Given

Operating Frequency – 95MHz

Necessary design parameters

Length of the monopole - λ/4

If frequency is 95MHz, then λ = c/f = 316 cm

Then L = 316 cm

So, L/4 = 79 cm

Consider radius of the monopole = 1 cm

Radius of finite circular ground plane = 3 λ /2 π = (3 * 316) / (2 * 3.142) = 150 cm (approximately)

Note: The radius and length of monopole can be changed slightly to obtain the desired performance

Steps for simulation

Step 1: Starting HFSS

Step 2: Creating the antenna model

Step 3: Configurations (Settings)

Step 4: Validation and analysis

Step 5: Results

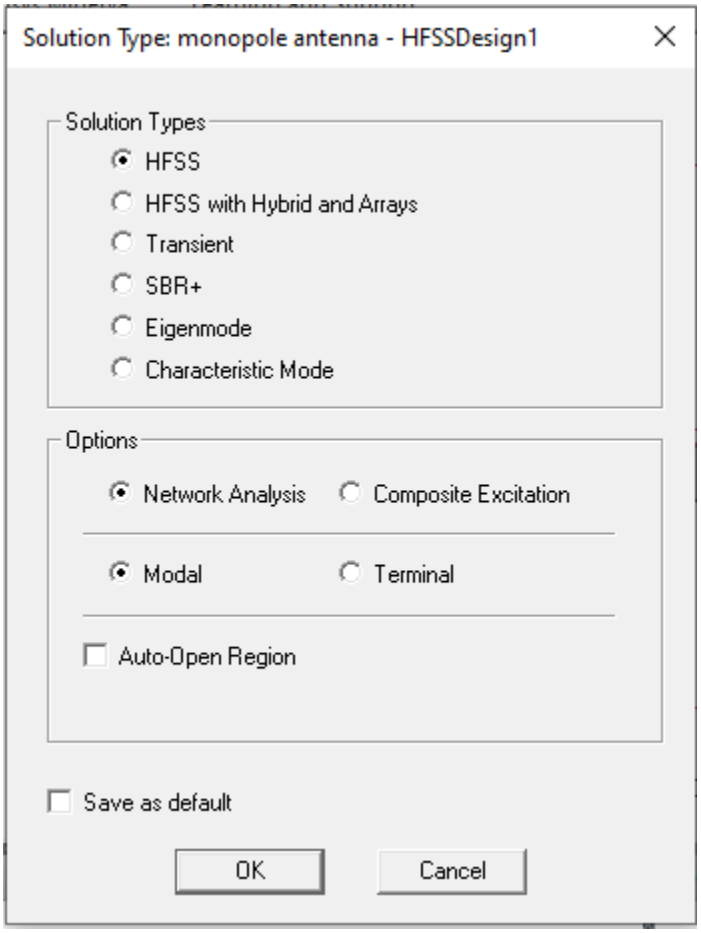
Step 1

Starting HFSS

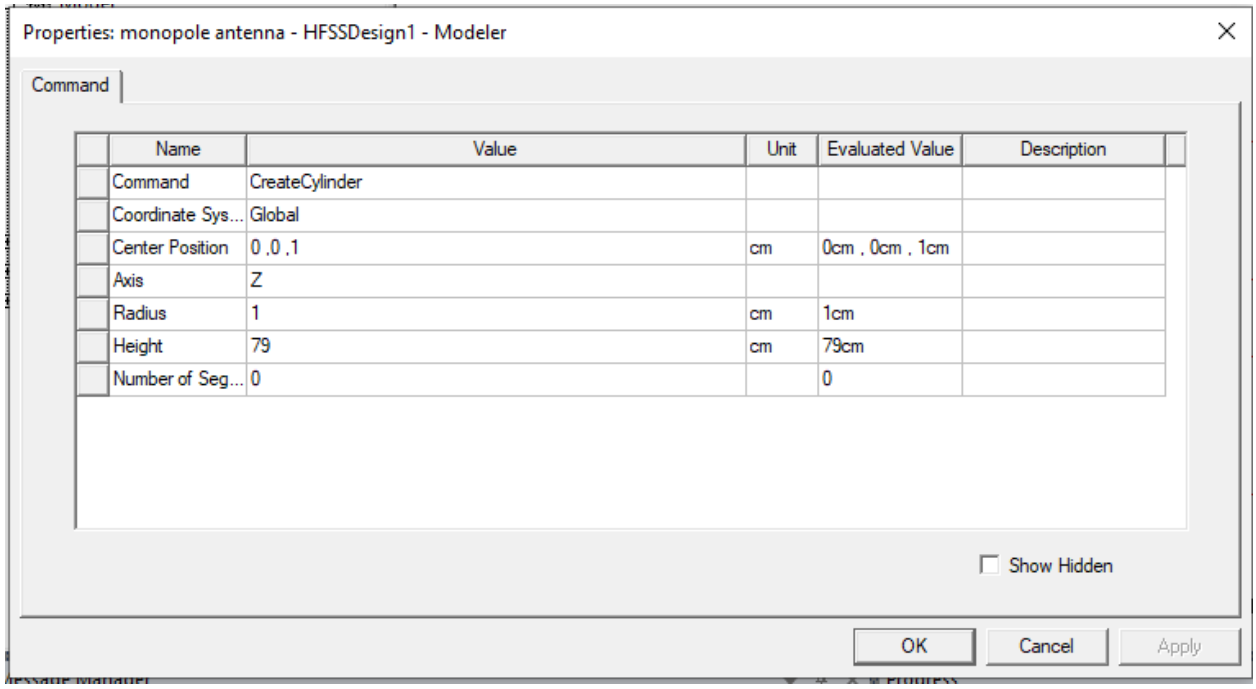
Step 2

Creating the antenna model

- From the HFSS menu select **Solution Type** as given below.

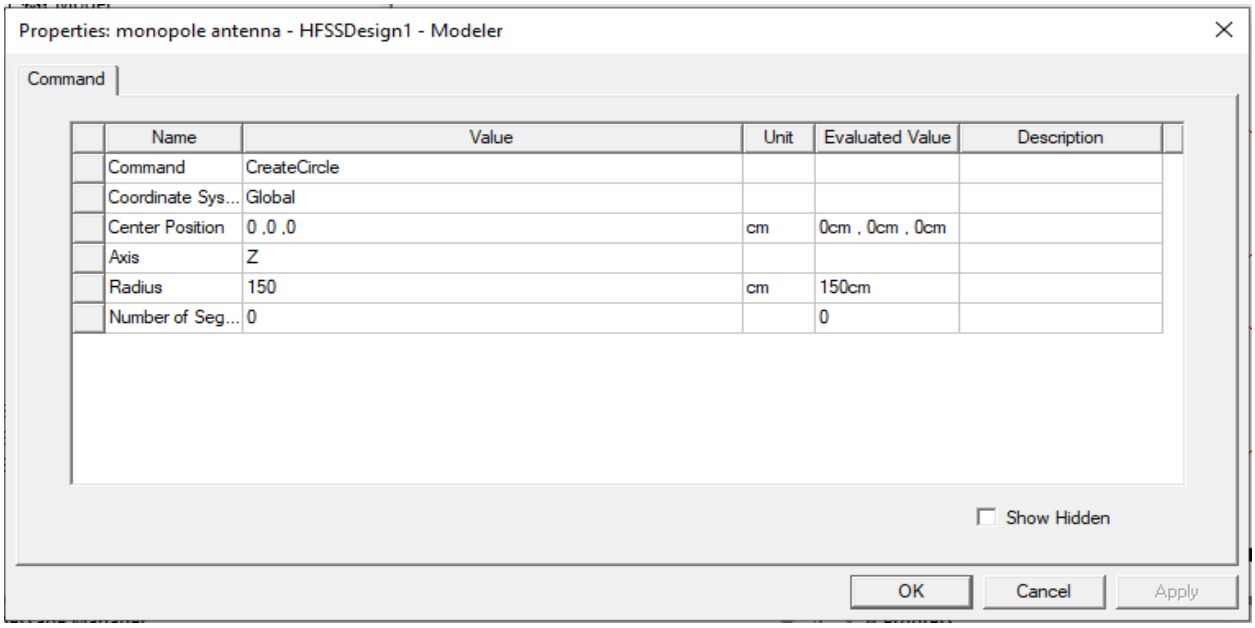


- Draw a cylinder with following coordinates (Units - cm),



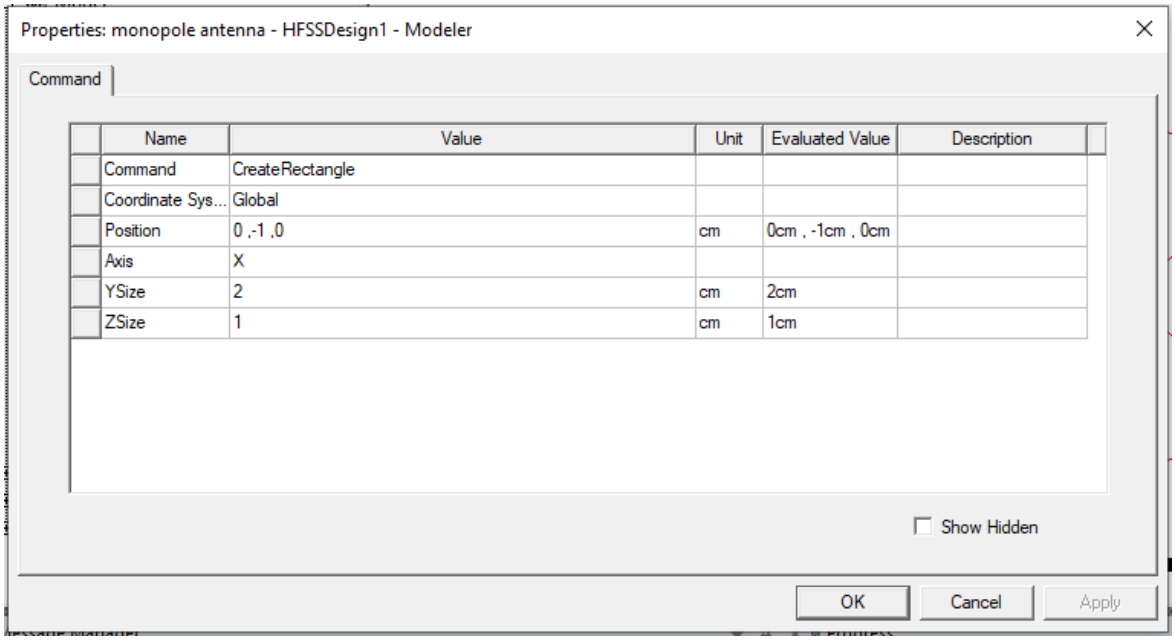
- Right click on cylinder1 in model section
- Select Assign material
- Type pec in the search by name space
- Select pec
- Click ok

- Draw a circle (i.e. ground plane) using the following co-ordinates,



- Right click on Circle1 in model section, assign boundary, perfect E

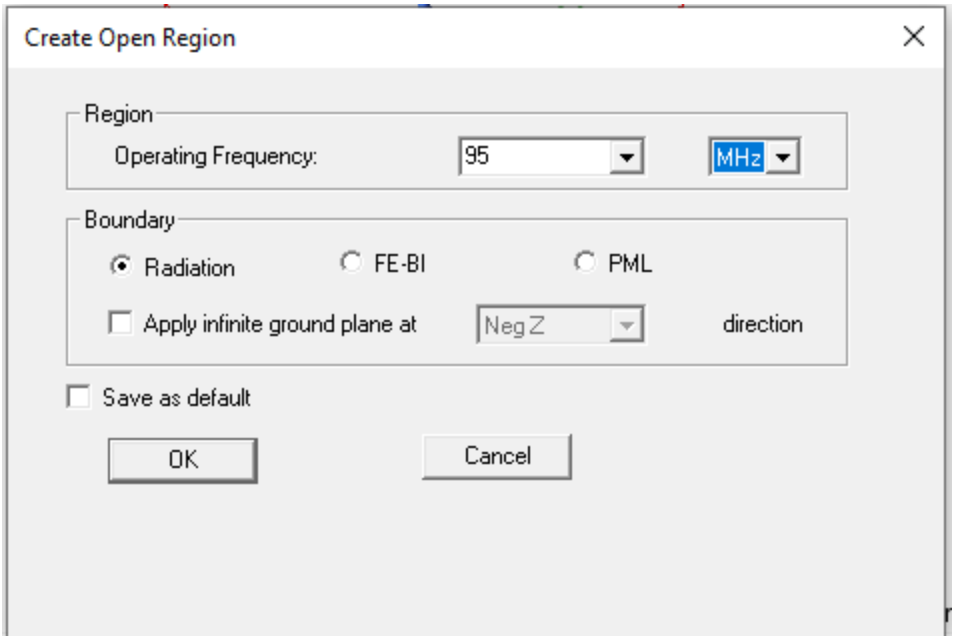
- Click on visibility tab on the right corner of the window, click on XY dropdown and select YZ
- Draw a rectangle (i.e. feed) with following co-ordinates,



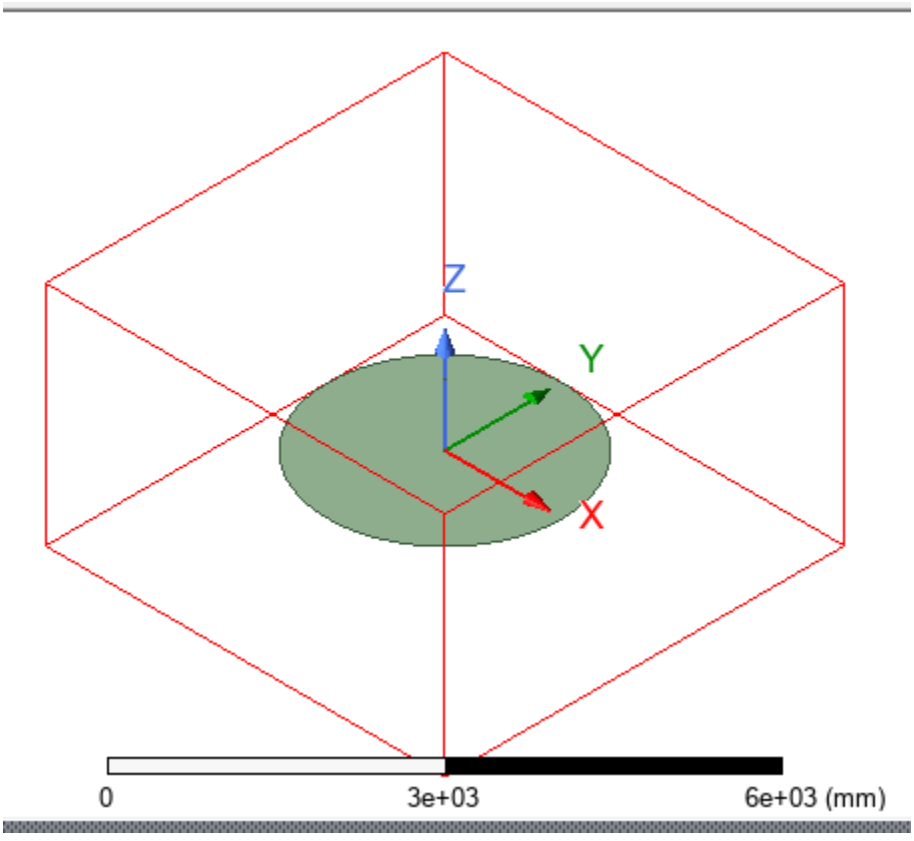
- Right click on rectangle1 in model section, assign excitation, port, lumped port, resistance 50 Ohms, next, click on dropdown under integration line and select choose new line
- Draw a line from bottom center to the top center of the rectangle, next, finish

- Click on visibility tab on the right corner of the window, click on YZ dropdown and select XY

- Right click in the work area, create open region, set operating frequency as 95MHz, select radiation boundary, press ok as given below



- Now your model will look like the figure below with dipole antenna in the middle and radiation box enclosing the antenna.

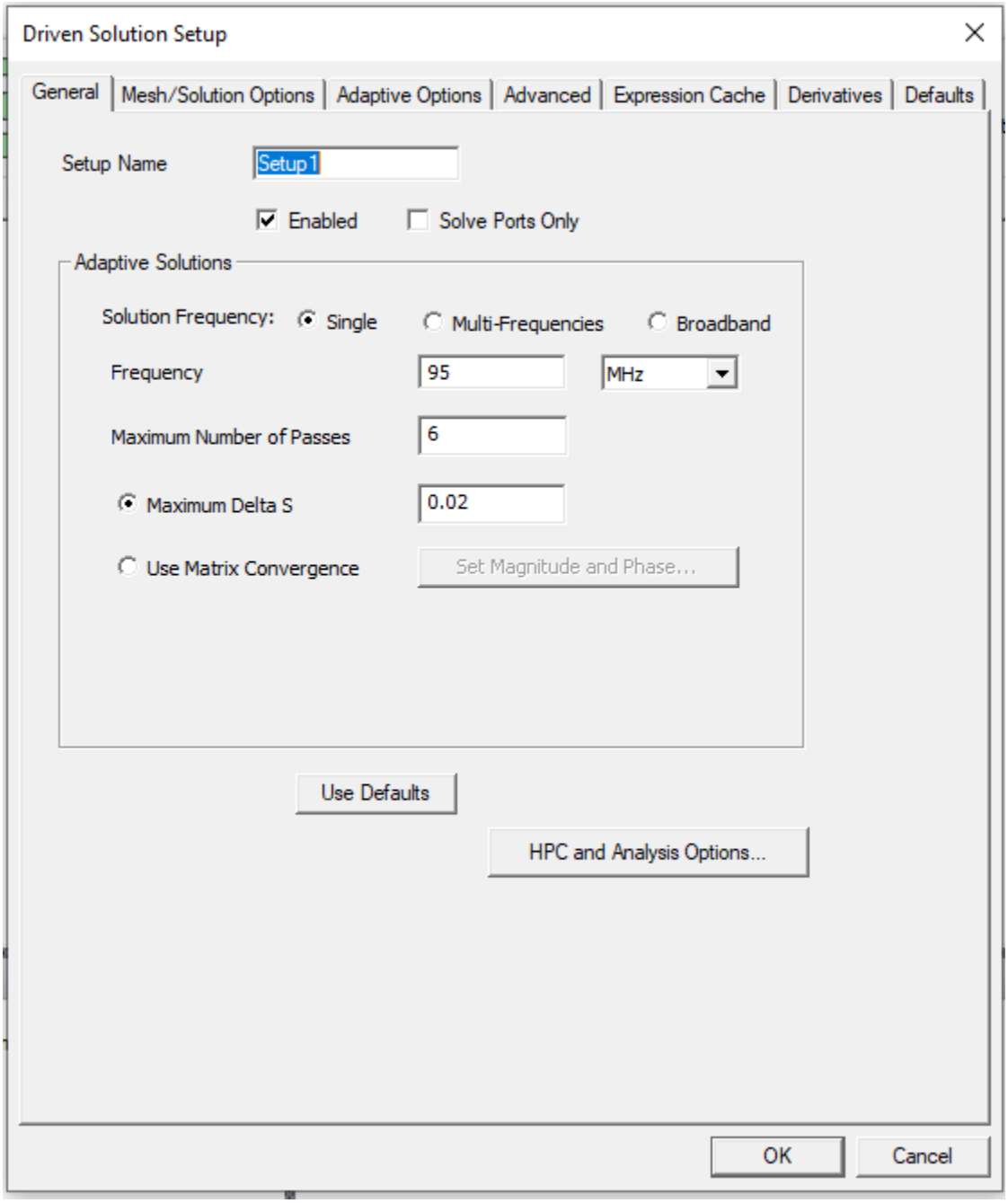


- Now your model is constructed along with radiation boundary.

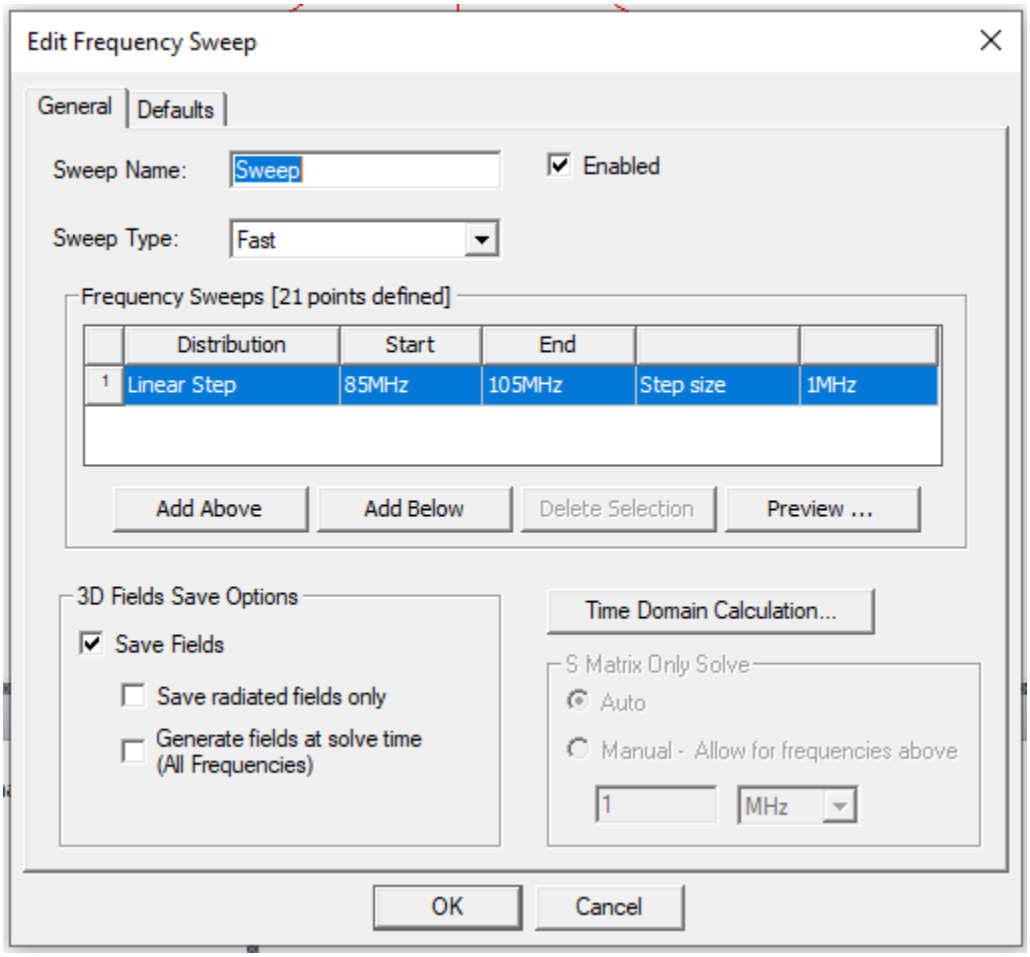
Step 3

Configurations (Settings)

- In the project manager section, Right click on analysis, add solution setup, advanced
- Enter the values as given below



- Click on ok
- After clicking ok, in the next window that appears enter the values as given below

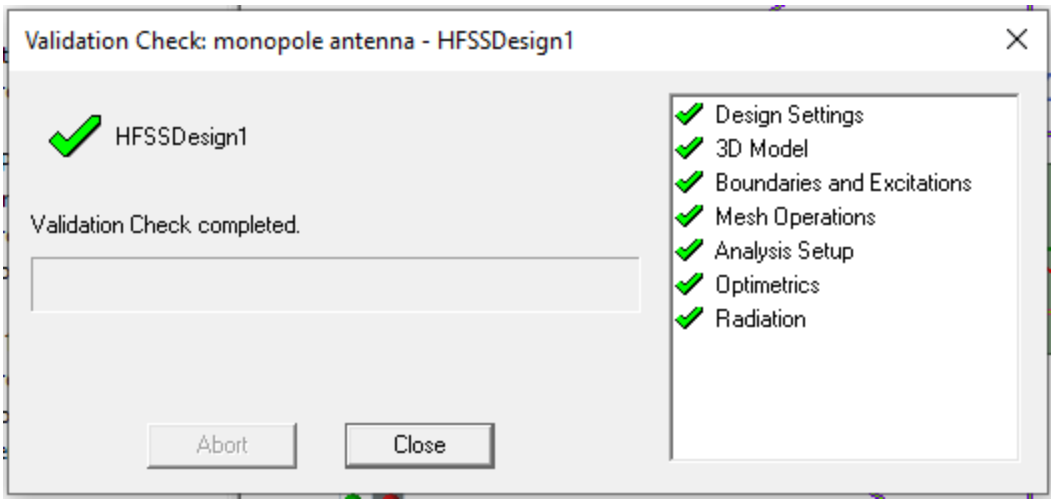


- Click ok

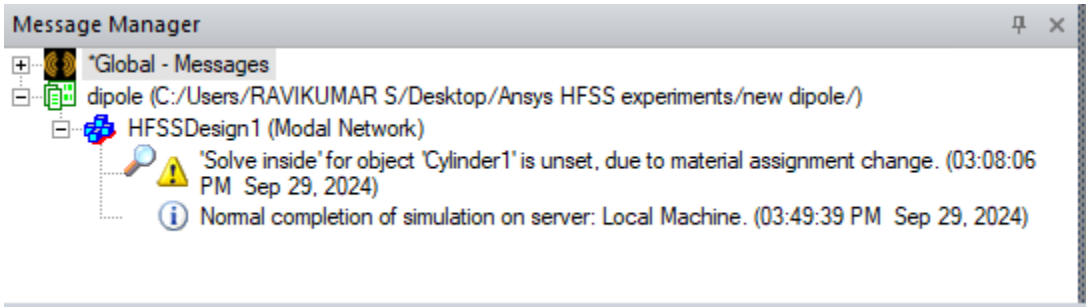
Step 4

Validation and Analysis

- Go to HFSS, Validation check
- The following window will appear if model created is correct



- Close the validation window
- Go to HFSS, Analyze all
- It will take some time to calculate results based on antenna complexity and computer configuration which is running the software
- If simulation goes well then you will receive a message in the message manager section as Normal completion of simulation on server:Local machine as given in figure below.

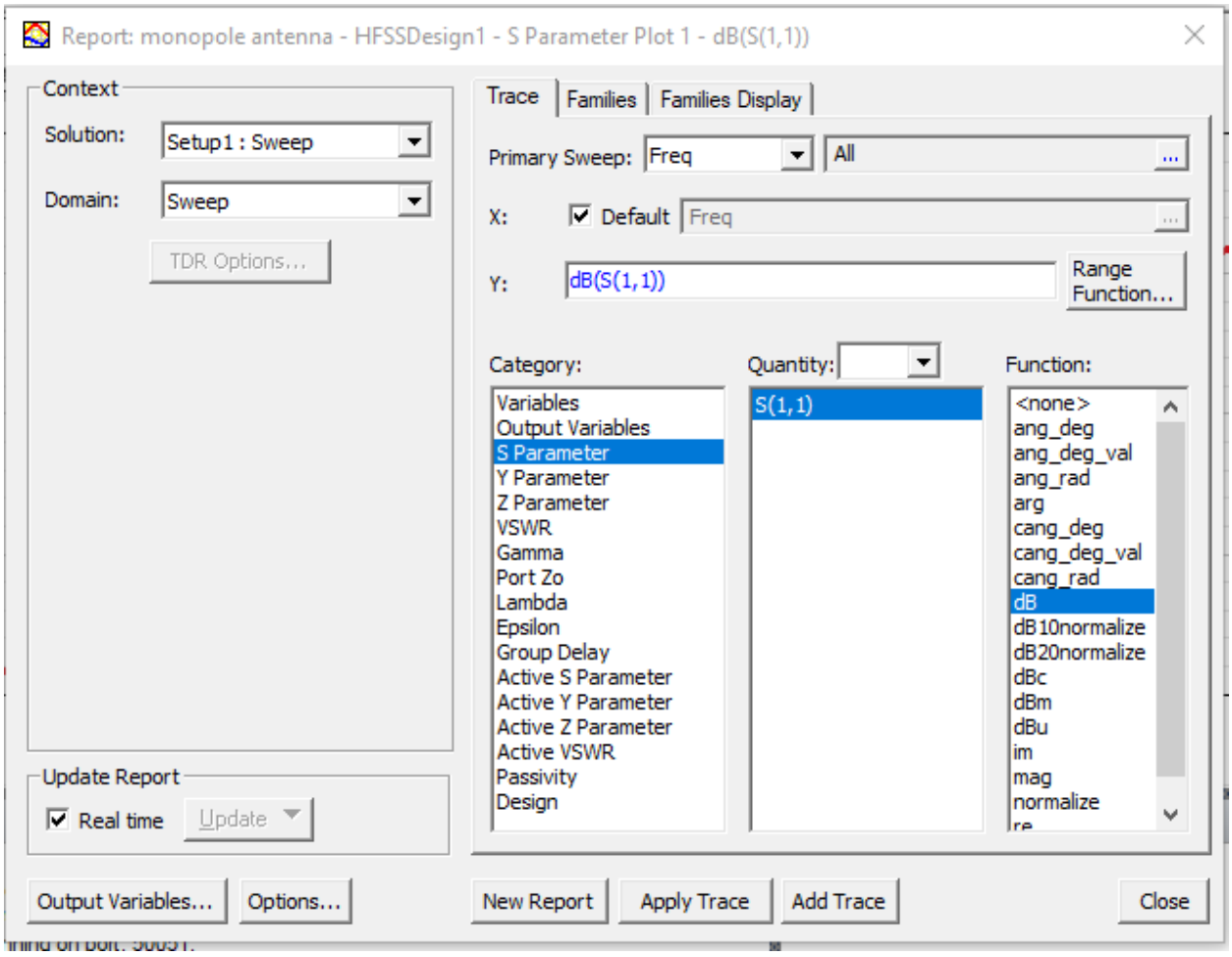


- If any errors are there then such error will be shown in message manager section. They have to be resolved.

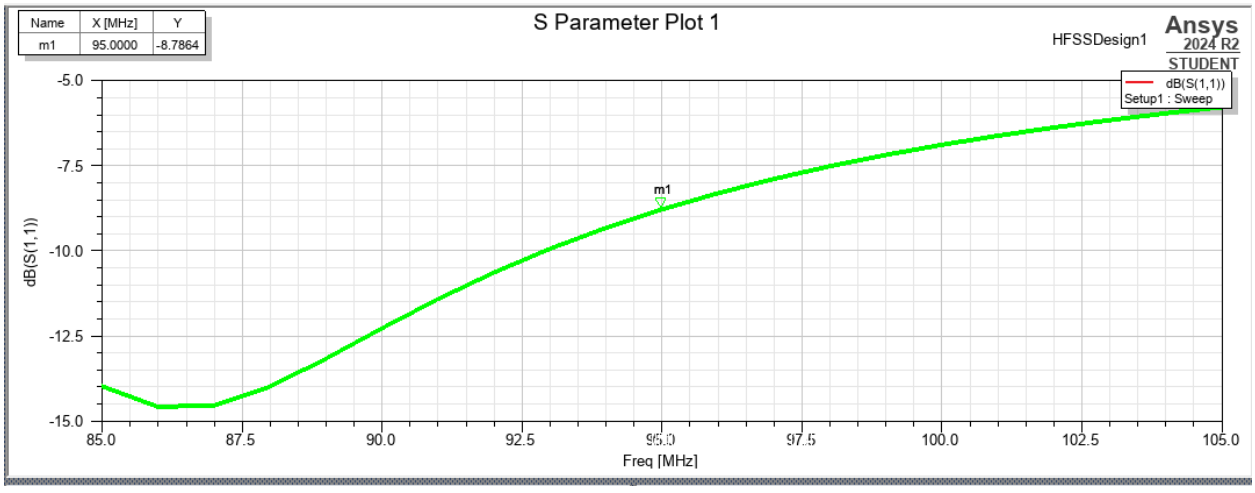
Step 5

Results

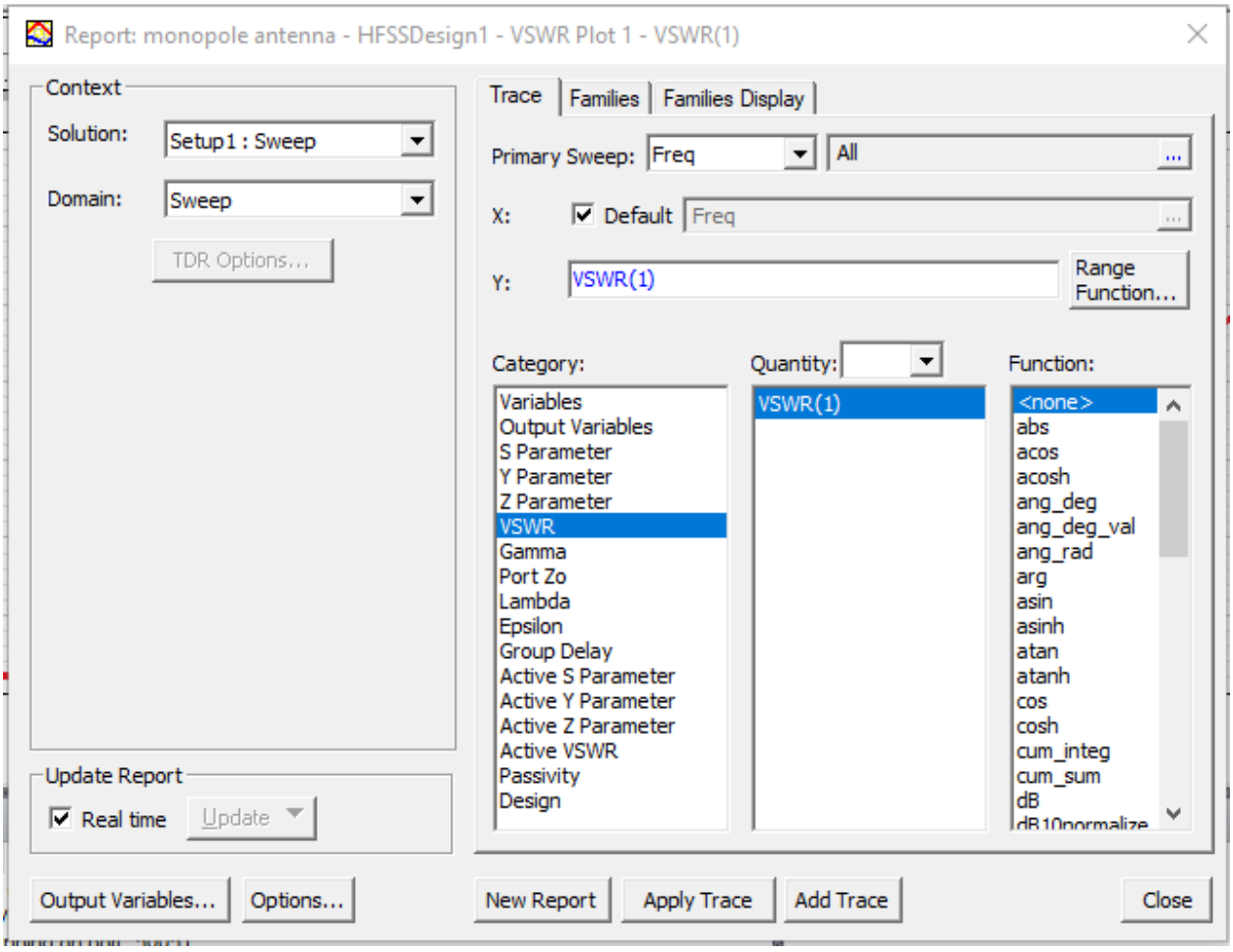
- Right click on results in project manager section, create modal solution data report, rectangular plot.
- The following window should appear,



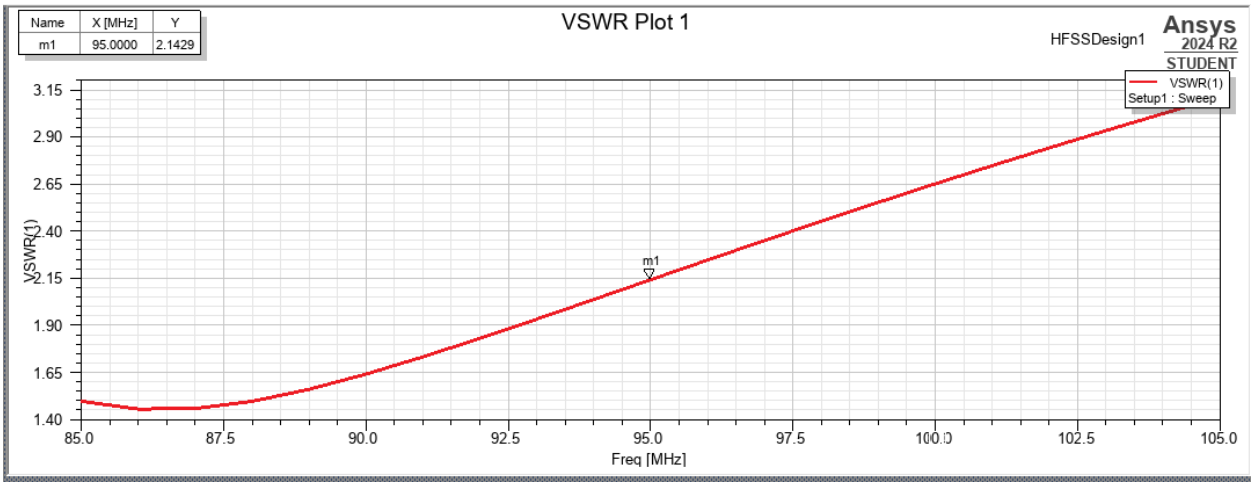
- Click on new report
- The following window will appear showing **S11 parameter (Input reflection coefficient)**



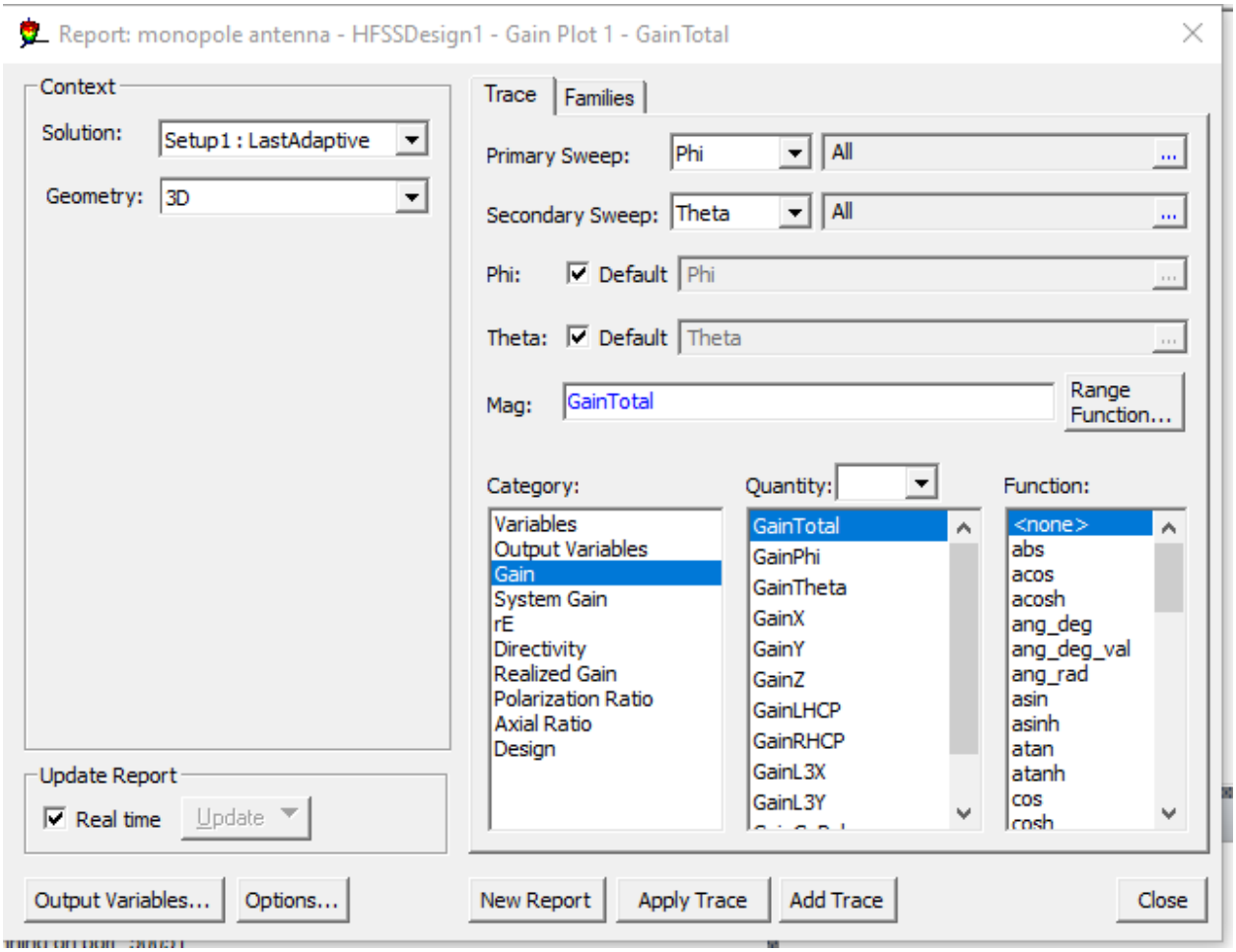
- In the same way Right click on results in project manager section, create modal solution data report, rectangular plot, select VSWR in category section, click on new report as given below



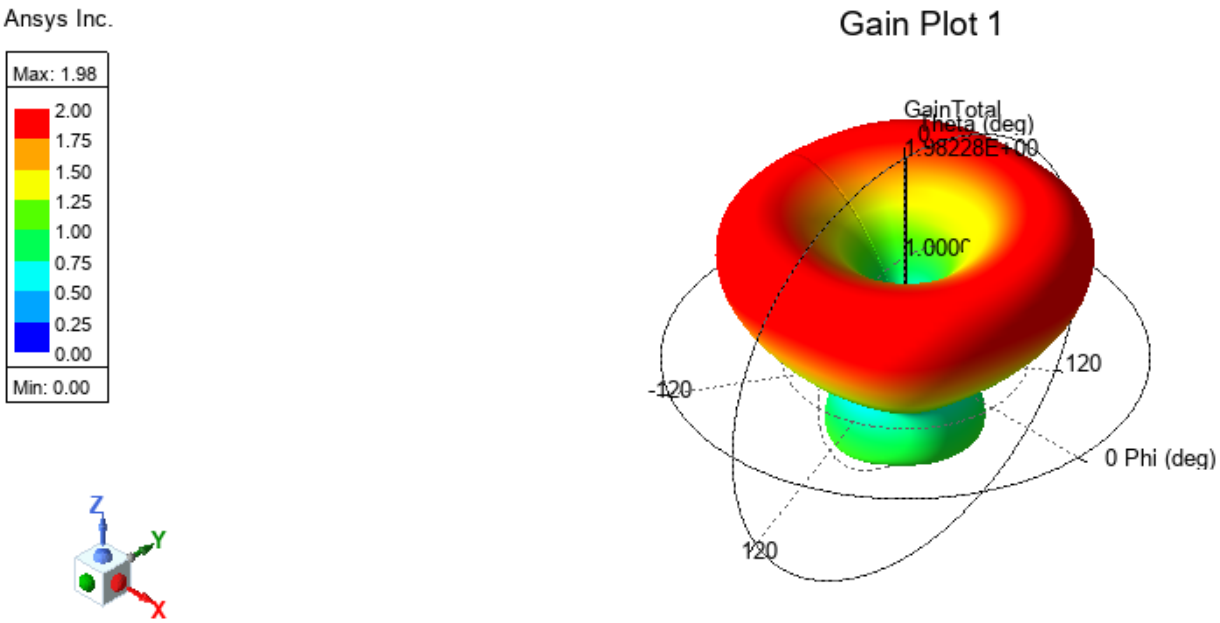
- The **VSWR** result will appear as given below,



- Right click on results, create far fields report, 3D polar plot
- The following window will open



- Click on new report
- The following 3D radiation pattern will appear,



As seen in the S parameter graph there is a slight deviation in the operating frequency of the antenna from desired. The desired operating frequency is 95MHz and the current operating frequency as indicated by the S parameter graph is 86.5MHz. So, by slightly adjusting design parameters such as length, radius of wire or length of feed the operating frequency can be bought closer to the desired frequency.

Tabular Column

Sl No	Sl No	Before Tuning	After Tuning
1	Length of monopole	79cm	
2	S11(in dB) at 95MHz		
3	VSWR at 95MHz		
4	Bandwidth in Hz (at -10dB points of S11)		
5	Gain		

Note:

Length of the monopole can be reduced to get lowest S parameter value at 95MHz thereby making antenna to operate at 95MHz.

Assignment

Assignment

Experiment 7

Probefed Rectangular Microstrip Patch Antenna

Aim:-

To design Coaxial Probefed rectangular microstrip patch antenna and study of its parameters.

Theory:-

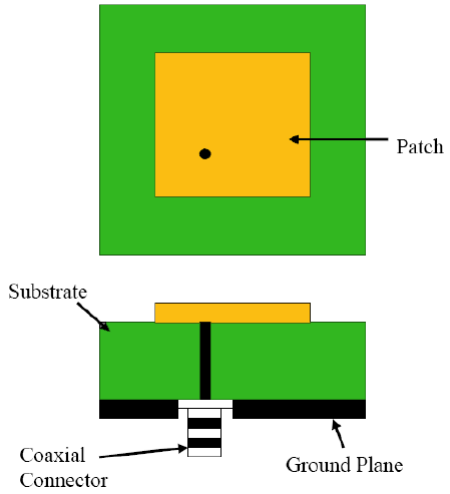
Microstrip patch antennas have gained a lot of popularity over the last three decades. This is due to their low-profile structures, cost effectiveness, ease of fabrication and ease of installation. It is easy to print microstrip antennas to a circuit board. In high performance systems with constraints of size, weight and cost, microstrip antennas prove to have numerous applications. Due to these advantages, microstrip patch antennas are becoming widespread in mobile radio and wireless communications.

Characteristics of Microstrip Patch Antennas

Consider the microstrip patch antenna shown in figure. The primary configuration consists of three structures:

- Patch
- Substrate
- Ground

The patch is a thin metallic strip that is placed a small fraction of a wavelength above the ground. The region between the patch and ground is called substrate. The substrate is ideally a dielectric material. Length of rectangular patch is less than λ/2. The microstrip patch antenna is designed such that it gives a broadside pattern i.e. maximum radiation in the direction normal to the patch. To achieve a broadside radiation pattern, the patch is excited from beneath.



Design Steps :-

Given

Resonant Frequency(fo) – 2.4GHz

Relative permittivity of substrate(εr) – 4.4

Thickness of substrate(h) – 1.6 mm

Necessary design parameters

The width and length of the patch is calculated by,

$$Width = \frac{c}{2f_o\sqrt{\frac{\varepsilon_R+1}{2}}}; \quad \varepsilon_{eff} = \frac{\varepsilon_R+1}{2} + \frac{\varepsilon_R-1}{2}\left[\frac{1}{\sqrt{1+12\left(\frac{h}{W}\right)}}\right]$$

$$Length = \frac{c}{2f_o\sqrt{\varepsilon_{eff}}} - 0.824h\left(\frac{(\varepsilon_{eff}+0.3)\left(\frac{W}{h}+0.264\right)}{(\varepsilon_{eff}-0.258)\left(\frac{W}{h}+0.8\right)}\right)$$

f_0 is the Resonance Frequency

W is the Width of the Patch

L is the Length of the Patch

h is the thickness of substrate

ε_r is the relative Permittivity of the dielectric substrate

c is the Speed of light

Calculation Result:-

By substituting the values of fo, εr and h we get length and width of the patch as given below,

Width: 38.01 mm ≈ 38 mm

Length: 29.42 mm ≈ 29.4 mm

Steps for simulation

Step 1: Starting HFSS

Step 2: Creating the antenna model

Step 3: Configurations (Settings)

Step 4: Validation and analysis

Step 5: Results

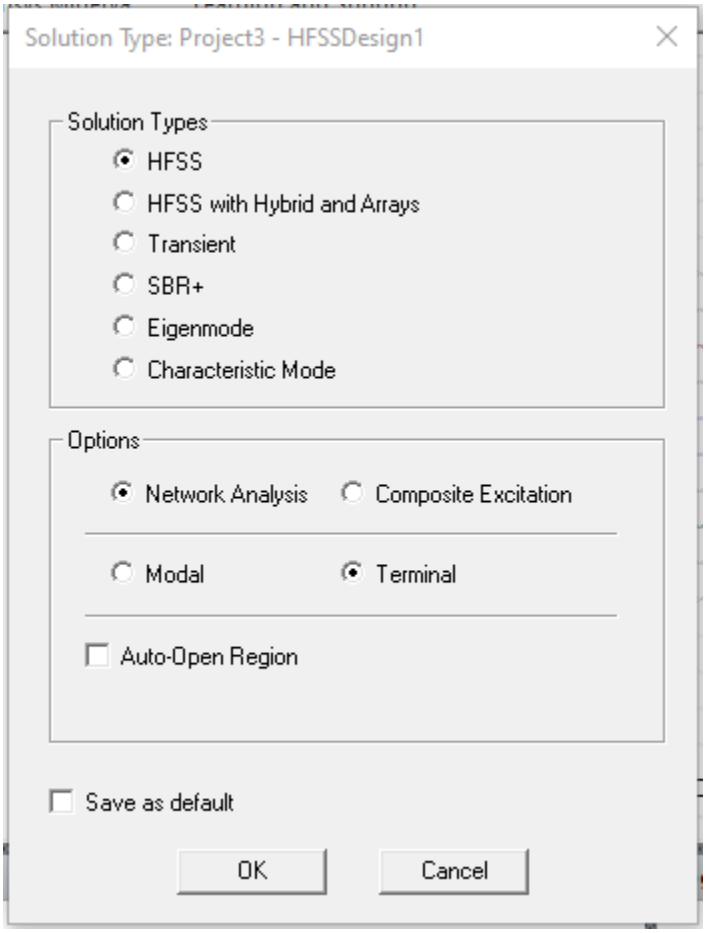
Step 1

Starting HFSS

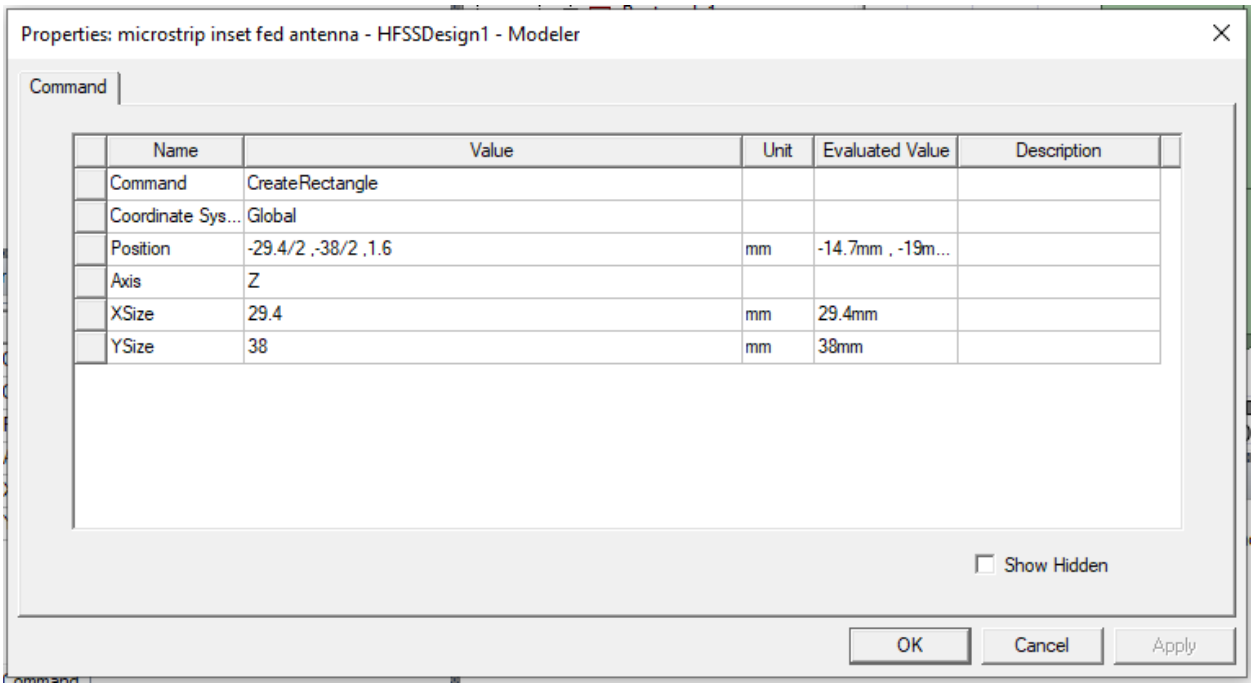
Step 2

Creating the antenna model

- From the HFSS menu select **Solution Type** as given below.

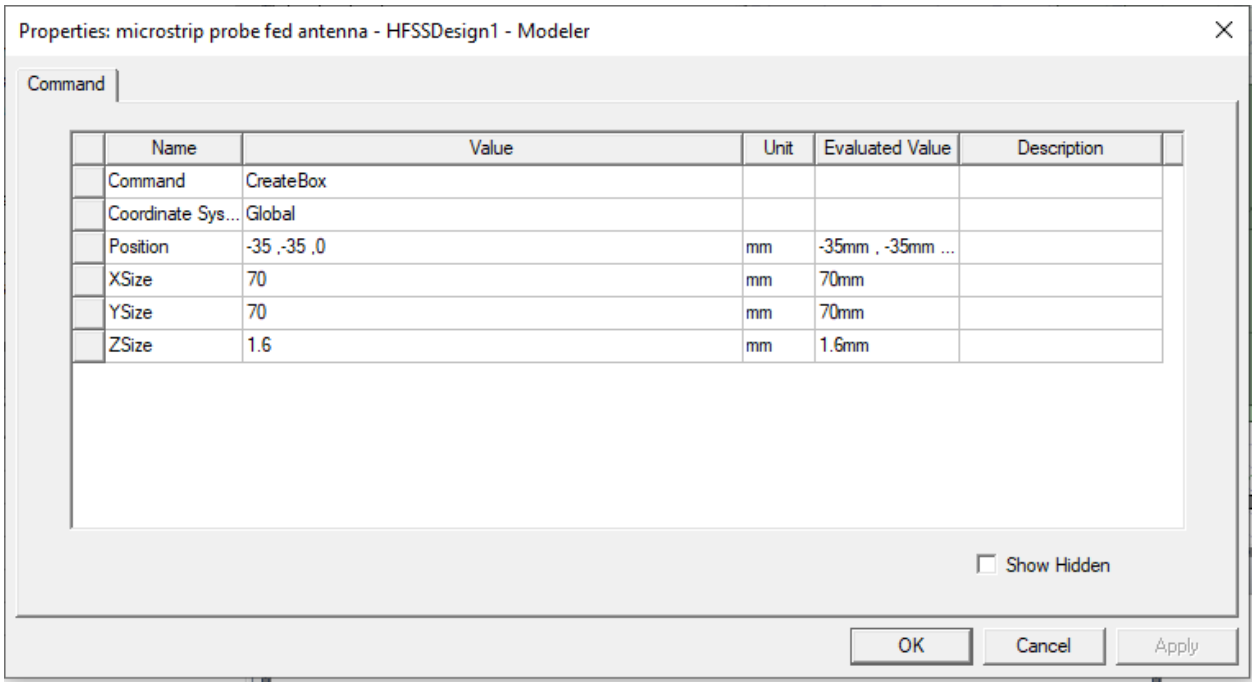


- Draw a rectangle(patch) with following coordinates,



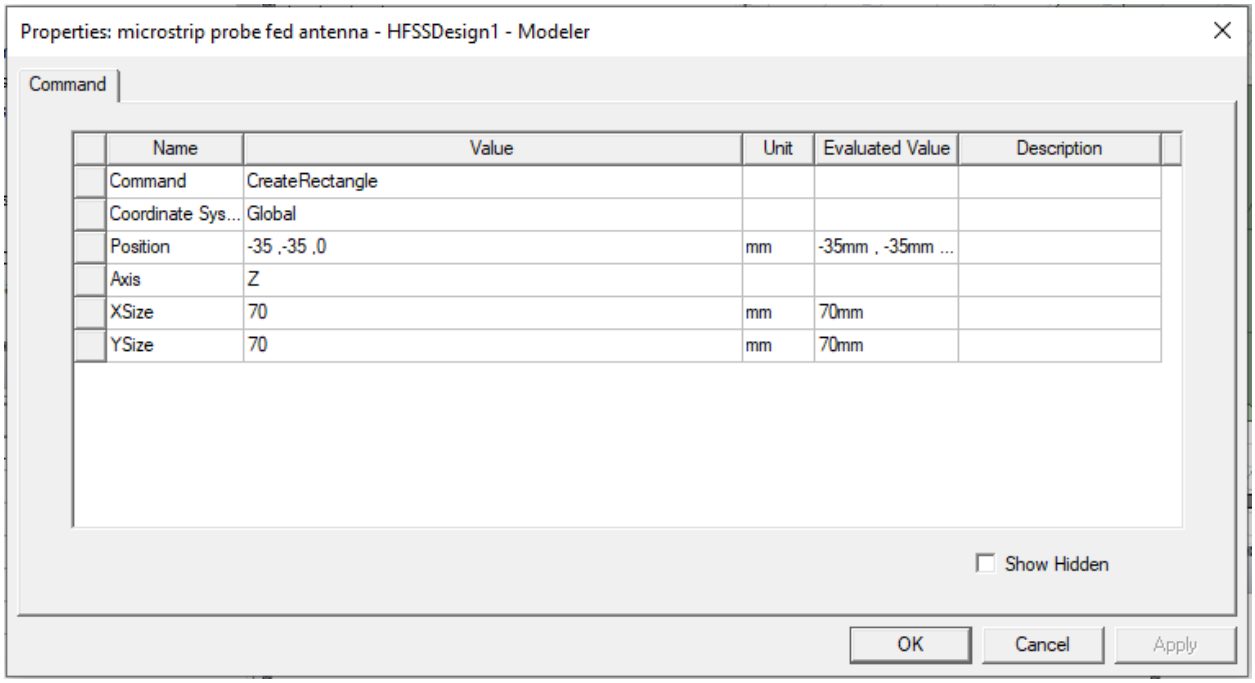
- Right click on rectangle1 in model section, Assign boundary, Perfect E
- Click ok

- Draw a box(substrate) with dimensions as given below,



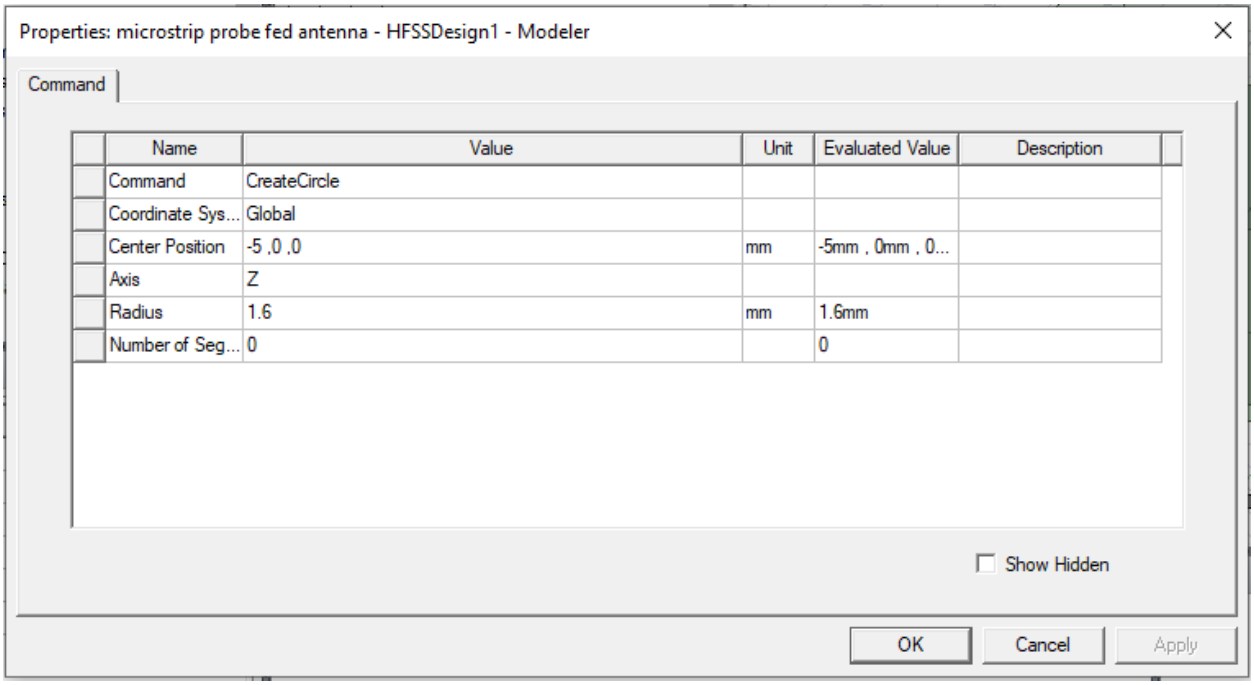
- Right click on box1 in model section, assign material, type fr4 in search section, select fr4, press ok

- Draw a rectangle (ground plane) with the following co-ordinates,

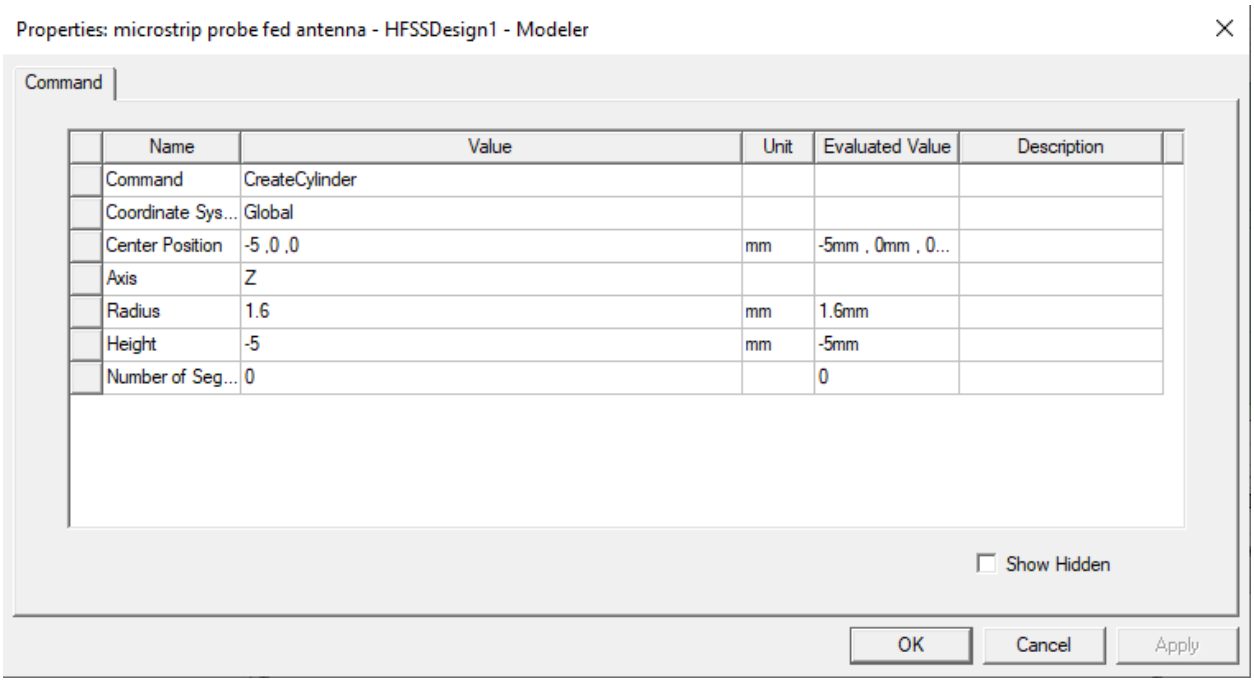


- Right click on rectangle2 in model section, Assign boundary, Perfect E, Click ok

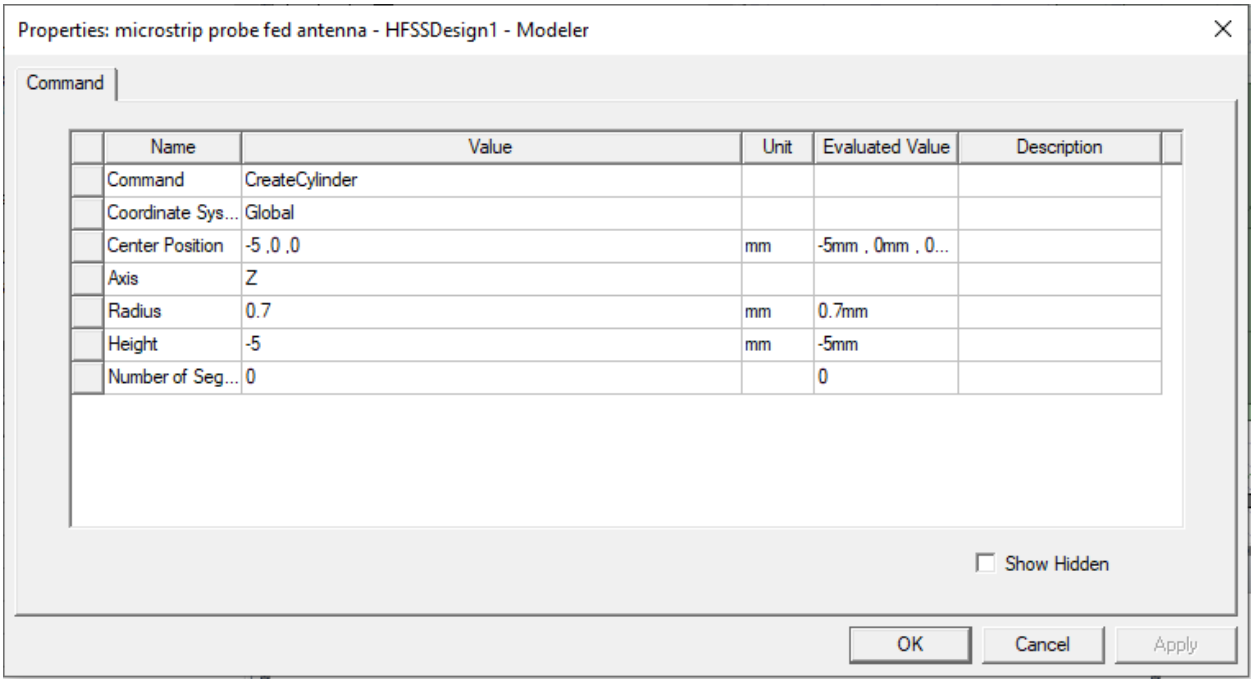
- Draw a circle with following dimensions,



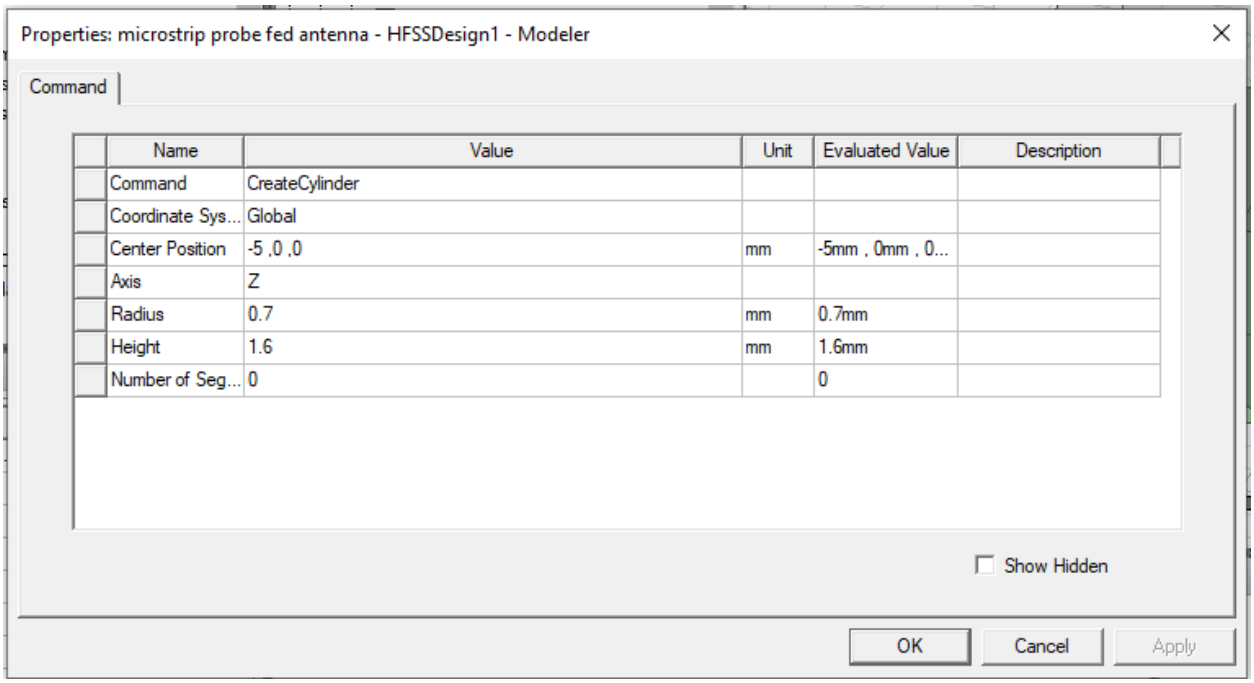
- Subtract rectangle2 and circle1
- Draw a cylinder (co-axial cable outer layer) with following dimensions,



- Draw one more cylinder(coaxial cable inner conductor) with following dimensions,

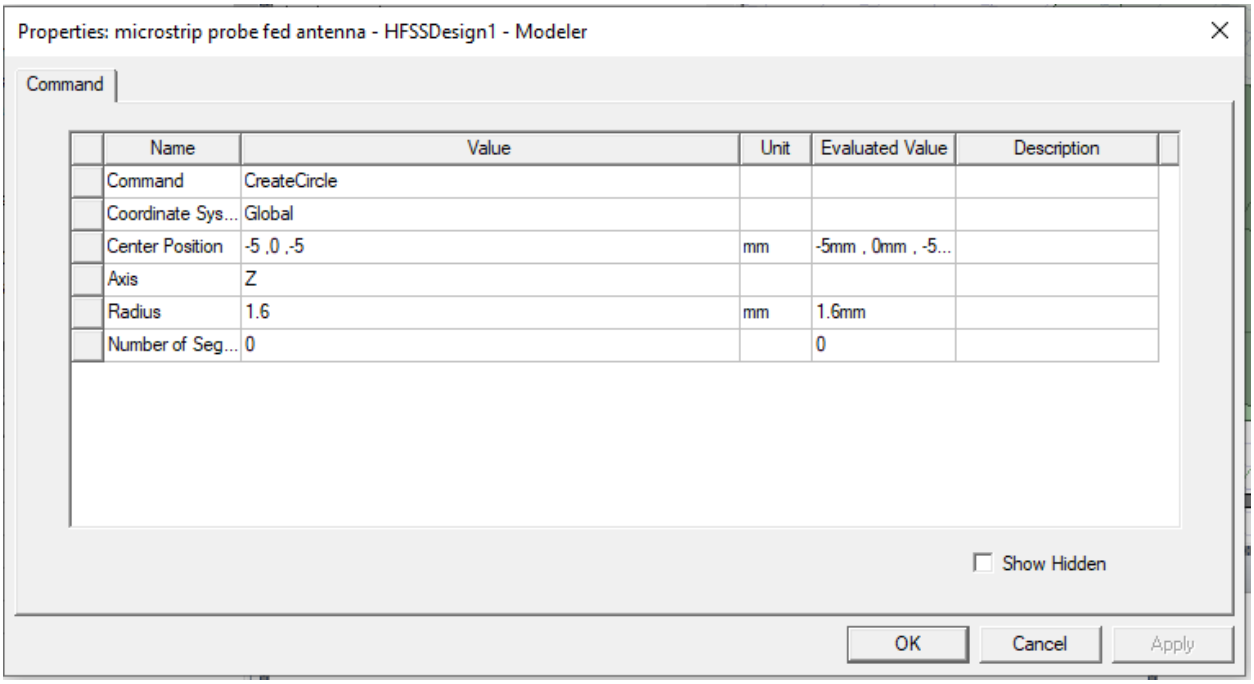


- Assign material as **pec** to cylinder2
- Draw one more cylinder(coaxial cable inner conductor-probe) with following dimensions,

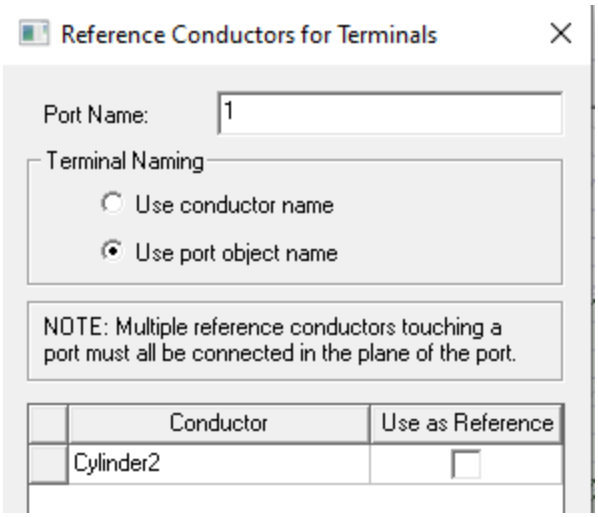


- Assign material as pec to cylinder3

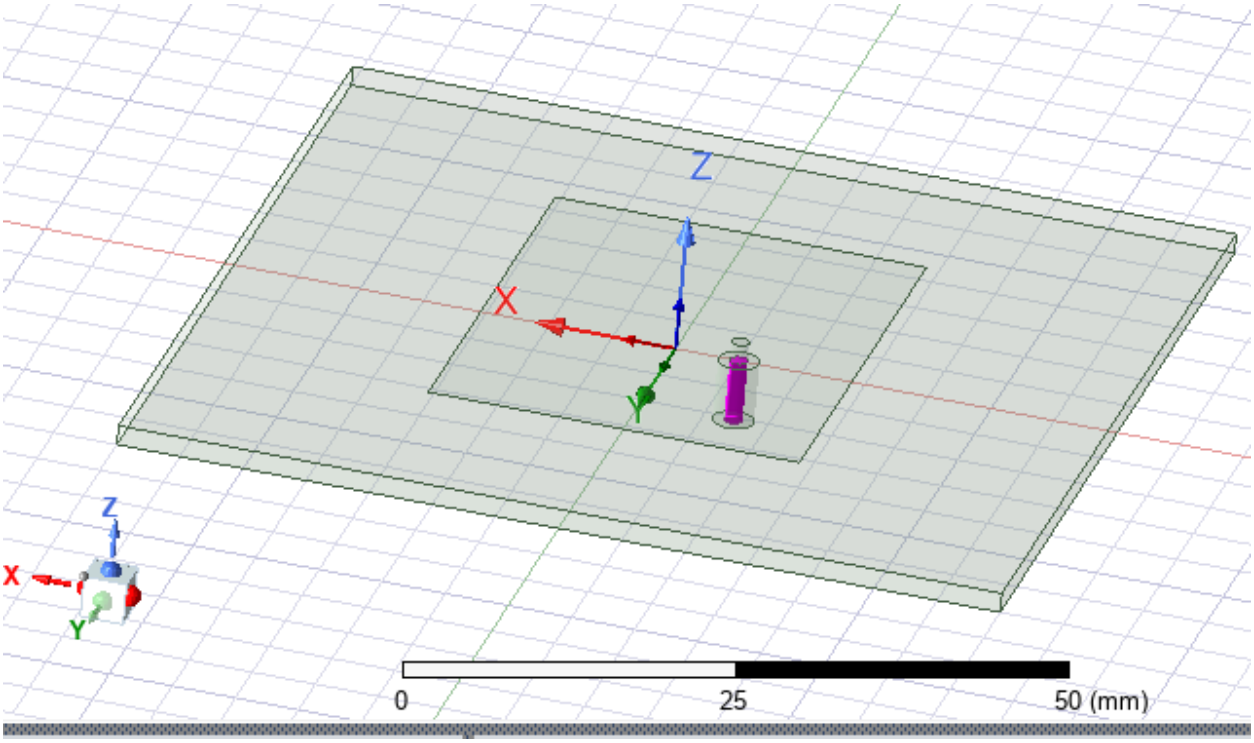
- Draw a circle with following dimensions,



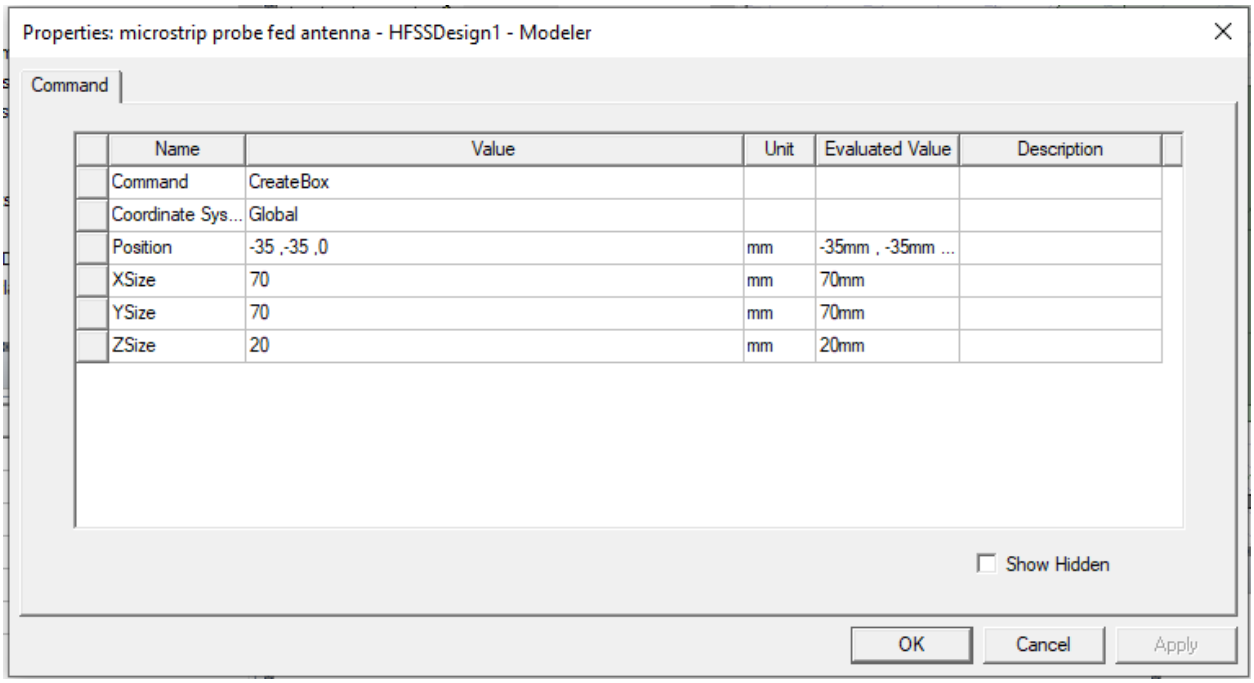
- Right click on circle2 in model section, assign excitation, port, terminal wave port
- Select option as given below and press ok



Now the model will look like as given below,



- Create a box(radiation box) with dimensions as given below,



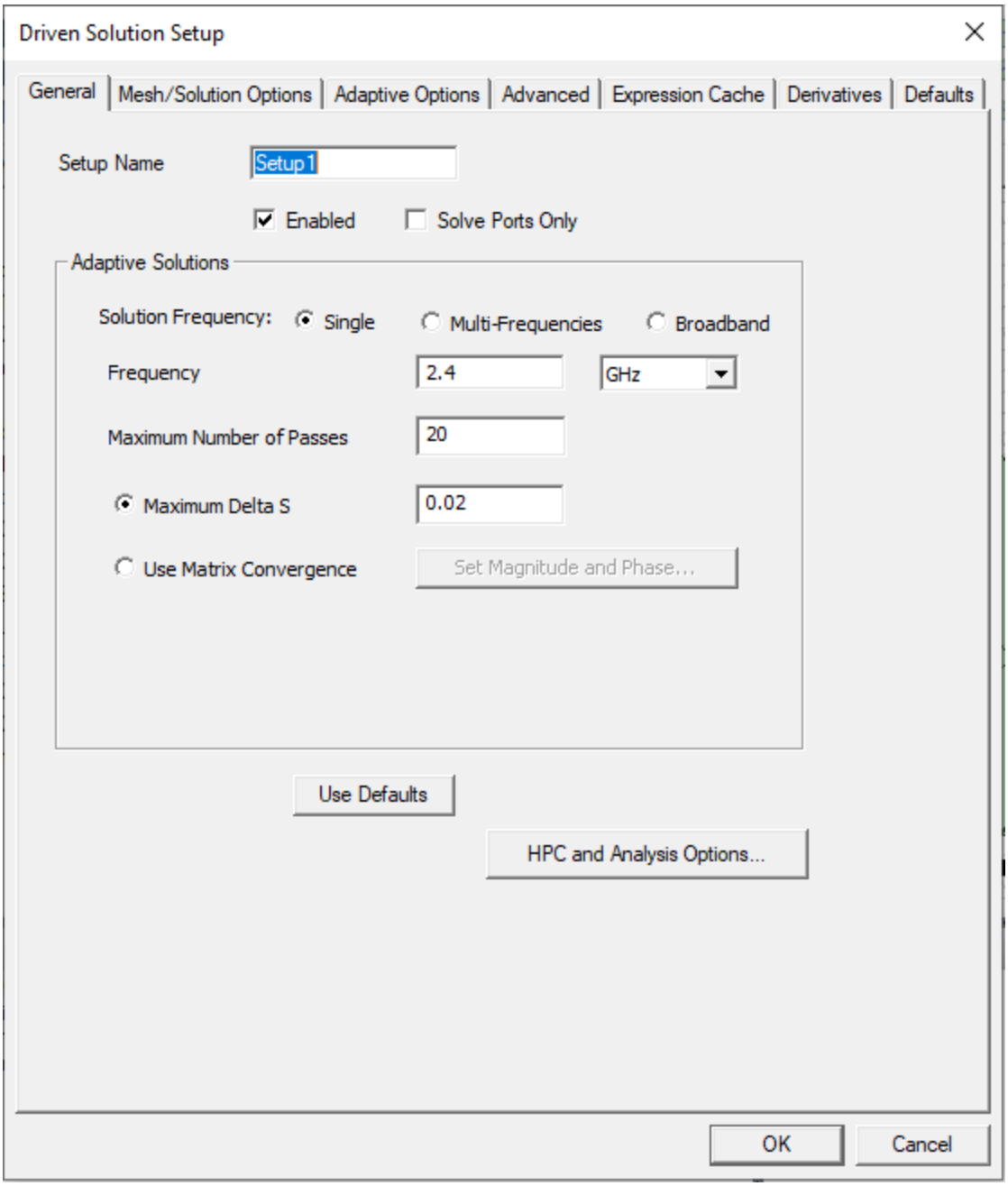
- Assign the material of box2 as “air”
- Right click in the work area, selection mode, faces
- Now you can select individual faces of an object instead of an entire object
- Now select 5 faces of box2 i.e. radiation box using control button except the bottom face
- Right click, assign boundary, radiation, ok

- Now your model is constructed along with radiation boundary.

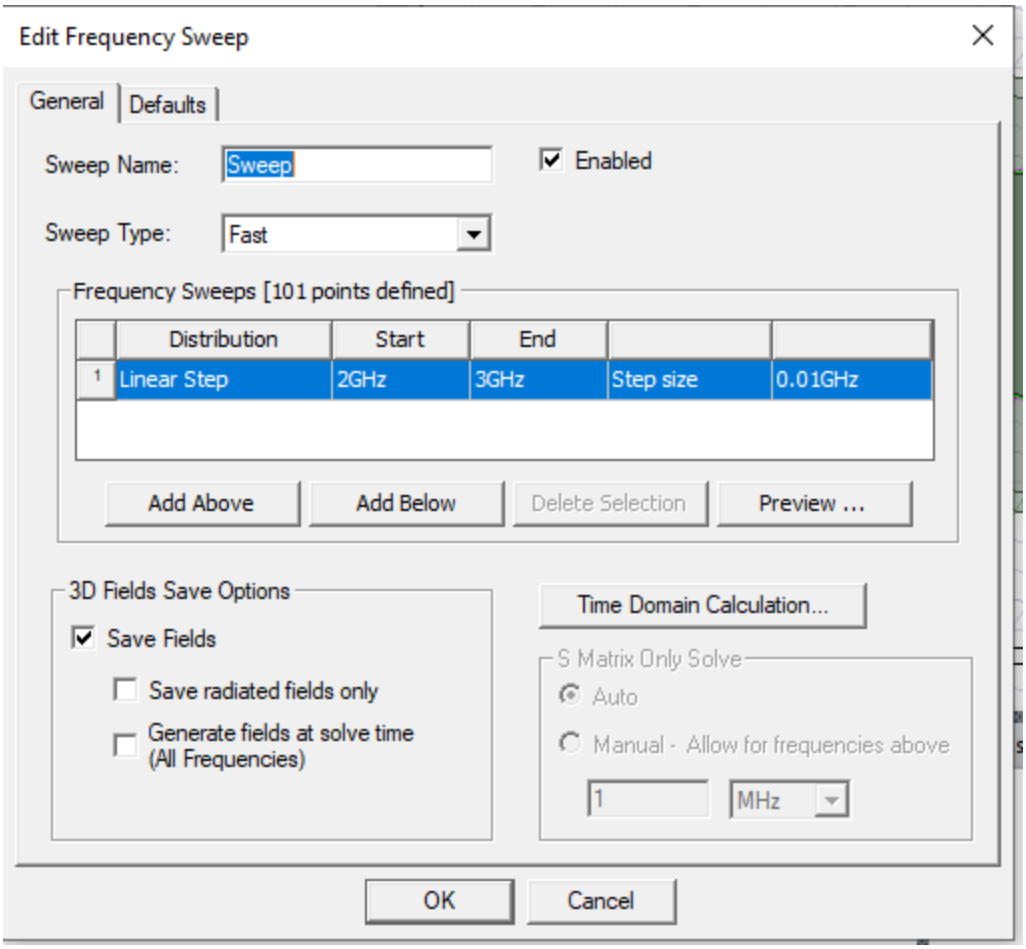
Step 3

Configurations (Settings)

- In the project manager section, Right click on analysis, add solution setup, advanced
- Enter the values as given below



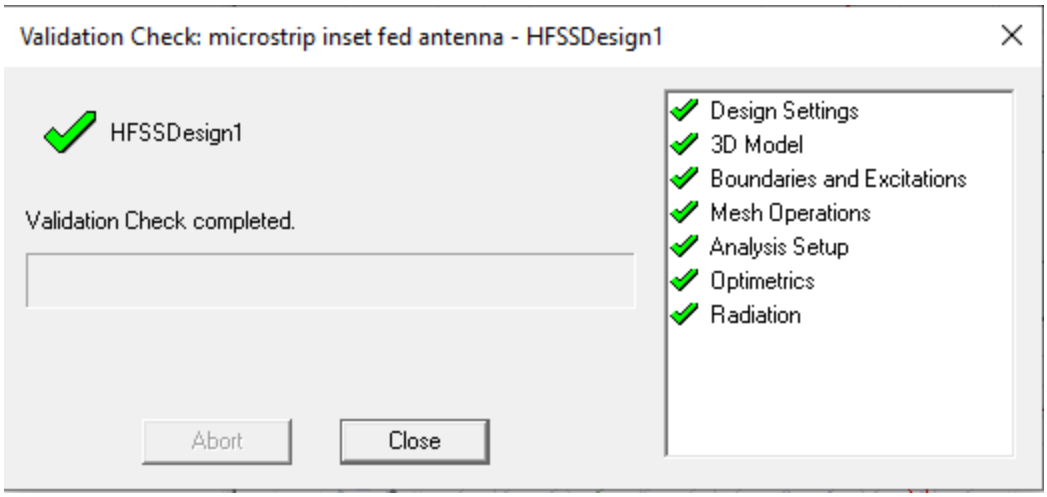
- Click on ok
- In the next window that appears after clicking ok, enter the values as given below



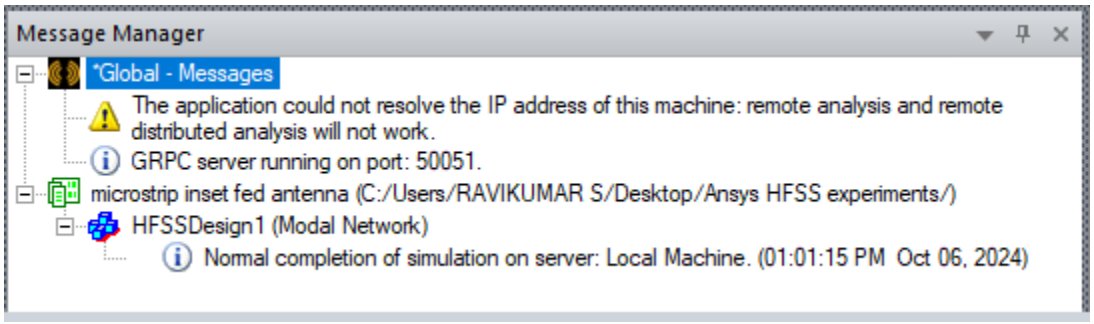
Step 4

Validation and Analysis

- Go to HFSS, Validation check
- The following window will appear if model created is correct



- Close the validation window
- Go to HFSS, Analyze all
- It will take some time to calculate results based on antenna complexity and computer configuration which is running the software
- If simulation goes well then you will receive a message in the message manager section as Normal completion of simulation on server:Local machine as given in figure below.

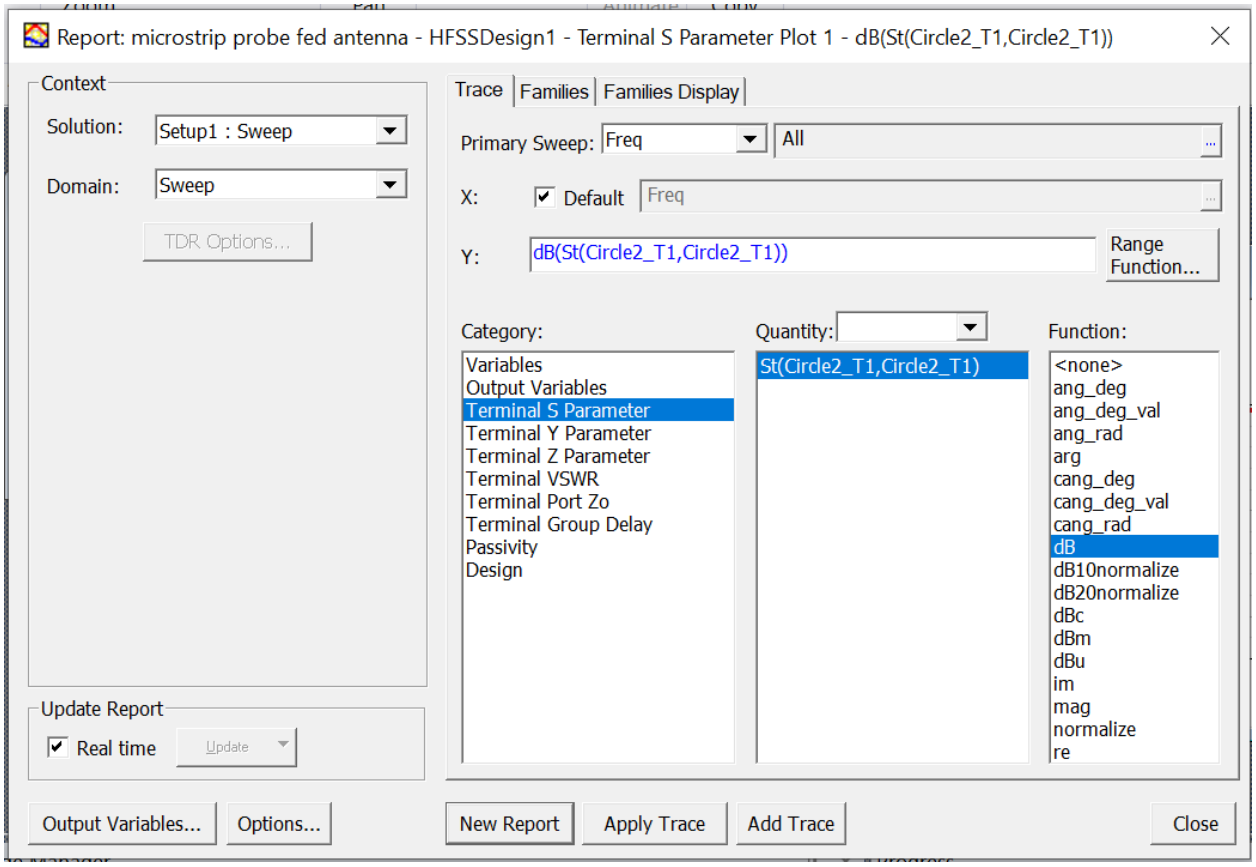


- If any errors are there then such error will be shown in message manager section. They have to be resolved.

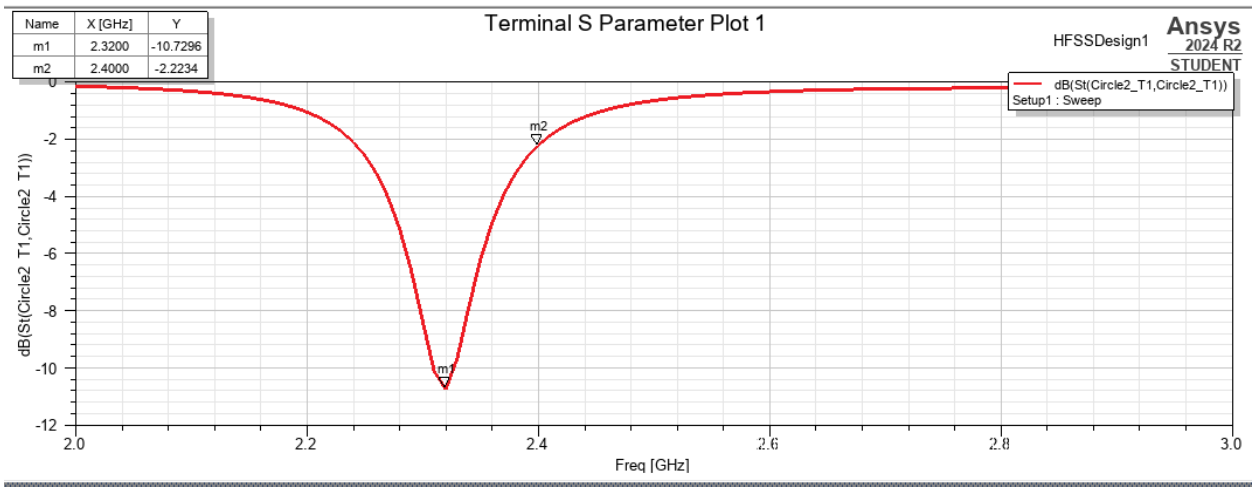
Step 5

Results

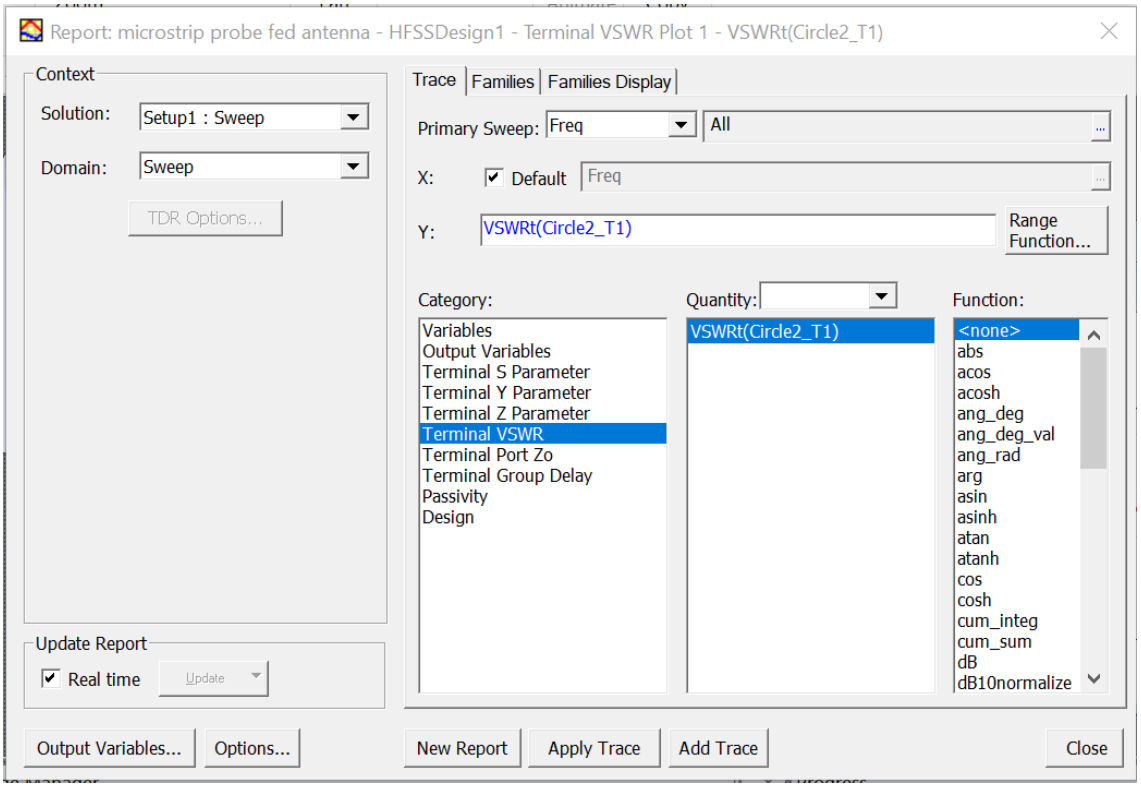
- Right click on results in project manager section, create terminal solution data report, rectangular plot,
- The following window should appear,



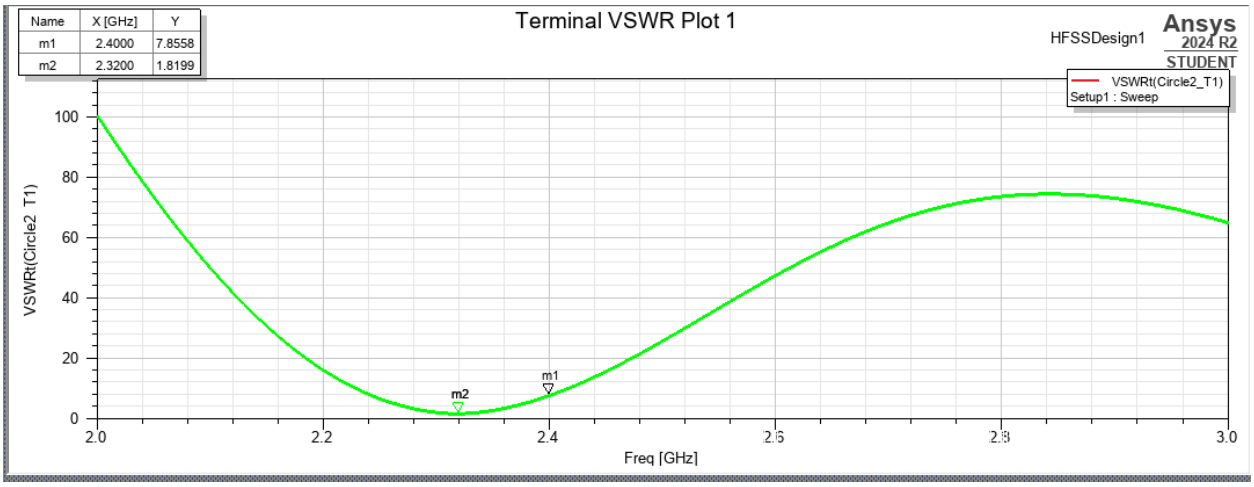
- Click on new report
- The following window will appear showing S11 parameter (Input reflection coefficient)



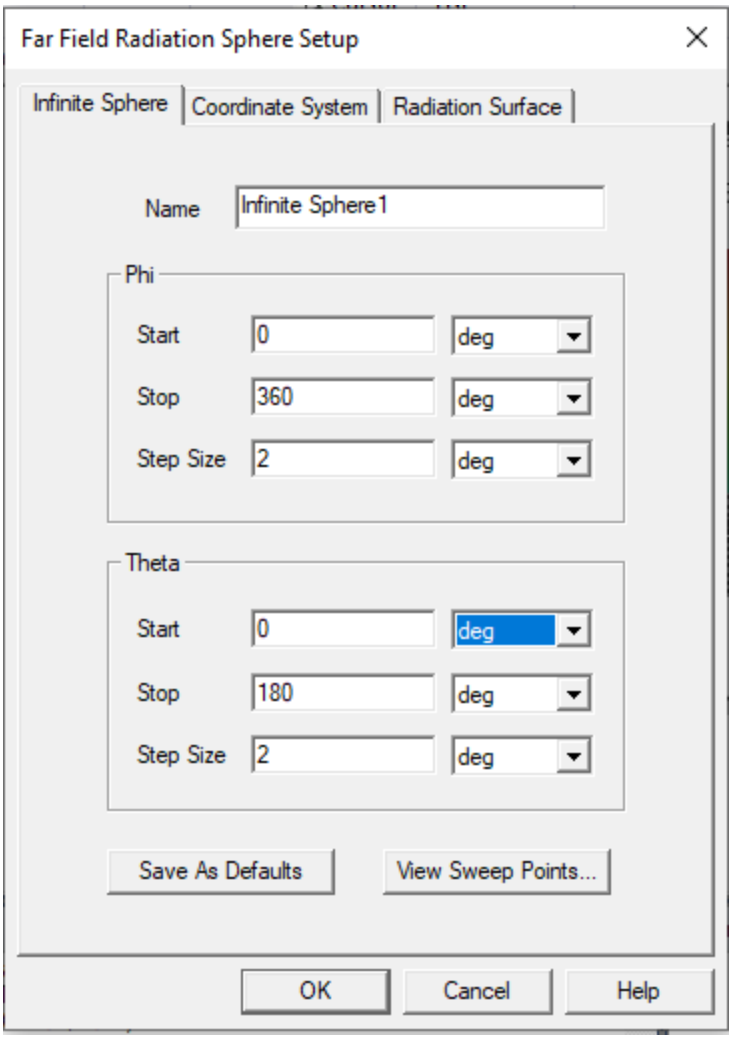
- In the same way Right click on results in project manager section, create terminal solution data report, rectangular plot, select VSWR in category section, click on new report as given below,



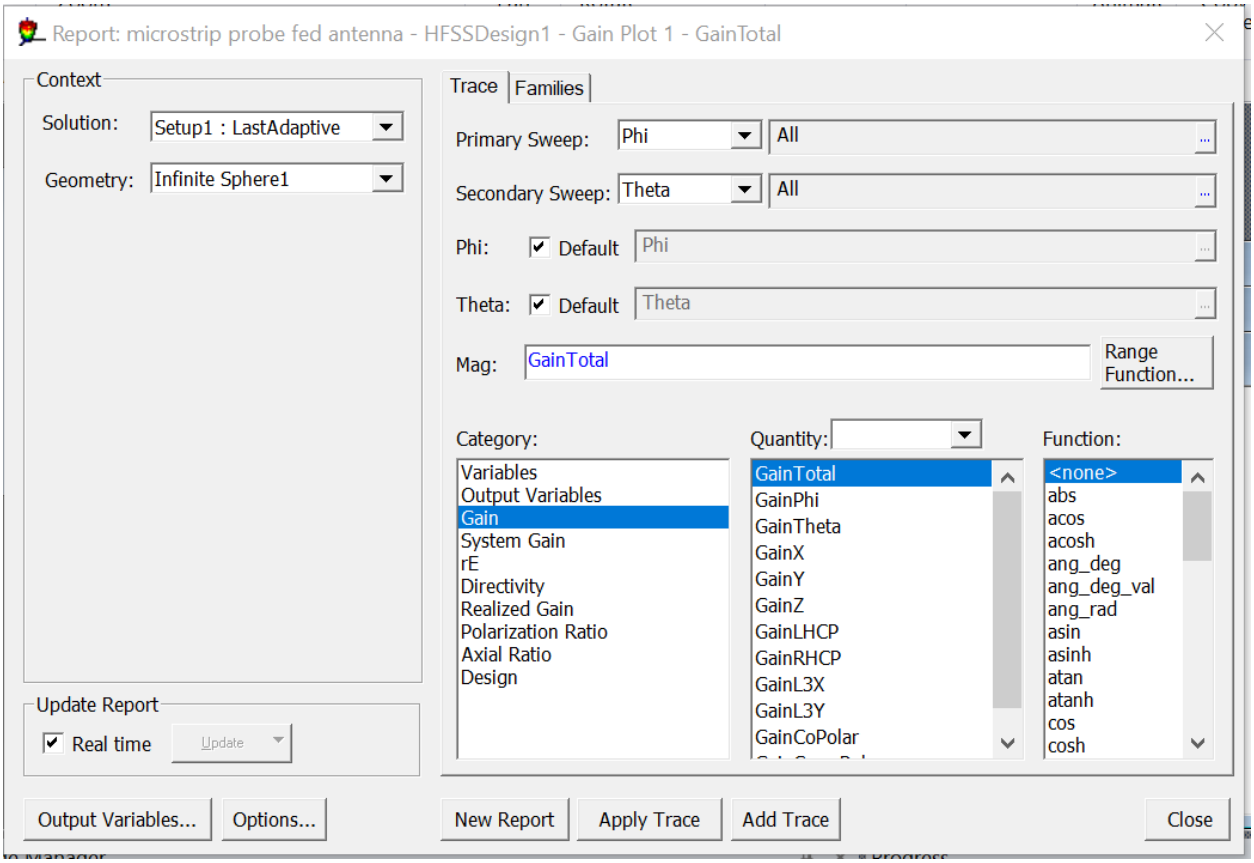
- The VSWR result will appear as given below,



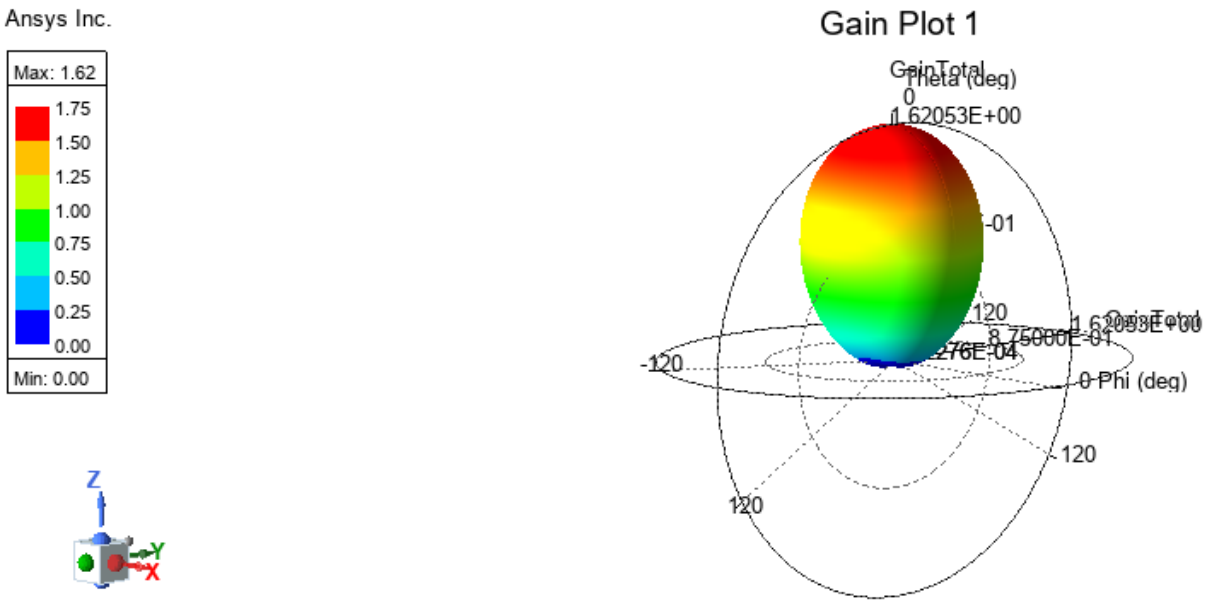
- Right click on radiation in project manager window, insert far field setup, infinite sphere
- Enter the values as given below,



- Right click on results, create far fields report, 3D polar plot
- The following window will open,



- Click on new report
- The following 3D radiation pattern will appear



Tabular Column

Sl No	Sl No	Before Tuning	After Tuning
1	Width of Patch	38 mm	
2	Length of Patch	29.4 mm	
3	S11(in dB) at 2.4GHz		
4	VSWR at 2.4GHz		
5	Bandwidth in Hz (at -10dB points of S11)		
6	Gain		

Note:

For tuning the antenna try to reduce the width and/or length of the patch by small amount to make the antenna to operate at the required frequency.

Assignment

Assignment

Experiment 8

Insetfed Rectangular Microstrip Patch Antenna

Aim:-

To design line fed rectangular microstrip patch antenna and study of its parameters.

Theory:-

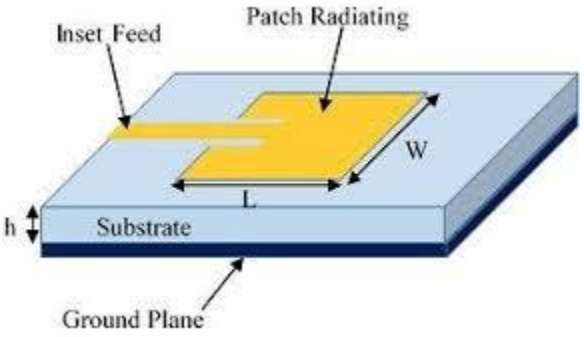
Microstrip patch antennas have gained a lot of popularity over the last three decades. This is due to their low-profile structures, cost effectiveness, ease of fabrication and ease of installation. It is easy to print microstrip antennas to a circuit board. In high performance systems with constraints of size, weight and cost, microstrip antennas prove to have numerous applications. Due to these advantages, microstrip patch antennas are becoming widespread in mobile radio and wireless communications.

Characteristics of Microstrip Patch Antennas

Consider the microstrip patch antenna shown in figure. The primary configuration consists of three structures:

- Patch
- Substrate
- Ground

The patch is a thin metallic strip that is placed a small fraction of a wavelength above the ground. The region between the patch and ground is called substrate. The substrate is ideally a dielectric material. Length of rectangular patch is less than λ/2. The microstrip patch antenna is designed such that it gives a broadside pattern i.e. maximum radiation in the direction normal to the patch. To achieve a broadside radiation pattern, the patch is excited from beneath.



The microstrip patch antenna can be fed in a number of ways. The two most common methods for feeding are coaxial probe, microstrip line.

Design Steps :-

Given

Resonant Frequency(fo) – 2.4GHz

Relative permittivity of substrate(εr) – 4.4

Thickness of substrate(h) – 1.6 mm

Necessary design parameters

The width and length of the patch is calculated by,

$$Width = \frac{c}{2f_o\sqrt{\frac{\varepsilon_R+1}{2}}}; \quad \varepsilon_{eff} = \frac{\varepsilon_R+1}{2} + \frac{\varepsilon_R-1}{2}\left[\frac{1}{\sqrt{1+12\left(\frac{h}{W}\right)}}\right]$$

$$Length = \frac{c}{2f_o\sqrt{\varepsilon_{eff}}} - 0.824h\left(\frac{(\varepsilon_{eff}+0.3)\left(\frac{W}{h}+0.264\right)}{(\varepsilon_{eff}-0.258)\left(\frac{W}{h}+0.8\right)}\right)$$

f_0 is the Resonance Frequency

W is the Width of the Patch

L is the Length of the Patch

h is the thickness of substrate

ε_r is the relative Permittivity of the dielectric substrate

c is the Speed of light

Calculation Result:-

By substituting the values of fo, εr and h we get length and width of the patch as given below,

Width: 38.01 mm ≈ 38 mm

Length: 29.42 mm ≈ 29.4 mm

Steps for simulation

Step 1: Starting HFSS

Step 2: Creating the antenna model

Step 3: Configurations (Settings)

Step 4: Validation and analysis

Step 5: Results

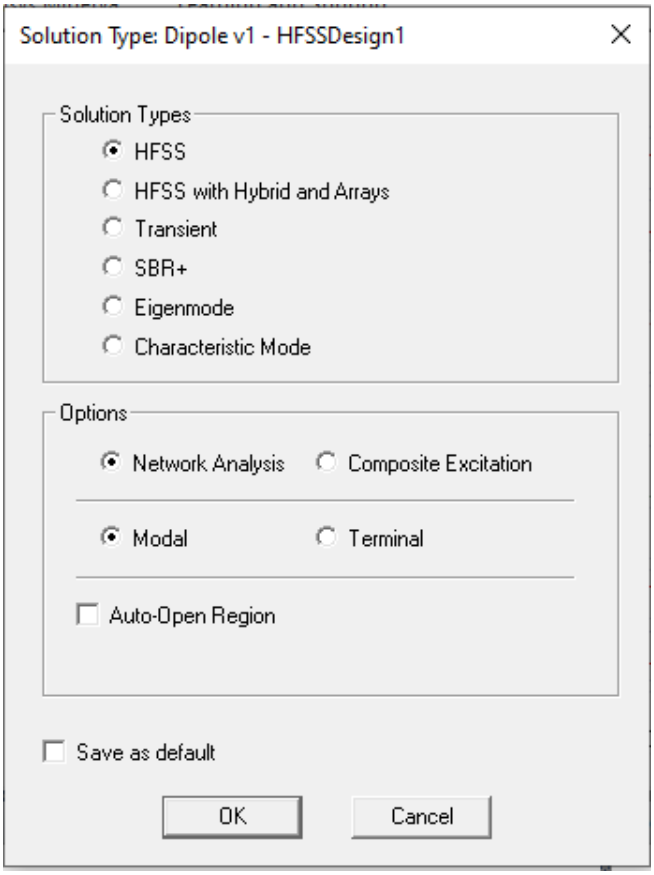
Step 1

Starting HFSS

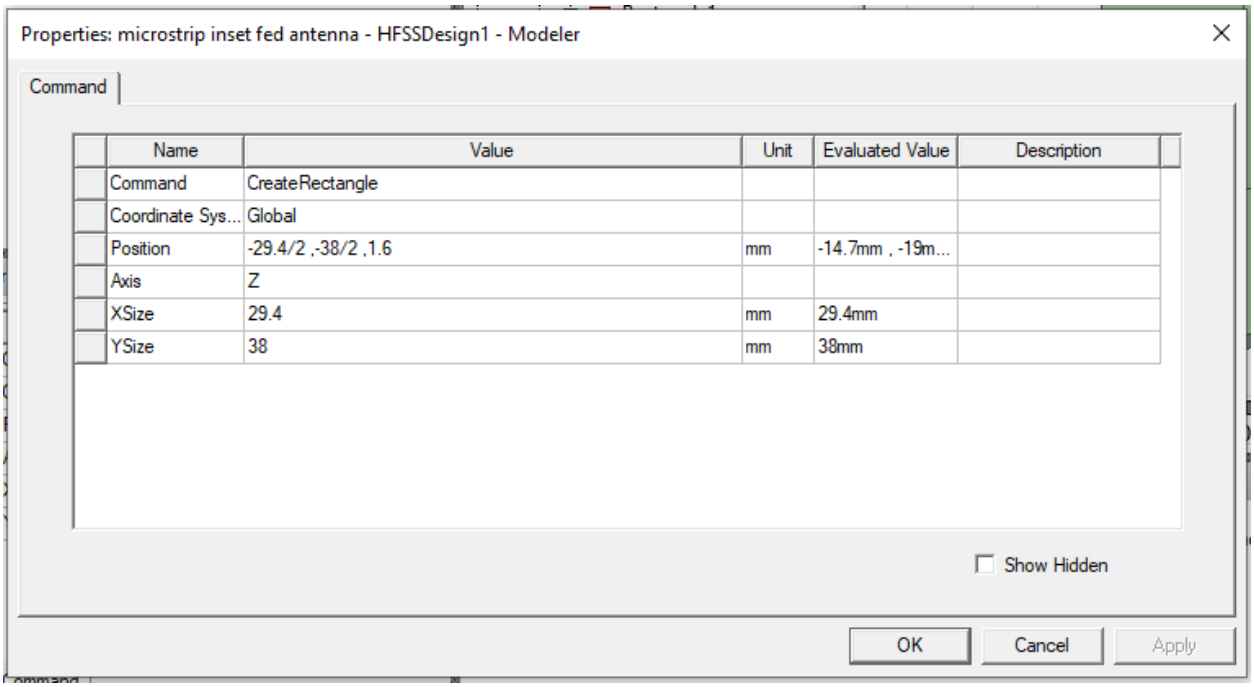
Step 2

Creating the antenna model

- From the HFSS menu select **Solution Type** as given below.

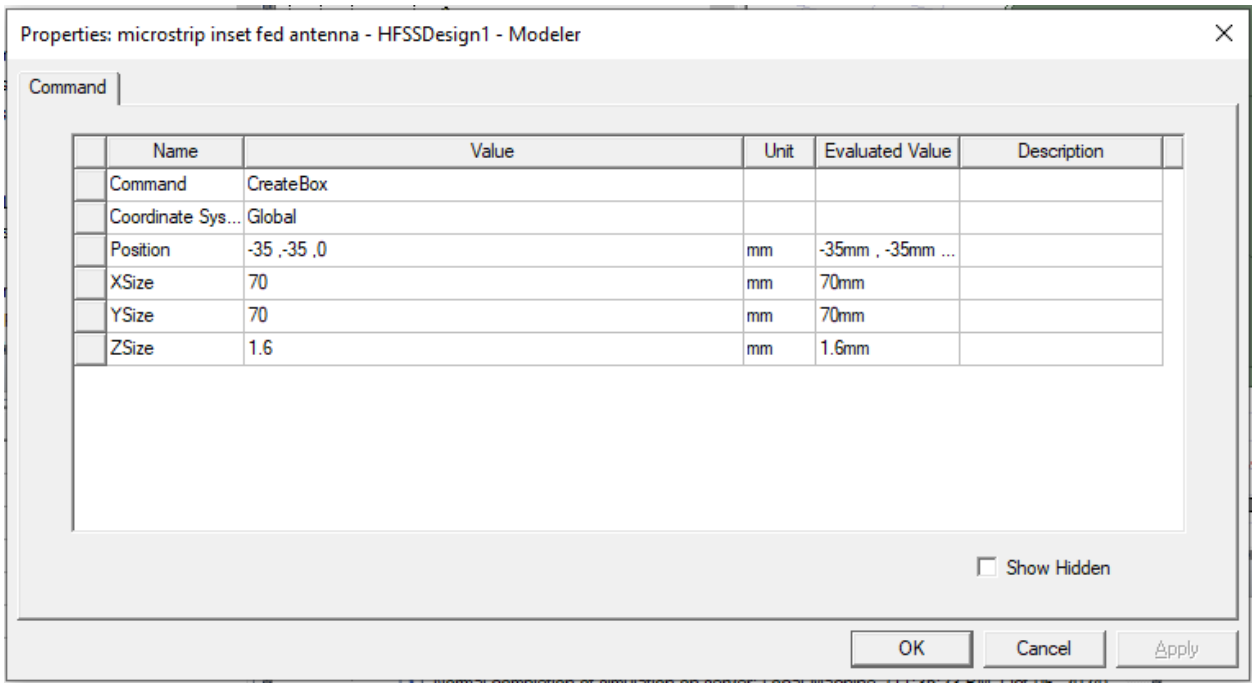


- Draw a rectangle(patch) with following coordinates,



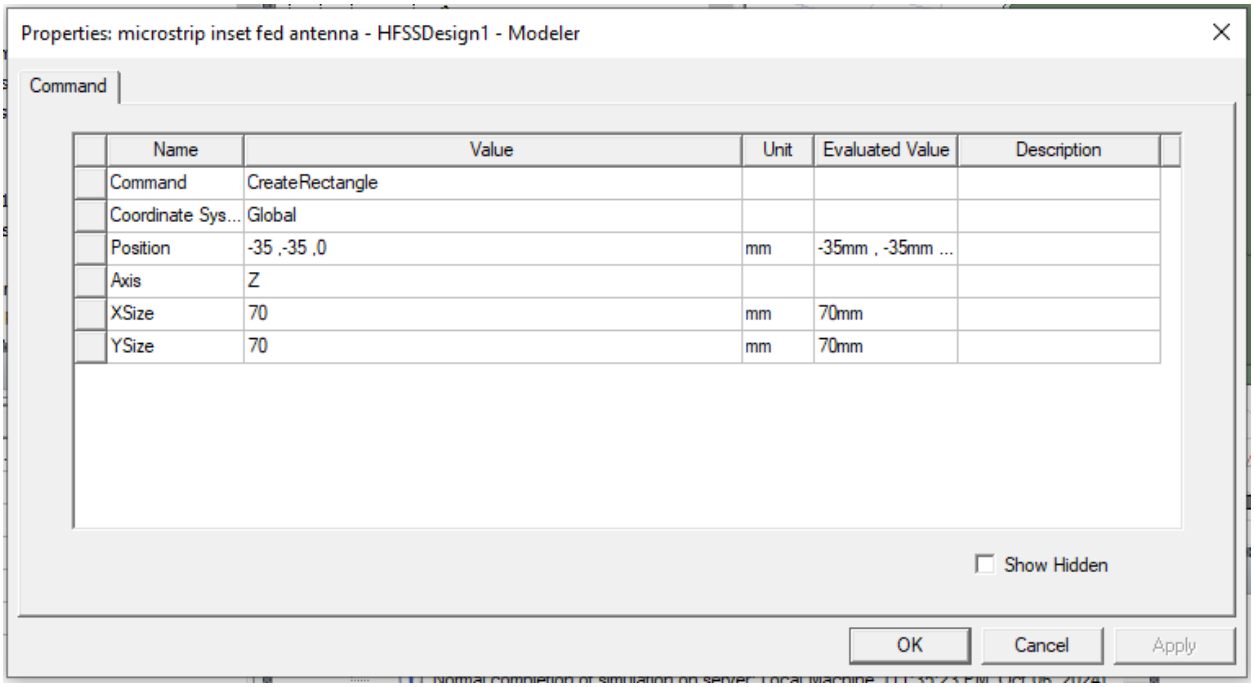
- Right click on rectangle1 in model section, Assign boundary, Perfect E
- Click ok

- Draw a box(substrate) with dimensions as given below,



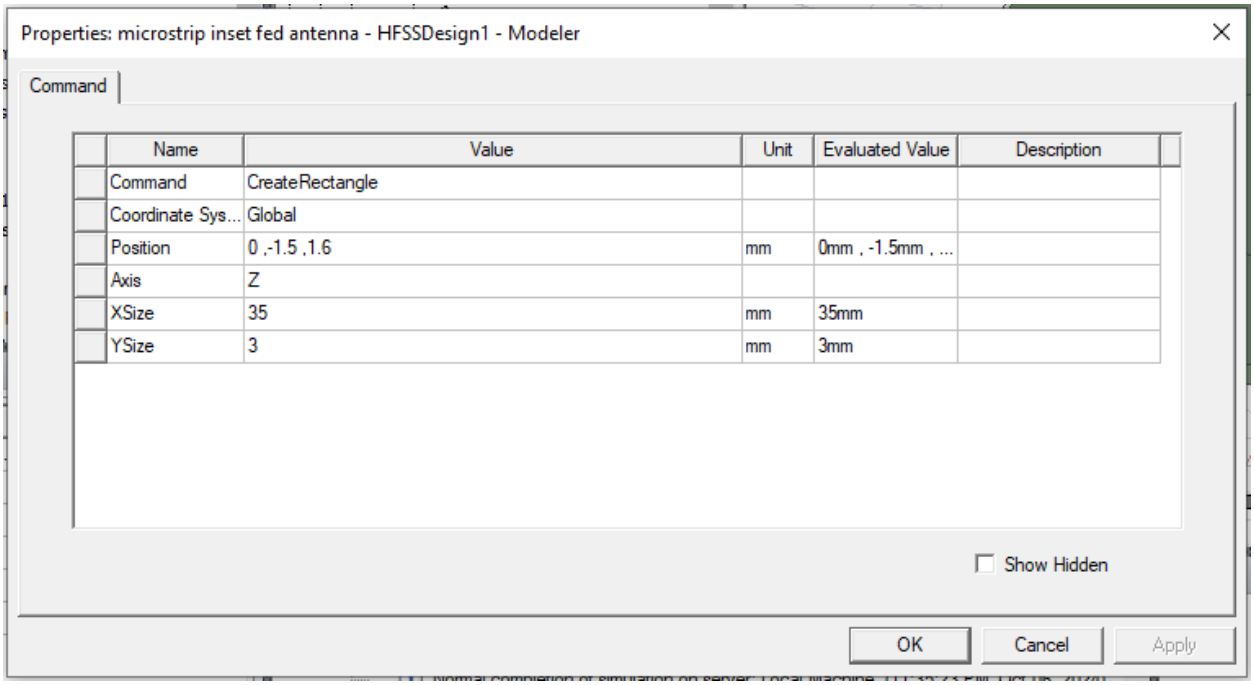
- Right click on box1 in model section, assign material, type fr4 in search section, select fr4, press ok

- Draw a rectangle (ground plane) with the following co-ordinates,

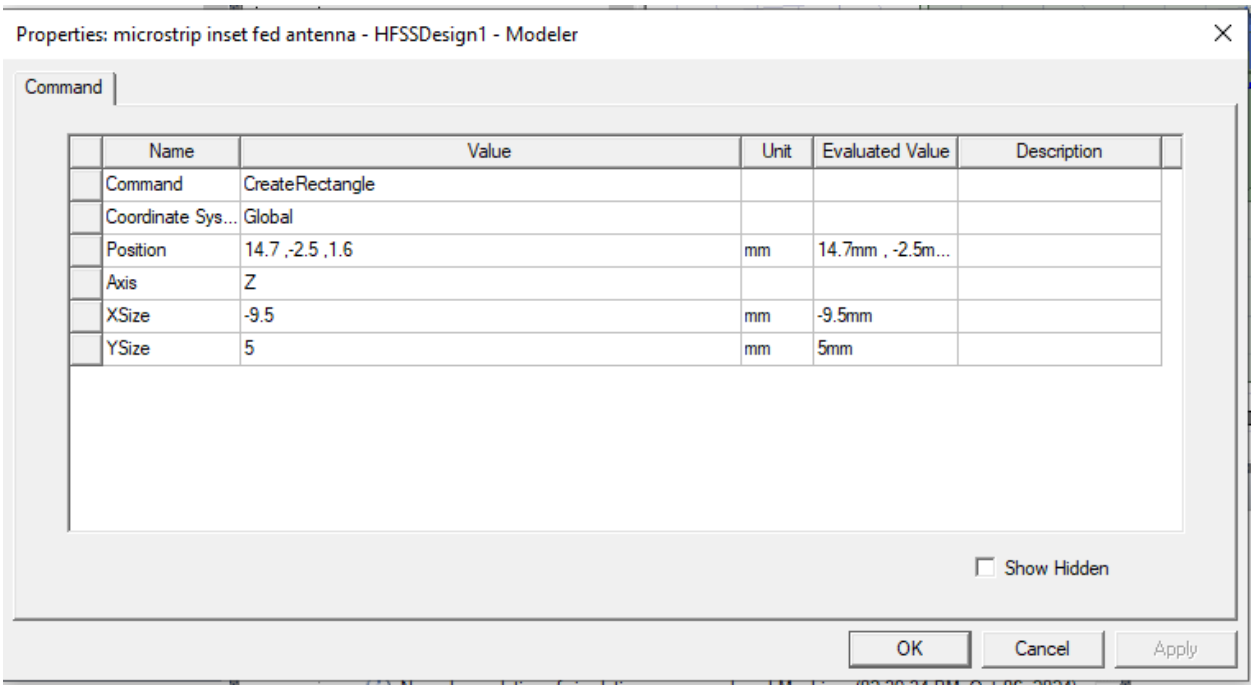


- Right click on rectangle2 in model section, Assign boundary, Perfect E, Click ok

- Draw one more rectangle(feedline) with the following dimensions

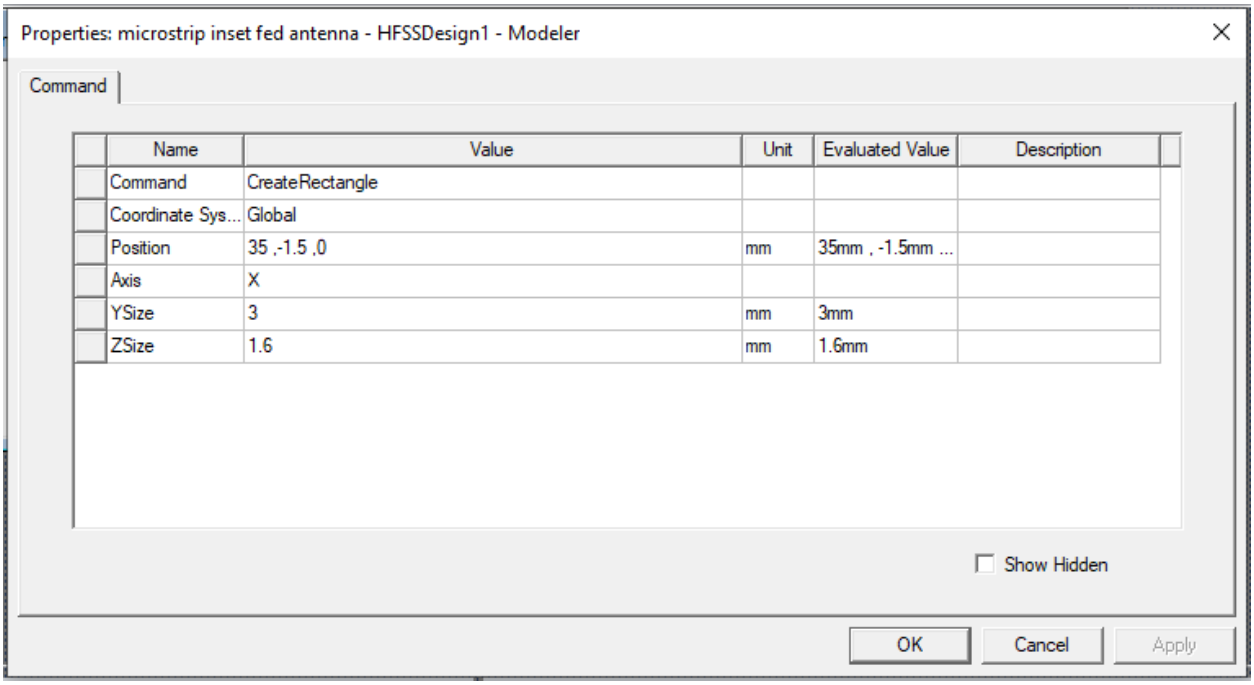


- Draw one more rectangle(cut) with the following dimensions

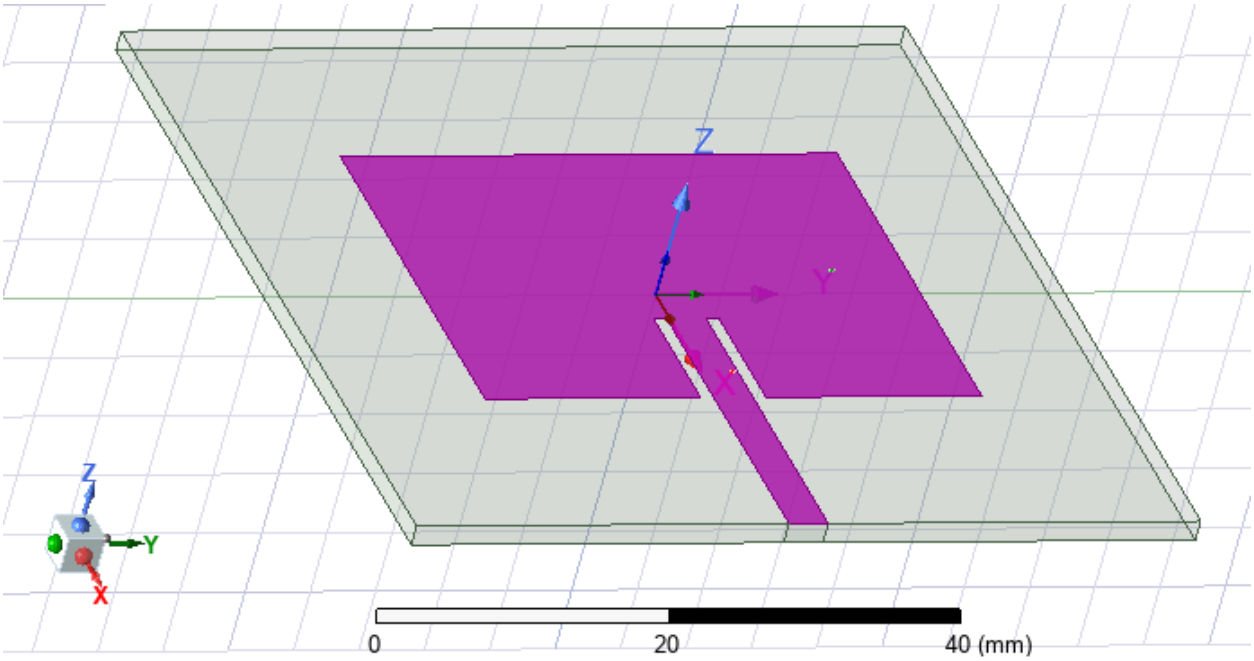


- By pressing down control button select rectangle1 and rectangle4, click subtract
- By pressing down control button select rectangle1 and rectangle3, click unite

- Click on visibility tab on the right corner of the window, click on XY dropdown and select YZ
- Draw rectangle with following dimensions



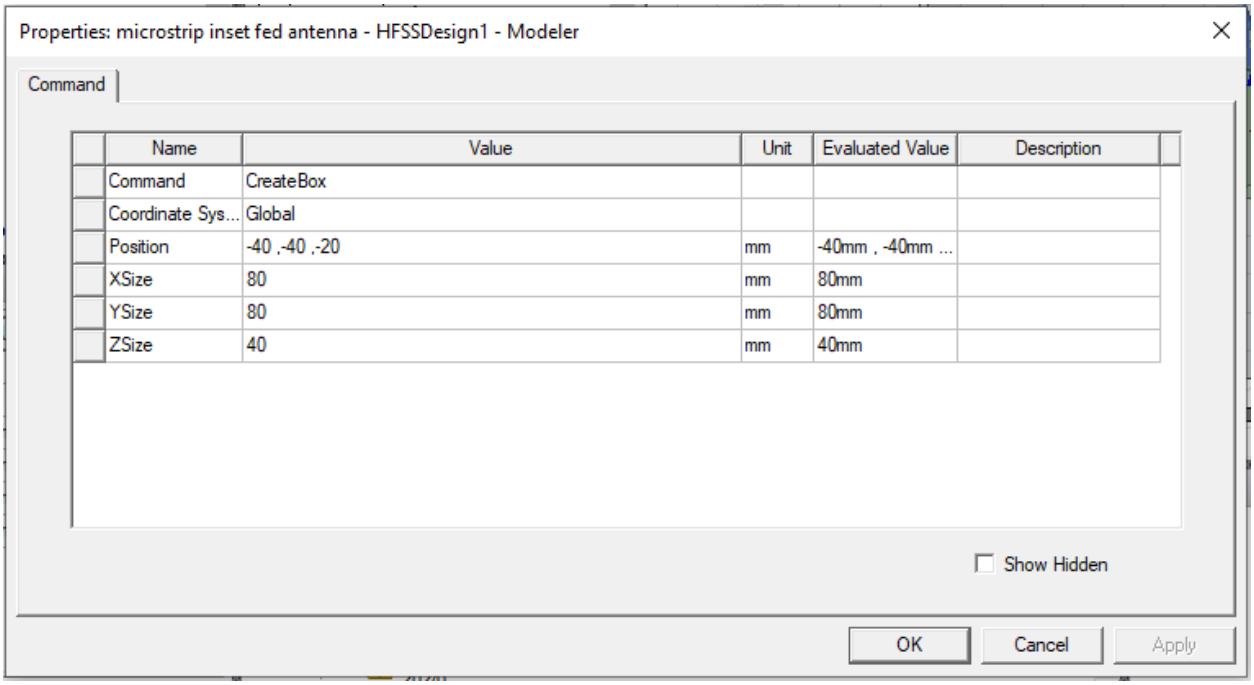
Now the model will look like as given below,



- Right click on rectangle5 in model section, assign excitation, port, lumped port, resistance 50 Ohms, next, click on dropdown under integration line and select choose new line
- Draw a line from bottom center to the top center of the rectangle5, next, finish

- Click on visibility tab on the right corner of the window, click on YZ dropdown and select XY

- Create a box with dimensions as given below,



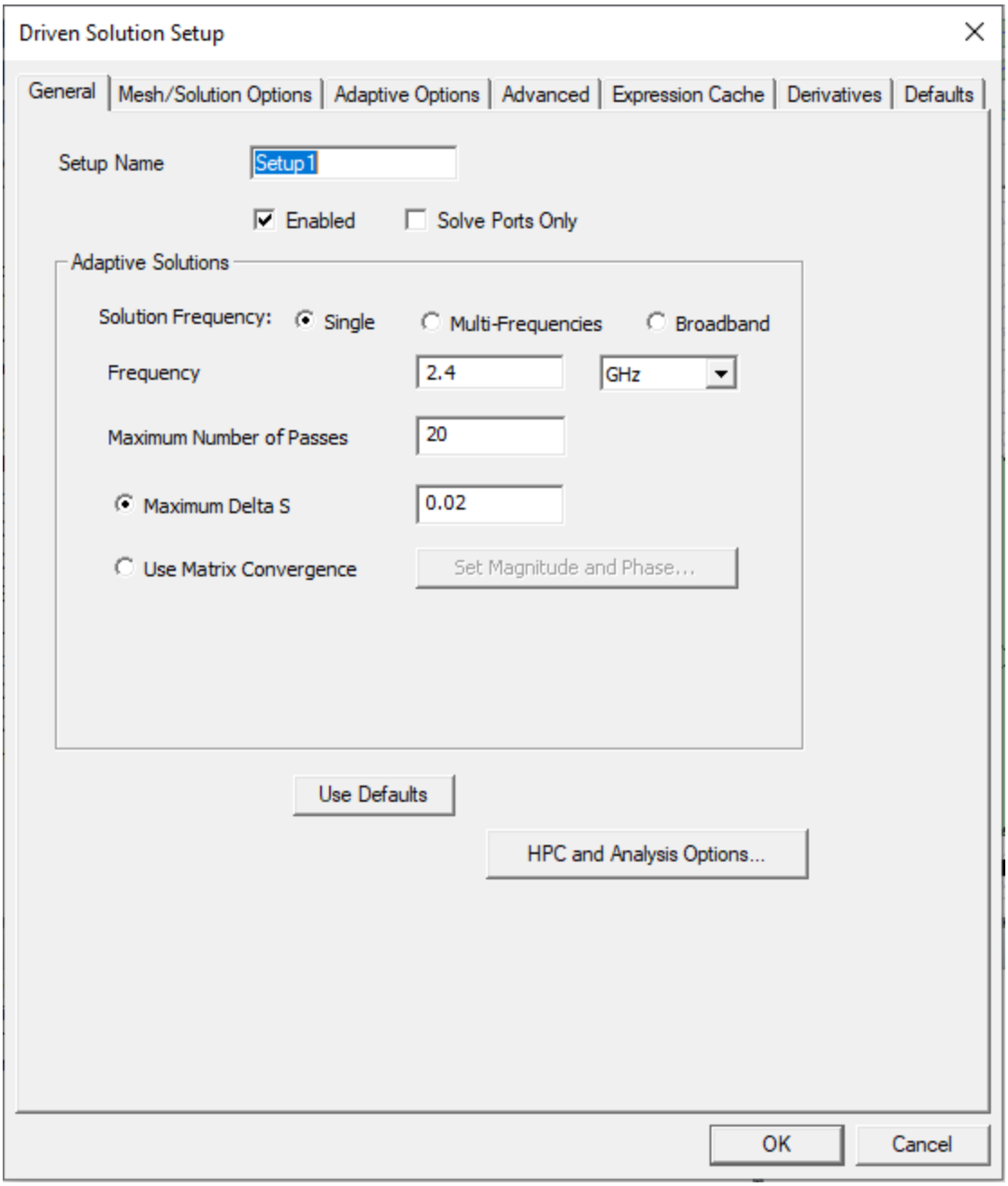
- Assign the material of box2 as “air”
- Right click on Box2 in model section, assign boundary, radiation, ok

- Now your model is constructed along with radiation boundary.

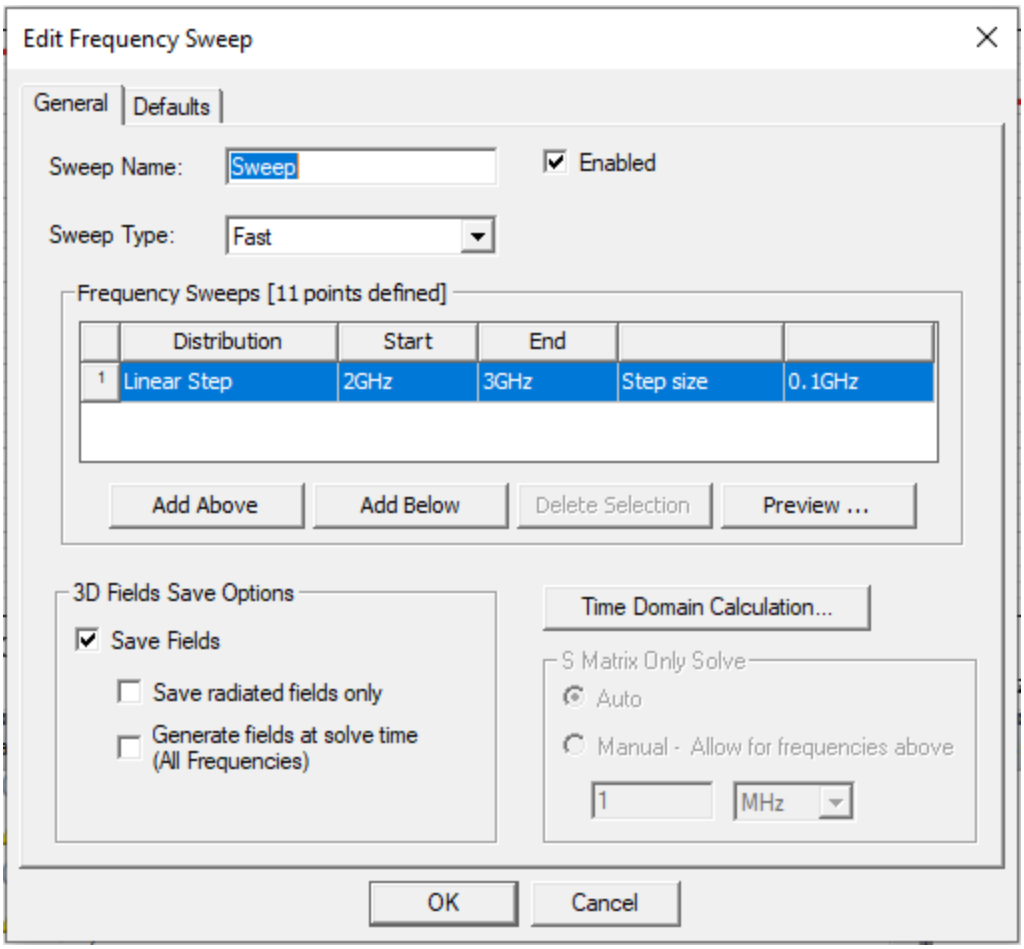
Step 3

Configurations (Settings)

- In the project manager section, Right click on analysis, add solution setup, advanced
- Enter the values as given below



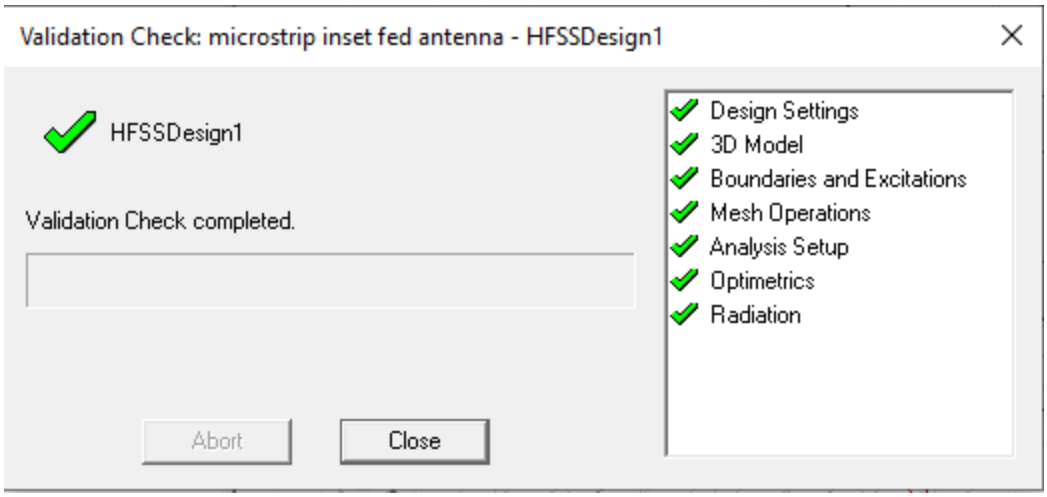
- Click on ok
- In the next window that appears after clicking ok, enter the values as given below



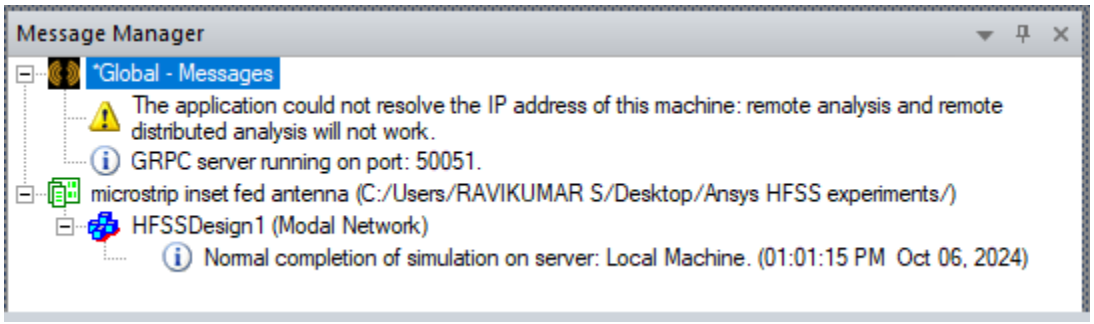
Step 4

Validation and Analysis

- Go to HFSS, Validation check
- The following window will appear if model created is correct



- Close the validation window
- Go to HFSS, Analyze all
- It will take some time to calculate results based on antenna complexity and computer configuration which is running the software
- If simulation goes well then you will receive a message in the message manager section as Normal completion of simulation on server:Local machine as given in figure below.

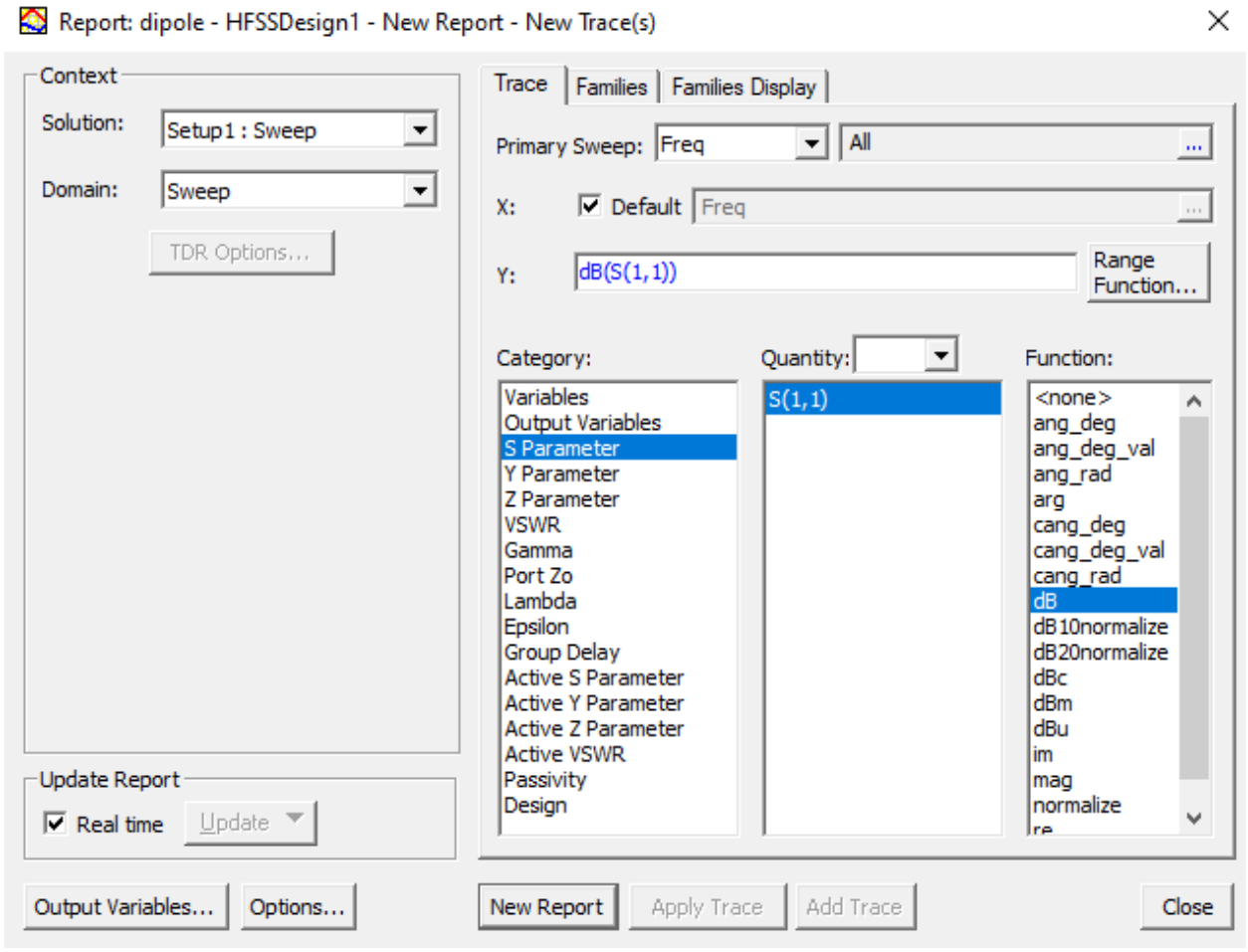


- If any errors are there then such error will be shown in message manager section. They have to be resolved.

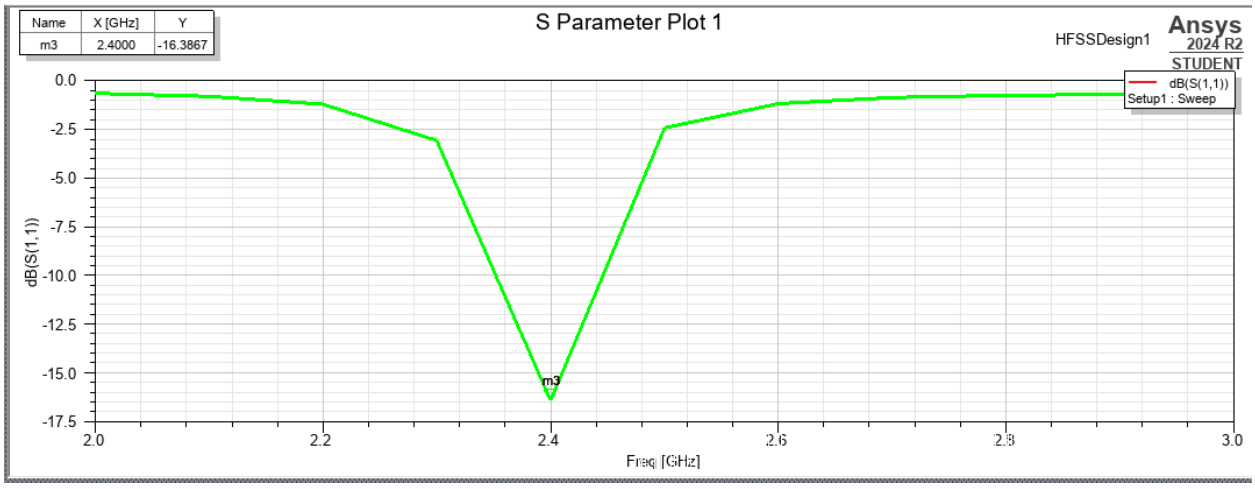
Step 5

Results

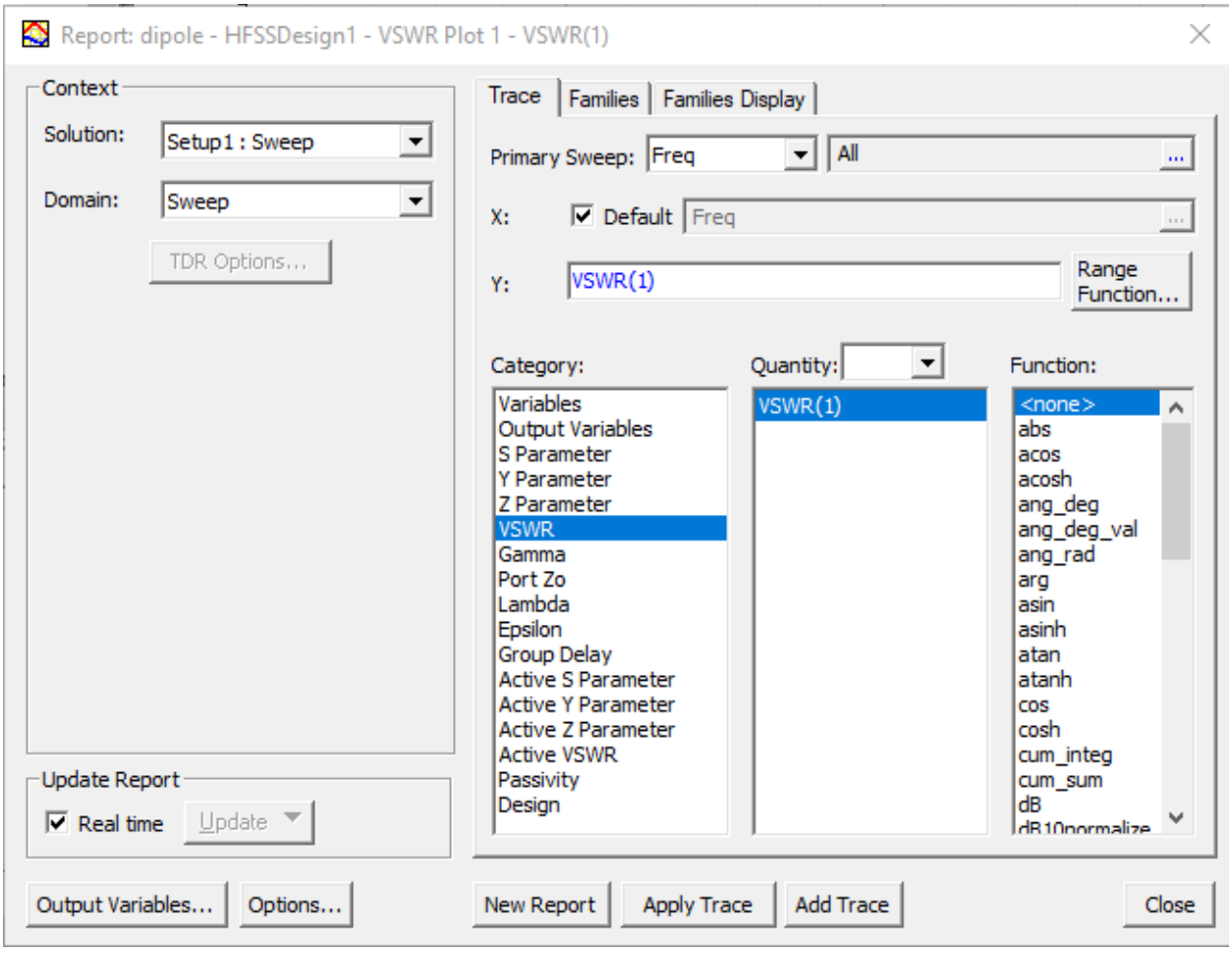
- Right click on results in project manager section, create modal solution data report, rectangular plot,
- The following window should appear



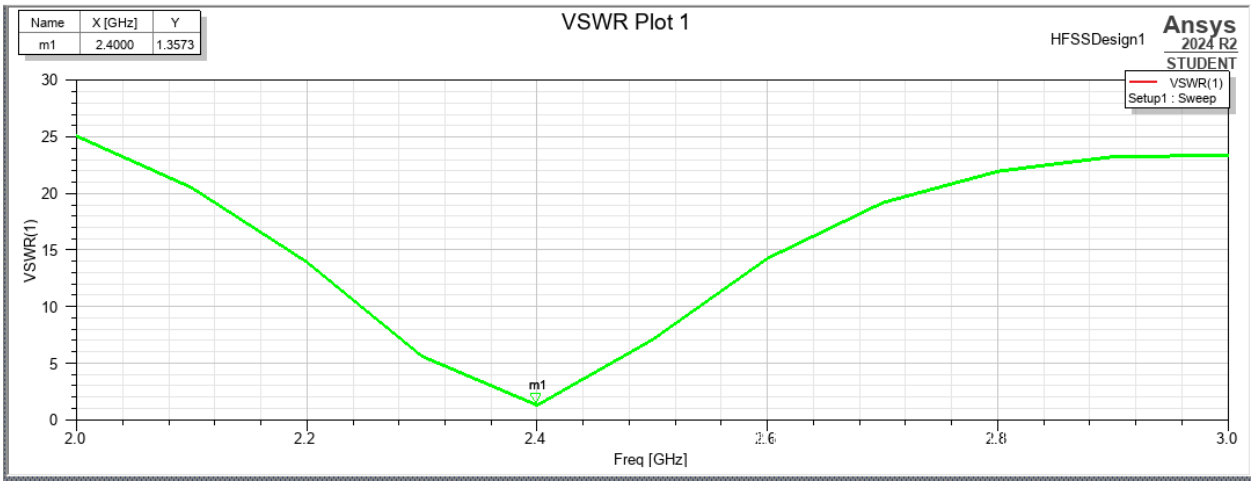
- Click on new report
- The following window will appear showing S11 parameter (Input reflection coefficient)



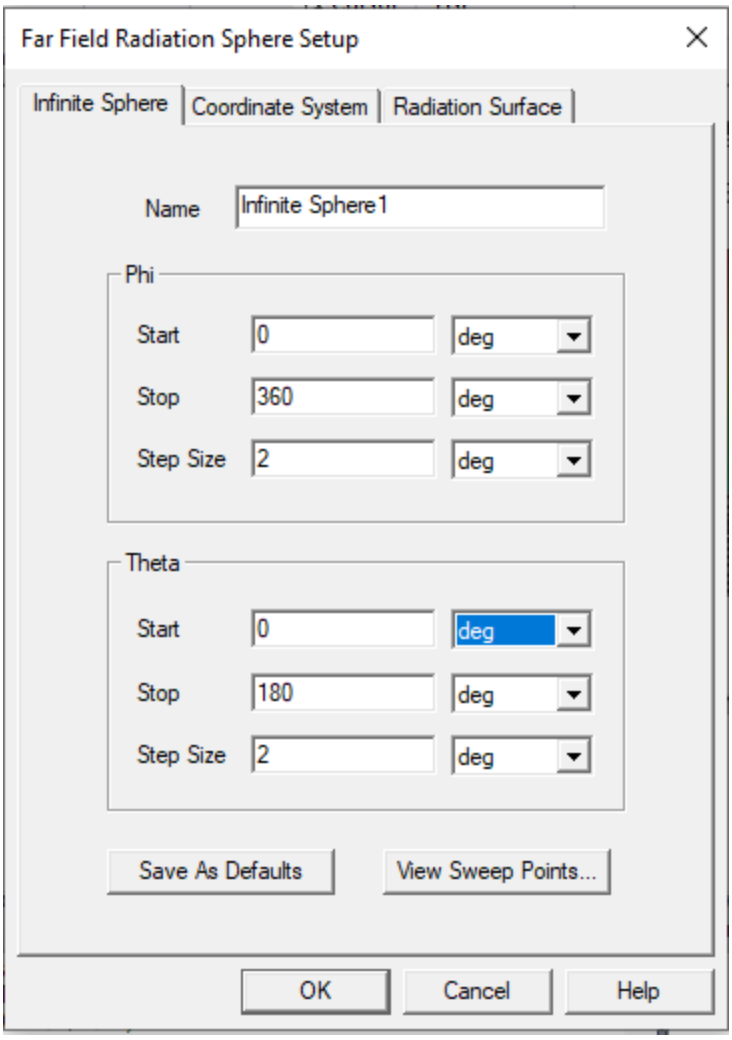
- In the same way Right click on results in project manager section, create modal solution data report, rectangular plot, select VSWR in category section, click on new report as given below



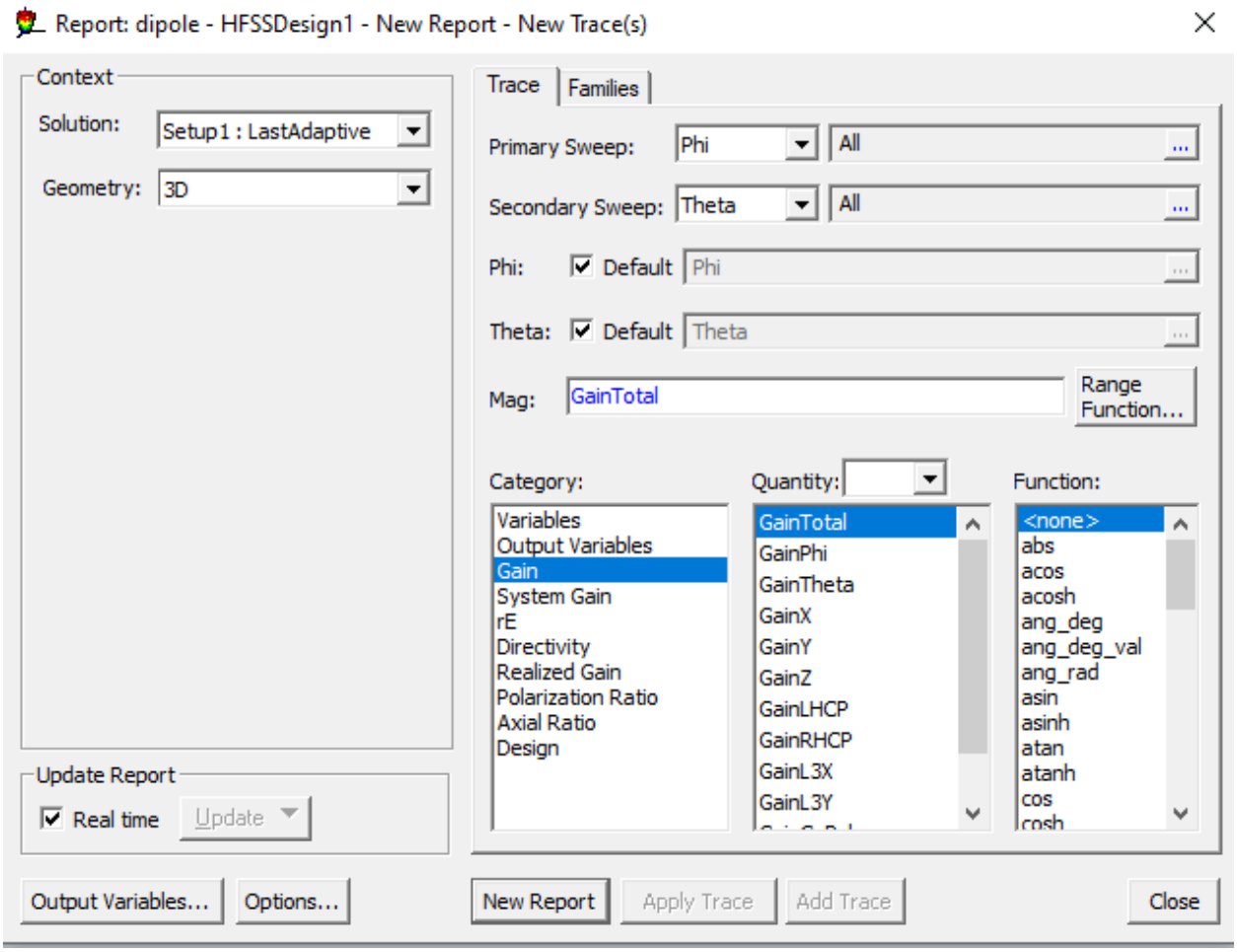
- The VSWR result will appear as given below,



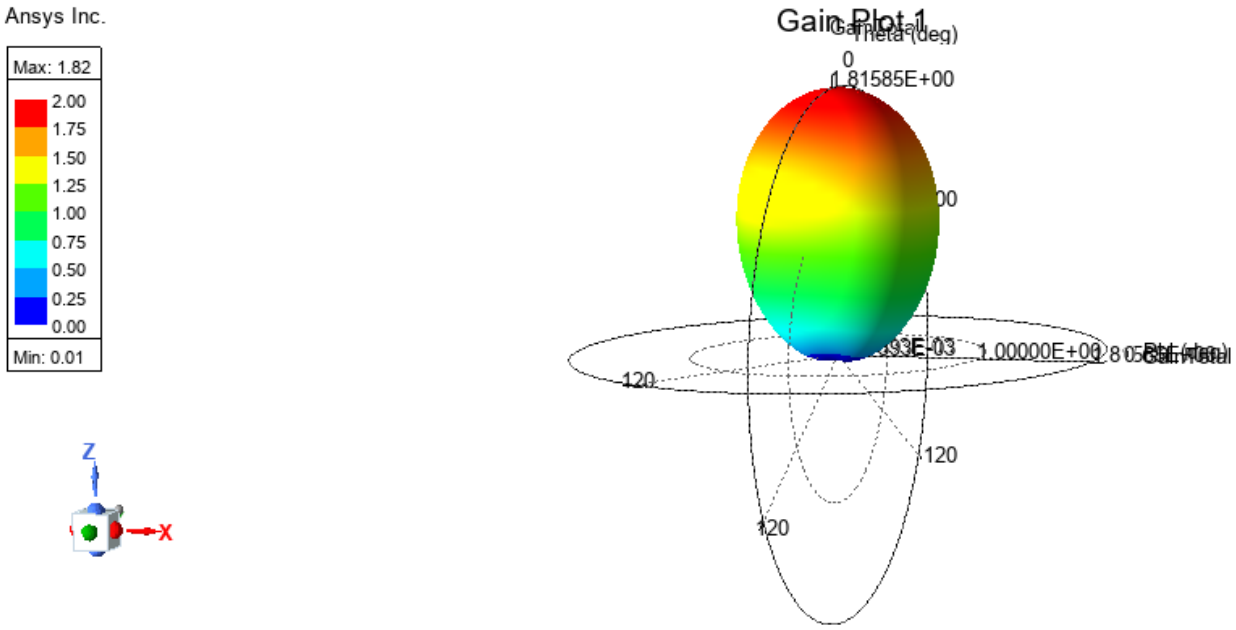
- Right click on radiation in project manager window, insert far field setup, infinite sphere
- Enter the values as given below



- Right click on results, create far fields report, 3D polar plot
- The following window will open,



- Click on new report
- The following 3D radiation pattern will appear



Tabular Column

Sl No	Sl No	Before Tuning	After Tuning
1	Width of Patch	38 mm	
2	Length of Patch	29.4 mm	
3	S11(in dB) at 2.4GHz		
4	VSWR at 2.4GHz		
5	Bandwidth in Hz (at -10dB points of S11)		
6	Gain		

Assignment

Assignment

Experiment 9

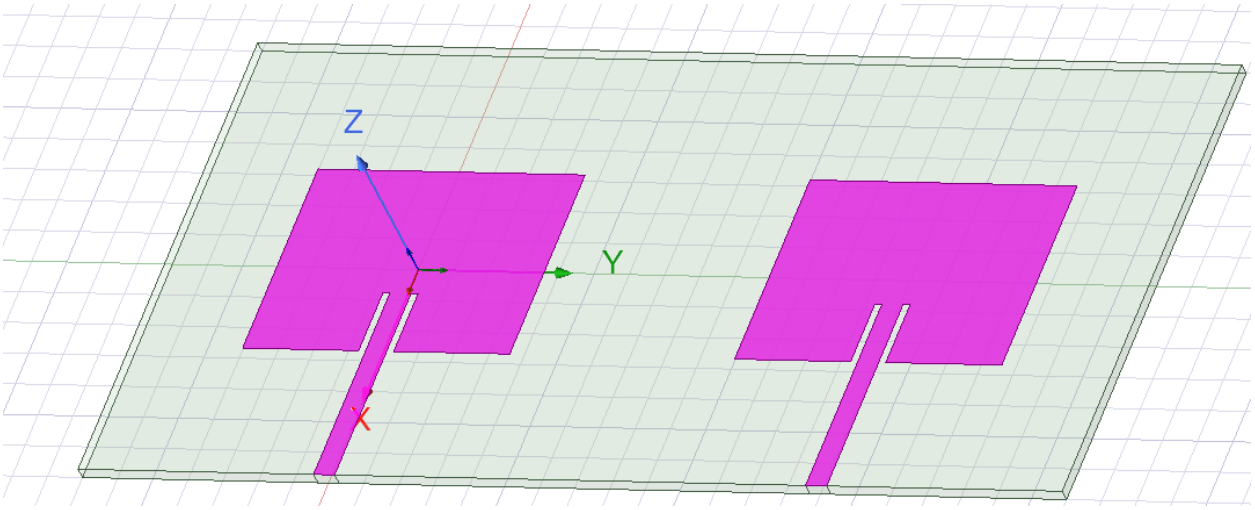
Rectangular Microstrip Patch Antenna Array

Aim:-

To design insetfed rectangular microstrip patch antenna array and study of its parameters.

Theory:-

A patch antenna array is a collection of patch antennas that are used to harvest energy from radio frequency (RF) waves. Patch antennas are made up of a metal patch mounted on a larger metal sheet called a ground plane. They are commonly used for microwave frequencies, particularly in the 2.4 GHz band and higher.



Design Steps :-

Given

Resonant Frequency(fo) – 2.4GHz

Relative permittivity of substrate(εr) – 4.4

Thickness of substrate(h) – 1.6 mm

Necessary design parameters

The width and length of the patch is calculated by,

$$Width = \frac{c}{2f_o\sqrt{\frac{\varepsilon_R+1}{2}}}; \quad \varepsilon_{eff} = \frac{\varepsilon_R+1}{2} + \frac{\varepsilon_R-1}{2}\left[\frac{1}{\sqrt{1+12\left(\frac{h}{W}\right)}}\right]$$

$$Length = \frac{c}{2f_o\sqrt{\varepsilon_{eff}}} - 0.824h\left(\frac{(\varepsilon_{eff}+0.3)\left(\frac{W}{h}+0.264\right)}{(\varepsilon_{eff}-0.258)\left(\frac{W}{h}+0.8\right)}\right)$$

f_0 is the Resonance Frequency

W is the Width of the Patch

L is the Length of the Patch

h is the thickness of substrate

ε_r is the relative Permittivity of the dielectric substrate

c is the Speed of light

Calculation Result:-

By substituting the values of fo, εr and h we get length and width of the patch as given below,

Width: 38.01 mm ≈ 38 mm

Length: 29.42 mm ≈ 29.4 mm

Steps for simulation

Step 1: Starting HFSS

Step 2: Creating the antenna model

Step 3: Configurations (Settings)

Step 4: Validation and analysis

Step 5: Results

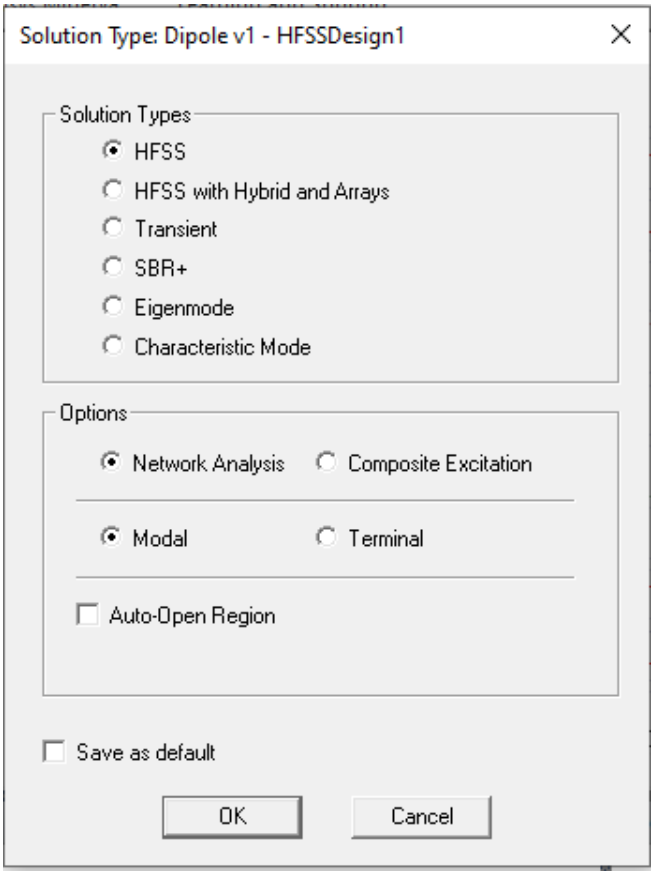
Step 1

Starting HFSS

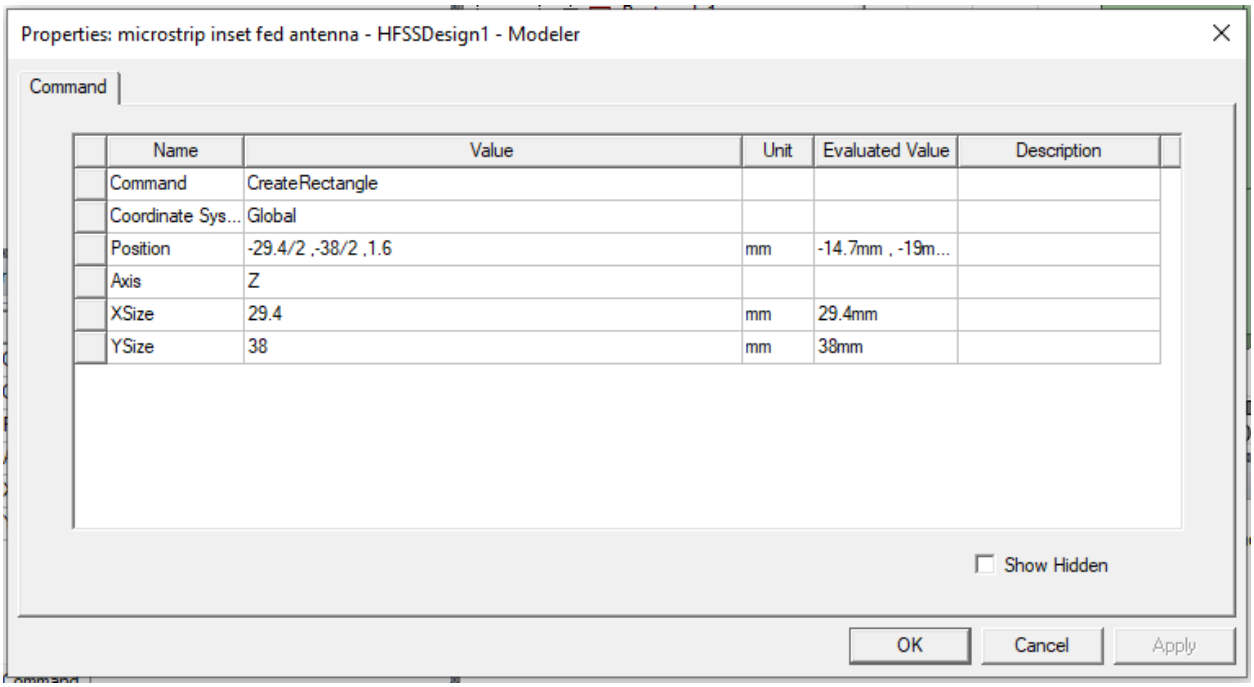
Step 2

Creating the antenna model

- From the HFSS menu select **Solution Type** as given below.

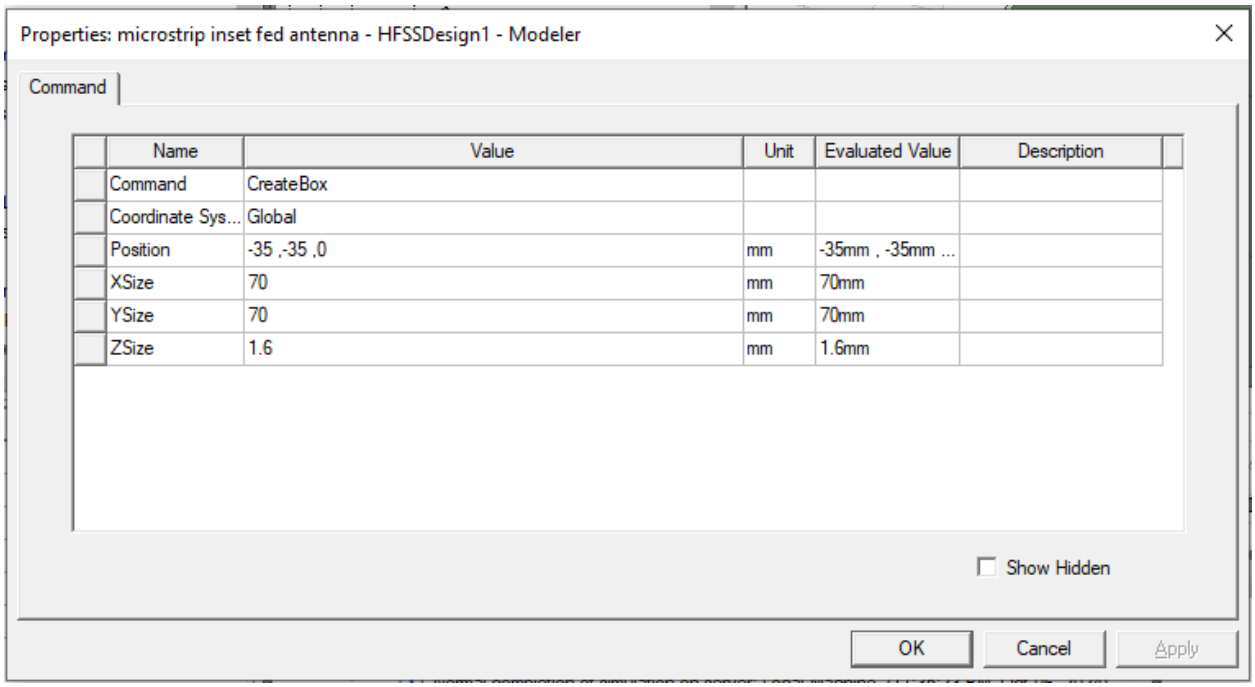


- Draw a rectangle(patch) with following coordinates,



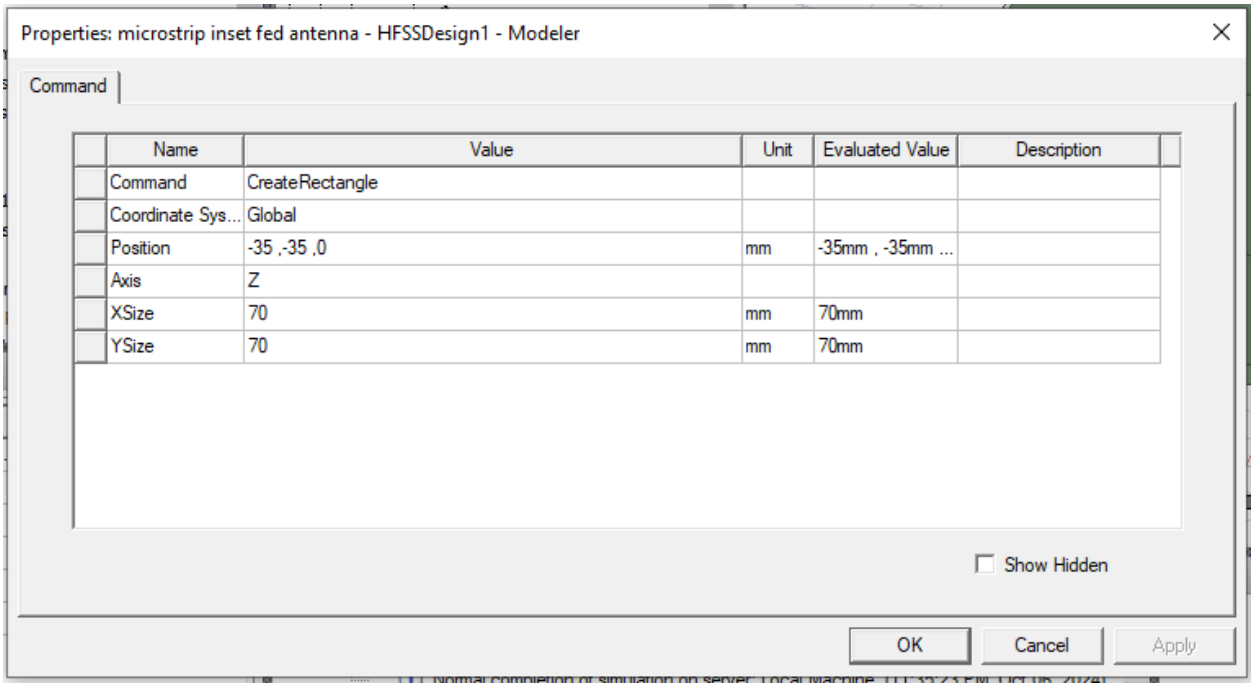
- Right click on rectangle1 in model section, Assign boundary, Perfect E
- Click ok

- Draw a box(substrate) with dimensions as given below,



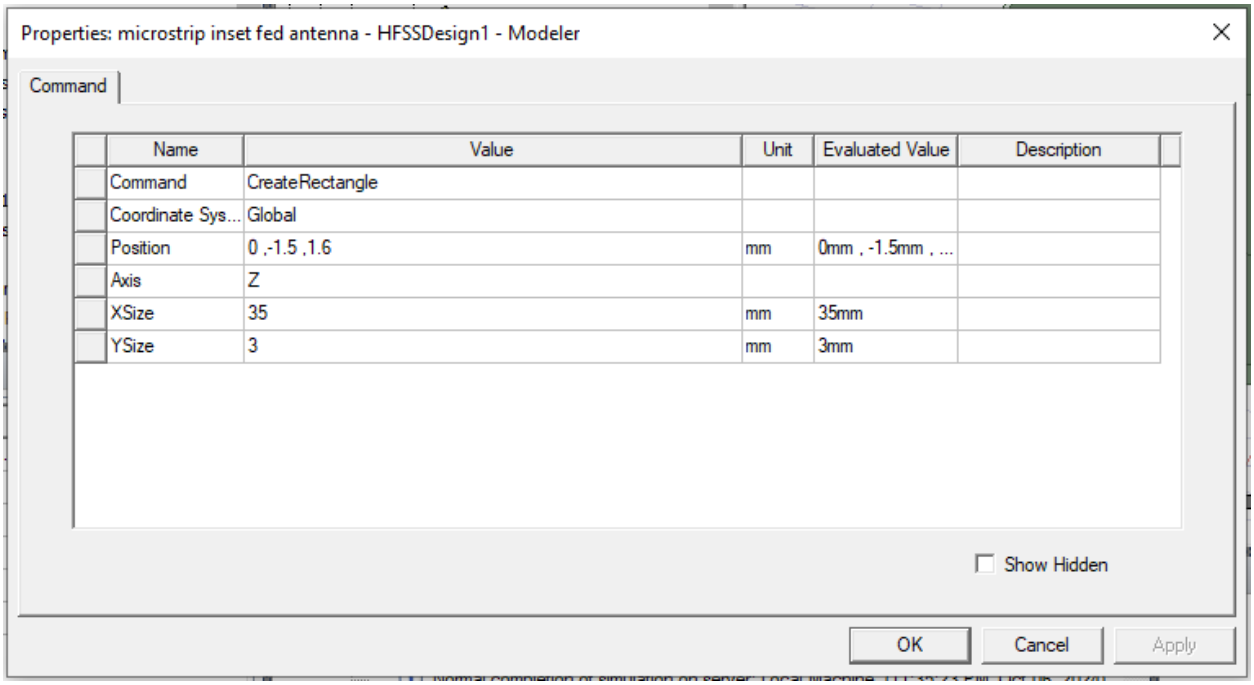
- Right click on box1 in model section, assign material, type fr4 in search section, select fr4, press ok

- Draw a rectangle (ground plane) with the following co-ordinates,

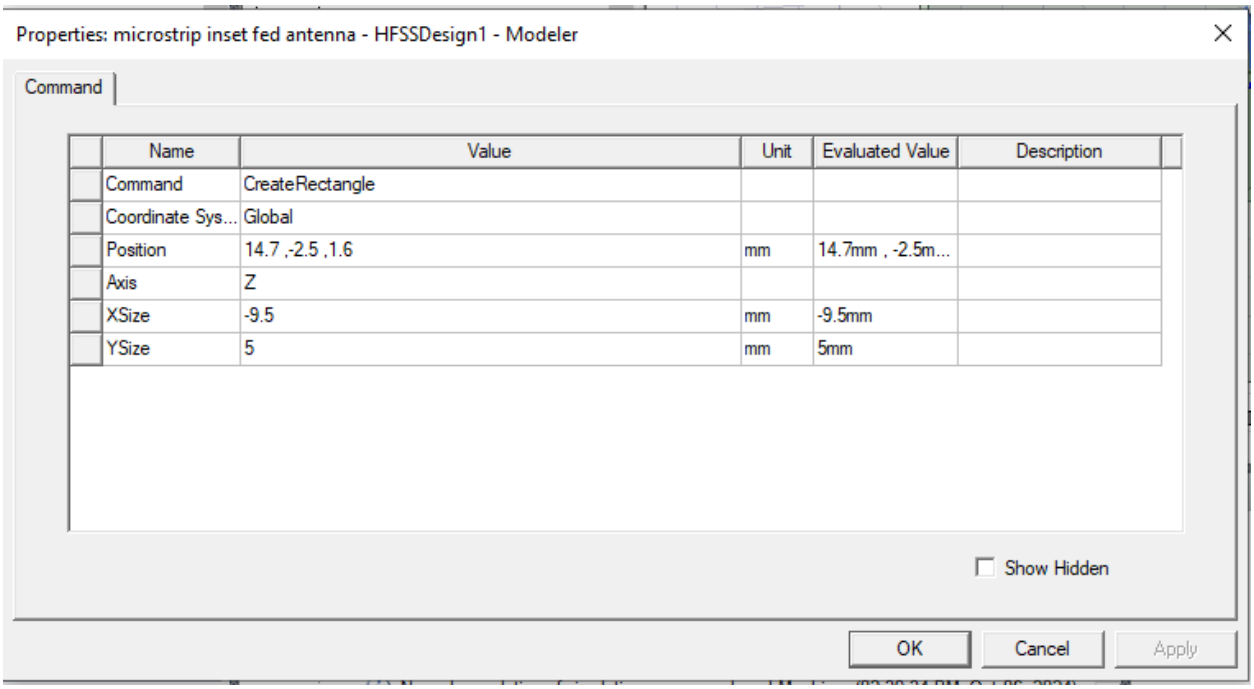


- Right click on rectangle2 in model section, Assign boundary, Perfect E, Click ok

- Draw one more rectangle(feedline) with the following dimensions

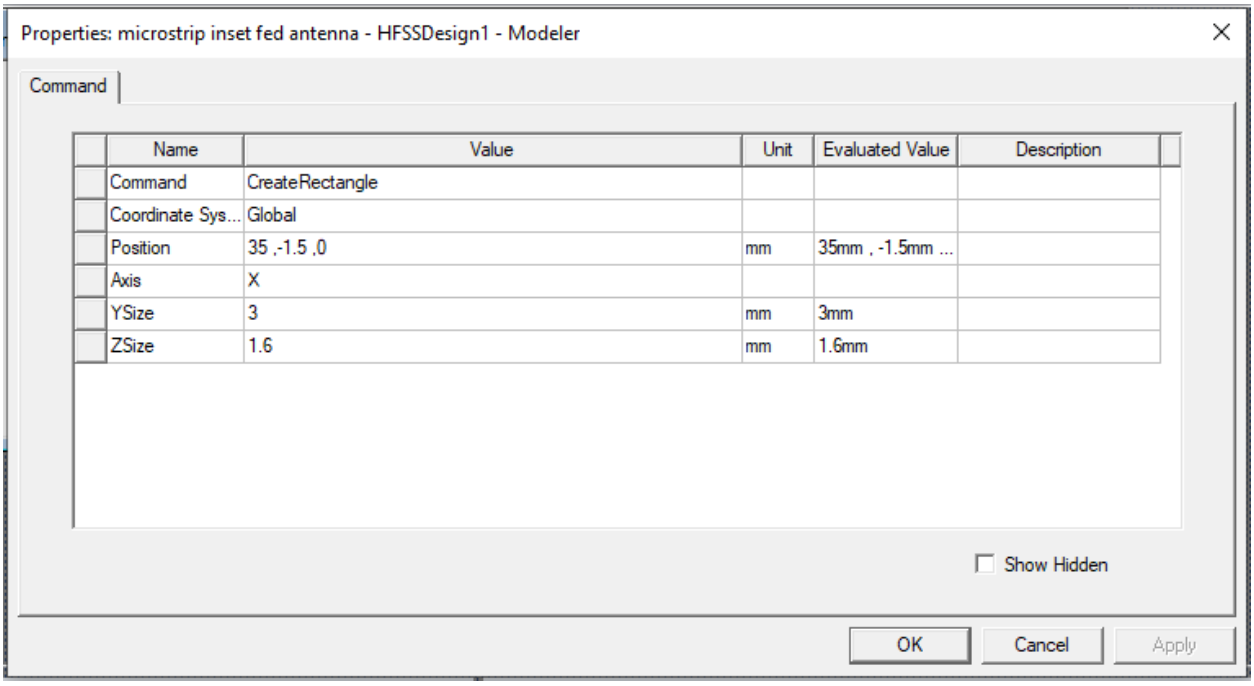


- Draw one more rectangle(cut) with the following dimensions

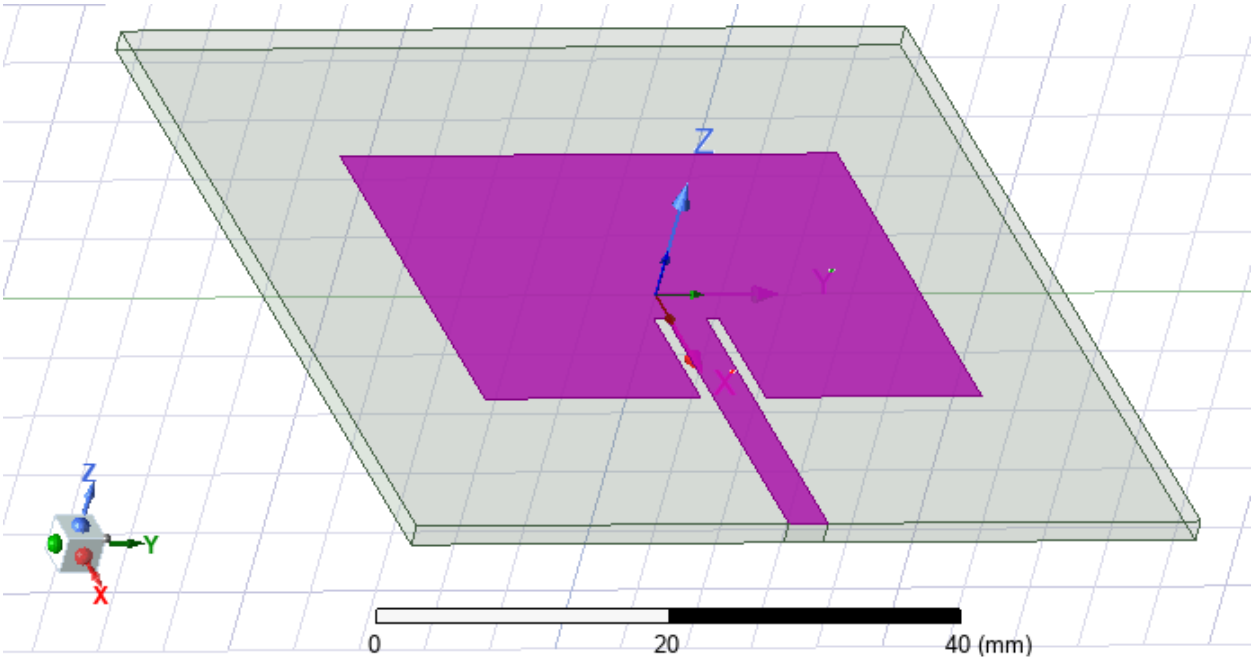


- By pressing down control button select rectangle1 and rectangle4, click subtract
- By pressing down control button select rectangle1 and rectangle3, click unite

- Click on visibility tab on the right corner of the window, click on XY dropdown and select YZ
- Draw rectangle with following dimensions

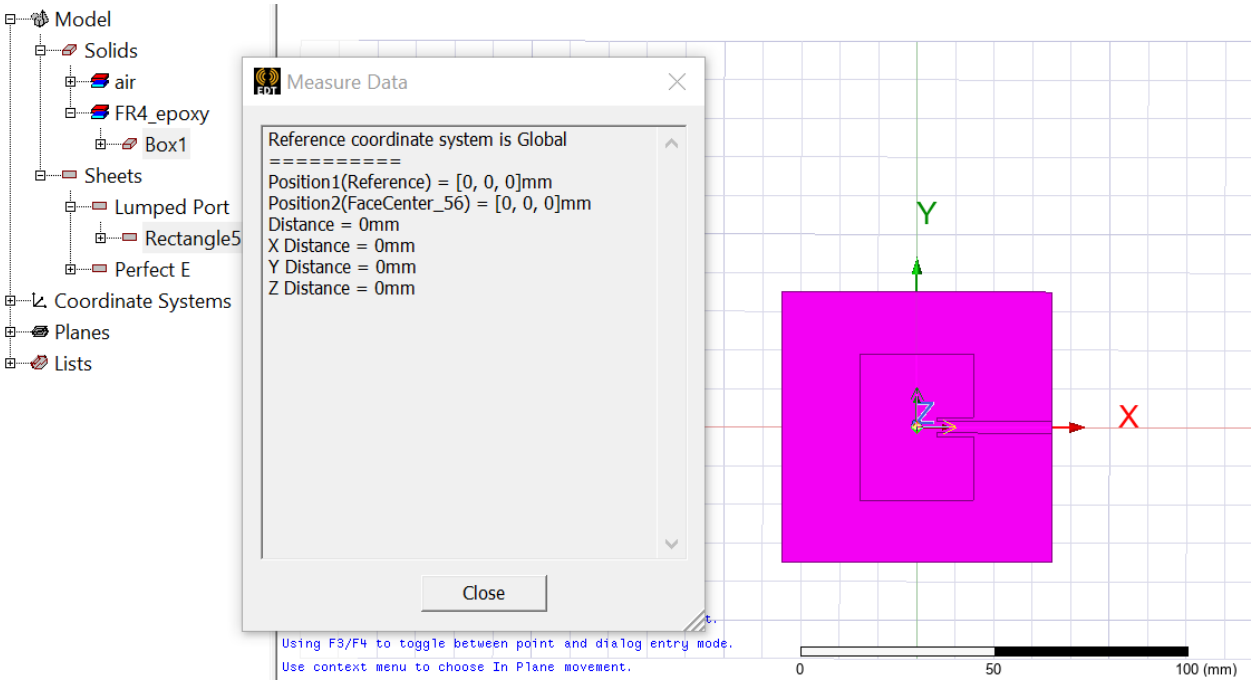


Now the model will look like as given below,

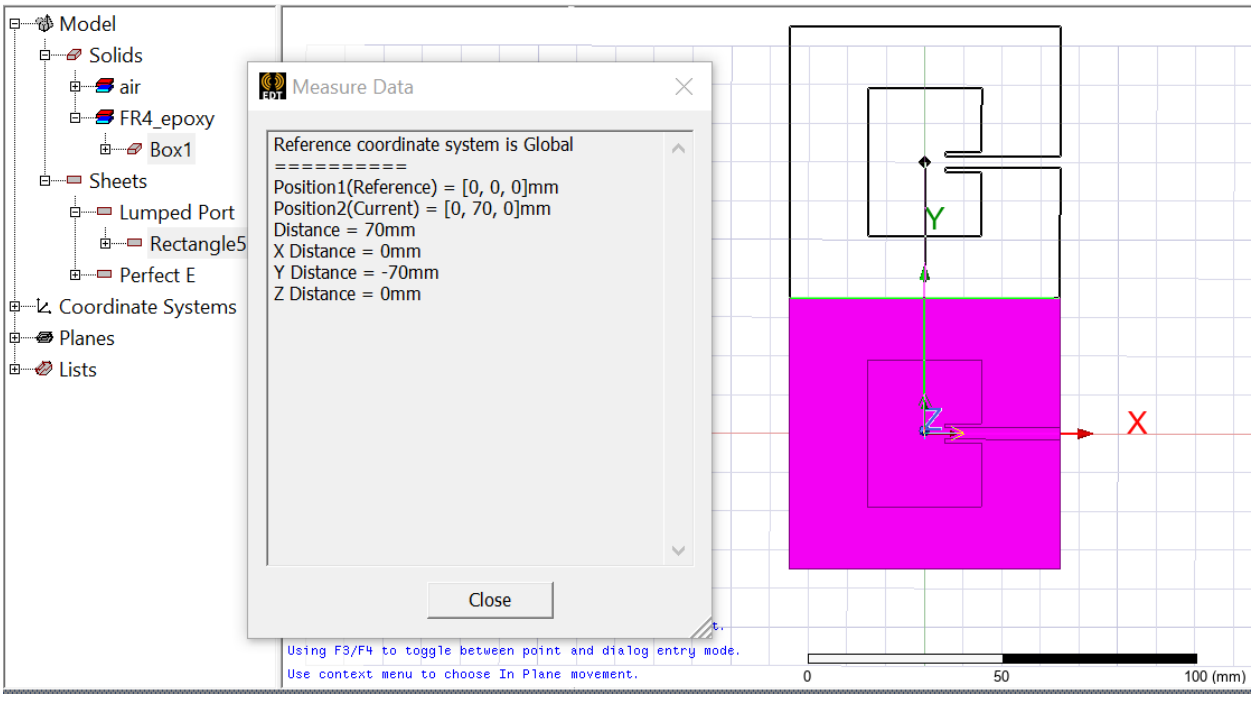


- Right click on rectangle5 in model section, assign excitation, port, lumped port, resistance 50 Ohms, next, click on dropdown under integration line and select choose new line
- Draw a line from bottom center to the top center of the rectangle5, next, finish

- Press control A (select all objects) in work area, right click on antenna, select edit, duplicate along line
- Press once at the origin as given below,

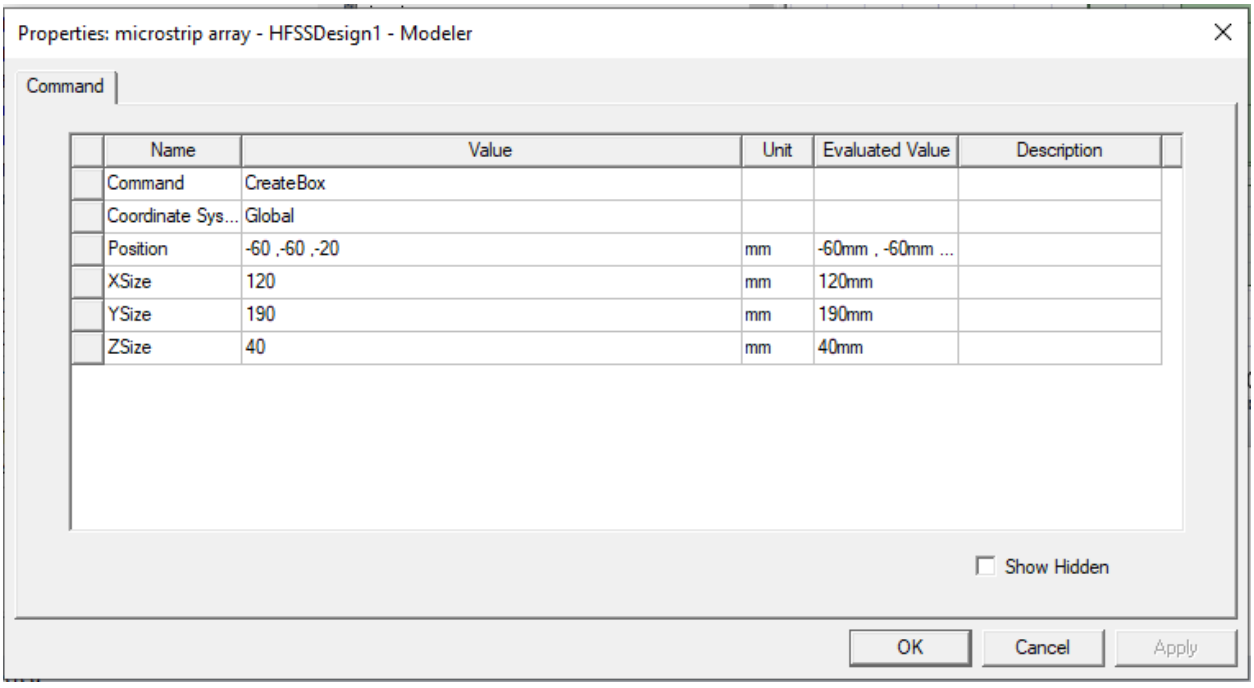


- Note that the Position1 and Position2 are both showing [0,0,0] coordinates
- Move the cursor by 70mm to the positive side of Y-axis(you can verify by seeing Position2 which should be [0,70,0]mm), then click once again as given below

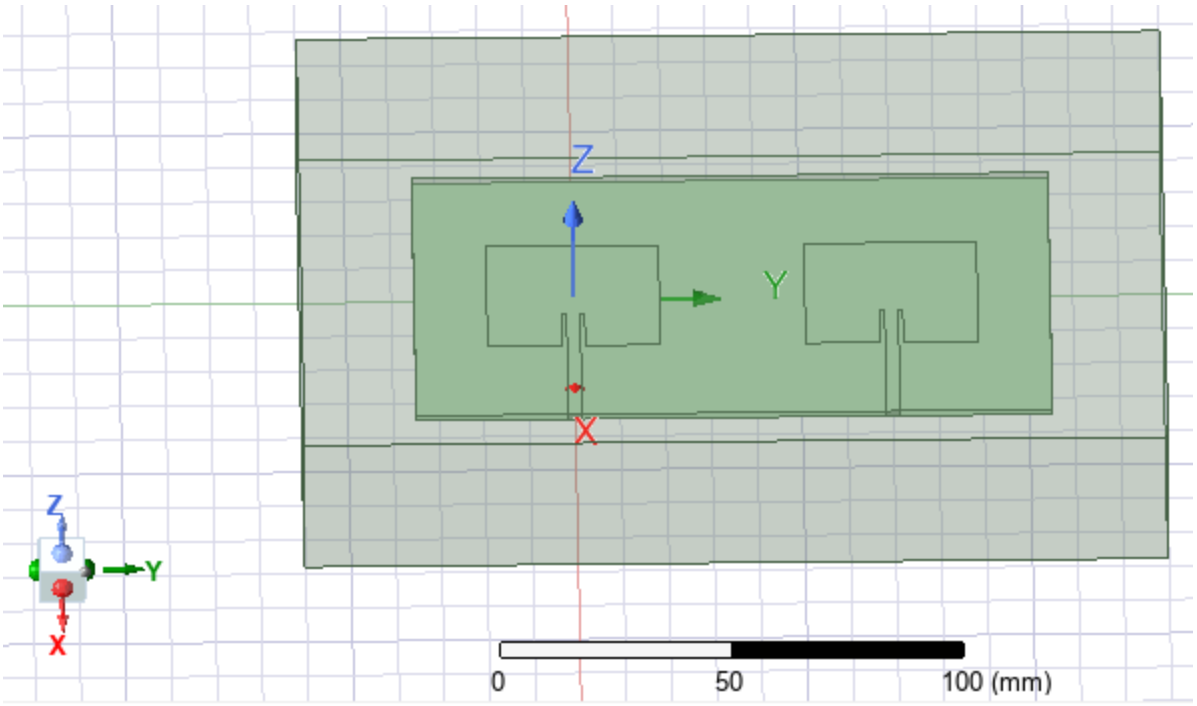


- Press ok in the next window that appears
- Now two separate antennas are created
- Select both the substrates of two antennas and click on unite

- Select both the ground planes of two antennas and click on unite
- Create a box(radiation boundary) with dimensions as given below,



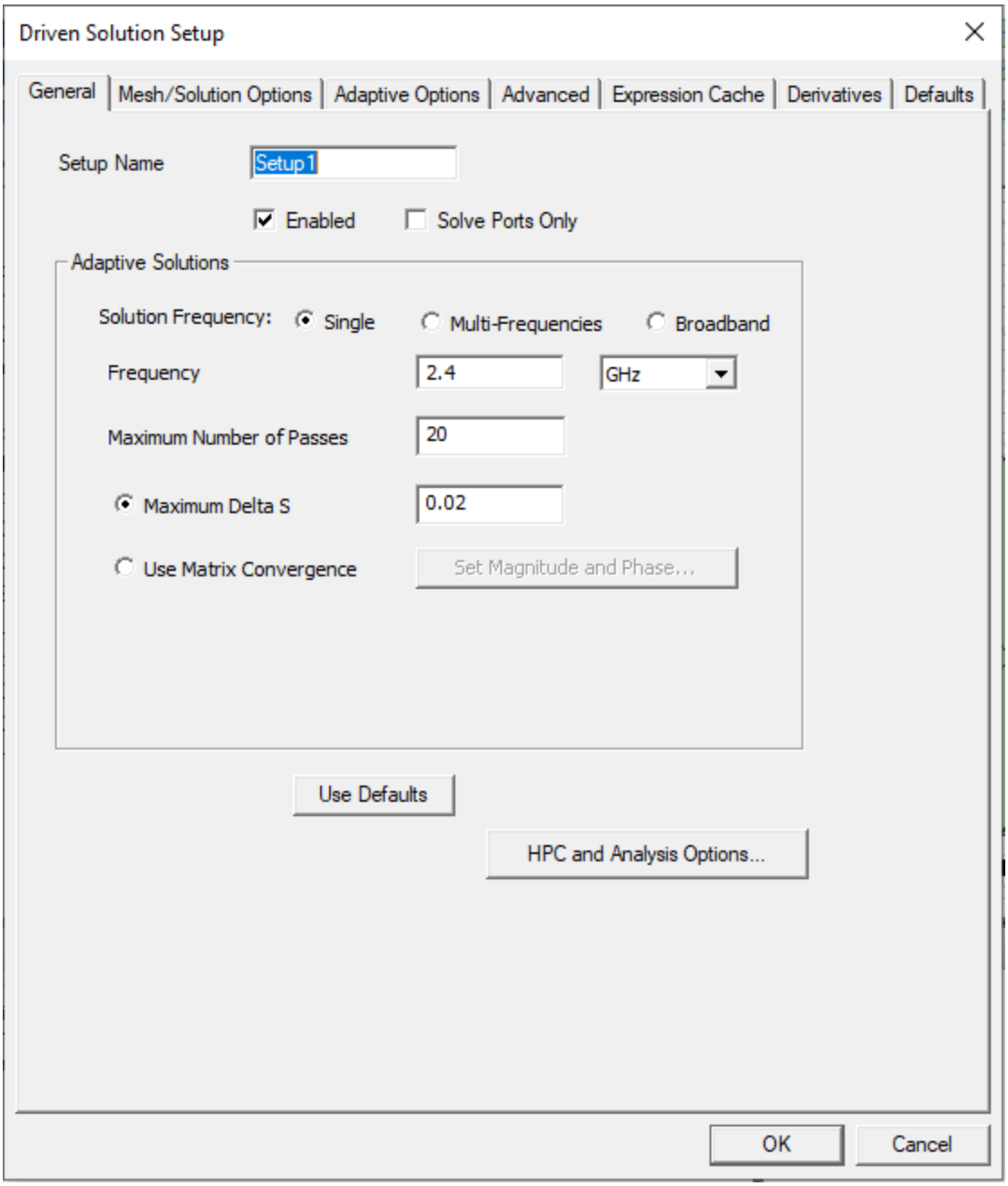
- Assign the material of box2 as “air”
- Right click on Box2 in model section, assign boundary, radiation, ok
- Now your model is constructed along with radiation boundary as given below,



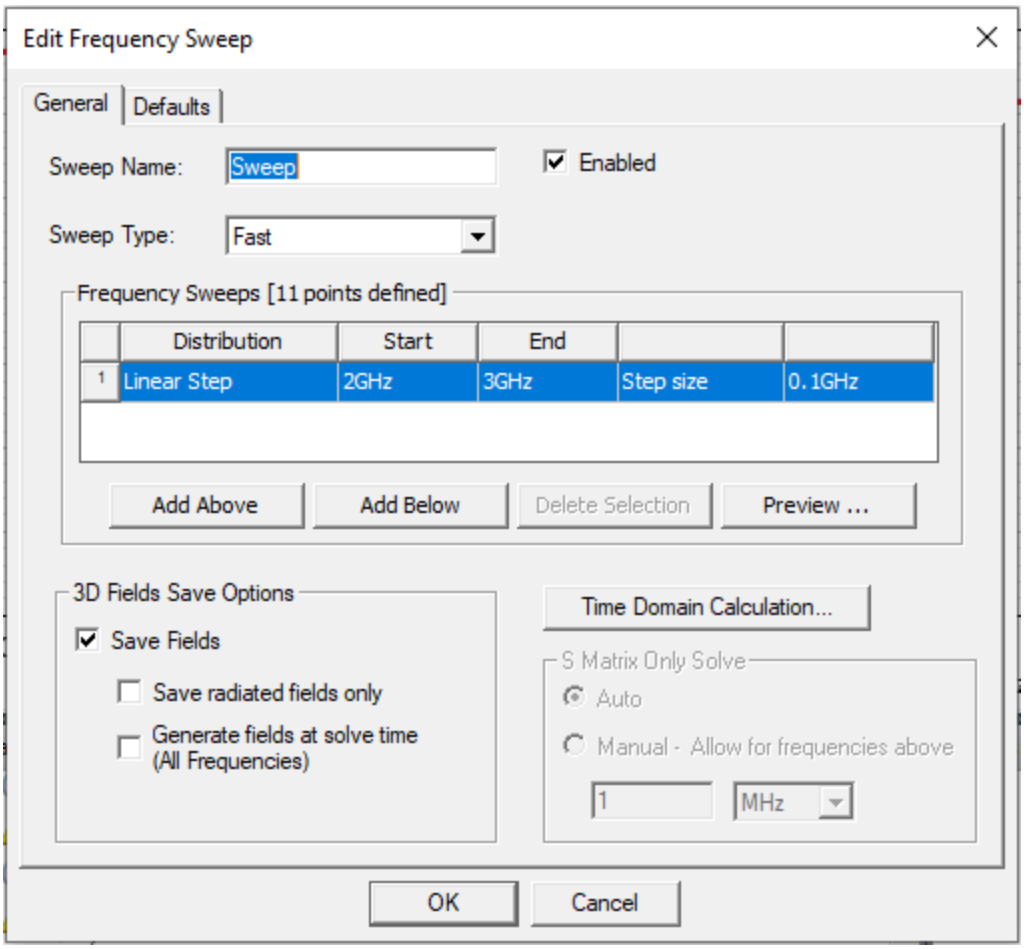
Step 3

Configurations (Settings)

- In the project manager section, Right click on analysis, add solution setup, advanced
- Enter the values as given below



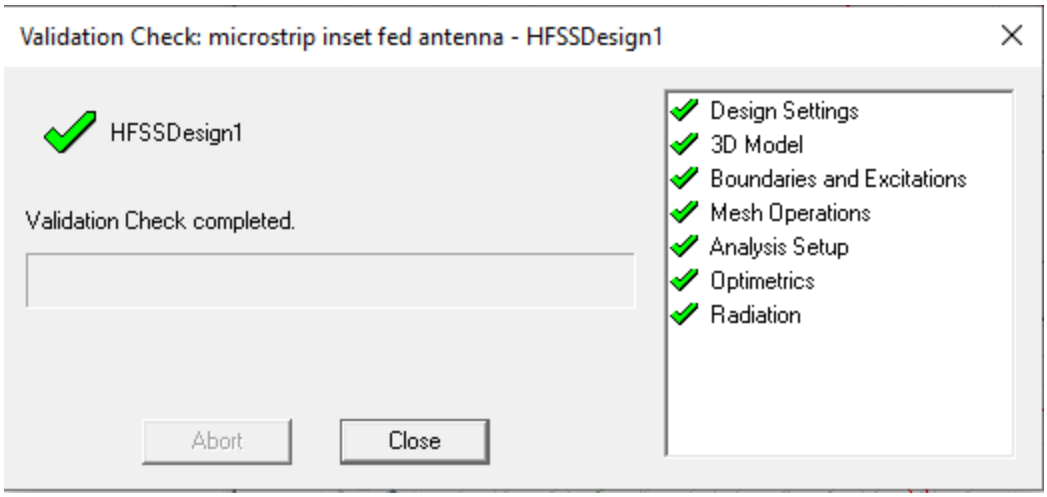
- Click on ok
- In the next window that appears after clicking ok, enter the values as given below



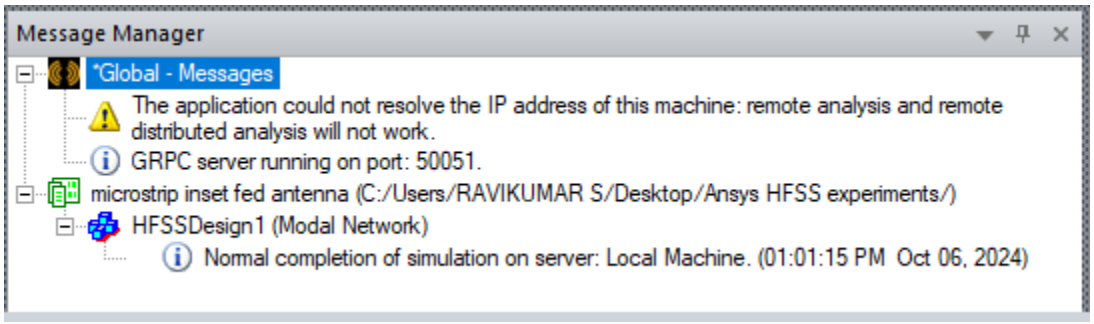
Step 4

Validation and Analysis

- Go to HFSS, Validation check
- The following window will appear if model created is correct



- Close the validation window
- Go to HFSS, Analyze all
- It will take some time to calculate results based on antenna complexity and computer configuration which is running the software
- If simulation goes well then you will receive a message in the message manager section as Normal completion of simulation on server:Local machine as given in figure below.

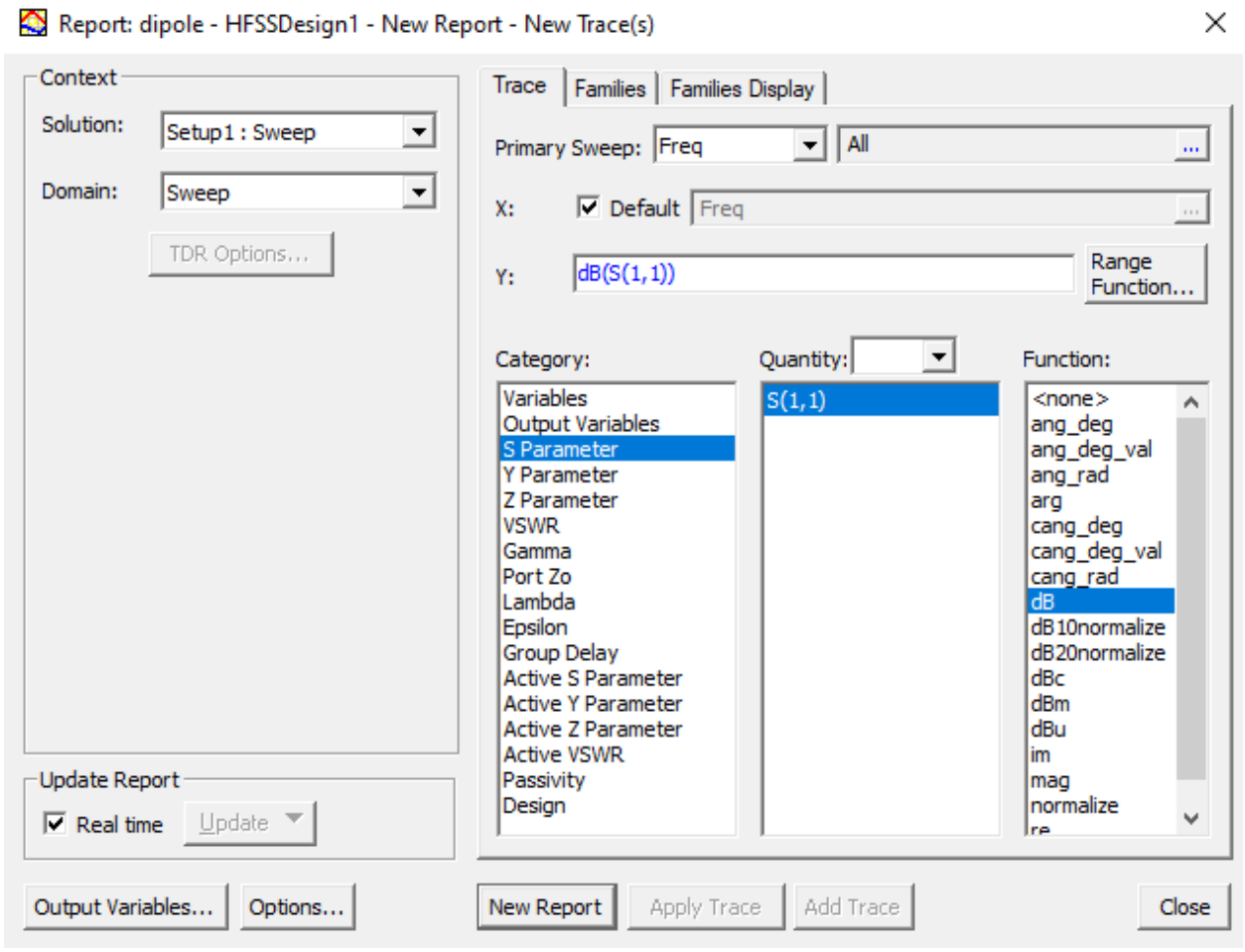


- If any errors are there then such error will be shown in message manager section. They have to be resolved.

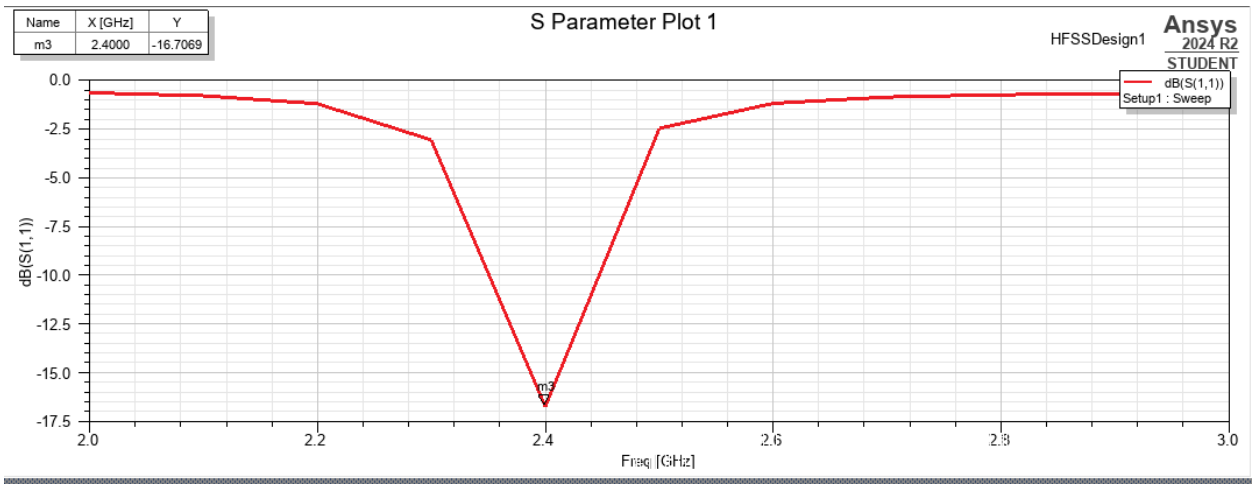
Step 5

Results

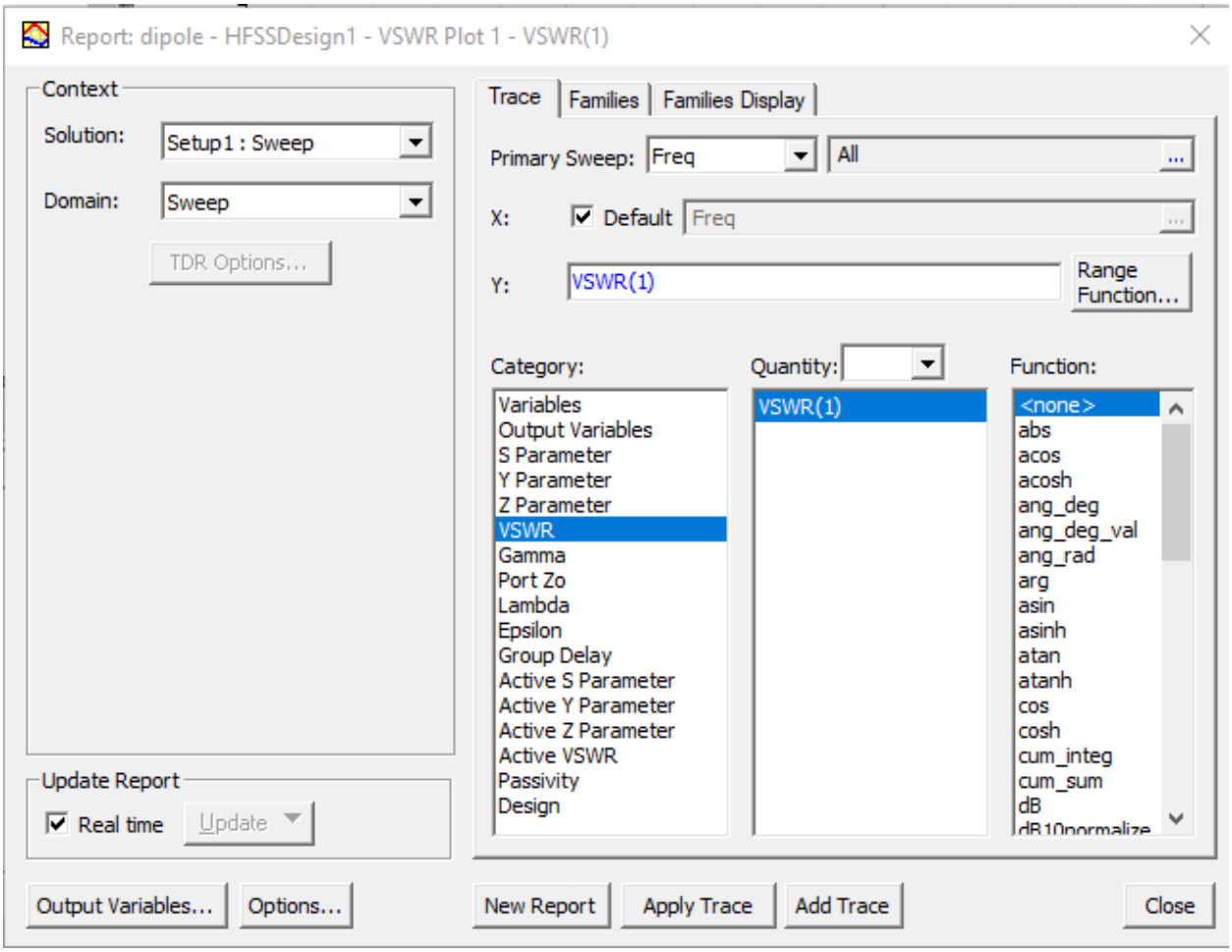
- Right click on results in project manager section, create modal solution data report, rectangular plot,
- The following window should appear



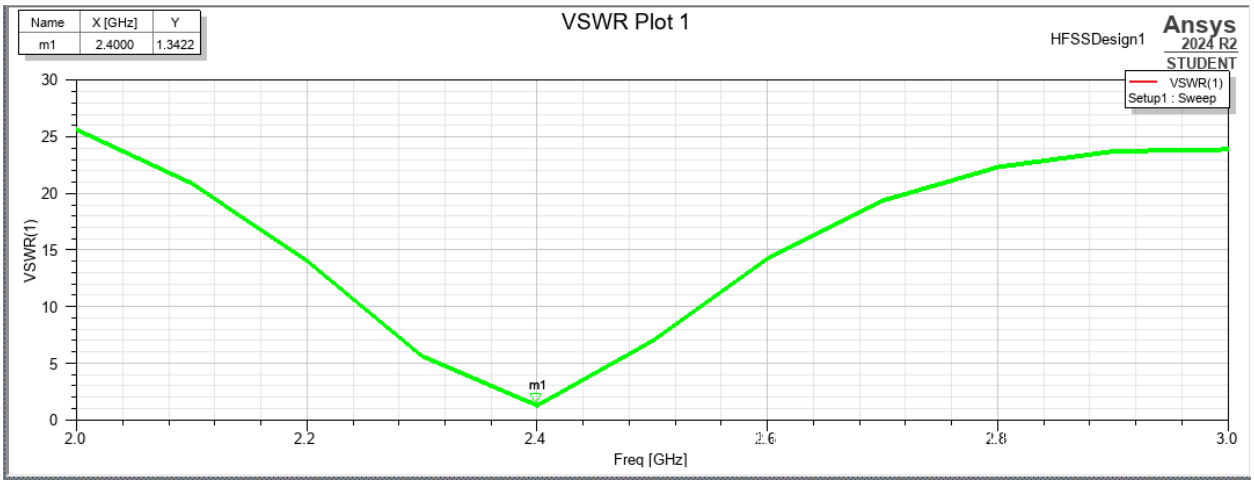
- Click on new report
- The following window will appear showing S11 parameter (Input reflection coefficient)



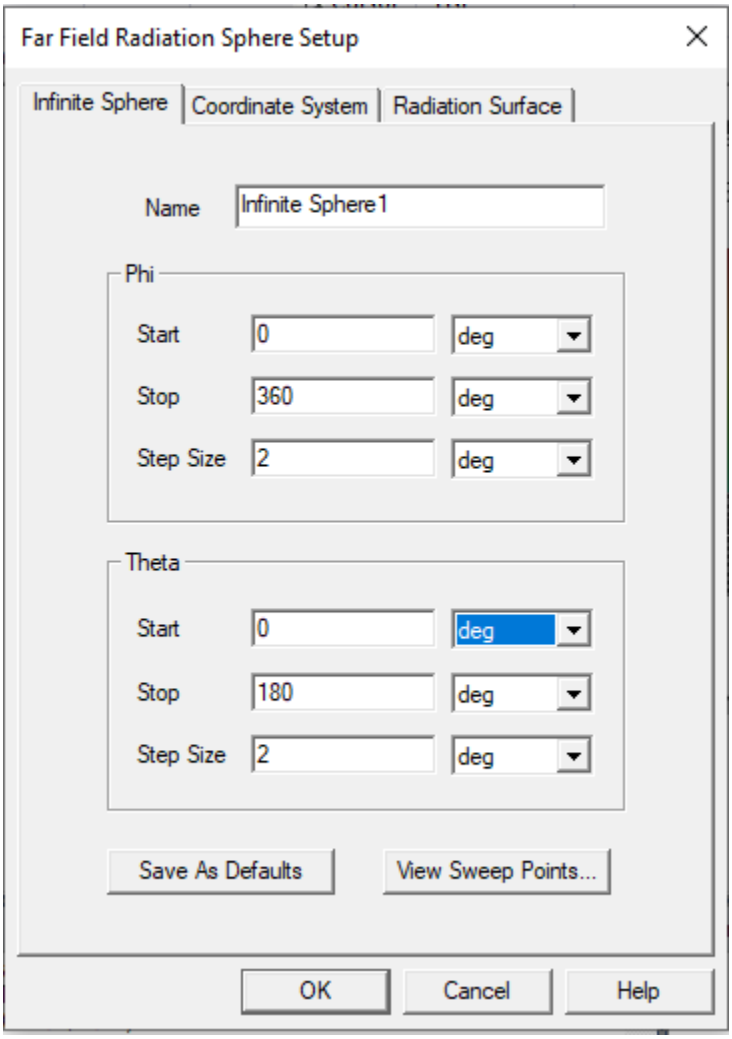
- In the same way Right click on results in project manager section, create modal solution data report, rectangular plot, select VSWR in category section, click on new report as given below



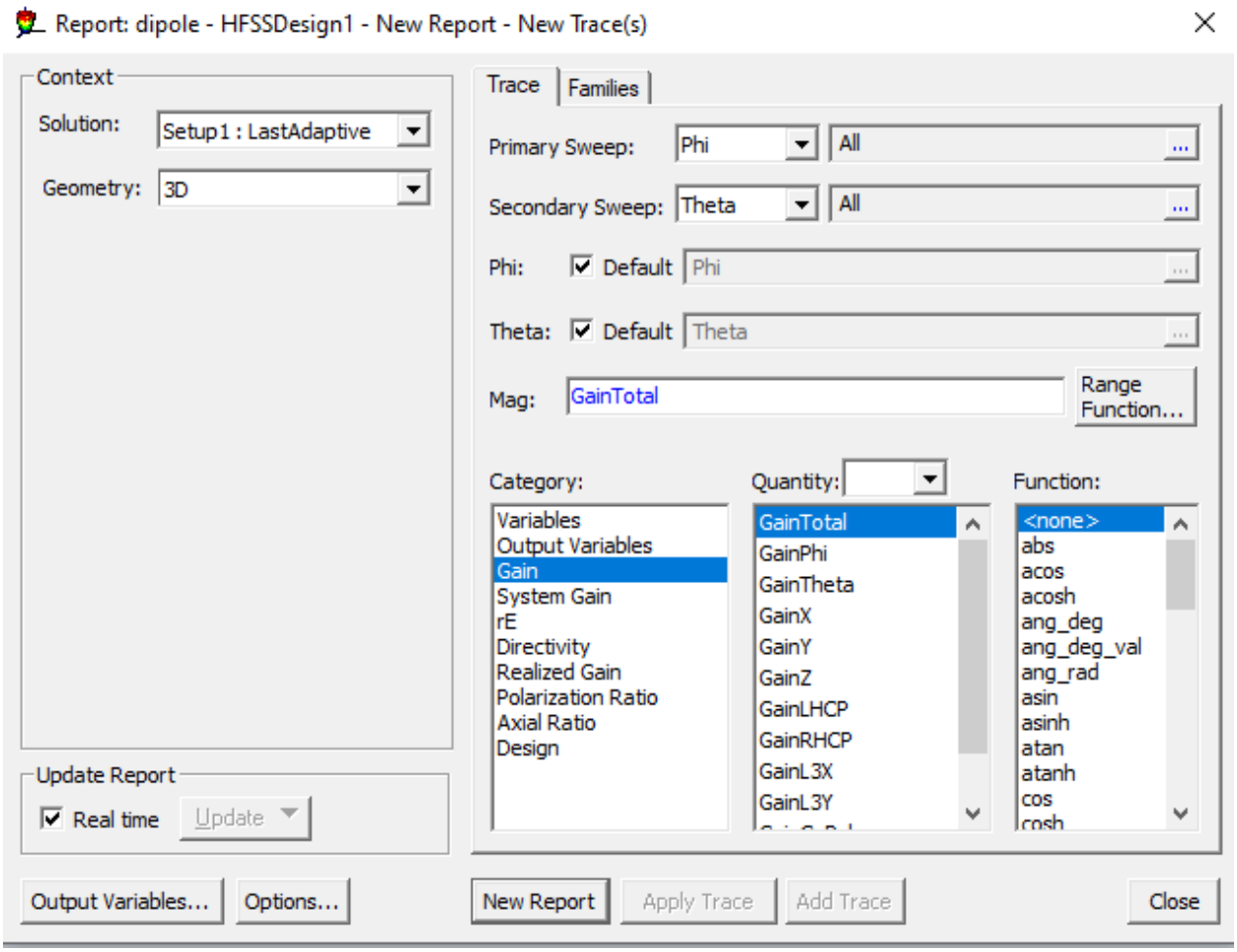
- The VSWR result will appear as given below,



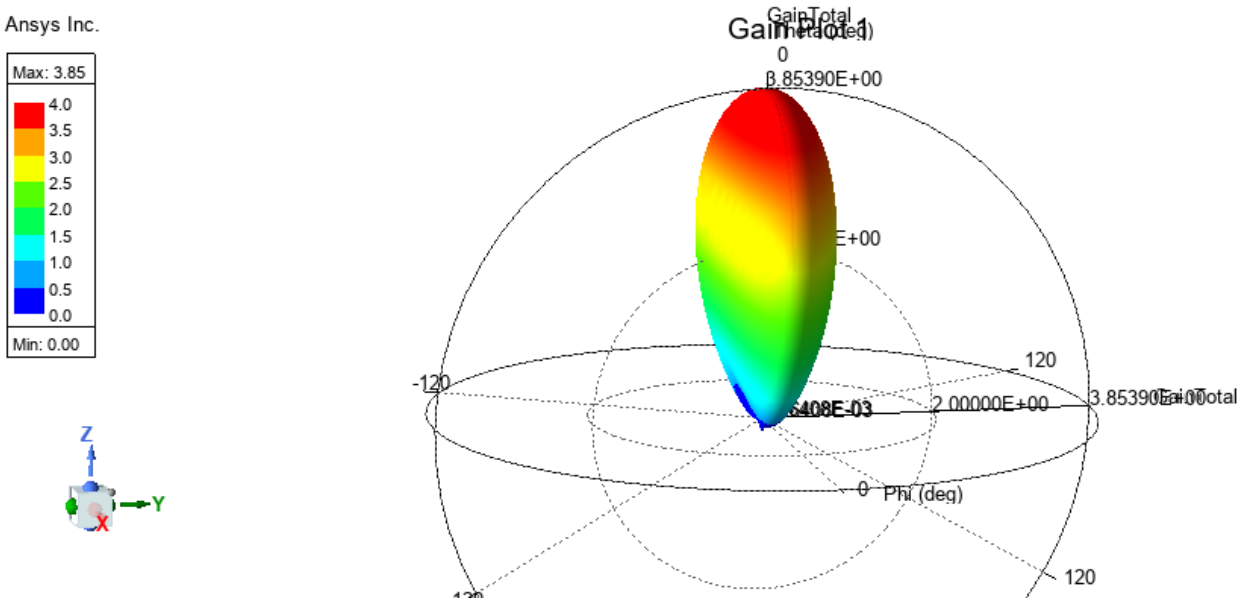
- Right click on radiation in project manager window, insert far field setup, infinite sphere
- Enter the values as given below



- Right click on results, create far fields report, 3D polar plot
- The following window will open,



- Click on new report
- The following 3D radiation pattern will appear,



Tabular Column

Sl No	Sl No	Before Tuning	After Tuning
1	Width of Patch	38 mm	
2	Length of Patch	29.4 mm	
3	S11(in dB) at 2.4GHz		
4	VSWR at 2.4GHz		
5	Bandwidth in Hz (at -10dB points of S11)		
6	Gain		

Assignment

Assignment

Experiment 10

Yagi-Uda Antenna

Aim:-

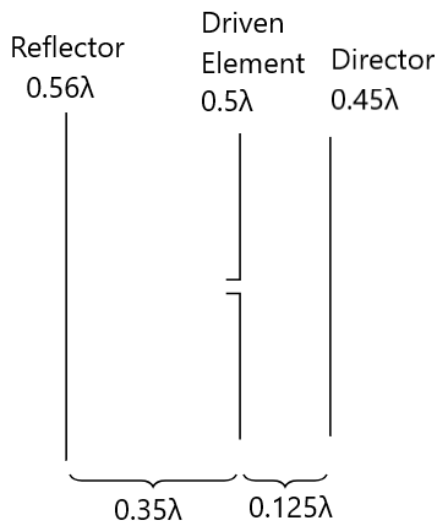
To design and analyze the performance of Yagi-Uda antenna.

Theory:-

The Yagi-Uda antenna, also known as Yagi antenna, catches TV and radio signals. Invented by Japanese scientists Hidetsugu Yagi and Shintaro Uda in the 1920s, it's great at sending signals in one direction. Useful for improving TV signals, ham radios, radar, and satellite communication. Works best in specific places needing strong and focused signal coverage.

Components of Yagi-Uda Antenna

- **Driven Element:** In the Yagi-Uda antenna, the driven element is vital for signal transmission and reception. It interacts with electromagnetic waves, facilitating effective communication.
- **Reflector:** Strategically placed behind the driven element, the reflector significantly enhances the antenna's performance. It acts passively, redirecting and focusing incoming signals toward the driven element to improve reception.
- **Directors:** Positioned in front of the driven element, directors are additional elements that collaborate to enhance the antenna's directional capabilities. By carefully adjusting the number and spacing of directors, the antenna efficiently amplifies signals in the desired direction, resulting in improved reception quality.



Yagi-Uda Antenna

Design Steps:-

Given

Operating Frequency – 0.9 GHz

Necessary design parameters

1. **Driven Element**

Length of the driven element(dipole) - λ/2

If frequency is 0.9 GHz, then λ = c/f = 333 mm

Then Length of the driven element(dipole) = 166.5 mm

Consider feed gap = 1mm

2. **Reflector**

Length of Reflector – 186.48 mm

Spacing between Reflector and Driven Element – 116.55 mm

3. **Director**

Length of Director – 149.85 mm

Spacing between Director and Driven Element – 41.625 mm

Consider radius of the all 3 elements = 1mm

Steps for simulation

Step 1: Starting HFSS

Step 2: Creating the antenna model

Step 3: Configurations (Settings)

Step 4: Validation and analysis

Step 5: Results

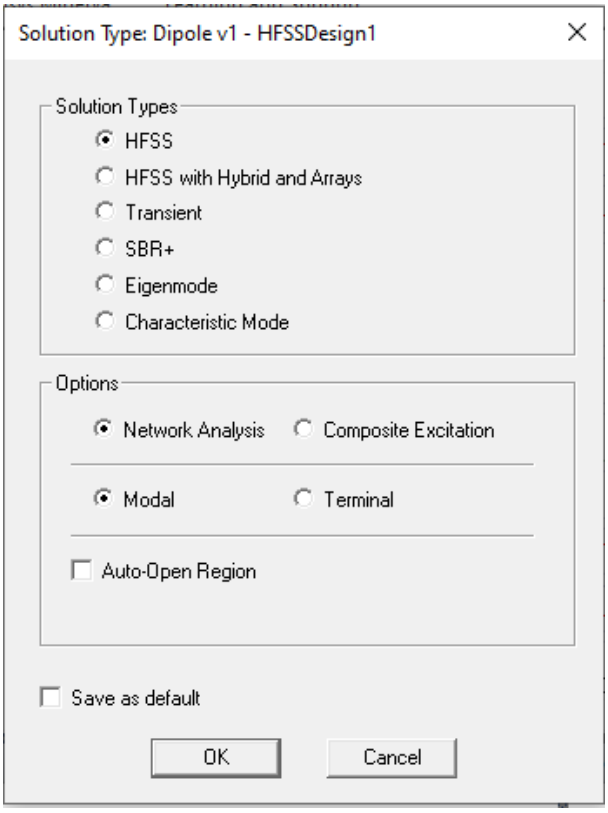
Step 1

Starting HFSS

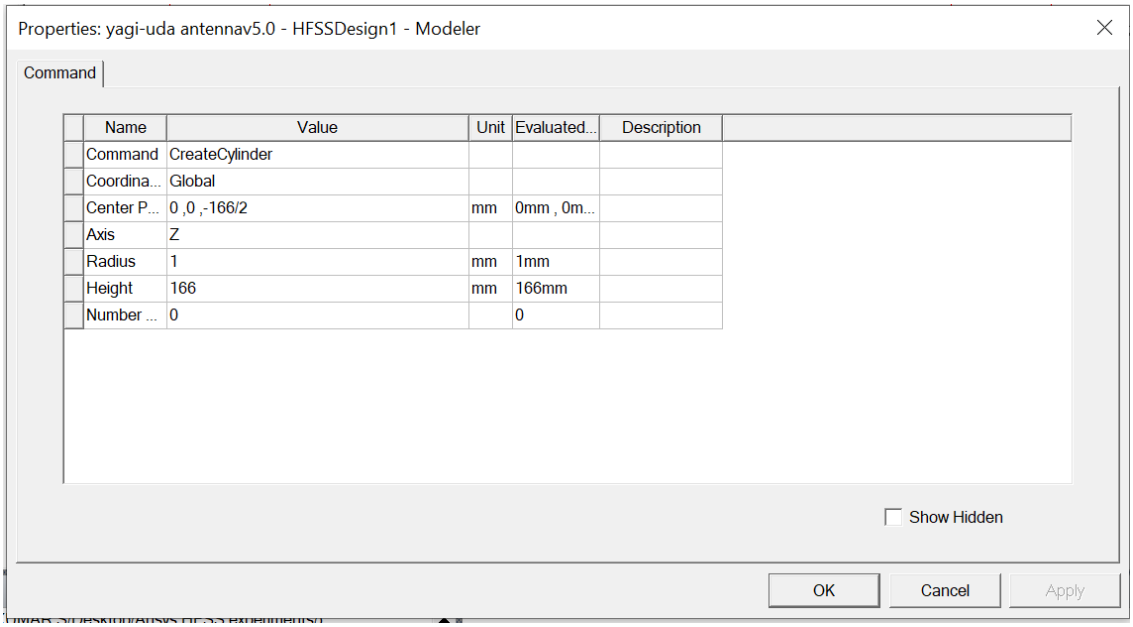
Step 2

Creating the antenna model

- From the HFSS menu select **Solution Type** as given below.

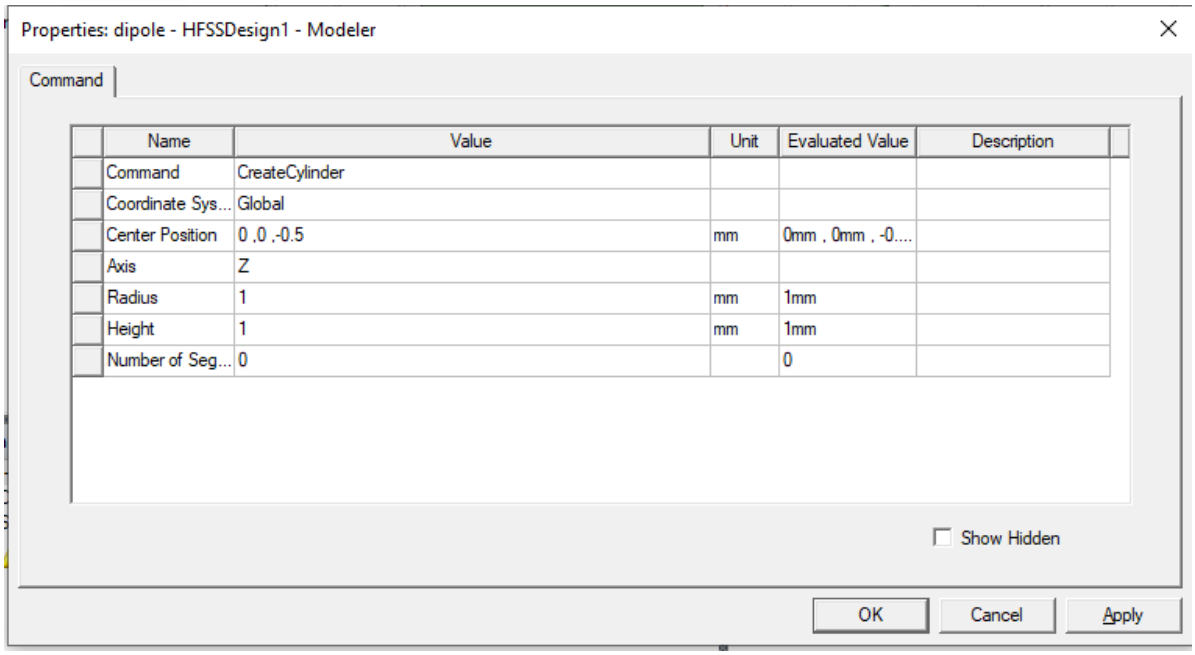


- Draw a cylinder with following coordinates,

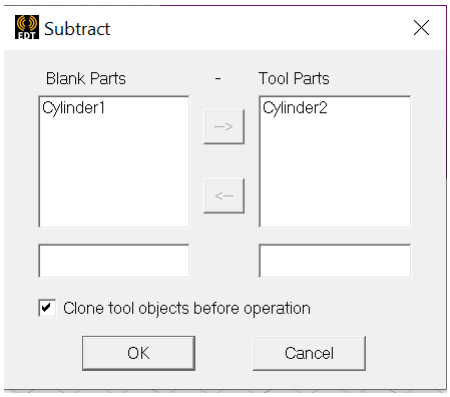


- Right click on cylinder1 in model section
- Select Assign material
- Type pec in the search by name space
- Select pec
- Click ok

- Draw another cylinder for creating a gap in the middle of the dipole with dimensions as given below,



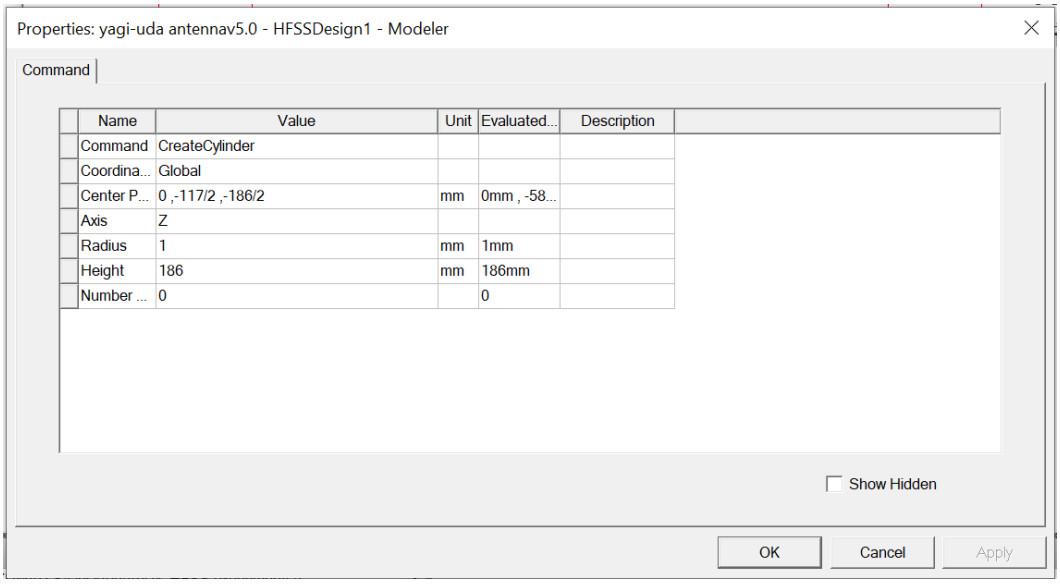
- To create space for giving excitation, select the two cylinders i.e. cylinder1(dipole) and cylinder2(feed)
- Go to Modeler, Boolean, Subtract
- The following window will appear,



- Check the box on clone tool objects before operation
- Click ok

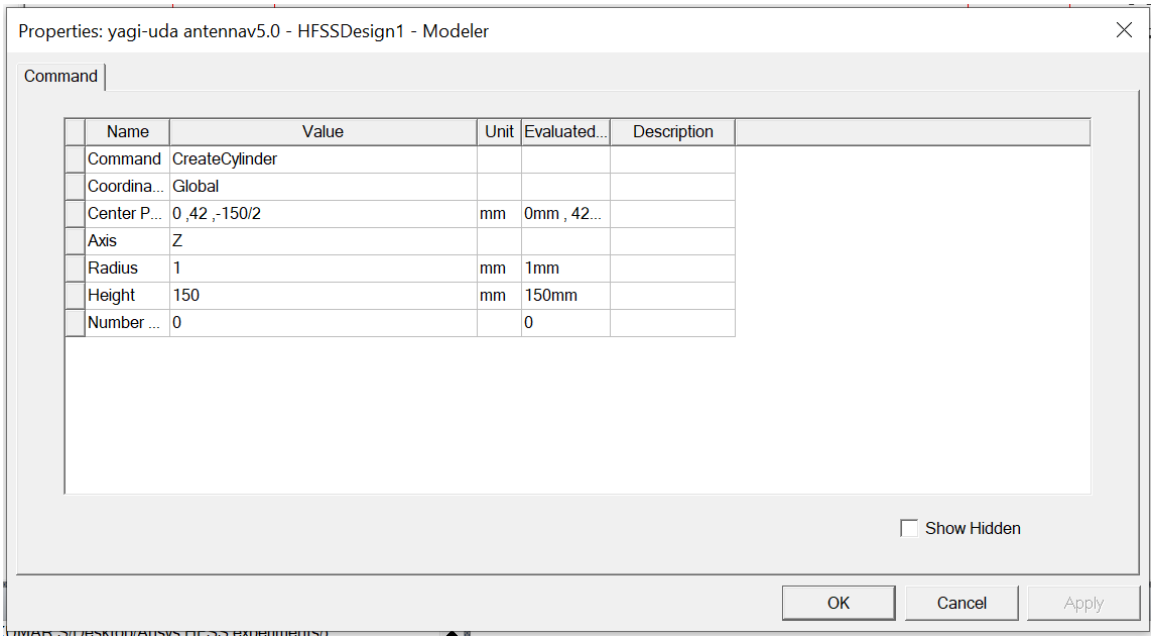
- Select cylinder2
- Goto Modeler, Surface, Section, select YZ plane and press ok.
- A rectangle is now created in YZ plane within cylinder2(i.e. feed)

- Right click on cylinder2_section1 in model section, assign excitation, port, lumped port, resistance 50 Ohms, next, click on dropdown under integration line and select choose new line
- Draw a line from bottom center to the top center of the rectangle, next, finish
- Draw a cylinder(reflector) with following coordinates



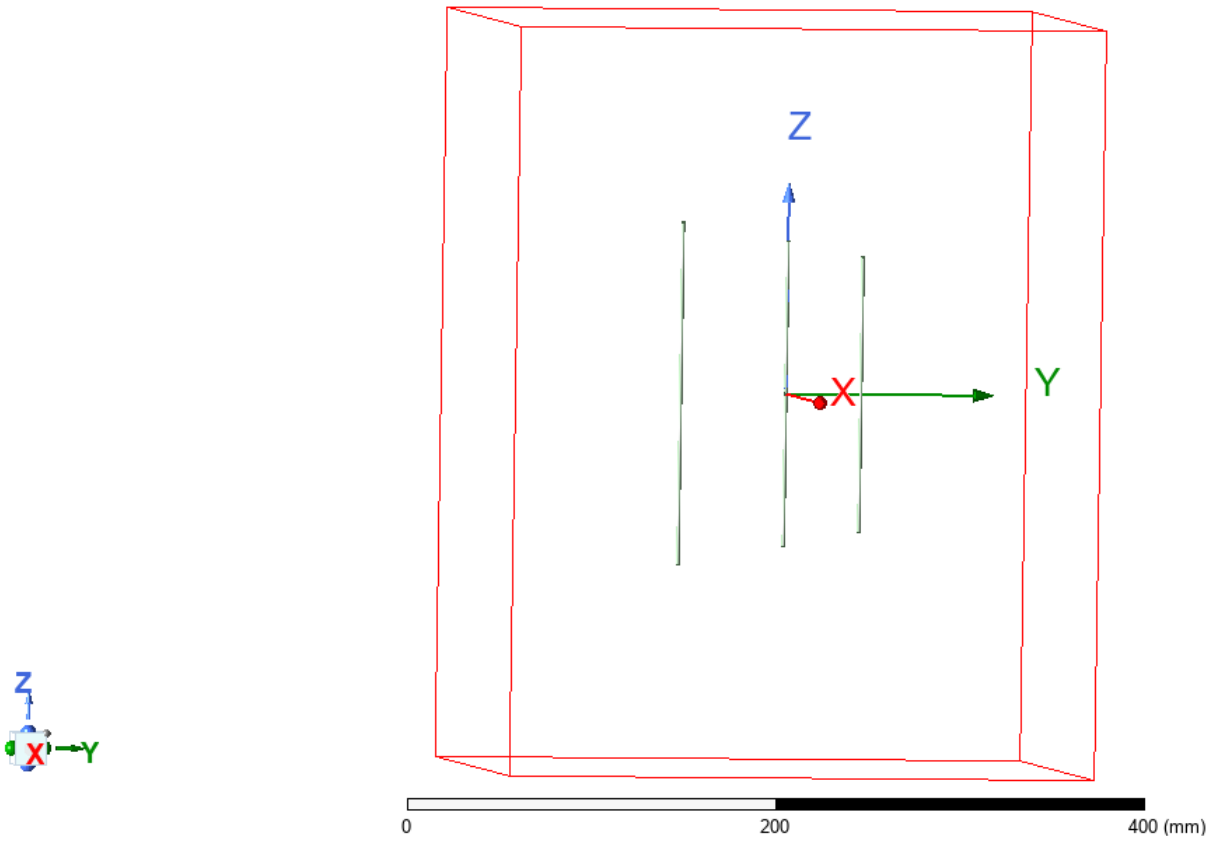
- Right click on cylinder3 in model section
- Select Assign material
- Type pec in the search by name space
- Select pec, ok

Draw a cylinder(director) with following coordinates



- Right click on cylinder4 in model section
- Select Assign material
- Type pec in the search by name space
- Select pec, ok

- Right click in the work area, create open region, set operating frequency as 0.9 GHz, select radiation boundary, ok
- Now your model will look like the figure below with dipole antenna in the middle, reflector and director on either side and radiation box enclosing the three.

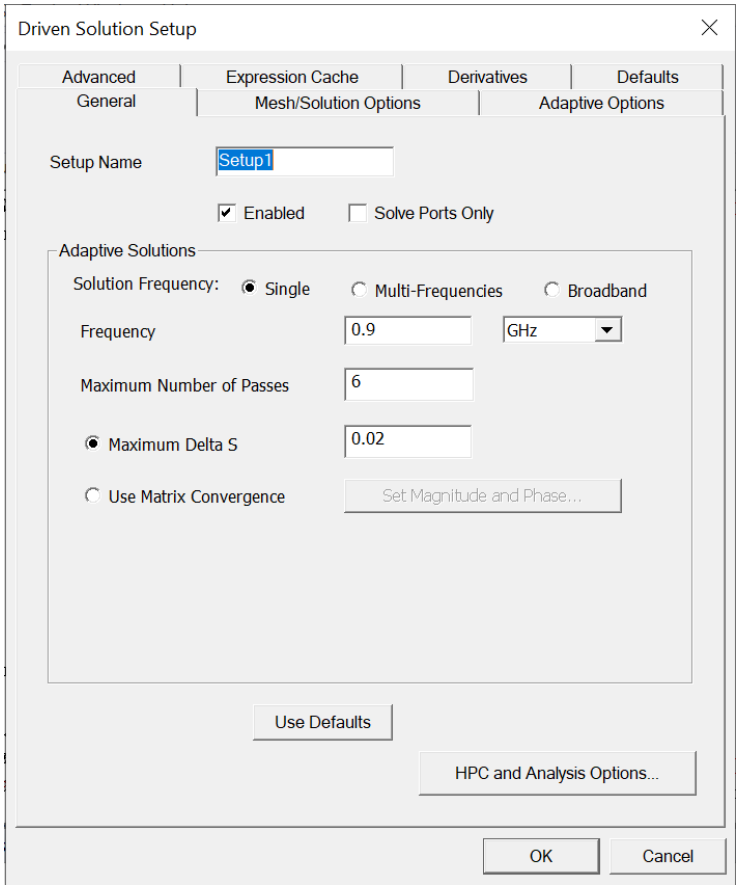


- Now your model is constructed along with radiation boundary.

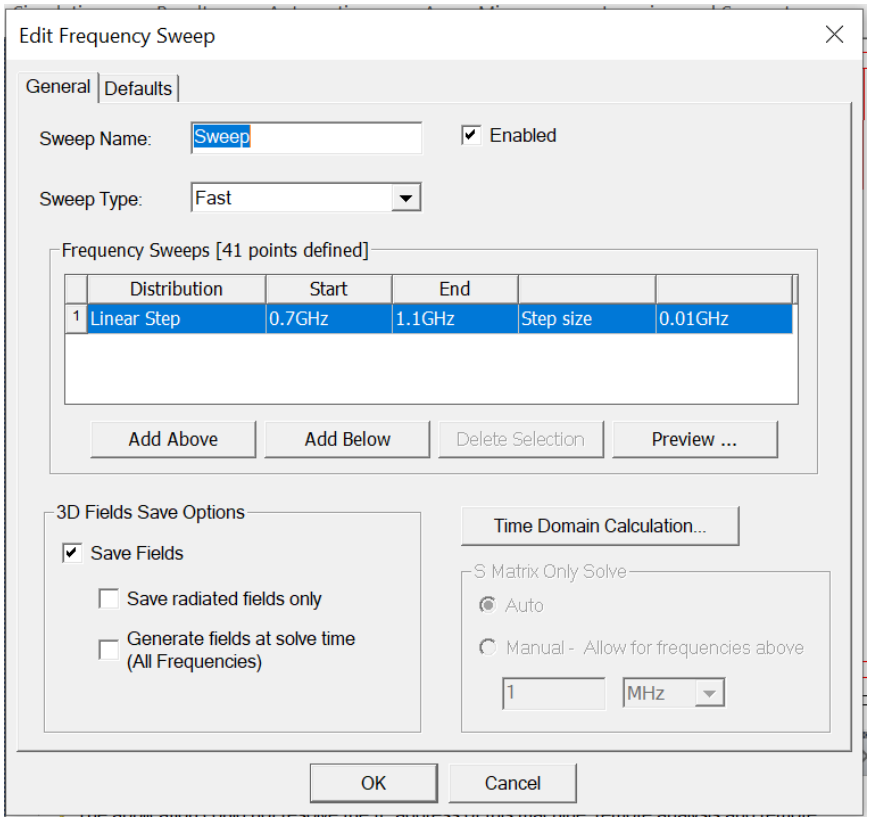
Step 3

Configurations (Settings)

- In the project manager section, Right click on analysis, add solution setup, advanced
- Enter the values as given below



- Click on ok
- In the next window that appears after clicking ok enter the values as given below,

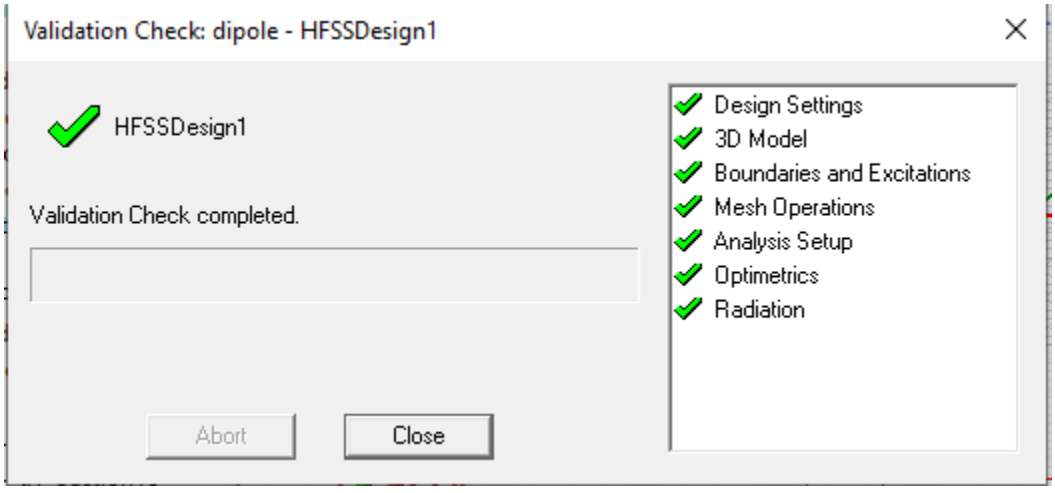


- Click ok

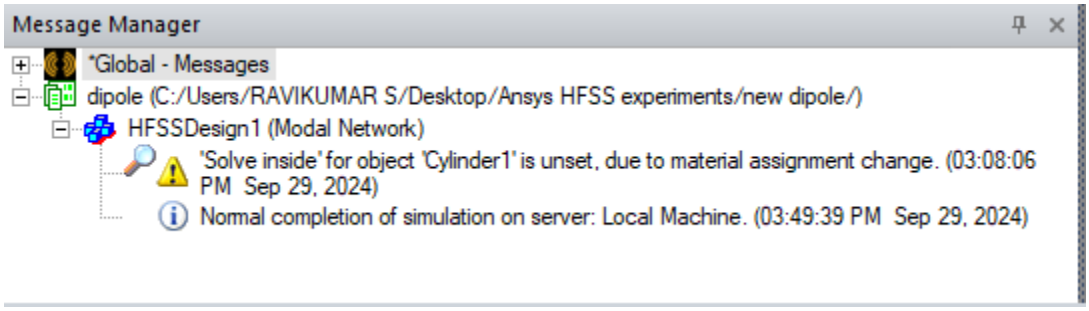
Step 4

Validation and Analysis

- Go to HFSS, Validation check
- The following window will appear if model created is correct



- Close the validation window
- Go to HFSS, Analyze all
- It will take some time to calculate results based on antenna complexity and computer configuration which is running the software
- If simulation goes well then you will receive a message in the message manager section as Normal completion of simulation on server:Local machine as given in figure below.

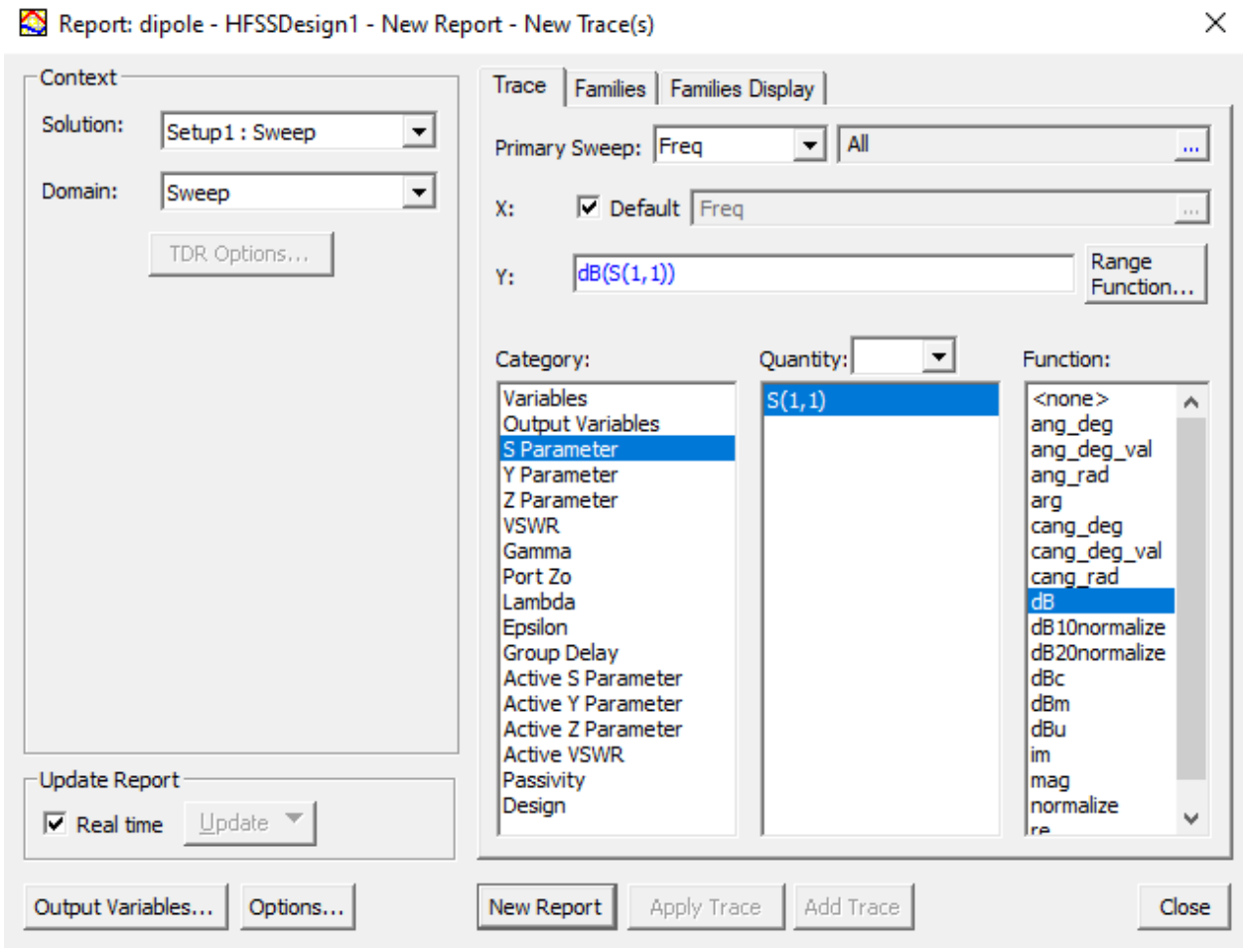


- If any errors are there then such error will be shown in message manager section. They have to be resolved.

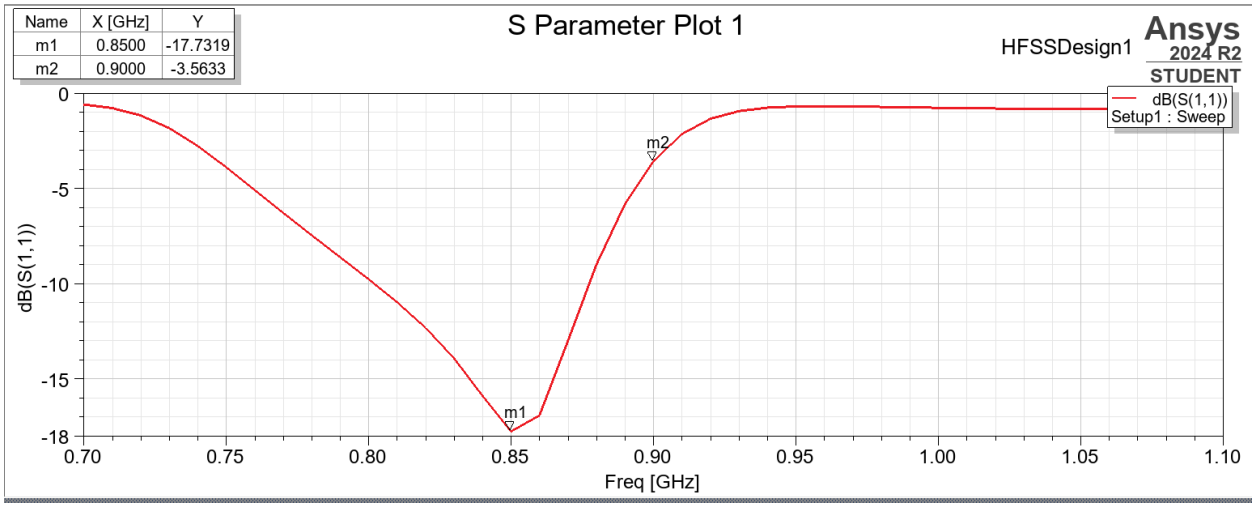
Step 5

Results

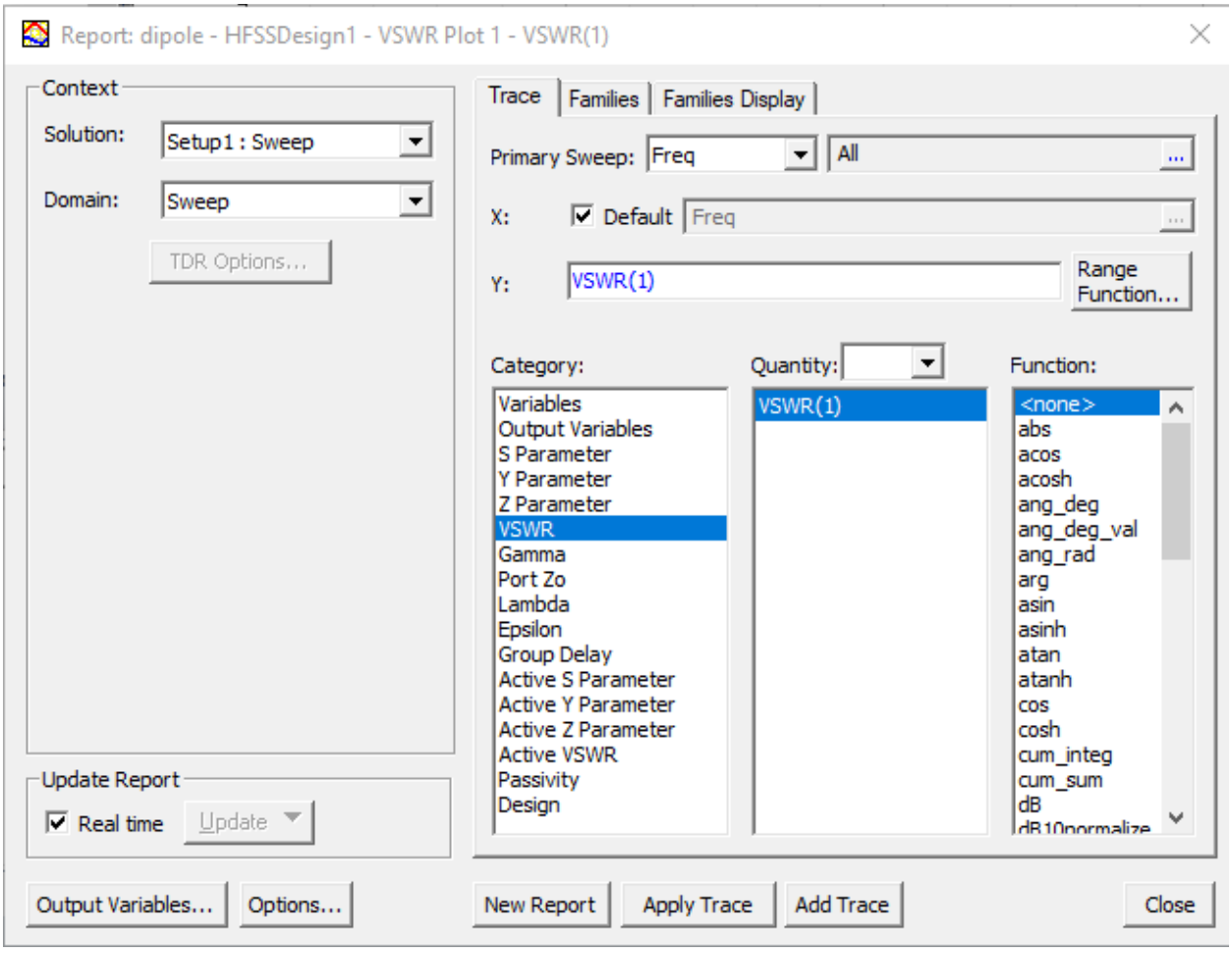
- Right click on results in project manager section, create modal solution data report, rectangular plot,
- The following window should appear,



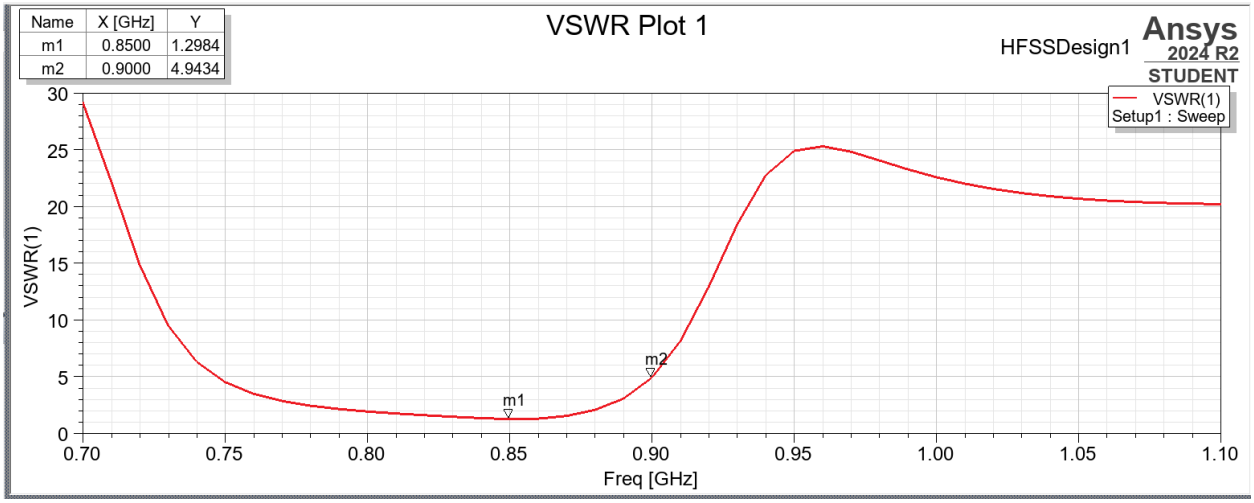
- Click on new report
- The following window will appear showing S11 parameter (Input reflection coefficient)



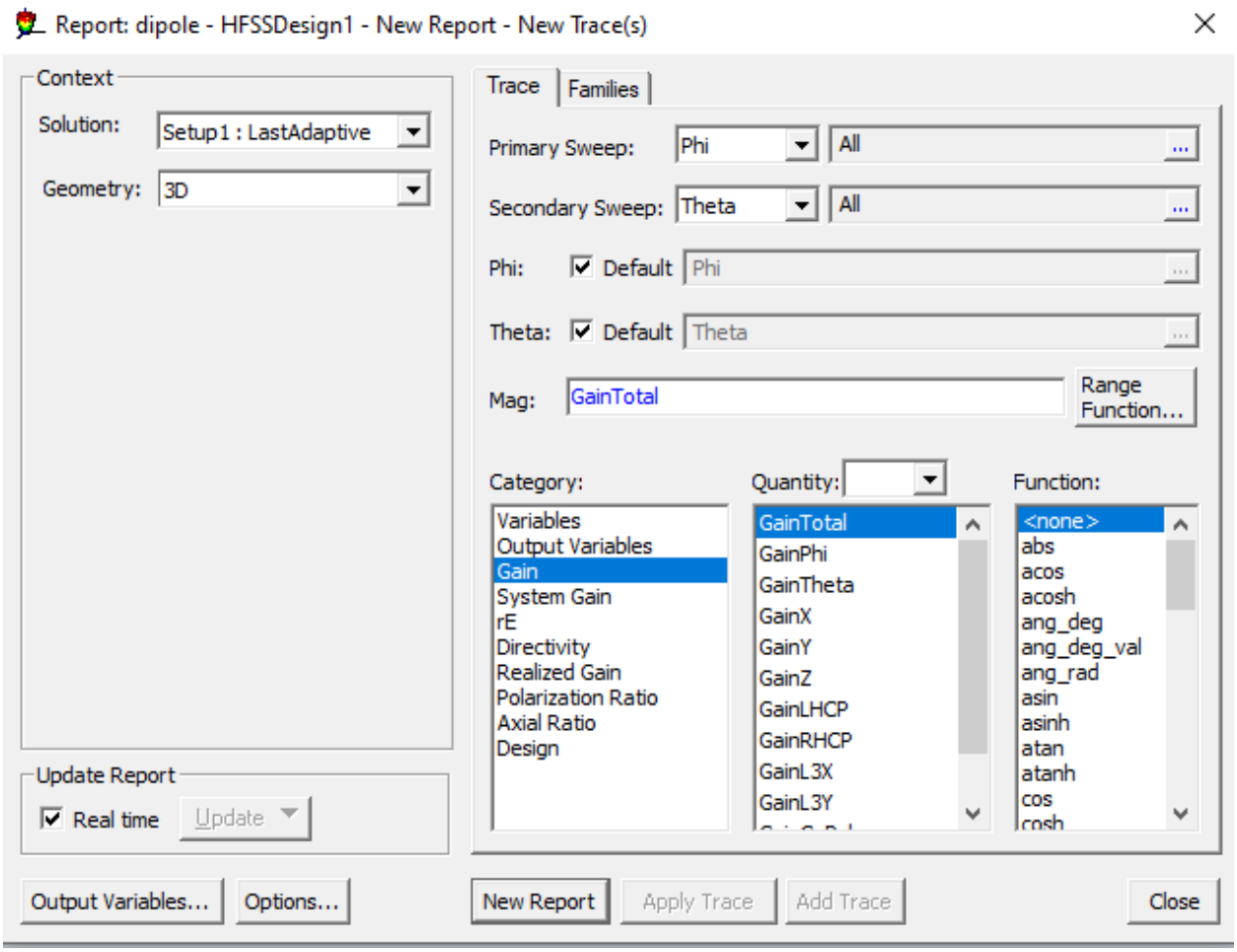
- In the same way Right click on results in project manager section, create modal solution data report, rectangular plot, select VSWR in category section, click on new report as given below



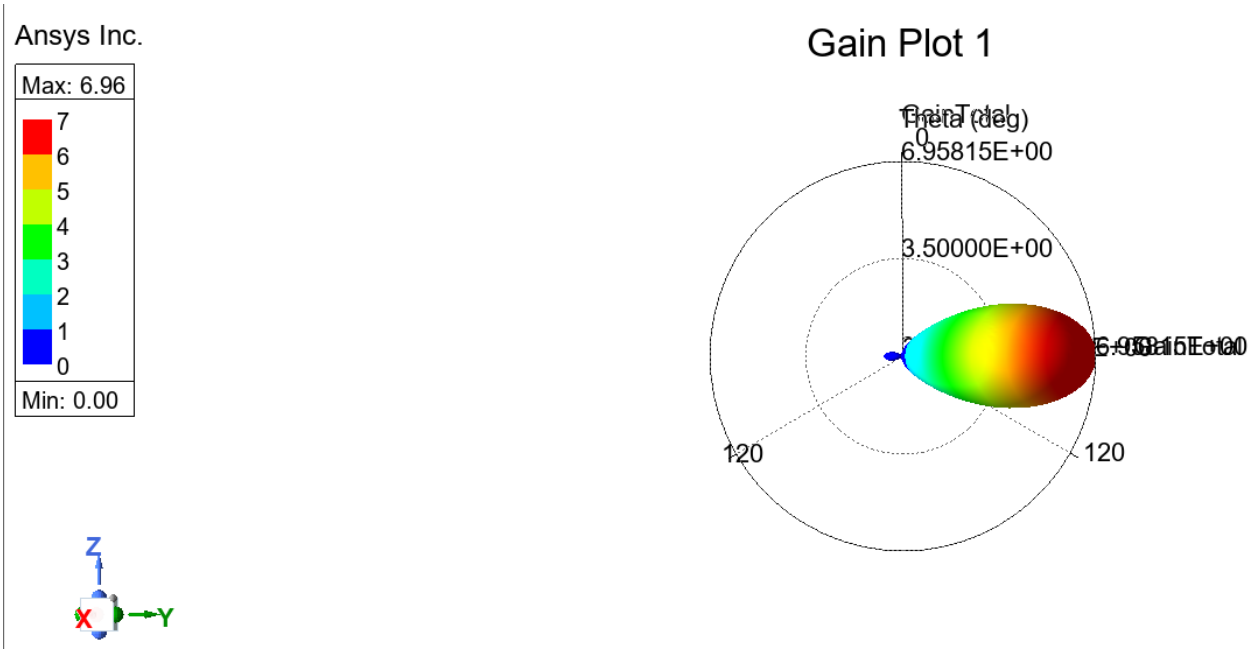
- The VSWR result will appear as given below,



- Right click on results, create far fields report, 3D polar plot
- The following window will open,



- Click on new report
- The following 3D radiation pattern will appear



As seen in the S parameter graph there is a slight deviation in the operating frequency of the antenna from desired. The desired operating frequency is 0.9 GHz and the current operating frequency as indicated by the S parameter graph is 0.85GHz. So, by slightly adjusting design parameters such as length of driven element and director desired performance can be obtained for 0.9GHz.

Tabular Column

Sl No	Sl No	Before Tuning	After Tuning
1	Length of Driven Element	166 mm	
2	Length of Director	150 mm	
3	S11(in dB) at 0.9GHz		
4	VSWR at 0.9GHz		
5	Bandwidth in Hz (at -10dB points of S11)		
6	Gain		

Assignment

Assignment

Experiment 11

Helical Antenna

Aim:-

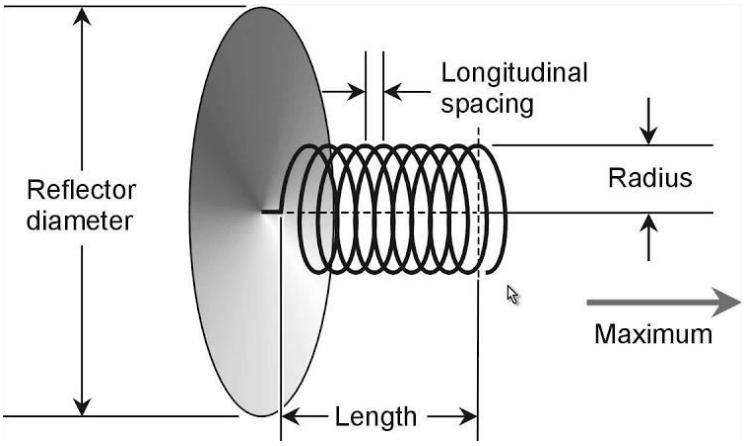
To design and analyze the performance of Helical antenna.

Theory:-

Helical antenna is an example of wire antenna and itself forms the shape of a helix. This is a broadband VHF and UHF antenna.

The frequency range of operation of helical antenna is around 30MHz to 3GHz. This antenna works in VHF and UHF ranges.

Helical antenna or helix antenna is the antenna in which the conducting wire is wound in helical shape and connected to the ground plate with a feeder line. It is the simplest antenna, which provides circularly polarized waves. It is used in extra-terrestrial communications in which satellite relays etc., are involved.



The above image shows a helical antenna system, which is used for satellite communications. These antennas require wider outdoor space.

It consists of a helix of thick copper wire or tubing wound in the shape of a screw thread used as an antenna in conjunction with a flat metal plate called a ground plate. One end of the helix is connected to one side of feed rectangle and the ground plane is connected to other side of feed rectangle.

The radiation of helical antenna depends on the diameter of helix, the turn spacing and the pitch angle.

Pitch angle is the angle between a line tangent to the helix wire and plane normal to the helix axis.

Specifications

Operating Frequency – 2 GHz

Radius of helix – 1.6 cm

Radius of Helix wire – 0.015cm

Pitch (Longitudinal spacing) – 2cm

Number of turns – 3.83

Radius of ground plane – 3.75 cm

Steps for simulation

Step 1: Starting HFSS

Step 2: Creating the antenna model

Step 3: Configurations (Settings)

Step 4: Validation and analysis

Step 5: Results

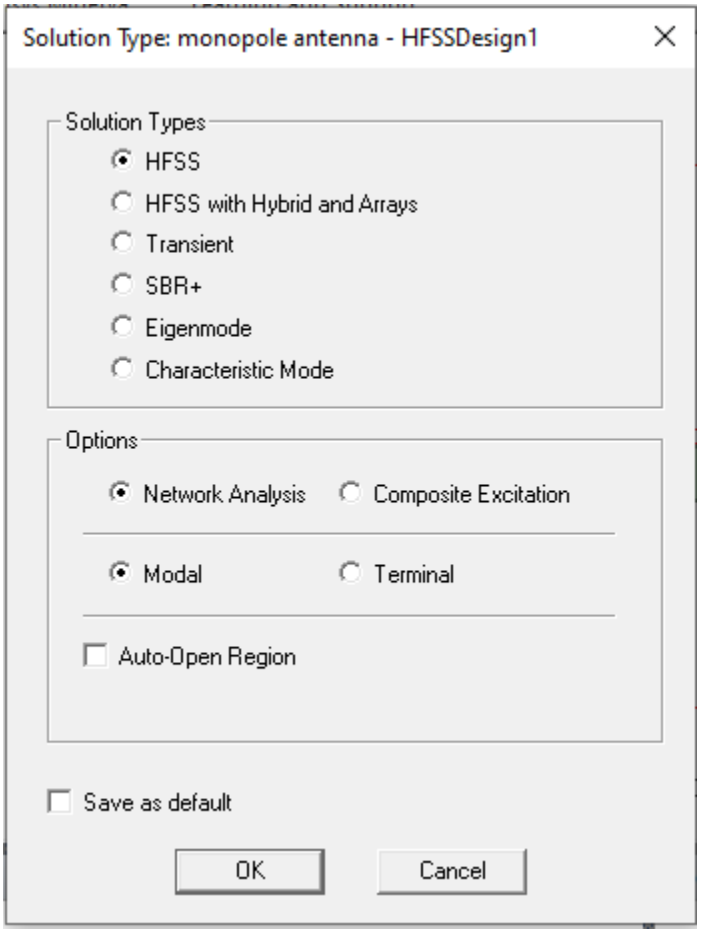
Step 1

Starting HFSS

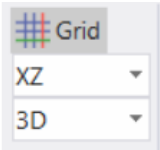
Step 2

Creating the antenna model

- From the HFSS menu select **Solution Type** as given below.

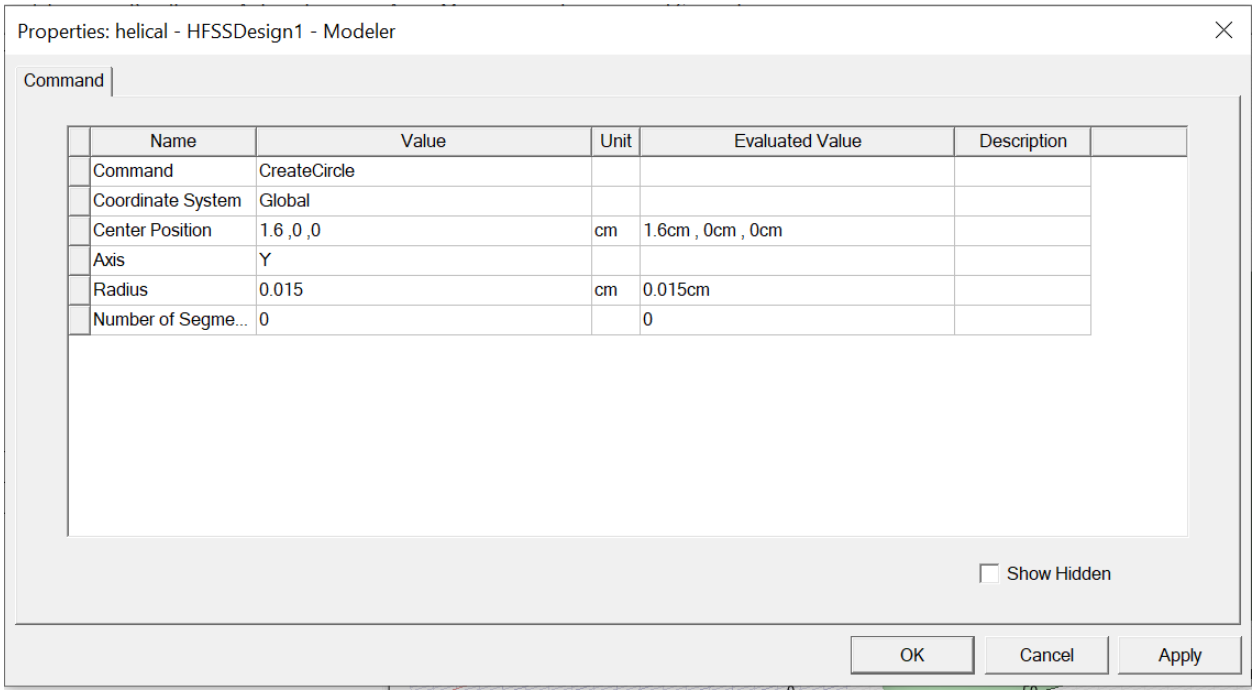


- Change the viewing plane to XZ

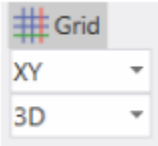


- Draw a circle on the XZ plane. This will be the cross section of the helix.

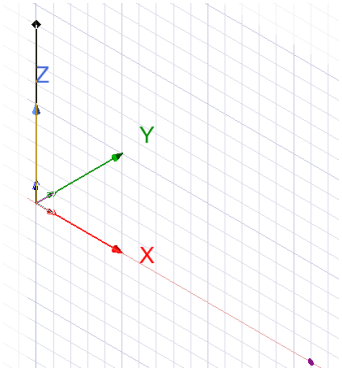
- Double click on CreateCircle and change its properties as follows,



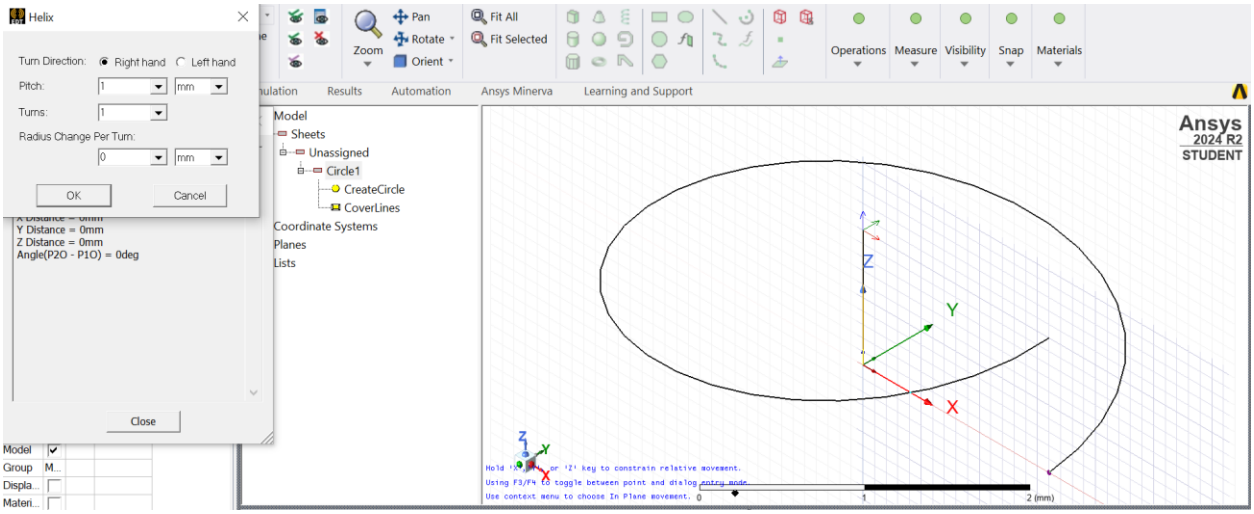
- Click OK.
- Change the viewing plane back to XY.



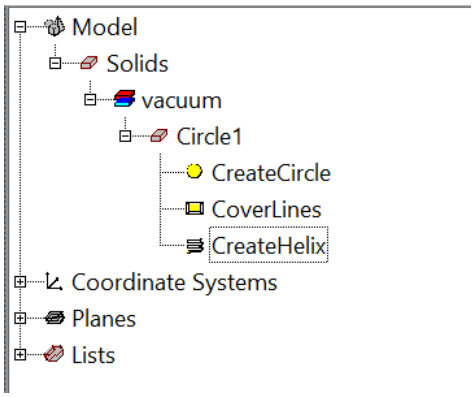
- click on the circle1 that we drew right now. You may see that the draw helix feature is available in the shape box.
- click on draw helix, click on the center of the coordinate system and draw the mouse along the z-axis and click once more as shown below



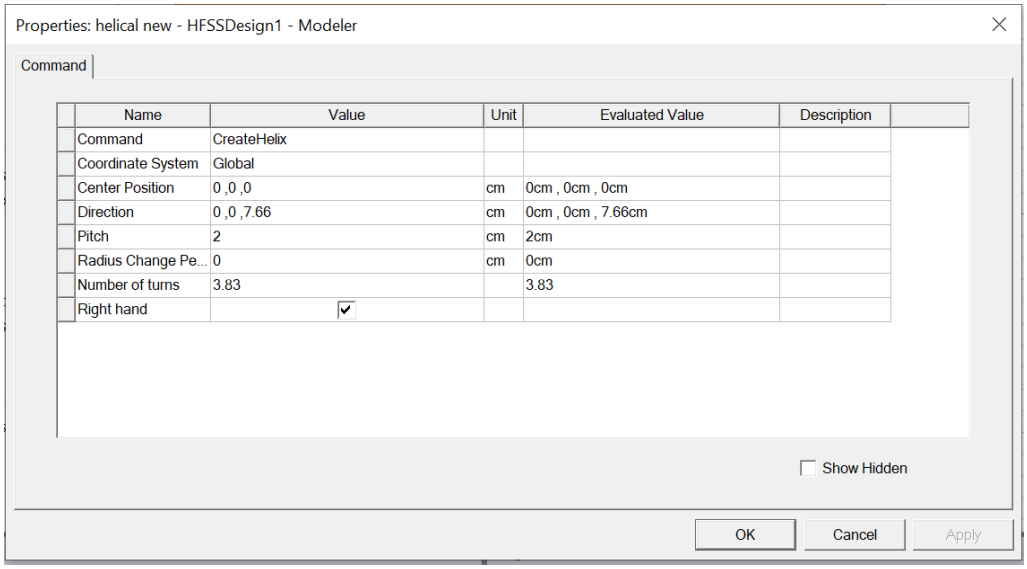
- This will also open a pop-up window and for now hit OK on that with default settings as given below



- Expand the Circle1 on the model pane.

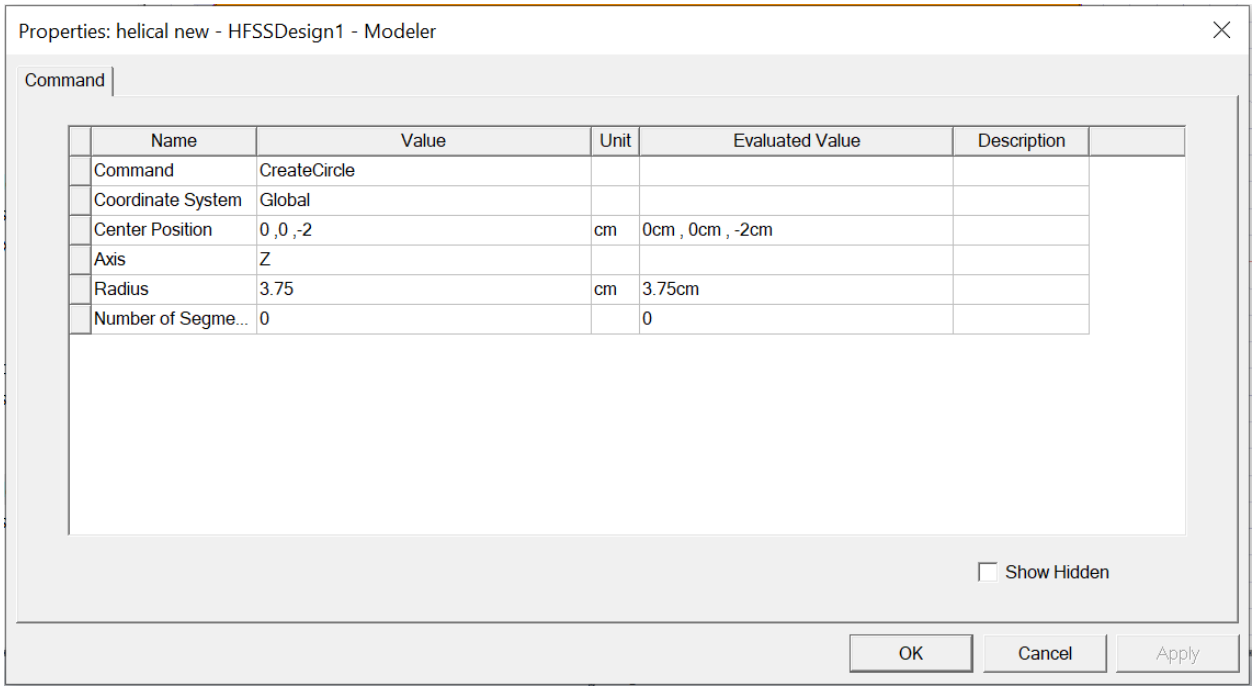


- Double click on the CreateHelix and change its properties as follows,



- Click Ok
- Now double click on Circle 1 and change its material property to pec (perfect electric conductor) and this will change the helix material to pec as well.

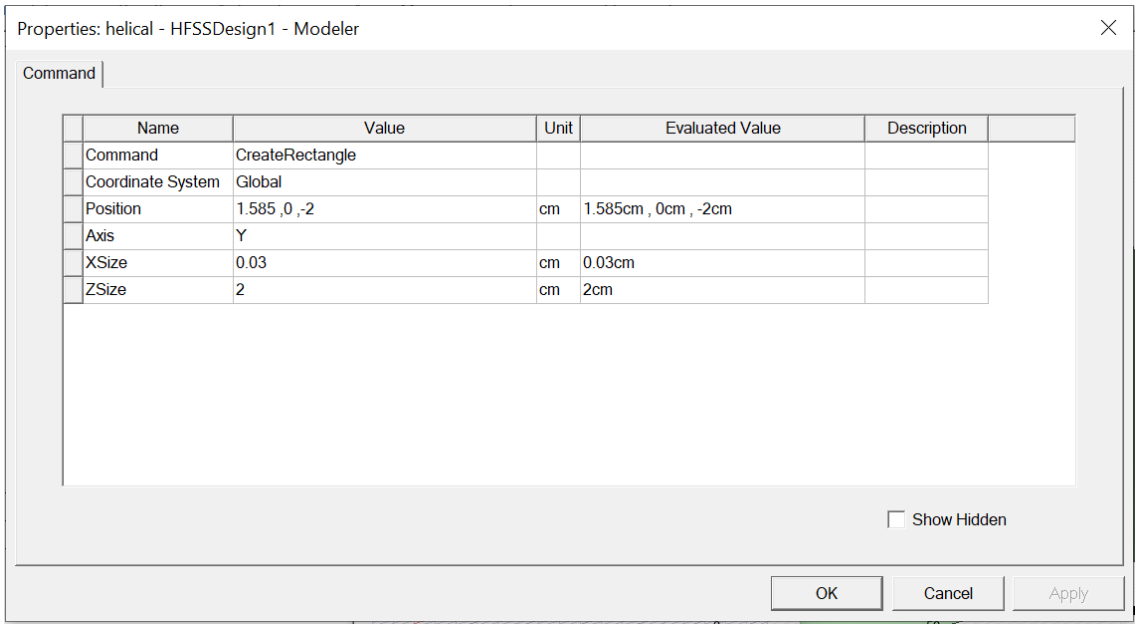
- Draw a circle(ground plane) on the XY plane.
- Double click on its create circle feature and change it properties as follows,



- Click Ok
- Right click on circle2 on model pane ->assign boundary ->perfect E

- Change the viewing plane back to XZ.
- Draw a rectangle.

- Double click on create rectangle feature and change its properties as follows,

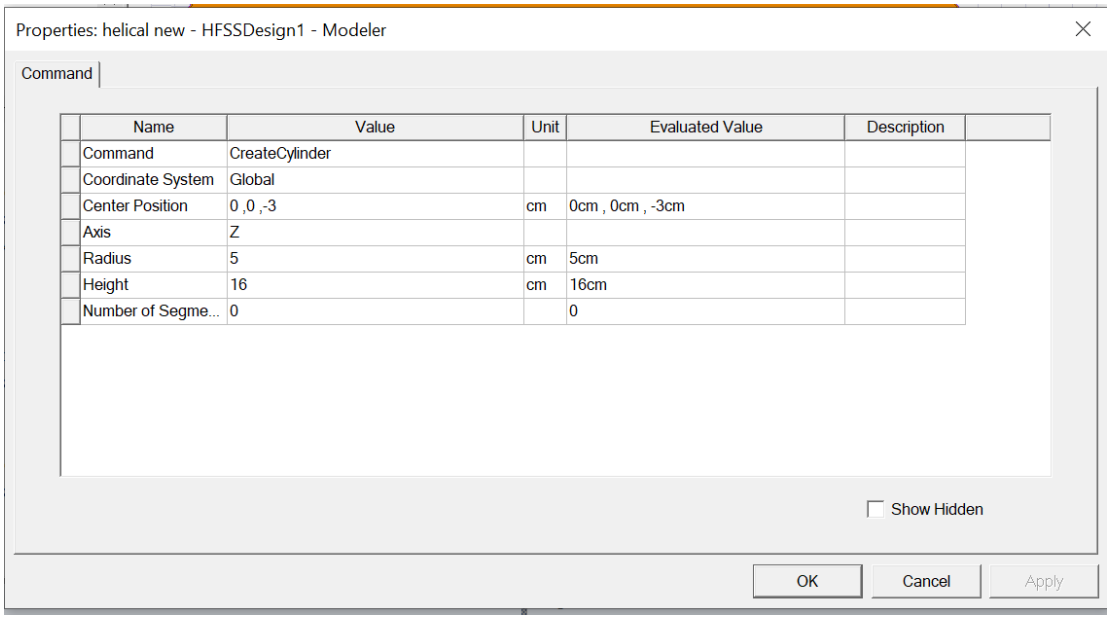


- Click Ok

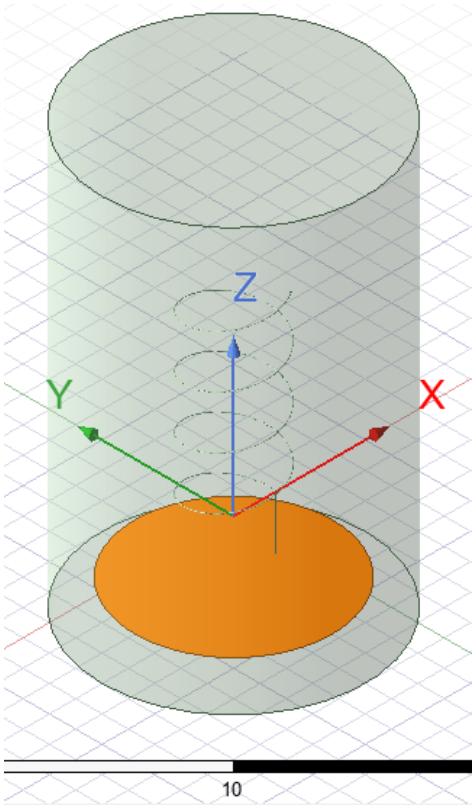
- Right click on rectangle1 in model section, assign excitation, port, lumped port, resistance 50 Ohms, next, click on dropdown under integration line and select choose new line
- Draw a line from bottom center to the top center of the rectangle, next, finish

- change the viewing plane back to XY.

- Draw a cylinder (radiation cylinder) and double click on cylinder 1 to change its properties as follows,



- right click on cylinder1 in the model pane ->assign boundary ->radiation.
- Now your model will look like the figure below with helix structure in the middle, circular ground plane in the bottom (assigning yellow colour to ground plane is not mandatory) and cylindrical radiation measuring surface enclosing the antenna.

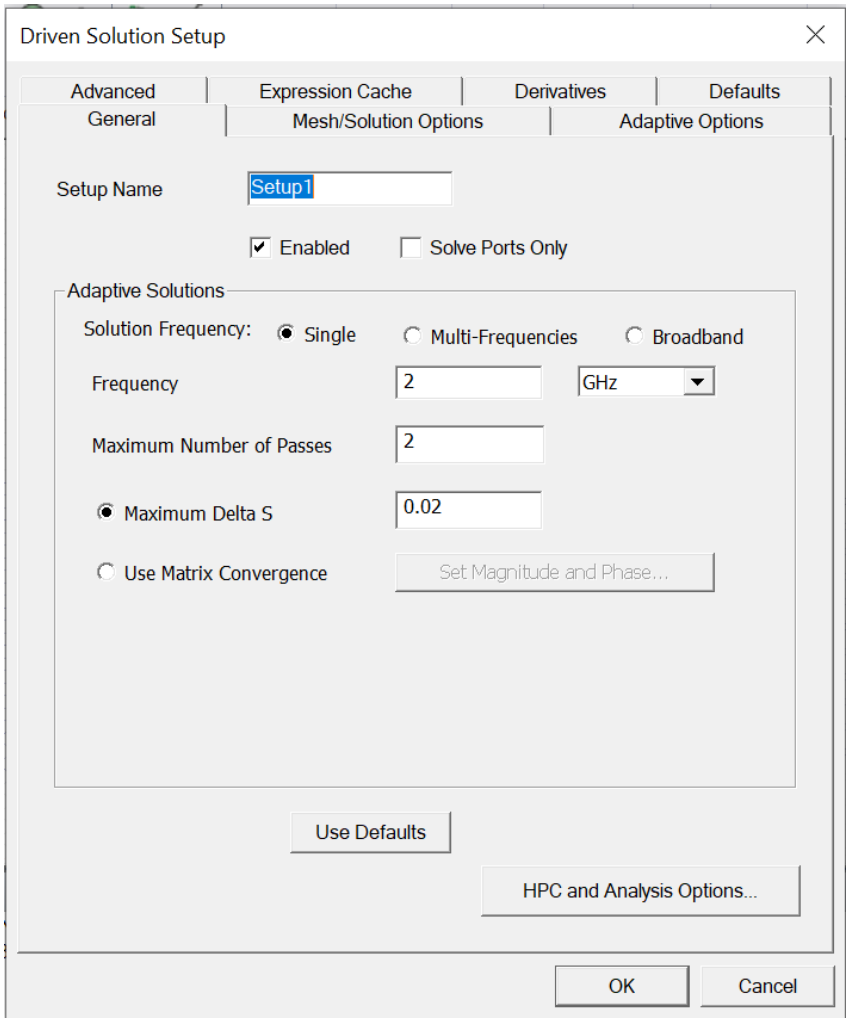


- Now your model is constructed along with radiation boundary.

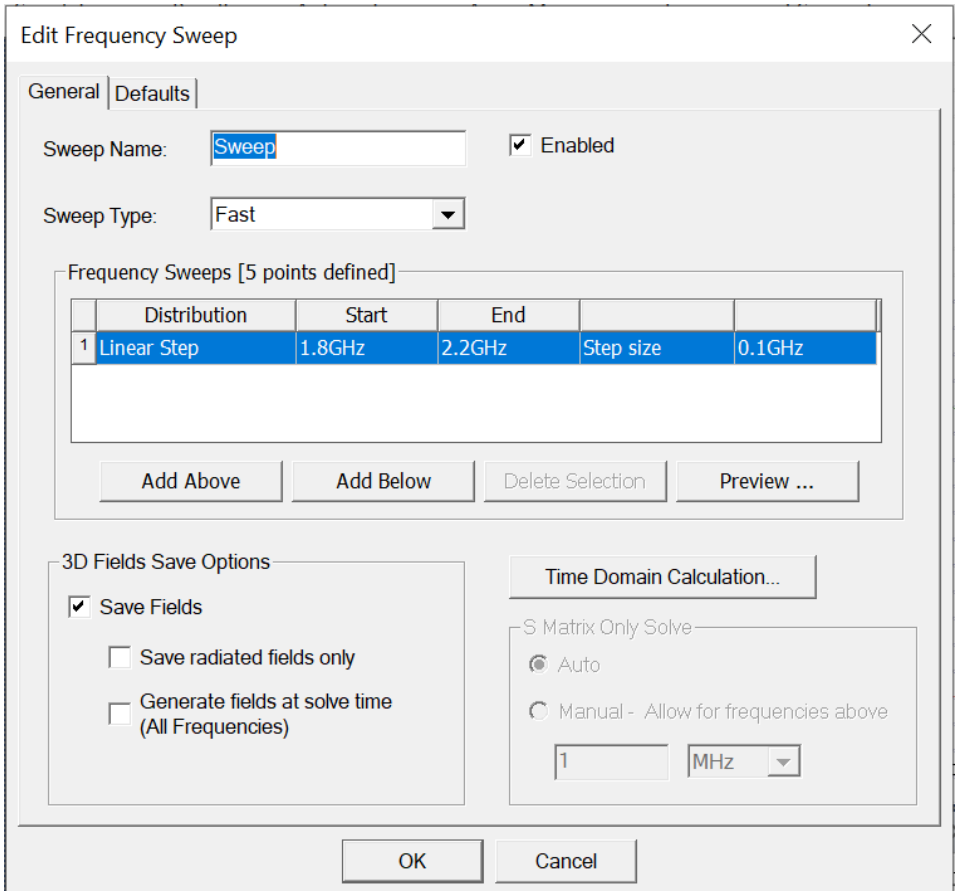
Step 3

Configurations (Settings)

- In the project manager section, Right click on analysis, add solution setup, advanced
- Enter the values as given below

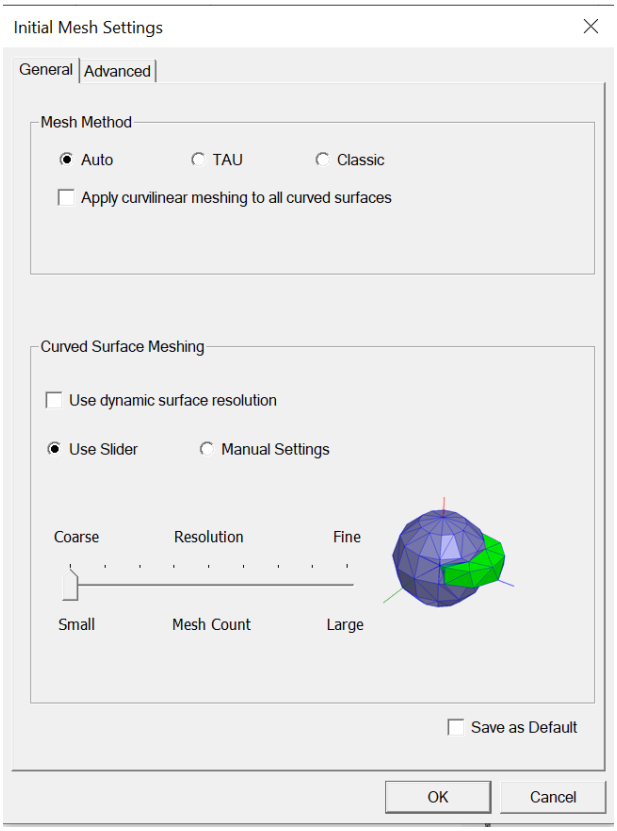


- Click on ok
- After clicking ok, in the next window that appears enter the values as given below



- Click ok

- Now right click on mesh in project manager window and select initial mesh settings
- Set the resolution to small

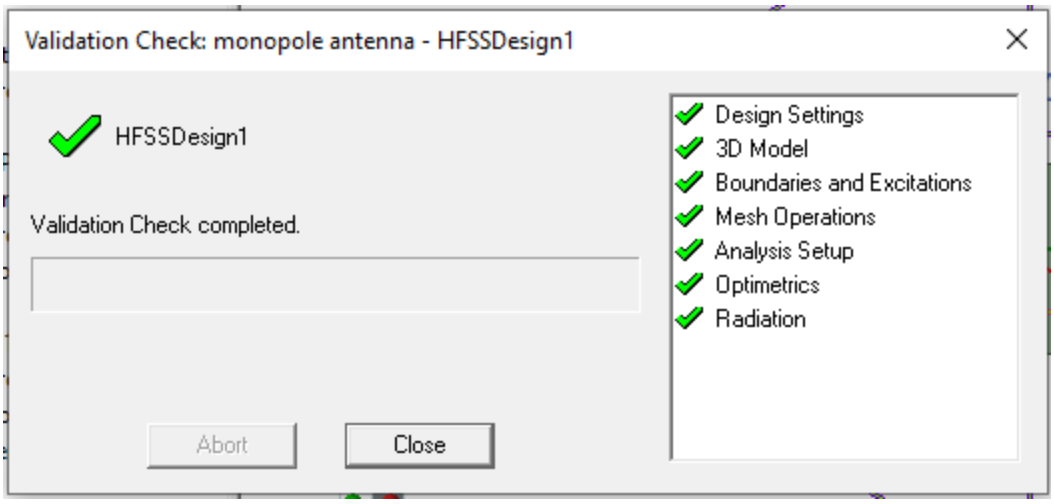


Click Ok

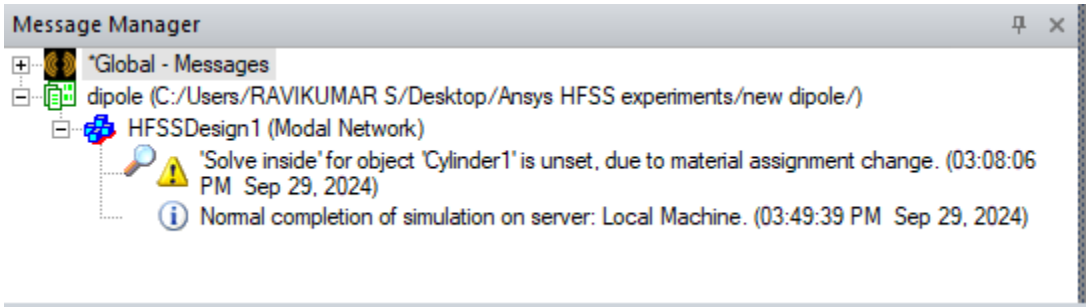
Step 4

Validation and Analysis

- Go to HFSS, Validation check
- The following window will appear if model created is correct



- Close the validation window
- Go to HFSS, Analyze all
- It will take some time to calculate results based on antenna complexity and computer configuration which is running the software
- If simulation goes well then you will receive a message in the message manager section as Normal completion of simulation on server:Local machine as given in figure below.

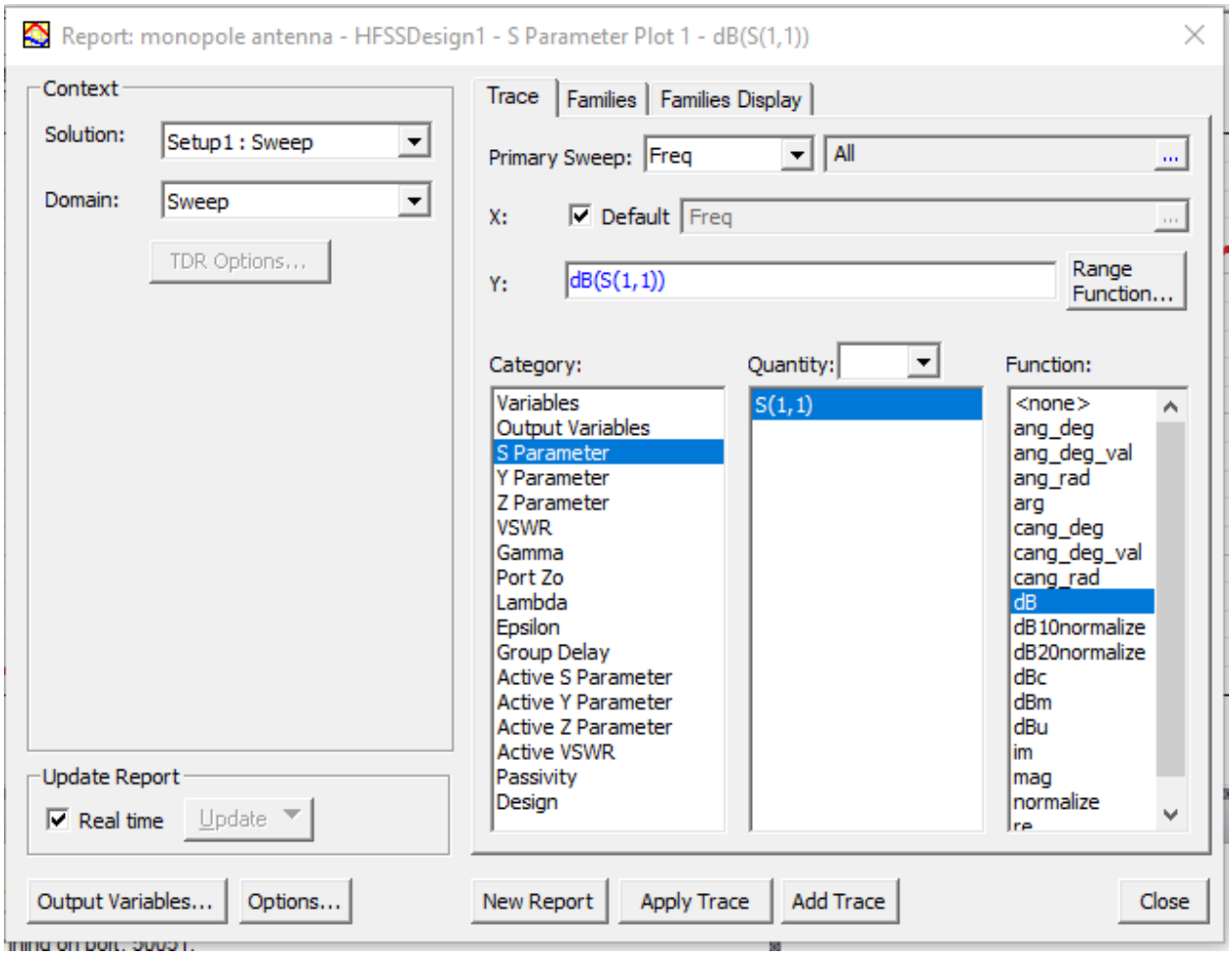


- If any errors are there then such error will be shown in message manager section. They have to be resolved.

Step 5

Results

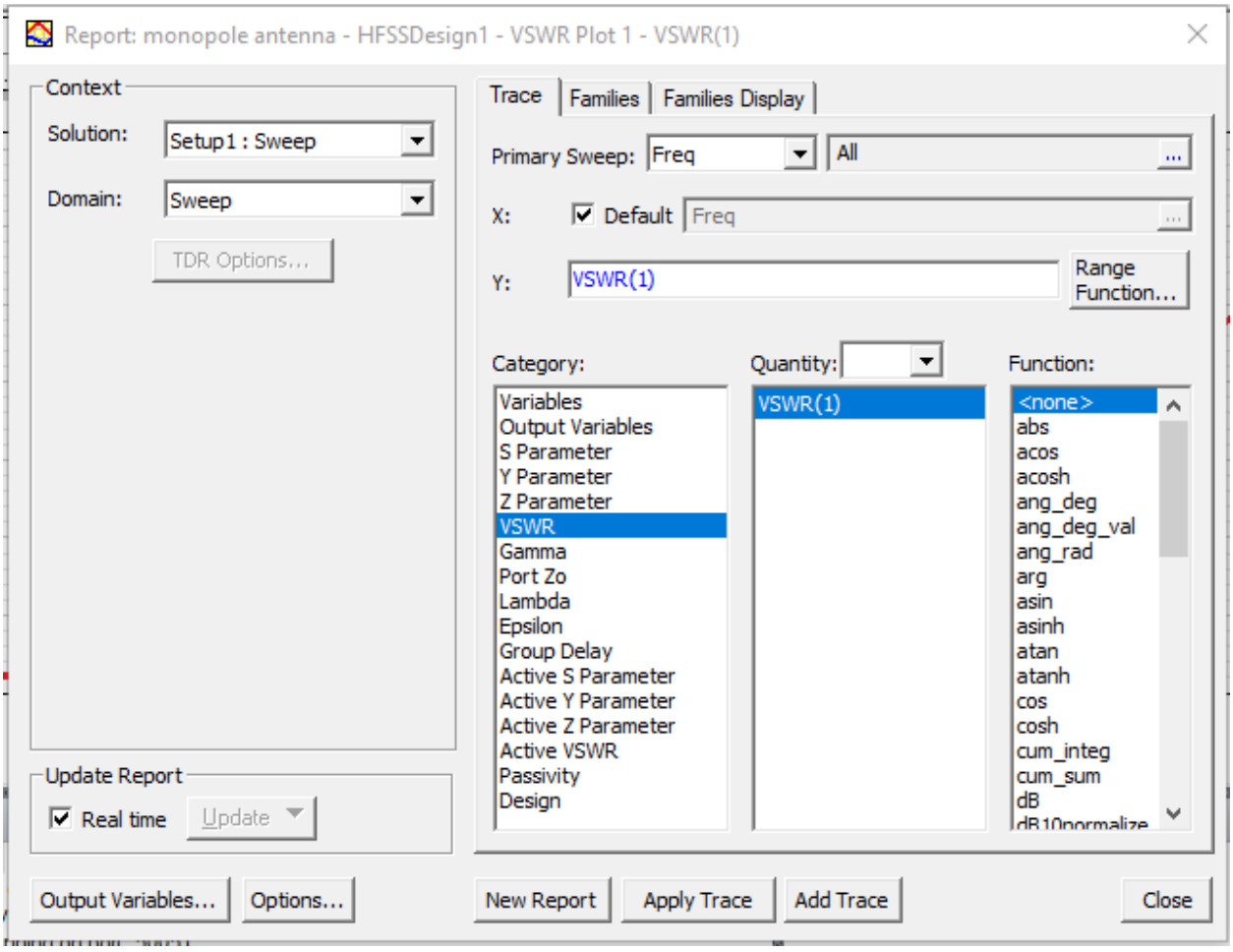
- Right click on results in project manager section, create modal solution data report, rectangular plot,
- The following window should appear



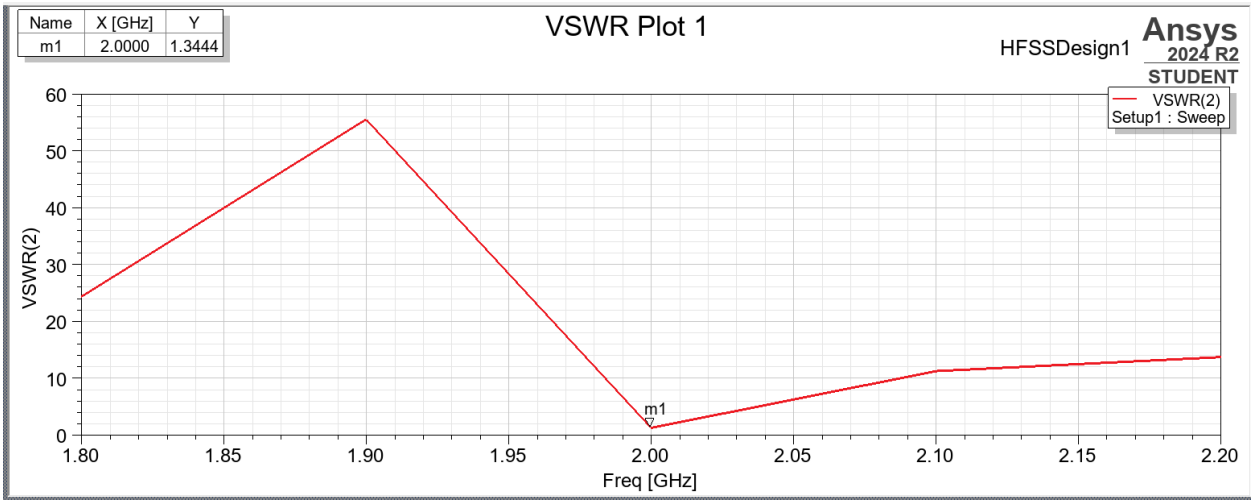
- Click on new report
- The following window will appear showing **S11 parameter (Input reflection coefficient)**



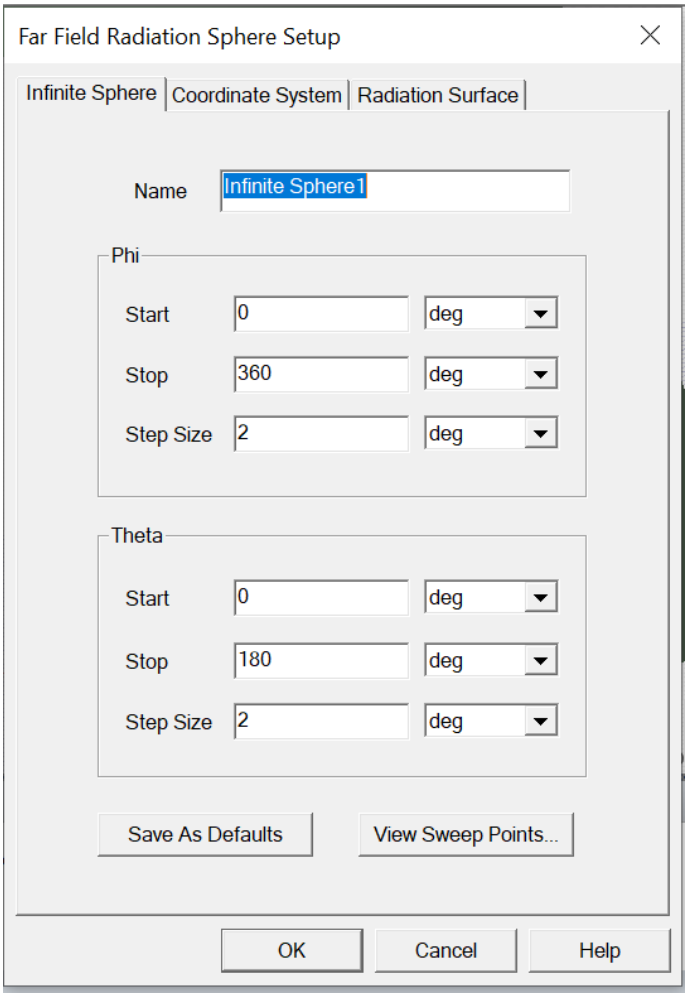
- In the same way Right click on results in project manager section, create modal solution data report, rectangular plot, select VSWR in category section, click on new report as given below



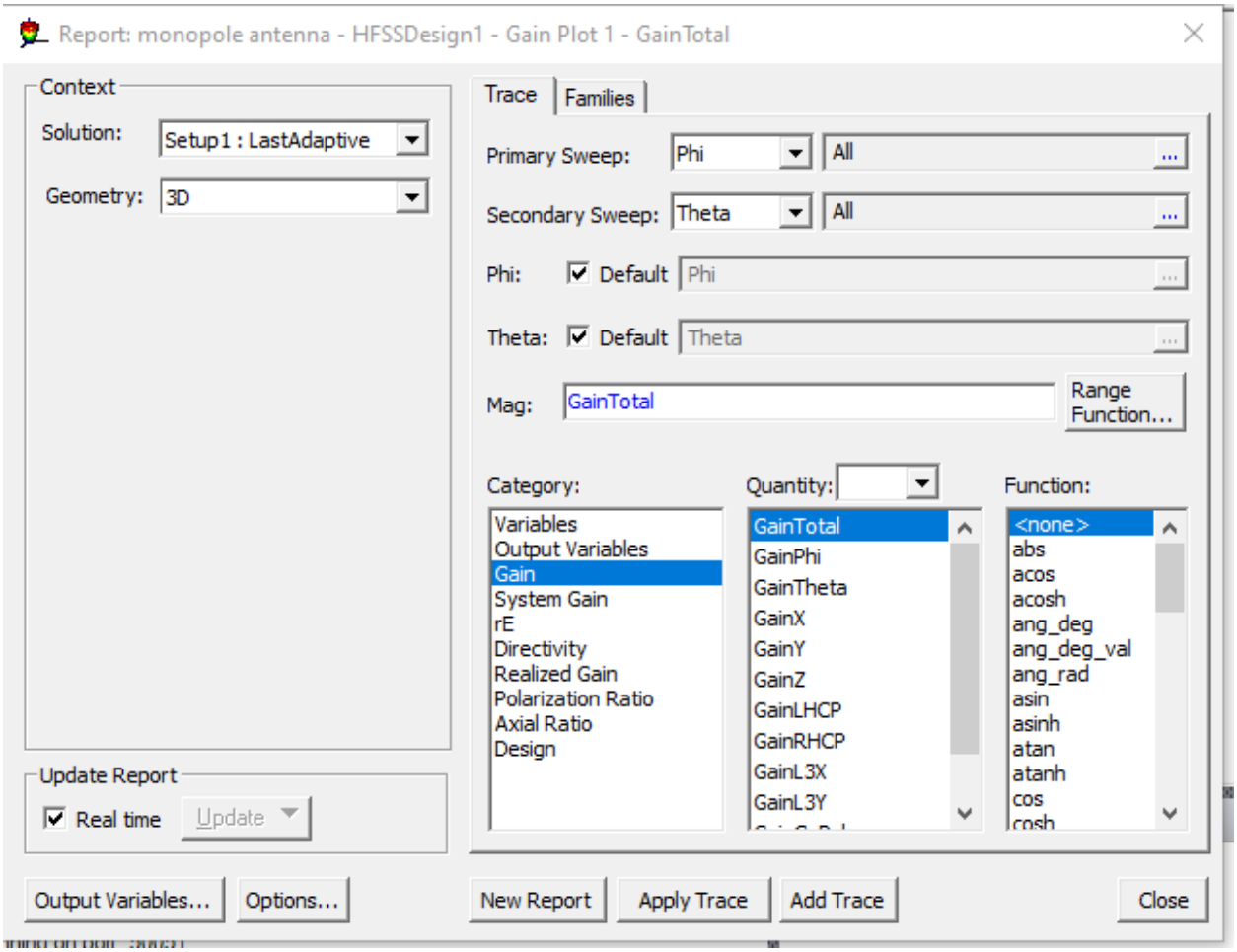
- The **VSWR** result will appear as given below,



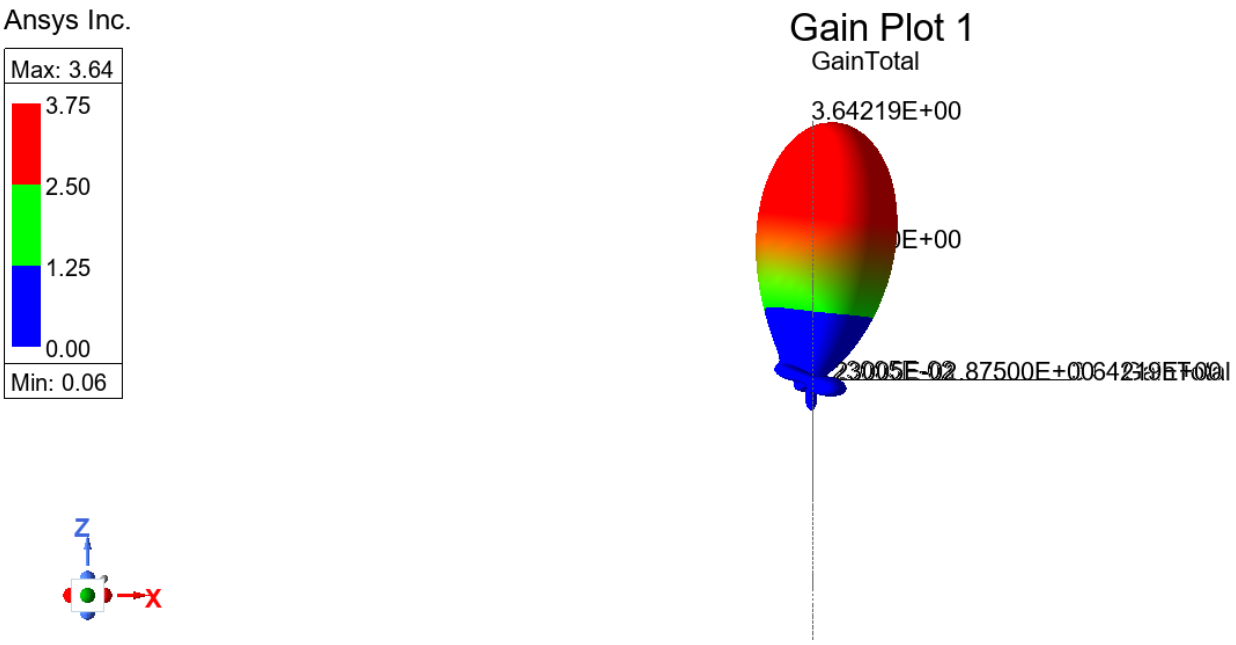
- Right click on radiation in project manager window, insert far field setup, infinite sphere
- Enter the values as given below



- Right click on results, create far fields report, 3D polar plot
- The following window will open



- Click on new report
- The following 3D radiation pattern will appear,



Tabular Column

Sl No	Sl No	Value
1	S11(in dB) at 2GHz	
2	VSWR at 2GHz	
3	Bandwidth in Hz (at -10dB points of S11)	
4	Gain	

Assignment

Assignment

Here are some fun facts about electromagnetic waves:

- **Transverse waves**

  All electromagnetic waves are transverse waves, meaning the electric and magnetic fields associated with the wave fluctuate in a direction perpendicular to the direction of the wave's propagation.

- **Perpendicular electric and magnetic fields**

  The electric and magnetic fields associated with electromagnetic waves are perpendicular to each other, forming a 90-degree angle.

- **Travel at the speed of light**

  All types of electromagnetic waves travel at the speed of light, which is the fastest speed known in the universe.

- **Have crests and troughs**

  Electromagnetic waves have crests and troughs, similar to ocean waves. The distance between the crests or troughs is called the wavelength.

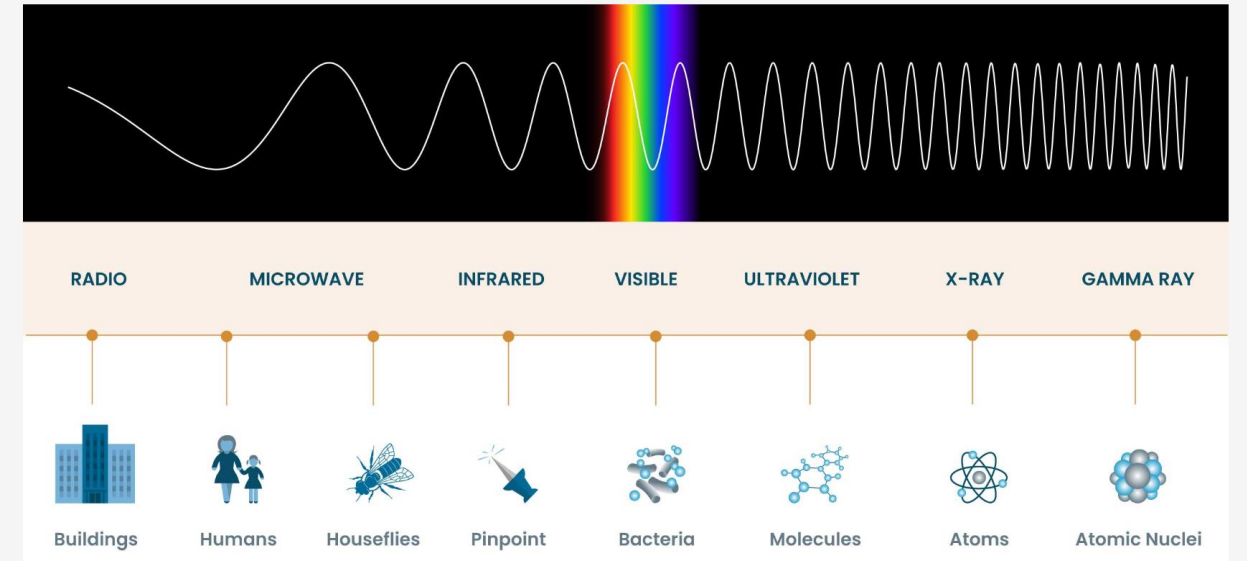
- **Travel through empty space**

  Electromagnetic waves can travel through empty space because they are waves of pure energy and don't require a medium to travel through.

- **Vary in size**

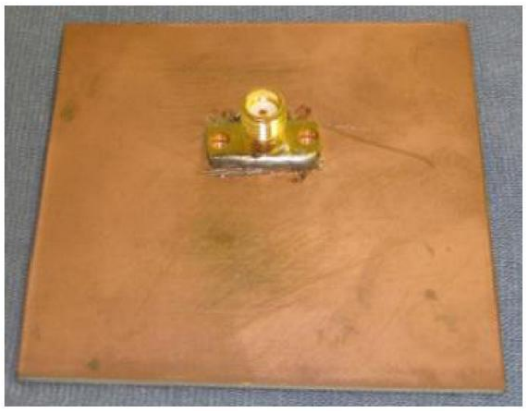
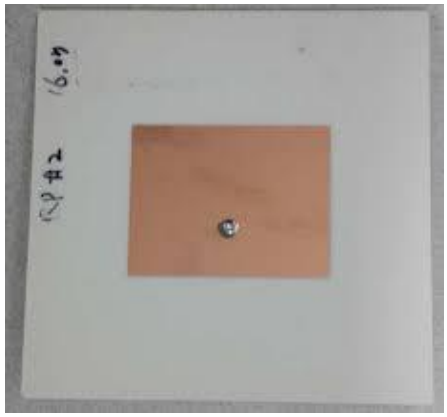
  Electromagnetic waves vary in size from very long radio waves to very short gamma-rays. The size of the wave is related to its energy, with smaller wavelengths having higher energy.

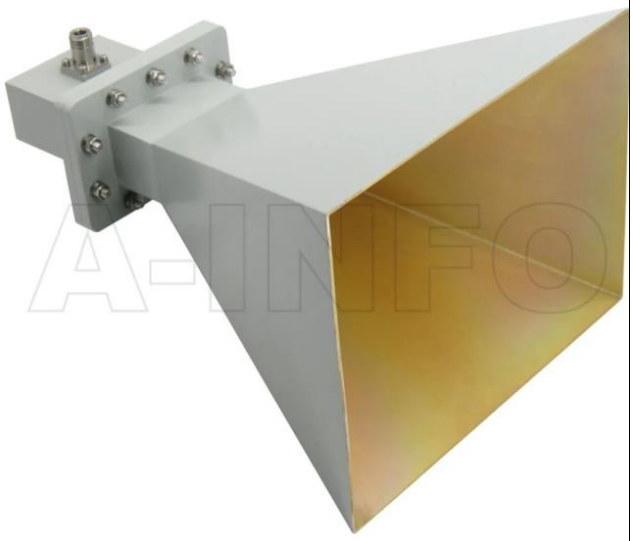
- **Polarized**
  Electromagnetic waves are polarized, and you can see this with a portable AM radio.



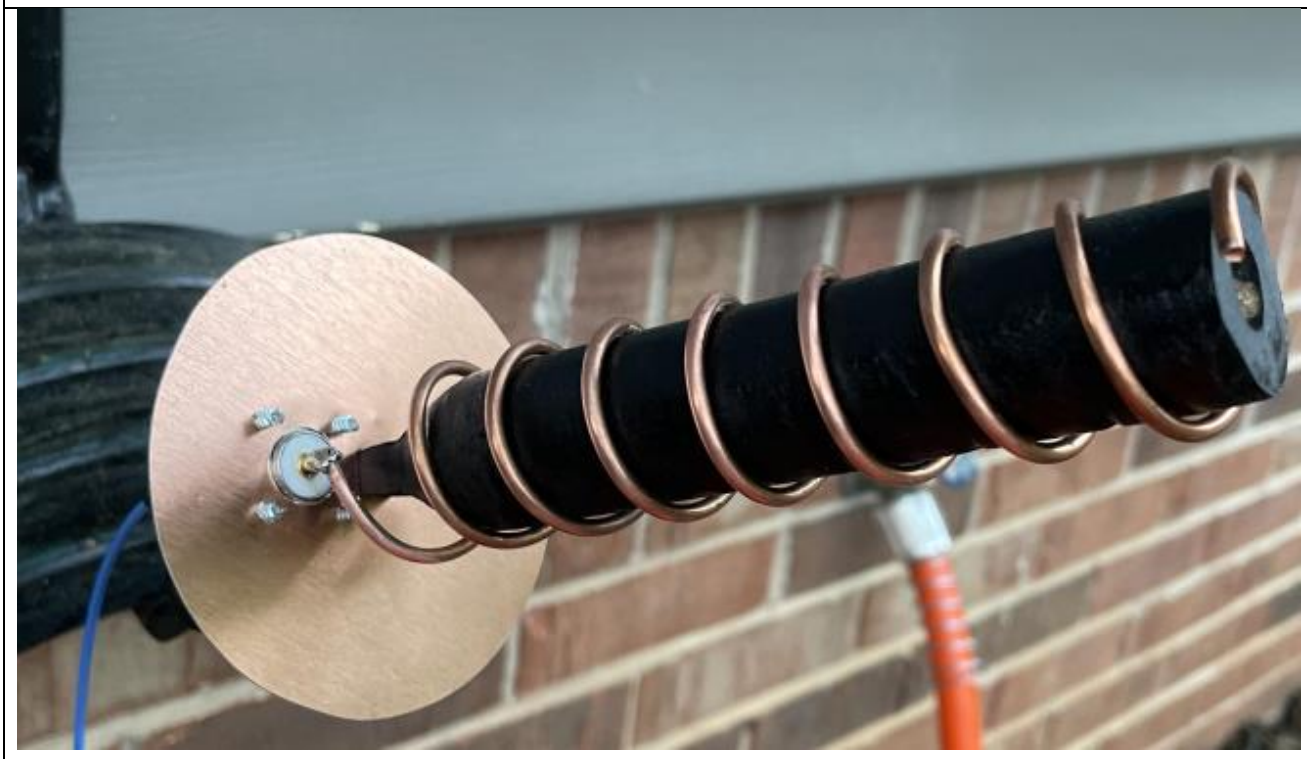
Comparison of different types of light, including wavelength size, and frequency.

Antennas Images

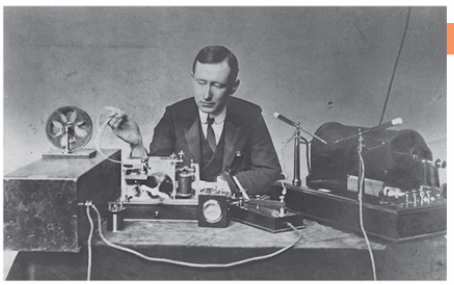
Dipole Antenna	Monopole Antenna
	
Insetfed Patch Antenna	Probefed Patch Antenna (Top and Bottom view)
	 

Yagi-Uda antenna	Horn Antenna
	

Helical Antenna



The history of wireless technology



19th century
Radio telegraphy enables the transmission of data over radio waves. (Photo of Gugliemo Marconi, who sent the world's first radio message across open water on May 13, 1897)

1963
Institute of Electrical and Electronics Engineers (IEEE) forms

1971
ALOHAnet uses ultra-high frequency signals in the 400 MHz band to broadcast data packets

1985
FCC opens the 900 MHz, 2.4 GHz and 5 GHz bands

1991
Cellular shifts from analog first-generation cellular to 2G digital services

1997
IEEE codifies the first 802.11 wireless standard, which supports speeds of 2 Mbps

1998
3rd Generation Partnership Project (3GPP) forms to create specifications for 3G

1999
- 802.11a offers up to 54 Mbps in the 5 GHz band
- 802.11b offers 11 Mbps in the 2.4 GHz band
- Vendors form Wireless Ethernet Compatibility Alliance (WECA)

Early 2000s
3G use increases with the popularity of smartphones

2002
WECA changes its name to Wi-Fi Alliance

2003
802.11g introduces speeds of 22 Mbps in 2.4 GHz band

2009
802.11n uses both the 2.4 GHz and 5 GHz bands and introduces multiple input, multiple output (MIMO)

2009 to early 2010s
4G and 4G LTE offer high cellular speeds and grow more popular

2013
802.11ac brings speeds up to 1 Gbps and introduces multi-user MIMO

2017
3GPP approves its Release 15 package, which introduces 5G



2019
802.11ax boosts efficiency and supports more devices

2024
IEEE expects to release 802.11be, which will operate in 2.4 GHz, 5 GHz and 6 GHz bands

2030
6G expected to launch

MARCONI PHOTO: PUBLIC DOMAIN/WIKIMEDIA COMMONS; 6G PHOTO: TANIT BOONRUEN/GETTY IMAGES

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VIVA Questions on Electromagnetics

1. Define Electromagnetics?
2. What is the difference between the scalar and vector?
3. Define scalar or dot product of vector.
4. Define vector or cross product of vectors.
5. How two vectors behave if their dot product is zero?
6. How two vectors behave if their cross product is zero?
7. State Divergence theorem.
8. Define Curl of a vector.
9. State Stokes theorem.
10. What is the use of divergence and curl operations?
11. State coulomb's law.
12. State Gauss's law for electric field.
13. In case of co-axial cable, what is the value of the electric field at the core of the cable and outside the conductor?
14. What will be the work done in moving the charge around any closed loop?
15. What do you mean by magnetostatics?
16. Which thumb rule is used to determine the direction of magnetic field around a conductor carrying a direct current?
17. State Biot-Savart law?
18. State Ampere's Circuital law.

VIVA Questions on Antennas

1. What is the difference between Directivity and Antenna Gain?
2. What are Maxwell's Equations?
3. What are the far field conditions for an antenna?
4. What is VSWR?
5. What is S11?
6. What is the difference between a monopole and dipole antenna?
7. What is structure Yagi-Uda Antenna?
8. What is a Phased Array Antenna?
9. What is Half Power Beam Width (HPBW)?
10. What is First Null Beam Width (FNBW)?
11. What is radiation pattern?
12. What is polarization?
13. Define radiation resistance.
14. What is radiation intensity?
15. Why array antennas are required?
16. What are the applications of dipole antenna?
17. What are the applications of monopole antenna?
18. What are the applications of microstrip patch antenna?
19. What are the applications of helical antenna?
20. What are the applications of yagi-uda antenna?